### F. Hemolytic Indicators

**293.** Which of these laboratory findings is associated with extravascular hemolysis?

**a** decreased bilirubin
**b** spherocytes
**c** microcytes
**d** hemoglobinuria

**294.** One of the major differences between extravascular and intravascular hemolysis is:

**a** free hemoglobin
**b** increased bilirubin
**c** decreased haptoglobin
**d** increased LDH

### G. Special Stains

**295.** A 56-year-old man is admitted to the hospital for treatment of a bleeding ulcer. The laboratory data shown in this table are obtained:

| test | result |
|---|---|
| RBC | $4.2 \times 10^6$/µL ($4.2 \times 10^{12}$/L) |
| WBC | $5.0 \times 10^3$/µL ($5.0 \times 10^6$/L) |
| HCT | 30% |
| HGB | 8.5 g/dL (85 g/L) |
| serum iron | 40 µg/dL (7.2 µmol/L) |
| TIBC | 460 µg/dL (82.3 µmol/L) |
| serum ferritin | 12 ng/mL (12 µg/L) |

Examination of the bone marrow reveals the absence of iron stores. This data is most consistent with which of these conditions?

**a** iron deficiency anemia
**b** anemia of chronic inflammation
**c** hemochromatosis
**d** acute blood loss


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**296.** A 40-year-old Caucasian male is admitted to the hospital for treatment of anemia, and presents with symptoms of lassitude, weight loss, and loss of libido. Admission laboratory data are shown in the table:

| test | result |
|---|---|
| WBC | $5.8 \times 10^3$/µL ($5.8 \times 10^9$/L) |
| RBC | $3.7 \times 10^6$/µL ($3.7 \times 10^{12}$/L) |
| HGB | 10.0 g/dL (100 g/L) |
| HCT | 32% |
| MCV | 86 $\mu m^3$ (86 fL) |
| MCH | 26 pg |
| MCHC | 32% |
| serum iron | 220 µg/dL (39.4 µmol/L) |
| TIBC | 300 µg/dL (53.7 µmol/L) |
| serum ferritin | 2800 ng/mL (2800 µg/L) |

Examination of the bone marrow reveals erythroid hyperplasia with a shift to the left of erythroid precursors. Prussian blue staining reveals markedly elevated iron stores noted with occasional sideroblasts seen. This data is most consistent with which of the following conditions?

**a** iron deficiency anemia
**b** anemia of chronic inflammation
**c** hemochromatosis
**d** acute blood loss

**297.** A screening procedure for detecting hemoglobin F is the:

**a** fluorescent spot test
**b** dithionite solubility test
**c** Kleihauer-Betke test
**d** heat instability test

**298.** The Prussian blue staining of peripheral blood identifies:

**a** Howell-Jolly bodies
**b** siderotic granules
**c** reticulocytes
**d** basophilic stippling

**299.** Which of the following stains is used to demonstrate iron, ferritin and hemosiderin?

**a** myeloperoxidase
**b** methylene blue
**c** specific esterase
**d** Prussian blue

**300.** Which of the following stains can be used to differentiate siderotic granules (Pappenheimer bodies) from basophilic stippling?

**a** Wright
**b** Prussian blue
**c** crystal violet
**d** myeloperoxidase

301. Which cell type shows the most intense staining with myeloperoxidase?

a neutrophil
b basophil
c lymphocyte
d monocyte

302. Which substrate is used for the detection of specific esterase?

a acetate
b chloroacetate
c pararosanilin acetate
d phenylene diacetate

303. A myeloperoxidase stain and immunophenotyping are performed on bone marrow smears from a patient diagnosed with an acute leukemia. The myeloperoxidase is positive and immunophenotyping shows positivity for CD4, CD13, CD11b, CD13, CD33 and CD64. What type of leukemia is indicated?

a lymphocytic
b myelogenous
c myelomonocytic
d erythroleukemia

304. Which of the following stains is closely associated with the lysosomal enzyme in primary (azurophilic) granules?

a myeloperoxidase
b nonspecific esterase
c methylene blue
d Prussian blue

### *H. Other Studies*

305. Heinz bodies are:

a readily identified with polychrome stains
b remnants of RNA
c closely associated with spherocytes
d denatured hemoglobin inclusions

306. Which of the following technical factors will cause a normal (low) erythrocyte sedimentation rate?

a gross hemolysis
b small fibrin clots in the sample
c increased room temperature
d tilting of the tube

307. The erythrocyte sedimentation rate (ESR) can be falsely elevated by:

a tilting the tube
b refrigerated blood
c air bubbles in the column
d specimen being too old

### *I. Flow Cytometry Immunophenotyping*

308. A confirmatory test for paroxysmal nocturnal hemoglobinuria is:

a heat instability test
b acid elution
c flow cytometric immunophenotyping
d dithionite solubility

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309. Which of the following test results are consistent with a diagnosis of paroxysmal nocturnal hemoglobinuria (PNH)? MLS ONLY

a decreased conversion of NADH to NAD
b increased production of globulin
c decreased hemolysis in acidified serum
d diminished CD55 on hematopoietic cells

310. Terminal deoxynucleotidyl transferase (TdT) is a marker found on:

a hairy cells
b myeloblasts
c monoblasts
d lymphoblasts

311. In which of the following types of cells is CD5 usually expressed?

a mature T cells
b pro B cells
c macrophages
d endothelial cells

312. Bone marrow examination reveals a hypercellular marrow consisting of probable lymphoblasts. The cells are positive for TdT, CD3 and CD7; however, the lymphoblasts are negative for surface immunoglobulins, CD19, and CD10 (CALLA). The most likely diagnosis is:

a large granular lymphocytosis
b chronic lymphocytic leukemia (CLL)
c T-cell leukemia (T-ALL)
d hairy cell leukemia

313. A cell surface marker that is expressed on neoplastic plasma cells and is helpful in the diagnosis of myeloma is:

a CD7
b CD19
c CD44
d CD138

314. Which of the following markers are usually negative in hairy cell leukemia?

a CD5
b CD11c
c CD25
d CD103

315. Which of the following markers is expressed in most cases of AML?

a CD2
b CD10
c CD11b
d CD117

316. What feature do CD markers define in a cell?

a surface membrane characteristics
b size
c cytoplasmic characteristics
d cell lineage

*J. Molecular and Cytogenetic Testing*

**317.** Which of the following conditions is most often associated with the V617F mutation of *JAK2*?

**a** chronic myelogenous leukemia
**b** acute monocytic leukemia
**c** MDS with multilineage dysplasia
**d** polycythemia vera

**318.** MLS ONLY The t(8;14) chromosomal translocation brings which of the following 2 genes in close proximity?

**a** core binding factor alpha and the retinoic acid receptor
**b** the Abelson tyrosine kinase and breakpoint cluster region
**c** *c-myc* and the immunoglobulin heavy chain
**d** core binding factor beta and the myosin heavy chain

**319.** MLS ONLY Which condition is associated with the *BCR/ABL* gene rearrangement and responds well to treatment with tyrosine kinase inhibitors?

**a** chronic myeloid leukemia
**b** essential thrombocytosis
**c** chronic idiopathic myelofibrosis
**d** polycythemia vera

**320.** This AML with recurrent cytogenetic abnormality often presents with disseminated intravascular coagulation:

**a** inv(3)
**b** t(6;9)
**c** t(8;21)
**d** t(15;17)

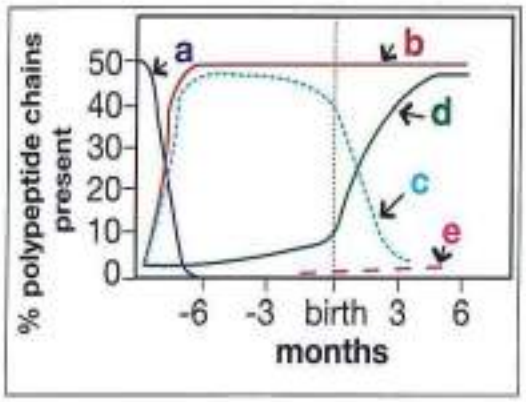

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## I. Physiology

### A. Production

1. **a** 2/3 of iron in the body is bound to hemoglobin. Myoglobin also has iron, but in lesser amounts.
[QCMLS 2021: 2.10.2.1.1]

***Please consult the diagram below for questions 2-4 regarding hemoglobin curves.***



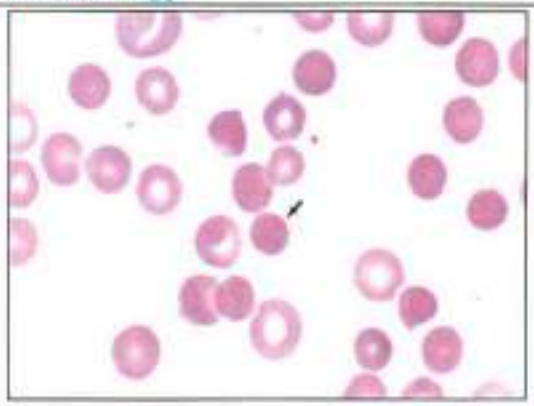

2. **b** Alpha chains, a constituent of embryonic Gower 2 hemoglobin, fetal hemoglobin, and adult hemoglobins A and A2, are synthesized early in gestation and maintained throughout embryonic, fetal, and infant development.
[McKenzie, 2020, p95-96]

3. **d** Beta chains, a constituent of adult hemoglobin A, are synthesized at a low level until about the 3rd trimester of pregnancy. Beta chain production increases from approximately gestational week 30.
[McKenzie 2020, p96]

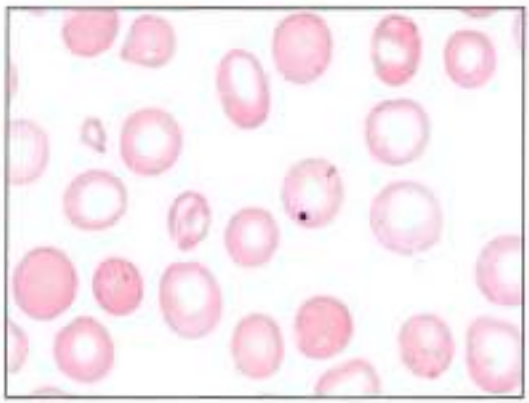
4. **c** Gamma chains, a constituent of fetal hemoglobin, are synthesized during liver and bone marrow erythropoiesis in the fetus, with production decreasing after birth.
[McKenzie, 2020, p95]

5. **d** The functional form of iron is the reduced state.
[QCMLS 2021: 4.4.1]

6. **c** The Donath-Landsteiner antibody is a biphasic, complement-fixing IgG antibody.
[QCMLS 2021: 1.4.2.4.1]



7. **a** This cell is a spherocyte and is most often associated with a deficiency or defect of membrane spectrin, though other proteins (band 3, ankyrin) may also be deficient or defective.
[QCMLS 2021: 4.7.1.3.1.1]



8. **a** Howell-Jolly bodies represent nuclear fragments remaining within the RBC cytoplasm after the nucleus has been extruded.
[QCMLS 2021: 4.10.4.2, t4.18]

9. **b** Methemoglobin results when iron is in the ferric ($Fe^{+3}$) state.
[QCMLS 2021: 4.4.1]

10. **b** An increased amount of cytoplasmic basophilia in a blood cell indicates decreased cytoplasmic maturation.
[QCMLS 2021: 4.10.2.1] [Glassy 2018, p184]

11. **a** Secondary granules first appear at the myelocyte stage, though primary granules are still retained.
[QCMLS 2021: 4.10.2.1, t4.16]

12. **d** Normal bone marrow has more metamyelocytes than myeloblasts, with even fewer numbers of basophils and eosinophils.
[Gulati 2021, p4 t2.1]

13. **d** Pluripotential hematopoietic stem cells are capable of producing cells of more than one lineage, specifically both lymphoid and myeloid.
[QCMLS 2021: 4.1.1]

14. **d** IL-5 released by T helper cells, mast cells, eosinophils and other lymphocytes has lineage specificity for eosinophils and is the major cytokine required for production and differentiation of eosinophils.
[QCMLS 2021: 4.1.4.2]



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15. **a** Auer rods are pinkish or reddish inclusions, often rod-shaped, in the cytoplasm of malignant myeloblasts. They originate from fused primary granules.
[McKenzie 2020, p506, 1161]

16. **c** Nuclear maturation and division occurs first and is generally complete before cytoplasmic maturation begins.
[McKenzie 2020, p166]

17. **a** Megakaryocytes are typically the largest hematopoietic cell in the bone marrow.
[Gulati 2021, p72-73]

18. **a** An average of 5-10 megakaryocytes are normally found in each 10x (low power field).
[McKenzie 2020, p950]

### *B. Destruction*

19. **b** Normal degradation products of red blood cells include iron, heme, and globin.
[QCMLS 2021: 4.6.1]

### *C. Function*

20. **b** The hexose monophosphate shunt prevents hemoglobin degradation by producing reduced glutathione.
[QCMLS 2021: 4.3.3]

21. **d** The major function of the spleen is to remove defective and aging cells, especially RBCs, in the process called culling.
[QCMLS 2021: 4.2]

22. **a** The major function of red blood cells is the passive transport of gases.
[QCMLS 2021: 4.3.2]

23. **b** Erythropoietin is a hormone that specifically targets the synthesis of red blood cells.
[QCMLS 2021: 4.1.3]

24. **c** Lymphocytes are a diverse type of cell with several functions. B cells produce antibodies, while NK cells can directly lyse other cell types.
[QCMLS 2021: 2.5.2]

25. **b** Phagocytosis is performed by neutrophils, a type of granulocyte.
[QCMLS 2021: 4.5.1]

26. **b** Basophils function as mediators of immediate type hypersensitivity (type 1) reactions.
[QCMLS 2021: 4.5.1]

27. **a** The acetylation of aspirin inactivates cyclooxygenase, which blocks thromboxane A2 production, which causes impaired platelet function.
[QCMLS 2021: 5.7.5.2.1, t5.13]

28. **a** In infants, the thymus is located above the heart and gives rise to lymphocytes. This gland is not active in adults.
[QCMLS 2021: 4.1.4.3]

29. **b** Abnormal hemoglobins include carboxyhemoglobin, methemoglobin, and sulfhemoglobin. Methemoglobin has iron in the +3 state and cannot bind oxygen.
[QCMLS 2021: 4.4.4]

30. **d** Phagocytosis is an essential function of defense for neutrophils. Bacteria can be recognized, ingested, degranulated and killed by this process.
[QCMLS 2021: 4.5.1]

## II. Disease States

### A. Erythrocytes

31. **d** Pernicious anemia is a type of vitamin $B_{12}$ deficiency resulting in impaired DNA synthesis and macrocytosis.
[QCMLS 2021: 4.7.1.2.1]

32. **d** RBC membrane defect increases susceptibility to complement mediated lysis.
[QCMLS 2021: 4.7.1.3.1.1]

33. **a** Hemochromatosis results in iron deposited in tissues.
[QCMLS 2021: 2.10.2.1.3]

34. **a** Iron is lost during therapeutic phlebotomy and, although this slows RBC production, iron deficiency could also develop.
[McKenzie 2020, p538]

35. **a** The direct antiglobulin test (DAT) is useful to distinguish inherited spherocytosis (a negative result) from acquired causes of spherocytosis resulting from immune mechanisms (positive DAT).
[QCMLS 2021: 4.7.1.3.1.1]

36. **b** Hepcidin causes iron to be trapped in macrophages and unavailable to developing RBCs, subsequently decreasing iron levels.
[QCMLS 2021: 4.7.1.1.4]

37. **b** Erythropoietin production is decreased and results in decreased production of RBCs.
[QCMLS 2021: 4.7.1.3.8.1]

38. **b** Aplastic crisis is a severe complication of sickle cell disease, causing a temporary pause in RBC production, which dramatically worsens the anemia.
[McKenzie 2020, p274]

39. **b** Anemia results during uremia because of decreased erythropoietin and subsequent decreased production of RBCs.
[McKenzie 2020, p358]

40. **b** RBC membrane defects are common to these disorders.
[QCMLS 2021: 4.7.1.3.1.1]

41. **b** In anemia of chronic inflammation, iron is present, but it is trapped in macrophages and unavailable to developing RBCs. Therefore, iron levels can be decreased, but the ability to bind iron (TIBC) will also be decreased.
[QCMLS 2021: 4.7.1.1.4]

42. **c** Iron deficiency anemia is associated with decreased serum iron and storage iron with a greater capacity to bind iron, as indicated by an elevated TIBC.
[QCMLS 2021: 4.7.1.1.1, f4.3]

43. **d** Chronic blood loss frequently results in iron deficiency anemia.
[QCMLS 2021: 4.7.1.1.1]

44. **b** The mechanism of the genetic abnormality in thalassemia reduces globin chain production and does not affect the structure of the globin molecule or synthesis of heme.
[QCMLS 2021: 4.7.1.1.2]

45. **c** Heterozygous delta-beta thalassemia is usually associated with normal $HbA_2$ levels and slightly elevated HbF levels.
[McKenzie 2020, p309]

46. **c** Anemia related to uremia and chronic renal disease has normal-sized, normochromic cells; it is the number of RBCs that is decreased because of reduced erythropoietin levels.
[QCMLS 2021: 4.7.1.3.8]



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47. **b** Hemolytic anemia is associated with the breakdown of heme and lysis of RBCs, which have high levels of lactate dehydrogenase.
[QCMLS 2021: 4.10.9, 4.6.3]

48. **b** G6PD deficiency is associated with the precipitation of oxidized and denatured hemoglobin as Heinz bodies.
[QCMLS 2021: 4.7.1.3.1.2]

49. **d** G6PD-deficient RBCs are sensitive to oxidant stress induced by medications and several other drugs, fava beans, and infection.
[QCMLS 2021: 4.7.1.3.1.2]

50. **a** G6PD-deficient RBCs are sensitive to the anti-malarial drug primaquine and oxidized and denatured hemoglobin can precipitate as Heinz bodies.
[QCMLS 2021: 4.7.1.3.1.2]

51. **d** Autoimmune hemolytic anemias result from the abnormal production of autoantibodies, which will be detected as a characteristic positive DAT.
[QCMLS 2021: 4.7.1.3.3]

52. **a** The impaired DNA synthesis associated with pernicious anemia causes decreased production of all blood cells as well as abnormally large, macrocytic RBCs.
[QCMLS 2021: 4.7.1.2.1, t4.6]

53. **b** The impaired DNA synthesis associated with a vitamin $B_{12}$ deficiency causes decreased production of all blood cells as well as abnormally large, macrocytic RBCs.
[QCMLS 2021: 4.7.1.2]

54. **c** Deficiencies of vitamin $B_{12}$ and folate result in pancytopenia and macrocytosis.
[QCMLS 2021: 4.7.1.2, t4.6]

55. **a** In folate deficiency (megaloblastic anemia), the peripheral blood smear will characteristically show abnormally large RBCs.
[QCMLS 2021: 4.7.1.2.1, t4.6]

56. **d** Pernicious anemia results in pancytopenia.
[QCMLS 2021: 4.7.1.2.1, t4.6]

57. **b** Megaloblastic anemia is caused by impaired DNA synthesis and results in asynchronous development of blood cells in the bone marrow.
[QCMLS 2021: 4.7.1.2.1]

58. **a** Megaloblastic anemias result in pancytopenia, which includes neutropenia and thrombocytopenia.
[QCMLS 2021: 4.7.1.2.1, t4.6]

59. **d** In anemia of chronic inflammation, iron is present, but it is trapped in macrophages and unavailable to developing RBCs. Therefore, iron levels can be decreased, but the ability to bind iron (TIBC) will also be decreased. Transferrin saturation may also be low or normal.
[QCMLS 2021: 4.7.1.2.1, t4.4]

60. **d** Peripheral blood cell morphology in HbC disease characteristically includes target cells and hexagonal or rod-shaped crystals (hemoglobin C crystals).
[QCMLS 2021: 4.7.1.3.7.3]

61. **d** Thalassemias are quantitative decreases in the production of globin chains needed for hemoglobin synthesis.
[QCMLS 2021: 4.7.1.1.2]

62. **c** Thalassemias are characterized by increased RBC counts, and microcytic, hypochromic red blood cells.
[QCMLS 2021: 4.7.1.1.2]

63. **b** The band 3 reduction test (also called the eosin-5-maleimide binding test or EMA) is a more sensitive procedure for confirming hereditary spherocytosis.
[QCMLS 2021: 4.7.1.3.1.1]

64. **c** Burn patients can develop relative polycythemia that results from loss of body fluid and subsequent dehydration.
[QCMLS 2021: 4.7.2.2]

65. **c** Basophilic stippling is a classic laboratory finding in lead poisoning.
[QCMLS 2021: 4.7.1.1.3]

66. **c** Hypersegmented neutrophils are the most prominent WBC feature associated with pernicious anemia and megaloblastic anemias in general.
[QCMLS 2021: 4.7.1.2.1]

67. **d** RBC indices are helpful in diagnosing hereditary spherocytosis, as the MCV and MCH are usually normal and the MCHC is increased. Spherocytes are the only erythrocytes with an elevated MCHC.
[QCMLS 2021: 4.7.1.3.1.1]

68. **b** Hereditary elliptocytosis may be linked to defects in several different RBC membrane proteins including spectrin.
[QCMLS 2021: 4.7.1.3.1.1]

69. **a** Hereditary stomatocytosis includes a group of disorders in which the RBC membrane displays abnormalities in cation permeability.
[QCMLS 2021: 4.7.1.3.1.1]

70. **a** Sideroblastic anemias may be inherited or acquired and caused by either mutations in heme synthetic enzymes or impaired activity of heme. The enzymatic defects or impaired activity result in excess iron accumulation in mitochondria and developing RBCs.
[QCMLS 2021: 4.7.1.1.3]

71. **a** Fanconi anemia is an autosomal recessive disorder showing normocytic anemia. Physically, patients show short stature, microcephaly and hyperpigmentation. HbF is elevated and patient survival is limited.
[Ciesla 2017, p109]

72. **a** An inappropriate increase in erythropoietin has been associated with certain tumors that secrete erythropoietin or an erythropoietin-like substance. Tumors are often associated with renal disease.
[QCMLS 2021: 4.7.2.1]

73. **c** During or immediately following a hemolytic episode related to a G6PD deficiency, abnormalities in erythrocytes may include poikilocytosis, polychromasia, some spherocytes, and bite cells. Heinz bodies are only visible with a supravital stain.
[QCMLS 2021: 4.7.1.3.1.2]

74. **c** The peripheral blood smear in pyruvate kinase deficiency may show echinocytes and sometimes irregularly contracted cells.
[QCMLS 2021: 4.7.3.1.2]

75. **b** In hemoglobin C disease, hexagonal or rod-shaped crystals (hemoglobin C crystals) may be seen in RBCs, especially in patients who have received a splenectomy.
[QCMLS 2021: 4.7.1.3.7.3]

76. **b** HbD and HbG are abnormal variants that migrate at the same position as HbS on alkaline electrophoresis.
[QCMLS 2021: 4.10.6.2]

77. **c** HbLepore is a variant formed from the fusion of delta and beta genes that occurs from abnormal crossing-over during meiosis. The resulting fusion gene produces hybrid delta/beta globin chains that combine with 2 alpha chains to synthesize HbLepore.
[QCMLS 2021: 4.7.1.3.7.10]

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78. **a** The RBC count is increased in heterozygous beta thalassemia more than expected for the hemoglobin concentration.
[QCMLS 2021: 4.7.1.1.2]

79. **c** HPFH is caused by either the absence of beta and delta chain synthesis or a loss of suppression of the gamma globin gene through mutations of proteins inhibiting gene expression or in the gamma gene promoter region.
[QCMLS 2021: 4.7.1.3.7.9]

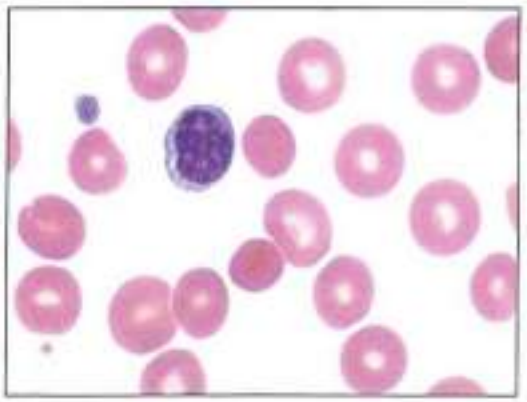
80. **d** The antibody produced in cold agglutinin disease is usually an autoantibody with anti-I specificity.
[QCMLS 2021: 4.7.1.3.3]

81. **b** Paroxysmal cold hemoglobinuria is most often seen in children with viral infections.
[QCMLS 2021: 4.7.1.3.3]

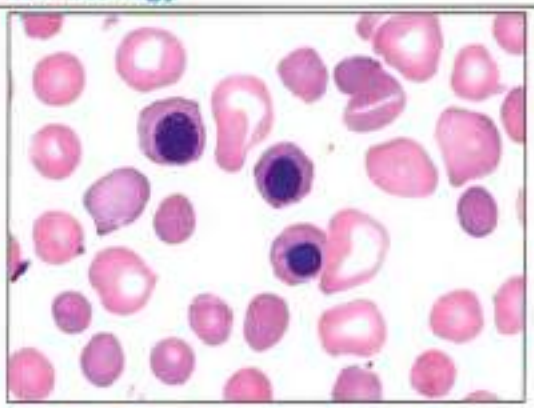
82. **d** Anemia of chronic inflammation (disease) is a common cause of anemia in hospitalized patients. It is associated with infections and inflammatory or malignant conditions.
[QCMLS 2021: 4.7.1.1.4]

83. **c** In 95-100% of polycythemia vera patients the *JAK2* V617 mutation is present.
[QCMLS 2021: 4.9.3.1.2, t4.11]

84. **a** The MCV and MCH are increased in megaloblastic anemia. Hypersegmented neutrophils are one of the distinguishing features of megaloblastic anemia. Anisocytosis (elevated RDW) is also present. Hypersegmented neutrophils may be seen before the megaloblastic anemia is fully developed.
[QCMLS 2021: 4.7.1.2.1]



85. **b** Nonmegaloblastic macrocytic anemias are characterized by erythrocytes that are round and macrocytic. This is in contrast to megaloblastic anemias in which macro-ovalocytes are present.
[QCMLS 2021: 4.7.1.2.2]



86. **d** Spherocytes and polychromasia are diagnostically important findings in RBCs in hemolytic disease of the newborn due to ABO incompatibility. Nucleated RBCs may also be seen in the peripheral blood of newborns.
[QCMLS 2021: 4.7.1.3.2]

87. **a** HbH disease occurs when 3 of 4 alpha genes are deleted.
[QCMLS 2021: 4.7.1.1.2]

88. **a** Granulopoiesis and erythropoiesis are both often increased in polycythemia vera, resulting in a normal M:E ratio.
[QCMLS 2021: 4.9.3.1.2, t4.11] [McKenzie 2020, p538] delete QC reference and update McK to McKenzie 2020, p538

89. **c** Iron deficiency anemia laboratory features include: decreased serum iron, serum ferritin, percent saturation, bone marrow iron stores and increased TIBC.
[QCMLS 2021: 4.7.1.1.1; t4.3]

90. **d** Secondary polycythemia can occur inappropriately when renal tumors secrete erythropoietin.
[McKenzie 2020, p539]

91. **b** Lung disease causes tissue hypoxia and results in secondary polycythemia.
[QCMLS 2021: 4.7.2.1]

92. **d** Some drugs interfere with DNA synthesis and result in megaloblastic anemia. Peripheral blood findings reflect the megaloblastic process and include characteristic cellular changes to include oval macrocytes, RBC inclusions, hypersegmented neutrophils, and platelet abnormalities.
[QCMLS 2021: 4.7.1.2.1, t4.5]

93. **a** When the bone marrow senses anemia, it increases production of red cells in a premature fashion, releasing nucleated red blood cells. Polychromasia will also be seen on the peripheral blood smear.
[Ciesla 2017, p27]

94. **c** Favism (consumption of fava beans) is the second most severe clinical condition of G6PD deficiency. After ingestion of fava beans an individual may experience hemoglobinuria, lethargy and abdominal pain.
[Ciesla 2017, p107]

### *B. Leukocytes*

95. **a** Myelofibrosis is often accompanied by a folate deficiency, which develops because of increased consumption by the malignant clone and results in macrocytic anemia.
[McKenzie 2020, p542]

96. **c** RBC precursors in erythroleukemia display several dysplastic characteristics.
[QCMLS 2021: 4.9.1.2]



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97. **b** Megaloblastoid erythropoiesis is a morphologic feature characteristic of dyserythropoiesis.
[QCMLS 2021: 4.9.2]

98. **c** In erythroleukemia, 50% or more of nucleated bone marrow cells are normoblasts, which indicates a decrease in the myeloid component and results in a low M:E.
[QCMLS 2021: 4.8.1.1, t4.9]

99. **c** Autoimmune hemolytic anemia may be associated with an underlying lymphoproliferative disease such as chronic lymphocytic leukemia (CLL).
[QCMLS 2021: 4.7.1.3.3]

100. **d** An absolute neutrophilic leukocytosis occurs when the total granulocyte count (circulating neutrophils) is greater than $7.5 \times 10^9$/L.
[QCMLS 2021: 4.8.1.1]

101. **c** Leukocytosis occurs when the total leukocyte count is greater than $11.0 \times 10^3$/µL.
[McKenzie 2020, p452]

102. **a** A relative lymphocytosis is defined as more than 45% lymphocytes on a manual differential or automated cell count.
[QCMLS 2021: 4.8.3.1]

103. **a** The Philadelphia chromosome results when chromosomes 22 and 9 are reciprocally translocated.
[McKenzie 2020, p523]

104. **a** Catecholamines such as epinephrine cause demargination of neutrophils.
[Rodak 2016, p485]

105. **c** The malignant cells in CLL are more fragile than normal lymphocytes and disperse during smear preparation, producing smudge cells.
[McKenzie 2020, p626]

106. **d** In infectious mononucleosis, the leukocyte count is usually increased due to an absolute lymphocytosis and increase in reactive lymphocytes.
[QCMLS 2021: 4.8.3.1]

107. **b** CML is associated with a marked increase in the M:E, reflective of enhanced myelopoiesis.
[QCMLS 2021: 4.9.3.1.2.1]

108. **c** Acute myelomonocytic leukemia (AMML) is classified by the WHO in the category of acute myeloid leukemia (AML), not otherwise specified.This leukemia was previously classified by the French-America-British cooperative group as FAB M4.
[QCMLS 2021: 4.9.1.1, t4.9]

109. **c** Features of acute erythroid leukemia may include dysplastic changes in erythrocytes such as megaloblastoid development.
[McKenzie 2020, p600]

110. **b** Neutropenia is a result of many conditions, including viral and overwhelming bacterial infections. Viral infections typically cause a lymphocytosis and subsequent relative decrease in neutrophils.
[QCMLS 2021: 4.8.3.1] [McKenzie 2020, p456]

111. **d** Auer rods are associated with some cases of AML.
[QCMLS 2021: 4.9.1]

112. **a** The differential diagnosis of CML should include a leukemoid reaction. The negative chromosome and genetic tests as well as the presence of toxic granulation, Döhle bodies, and vacuoles support a leukemoid reaction in this patient, rather than leukemia.
[QCMLS 2021: 4.8.1.1; t4.11] [McKenzie 2020, p455]

113. **b** Lymphocytes are the predominant cells responding to a viral infection, such as infectious mononucleosis.
[QCMLS 2021: 4.8.3.1]

114. **a** Most reactive lymphocytes in infectious mononucleosis are cytotoxic, CD8 positive T cells.
[McKenzie 2020, p477]

115. **b** CMV is the most common cause of a heterophile-negative reactive lymphocytosis.
[McKenzie 2020, p480]

116. **c** Dysplastic platelets, including micromegakaryocytes, may be seen in the peripheral blood in primary myelofibrosis.
[QCMLS 2021: 4.10.4.1, t4.16] [McKenzie 2020, p543]

117. **c** Peripheral blood dysplasia, such as pseudo-Pelger-Huët neutrophils, may be seen in CML.
[McKenzie 2020, p526]

118. **b** Basophils may be increased in CML as well as immature neutrophils and eosinophils.
[QCMLS 2021: 4.9.3.1.2, t4.11] [McKenzie 2020, p526]

119. **b** The myeloid to erythroid ratio (M:E) relates the bone marrow numbers of all granulocytes and precursors to the numbers of all nucleated RBC precursors. A hypercellular marrow (M:E) of 6:1 indicates more myeloid than erythroid cells in the bone marrow.
[McKenzie 2020, p961]

120. **d** Peripheral blood findings in primary myelofibrosis are numerous and include abnormal RBC morphology, such as teardrop cells. Also note frequently decreased platelets and immature granulocytes in this condition.
[QCMLS 2021: 4.9.3.1.2, t4.11] [McKenzie 2020, p542]

121. **d** Some cases (30%) of CML in terminal phase (blast crisis) may present as an acute lymphoblastic leukemia.
[QCMLS 2021: 4.9.3.1.2.1]

122. **d** Laboratory findings in terminal phase (blast crisis) of CML may include basophilia, thrombocytopenia, and a progressive increase in blasts. Blasts may exceed 20% in the peripheral blood and bone marrow. A positive Philadelphia chromosome and *BCR/ABL* fusion gene indicate CML.
[QCMLS 2021: 4.9.3.1.2.1]

123. **a** Abnormal laboratory findings in polycythemia vera include increased RBC, hemoglobin, hematocrit, RBC mass and erythropoietin levels.
[QCMLS 2021: 4.7.2.1]

124. **d** Laboratory findings indicate >20% myeloblasts, which defines AML.
[QCMLS 2021: 4.9.1]

125. **b** A predominance of blasts in AML would increase the M:E ratio.
[QCMLS 2021: 4.9.1] [McKenzie 2020, p587]

126. **b** DIC is a distinguishing feature of acute promyelocytic leukemia (APL) and often a presenting condition in patients.
[QCMLS 2021: 4.9.1.2]

127. **d** Chronic leukemias such as CMML are defined by less than 20% blasts in the peripheral blood or bone marrow.
[QCMLS 2021: 4.9.4]

128. **d** An important feature of CML is the presence of the *BCR/ABL* fusion gene.
[QCMLS 2021: t4.11]

129. **c** Acute (pure) eythroid leukemia is characterized by >80% immature erythroid precursors with at least 30% proerythroblasts and myeloblasts.
[QCMLS 2021: t4.9, 4.9.1.2]

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130. **d** Acute myelocytic leukemia typically presents with an elevated WBC count because of numerous circulating blasts. Blasts >20% of all nucleated peripheral blood and bone marrow cells establishes a diagnosis.
[QCMLS 2021: 4.9.1]

131. **a** A classic feature of CML is the presence of all stages of granulocyte maturation, as well as eosinophils and basophils.
[QCMLS 2021: t4.11]

132. **d** Classification of myeloproliferative neoplasms.This category includes CML, CNL, PV, PMF, ET, chronic eosinophilic leukemia, not otherwise specified, and MPN, unclassifiable.
[QCMLS 2021: 4.9.3.1.1]

133. **b** Patients with polycythemia vera may have headaches and blurred vision associated with an increased RBC mass. Peripheral blood findings may include markedly elevated RBC (and hemoglobin), leukocytosis with granulocytosis and lymphopenia, thrombocytosis, and elevated uric acid levels. Oxygen saturation is normal.
[QCMLS 2021: 4.9.3.1.2.2, t4.11] [McKenzie 2020, p537-538]

134. **d** Reactive lymphocytes are variable in morphology, including variations in size, nuclear shape and chromatin structure, as well as cytoplasm amount and color.
[QCMLS 2021: 4.8.3.1]

135. **c** CLL and SLL represent different clinical presentations of the same disease.
[QCMLS 2021: 4.9.6.1.1]

136. **c** TdT is a marker for primitive or immature lymphoid cells and is generally negative in acute myeloid leukemia.
[QCMLS 2021: 4.9.1]

137. **c** Characteristics of ALL include that it is a disease of both young children and adults, some patients are symptomatic with CNS involvement, the WBC is variable with anemia and thrombocytopenia typical, the malignant cells are lymphoblasts. Prognosis in children is good, and fair in adults.
[QCMLS 2021: 4.9.5]

138. **a** ALL, notably B cell ALL, is the most common leukemia in children.
[QCMLS 2021: 4.9.5.1] [McKenzie 2020, p613]

139. **d** CLL is a B cell malignancy of bone marrow and peripheral blood .
[QCMLS 2021: 4.9.6.1.1]

140. **c** Hairy cell leukemia is a rare condition of lymphocytic origin in which the lymphoid cells appear to have hair-like projections.
[QCMLS 2021: 4.9.6.1.3]

141. **b** The malignant lymphocytes in hairy cell leukemia characteristically have pale nuclei with fine chromatin and a pale blue cytoplasm with projections that resemble hairs or may appear ruffled or frayed.
[QCMLS 2021: 4.9.6.1.3]

142. **a** Flame cells are abnormal variants of plasma cells with bright red cytoplasm. They are associated with plasma cell neoplasms.
[McKenzie 2020, p635]

143. **b** Sheets or clusters of plasma cells may be seen in the bone marrow in multiple myeloma.
[QCMLS 2021: 4.9.6.2.4]

144. **d** Waldenström macroglobulinemia is a B cell malignancy.
[QCMLS 2021: 4.9.6.2.5]

145. **a** Mycosis fungoides is related to Sézary syndrome and the neoplastic cells are of T cell origin.
[QCMLS 2021: 4.9.6.2.3]

146. **a** Chédiak-Higashi syndrome is a disorder of granulocytes that results in the abnormal fusion of primary and secondary granules.
[QCMLS 2021: 4.8.2]

147. **d** Peripheral blood findings in May-Hegglin anomaly include irregularly sized inclusions in (primarily) neutrophils that resemble Döhle bodies, giant or large platelets, and thrombocytopenia.
[QCMLS 2021: 4.8.2]

148. **b** Pelger-Huët anomaly is a benign disorder in which a morphologic alteration of neutrophils occurs; the majority of nuclei are bilobed or single and round (nonsegmented or unilobed).
[QCMLS 2021: 4.8.2]

149. **a** Chédiak-Higashi syndrome is a disorder of granulocyte membranes that results in the abnormal fusion of primary (lysosomal) and secondary granules.
[QCMLS 2021: 4.8.2]

150. **d** Alder-Reilly anomaly is an inherited condition associated with the mucopolysaccharidoses and results in abnormal, large azurophilic granules in the cytoplasm that resemble toxic granulation. These abnormal granules may be seen in all leukocytes but are most often present in neutrophils.
[QCMLS 2021: 4.8.2]

151. **b** May-Hegglin anomaly is an inherited disorder in which peripheral blood findings include irregularly-sized inclusions in (primarily) neutrophils that resemble Döhle bodies, giant or large platelets, and thrombocytopenia.
[QCMLS 2021: 4.8.2]

152. **a** The primary morphologic feature of neutrophils in the Pelger-Huët anomaly is hyposegmentation, which also may be seen as an acquired condition associated with MDS/MPN, in COVID-19, and with some drugs.
[QCMLS 2021: 4.8.2]

153. **c** CGD is a functional defect of leukocytes in which antimicrobial oxygen metabolites cannot be produced and intracellular destruction of ingested organisms (phagocytosis) is impaired.
[QCMLS 2021: 4.8.2]

154. **a** Causes of leukopenia and neutropenia include several drugs and medications.
[QCMLS 2021: 4.8.1.2]

155. **d** CLL/SLL represent different clinical manifestations of one disease entity. These conditions are characterized by aberrant expression of the T lymphocyte antigen CD5. Expression of CD23 is also seen.
[QCMLS 2021: 4.9.6.1.1]

156. **d** Hairy cells have characteristicly abundant, pale-staining cytoplasm, cytoplasmic projections around the circumference of the cell ("hairs"), oval or reniform nuclei, and relatively fine chromatin.
[QCMLS 2021: 4.9.6.1.3]

157. **c** Prolymphocytes represent >55% of the lymphocytes seen in prolymphocytic leukemia.
[QCMLS 2021: 4.9.6.1.2]

158. **a** Most cases of Burkitt lymphoma are associated with an isolated chromosome translocation leading to a rearrangement of the *MYC* and *IgH* genes, typically t(8;14).
[QCMLS 2021: 4.9.7.2, t4.14]

159. **d** The fusion protein produced by the t(12;21) translocation is associated with a good prognosis in cases of pre-B cell ALL, with survival rates approaching 90%.
[QCMLS 2021: 4.9.5.1, t4.13]

160. **a** In plasma cell neoplasms, the monoclonal protein is present in the incidence of IgG, followed by IgA, IgM, IgD and IgE.
[QCMLS 2021: 4.9.6.2.4]


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161. **a** Sézary cells are malignant T cells in the skin, lymph nodes and peripheral blood. These cells may also be seen in a primary cutaneous T cell lymphoma called mycosis fungoides.
[QCMLS 2021: 4.9.6.2.3]

162. **a** A bimodal age distribution is seen in Hodgkin lymphoma, with a peak incidence in 15-35 year olds and again in patients over 50.
[QCMLS 2021: 4.9.7.1]

163. **b** In the WHO classification system, myeloproliferative neoplasms (MPNs) are characterized by panmyelosis of the bone marrow and peripheral blood erythrocytosis, granulocytosis and/or thrombocytosis.
[QCMLS 2021: 4.9.3]

164. **b** Dysplastic features in one or more cell lines are typical in the myelodysplastic syndromes (MDS).
[QCMLS 2021: 4.9.2]

165. **b** The subgroups of MDS include MDS with single lineage dysplasia, MDS with single lineage dysplasia and ringeds sideroblasts, MDS with multilineage dysplasia, MDS with multilineage dysplasis and ringed sideroblasts, MDS with excess blasts, MDS with isolated del (5q), MDS, unclassifiable, and refractory cytopenia of childhood. Refractory thrombocytopenia is included in the category of refractory cytopenia with single lineage dysplasia.
[QCMLS 2021: 4.9.2, t4.10]

166. **b** Patients who have MDS with isolated deletion of chromosome 5q have long-term stable disease.
[QCMLS 2021: 4.9.2]

167. **c** The classification of myelodysplastic syndromes/myeloproliferative neoplasms comprises chronic myelomonocytic leukemia, atypical chronic myeloid leukemia, juvenile myelomonocytic leukemia and myelodysplastic/myeloproliferative neoplasm, unclassifiable.
[QCMLS 2021: 4.9.4]

168. **d** The Philadelphia chromosome is an acquired translocation between the long arms of chromosomes 9 and 22 and is associated with chronic myelogenous leukemia.
[QCMLS 2021: 4.9.8.1.2, t4.11]

169. **d** The 2008 WHO classification of AML. AML, NOS (not otherwise specified) is a category of AML as defined by the WHO.
[QCMLS 2021: 4.9.1.1]

170. **d** AML with inv(16)(p13;q22) generally presents with monocytic and granulocytic maturation along with abnormal eosinophils in the bone marrow.
[QCMLS 2021: 4.9.1.2, t4.9]

171. **d** WHO criteria for AML indicate that blasts must comprise ≥20% of nonerythroid nucleated cells in acute leukemia. AML with minimal differentiation is characterized by no evidence of morphologic myeloid maturation (no granules) and <3% of blasts positive for myeloperoxidase.
[QCMLS 2021: 4.9.1.2]

172. **a** There are 2 variants of acute erythroid leukemia distinguished by the presence or absence of myeloblasts. The erythroid/myeloid variant is seen when ≥50% of nucleated cells in the bone marrow are erythroid precursors and ≥20% of the nonerythroid cells are myeloblasts. Pure erythroid leukemia is confirmed when >80% on nucleated cells in the bone marrow are erythroid precursors. Pure erythroid leukemia is rare.
[QCMLS 2021: 4.9.1.2]

173. **a** Children with Down syndrome have a 50x greater risk of developing acute myeloid leukemia (AML). The abnormal cells are megakaryoblasts. This condition is a category in the WHO classification of AML designated as Myeloid Proliferations Related to Down Syndrome.
[QCMLS 2021: 4.9.12]



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174. **c** More aggressive cases of CLL are associated with higher levels of CD38.
[QCMLS 2021: 4.9.6.1.1]

175. **a** Sézary cells are neoplastic T lymphocytes associated with Sézary syndrome and primary cutaneous T cell lymphoma.
[QCMLS 2021: 4.9.6.2.3]

176. **d** Foam cells with sphingomyelin and PAS positivity mark the classic morphology of Niemann-Pick cells.
[McKenzie 2015, p403]

177. **d** TRAP stain positivity is characteristic of hairy cell leukemia.
[QCMLS 2021: 4.9.6.1.3] [Glassy 2018, p300]

178. **b** Classic, hypergranular APL usually presents with a low WBC and few leukemic cells in the peripheral blood. The abnormal promyelocytes have kidney-shaped or lobulated nuclei with intensely granulated cytoplasm. Bundles of Auer rods are common, and the condition is associated with the t(15; 17) translocation and strong myeloperoxidase activity.
[QCMLS 2021: 4.9.1.2, t4.9]

179. **c** CLL lymphocytes express the pan B cell markers as well as CD5, which is normally not expressed in normal B cells, but is on the surface of normal T cells.
[QCMLS 2021: 4.9.6.1.1]

180. **d** Hairy cell leukemia cells typically express CD11c.
[QCMLS 2021: 4.9.6.1.3]

*C. Platelets*

181. **d** vWF is a constituent of platelet alpha granules and is synthesized in the megakaryocyte as it develops.
[QCMLS 2021: 5.3.1]

182. **b** Average life span of platelets in peripheral blood is 9.5 days.
[QCMLS 2021: 5.3.1]

183. **a** Aspirin interferes with prostaglandin metabolism in the platelet by inhibiting cyclooxygenase, which participates in the conversion of arachidonic acid to protein G2; protein G2 is necessary to produce thromboxane, which stimulates secretion from the platelet granules.
[QCMLS 2021: 5.5.2.2]

184. **a** Certain characteristics of platelet-type bleeding disorders include mucosal bleeding, petechiae, and a female predominance. Other types of bleeding are due to defects in coagulation factors. Hemarthrosis and deep hematomas occur in Factor VIII deficiency (hemophilia), delayed bleeding occurs in Factor XIII deficiency, and deep hematomas present with the other choices in the question.
[QCMLS 2021: 5.5]

185. **c** Hirudin, argatroban and dabigatran are direct thrombin inhibitors. They are either isolated from the leech, *Hirudo medicinalis* (hirudin), or are recombinant forms of the same protein (lepirudin, argatroban, dabigatran). Tirofiban is not related and is a platelet GPIIb/IIIa inhibitor and is an anti-platelet drug.
[QCMLS 2021: 5.7.5.2.4]

186. **b** Ticlopidine and clopidogrel both inhibit ADP mediated aggregation and present with blunted ADP response with aggregometry studies, but normal results with other agonists. Another class of inhibitors, like abciximab, block platelet function through GPIIb/IIIa.
[QCMLS 2021: 5.7.5.2.4]



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187. **b** vWF binds platelets through their GP Ib/IX/V membrane receptor. The importance of platelet adhesion is demonstrated by bleeding disorders such as von Willebrand disease, in which vWF is missing or defective, and Bernard Soulier disease, in which the GP Ib/IX/V receptor is absent.
[QCMLS 2021: 5.3.3, t5.1]

188. **d** May-Hegglin anomaly is characterized by decreased platelet counts; vWD and hemophilia patients do not present with low platelet counts. Glanzmann patients present with normal platelet counts that do not function.
[QCMLS 2021: 5.5.2]

189. **b** Acute ITP typically resolves within weeks and is more frequently seen in children; it is not gender-dependent; platelet destruction is increased.
[QCMLS 2021: 5.5.1.2.1]

190. **d** A quantitative abnormality of platelets (thrombocytopenia) is the most common cause of excessive or abnormal bleeding. Fibrinogen deficiencies are rare.
[QCMLS 2021: 5.5]

191. **d** Abnormalities in coagulation factors and DIC are ruled out by the normal PT/aPTT and fibrinogen. The low platelet count is due to the dilutional effect of the bypass surgery.
[QCMLS 2021: 5.5.2.2]

192. **a** Patients with thrombotic thrombocytopenic purpura (TTP) present with platelet counts $<20 \times 10^3/\mu L$ ($<20 \times 10^9/L$). Platelet thrombi are dispersed throughout the arterioles and capillaries subsequent to the accumulation of large vWF multimers made by endothelial cells and platelets. This is related to a deficiency of ADAMTS-13. DIC is caused by inflammation, infections and cancer. HUS is related to a toxin producing organism. ITP is a result of antibodies to platelets.
[QCMLS 2021: 5.5.1.2.2]

193. **b** The pathogenesis of HIT is that antibodies are produced against heparin-platelet factor 4 complex. This complex binds to FC receptors causing platelet activation and the formation of platelet microparticles, thrombocytopenia and hypercoagulability. ACLA and B2GP1 are seen in anti-phospholipid syndrome.
[QCMLS 2021: 5.7.4]

194. **a** Polycythemia vera is characterized by increased WBC, RBC, and platelet counts.
[QCMLS 2021: 4.9.3.1.2.2, t4.11]

195. **d** Congenital spherocytosis is characterized by an increased MCHC and an increased reticulocyte count; reactive thrombocytosis is not usually accompanied by abnormal platelets; myelofibrosis is characterized by abnormal RBC morphology and decreased platelets and reticulocytes.
[QCMLS 2021: 5.5.2.2]

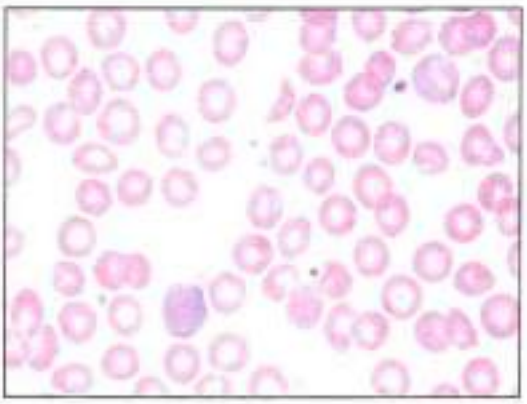
196. **b** May-Hegglin anomaly is a rare, autosomal dominant platelet disorder. Patients present with variable thrombocytopenia, giant platelets, and large Döhle body-like inclusions in the neutrophils, eosinophils, basophils, and monocytes. The majority of patients are asymptomatic, but a few have shown mild bleeding tendencies related to the degree of thrombocytopenia. Wiskott-Aldrich syndrome and Hĕrmanský-Pudlák syndrome are both disorders of platelet secretion, specifically storage pool disorders. Ehlers-Danlos syndrome is a vascular disorder in which a bleeding tendency is manifested primarily in subcutaneous hematoma formation.
[QCMLS 2021: 4.8.2]

197. **b** The 9:1 blood-to-anticoagulant ratio is effective, provided the patient's hematocrit is 55% or less. In polycythemia, the decrease in plasma volume relative to whole blood unacceptably raises the anticoagulant-to-plasma ratio, which causes falsely prolonged results, due to the increased level of anticoagulant in the tube, for clot-based coagulation tests.
[Rodak 2016, p764]

## III. Laboratory Testing

### A. Cell Counts

198. **d** Percent of cells counted × total count is the formula to calculate an absolute cell count.
[QCMLS 2021: 4.10.2.1]



199. **b** Reticulocytes appear as polychromatophilic red blood cells on a Romanowsky-stained blood smear. These cells are usually larger than normal cells with a blue tinge. The blue tinge is caused by the presence of residual RNA in the cytoplasm.
[QCMLS 2021: 4.10.2.1]

200. **c** Newborn infants have elevated reticulocyte counts.
[McKenzie 2015, front inside cover]

201. **c** In patients with cold agglutinins, the automated cell counters show an erroneously elevated MCV and an erroneously decreased RBC count due to clumping of the red cells.
[QCMLS 2021: 4.10.2.1]

202. **d** Autoagglutination of anticoagulated blood can occur at room temperature (22°C) in patients with a cold autoagglutinin. The MCV will be falsely elevated and the RBC count falsely decreased, resulting in an elevated MCHC. The blood sample should be warmed to 37°C and rerun.
[Harmening 2009, p261]

203. **a** Using a supravital stain, residual ribosomal RNA is precipitated within the reticulocytes.
[QCMLS 2021: 4.10.2.1]

204. **c** Using a supravital stain (new methylene blue), residual ribosomal RNA is precipitated within the reticulocytes.
[QCMLS 2021: 4.10.2.1]

205. **b** The calculated RBC parameters include the hematocrit, derived from the MCV and RBC, MCH (mean corpuscular hemoglobin), which is calculated from the hemoglobin concentration and the RBC, and the mean corpuscular hemoglobin concentration (MCHC), which is determined from the hemoglobin value and the hematocrit.
[QCMLS 2021: 4.10.7.2]

206. **a** Most hematology analyzers use 1 of 2 basic principles of operation: electronic impedance or optical scatter.
[QCMLS 2021: 4.10.1.1]

207. **b** The RDW is a quantitative estimate of erythrocyte anisocytosis.
[QCMLS 2021: 4.10.8.1]



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**208.** **a** The results show the effect of conventional chemotherapy on WBC count.
[McKenzie 2015, p393]

**209.** **c** The data reflect the definition of absolute neutropenia.
[QCMLS 2021: 4.8.1.2]

**210.** **a** Reference interval for percent lymphocytes indicates a relative lymphocytosis.
[McKenzie 2015, inside front cover]

**211.** **c** Calculation of absolute from relative % and WBC.
[QCMLS 2021: 4.10.2.1]

**212.** **a** The formula (number of cells counted × dilution × 10)/number of squares counted shows the calculation for a hemacytometer cell count.
[QCMLS 2021: 4.10.1.1]

**213.** **b** The formula for the calculation for a hemacytometer cell count is (number of cells counted × dilution × 10)/number of squares counted, so (50 × 100 ×10) / 8 = 6250.
[QCMLS 2021: 4.10.1.1]

**214.** **b** The WBC count can be corrected for NRBCs using the formula (uncorrect WBC count × 100) / (number of NRBCs per 100 WBCs + 100), so (10,000 × 100) / (25 +100) = 1,000,000 / 125 = 8000.
[QCMLS 2021: 4.10.1.1]

**215.** **c** Any automated differential result flag (to include an IG flag) should be confirmed with a manual differential.
[McKenzie 2015, p915]

**216.** **b** +++ is an indicator that the WBC count exceeds the upper limit of established reportable range.
[McKenzie 2015, p915]

**217.** **c** Low platelet counts should be verified with a review and estimate from a stained peripheral blood smear.
[McKenzie 2020, p1070]

**218.** **b** Coulter principle of particle counting.
[QCMLS 2021: 4.10.1.2]

**219.** **b** RBC × 3 should approximately equal the HGB; this HGB value is likely to be falsely high; common causes include lipemia.
[QCMLS 2021: 4.10.1.2]

**220.** **a** A falsely high hemoglobin is indicated because the hemoglobin does not match the hematocrit (HGB x 3 = HCT) and the MCHC is high. These results are associated with cold agglutinins, lipemia, and less often icterus. If cold agglutinins are present, the MCV would also be falsely increased and the RBC and HCT falsely decreased. Performing a manual hematocrit would allow a check for lipemia and verification of the automated hematocrit.
[McKenzie 2020, p1070]

**221.** **c** The formula for calculating the MCH is hemoglobin (g/dL) × 10 divided by the RBC count. The MCH indicates the average weight of hemoglobin in the red blood cell and is measured in picograms.
[QCMLS 2021: 4.10.8.1] [Rodak 2020, p164]

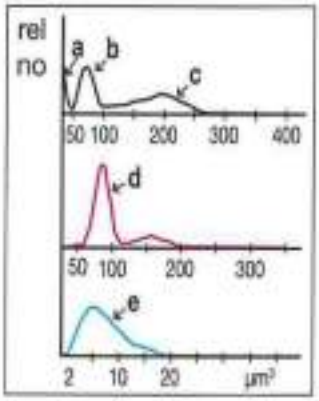
**222.** **d** Westgard rules apply to hematology and coagulation. A single control value is outside the +/– SD limit; the course of action is to reassay the control. Repeat of 1 out-of-range control is the first appropriate course of action.
[Rodak 2020, p21]

**223.** **a** Clumped platelets may cause an increase in the WBC and a decrease in the platelet count. Large clumps of platelets are counted as WBCs and not platelets.
[QCMLS 2021: 4.10.1.2] [Rodak 2020, p191]

224. **b** Platelets that are aggregated will affect a manual platelet count. Platelets adhere to foreign objects and to each other.
[Rodak 2020, p158-159]

225. **b** Combination of decreased RBC, increased MCV and increased MCHC is likely to be due to cold agglutinins. The corrective action for cold agglutins is to warm the specimen at 37°C.
[QCMLS 2021 4.10.1.2] [Rodak 2020 p191]

***Please consult the diagram below for questions 226 & 230 regarding automated instrument histograms.***




226. **d** The RBC is measured directly by impedance as the cells pass through the aperture. Pulse height is measured and volume histograms are generated.
[Rodak 2020, p177-179]

227. **a** Tubes that remain at room temperature for >5 hours have unacceptable blood cell artifacts. This will affect the HCT and the MCV. If the specimen is allowed to sit at room temperature for more than 4 hours the RBCs become spherical, which can inhibit the formation of rouleaux.
[Rodak 2020, p168-169]

228. **a** Instruments count particles within defined size limits. The upper limit separates large platelets from erythrocytes. Small or fragmented RBCs may be counted as platelets. RBCs and platelets are counted by impedance in the same chamber. WBCs are counted in a separate chamber.
[QCMLS 2021: 4.10.1.2] [Rodak 2020, p177-178]

229. **d** Whole calibration using fresh whole blood specimens that have been assayed using reference methods is the method of choice for validation.
[Rodak 2016, p224]

230. **d** Platelets are counted by impedance using lower thresholds (2-30 femtoliters) and the volume distribution to create a histogram.
[Rodak 2020, p179-180]

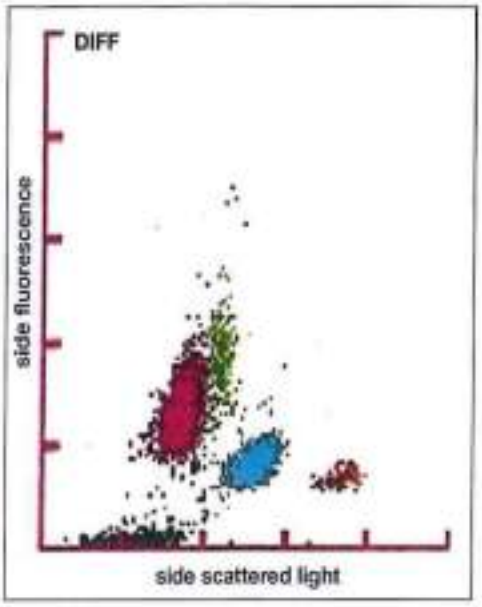
231. **d** Lipemia interferes with hemoglobin (as well as MCH and MCHC) by falsely elevating the results. MCH and MCH calculations are associated with the hemoglobin value. Turbidity affects spectrophotometric readings of how the hemoglobin value is determined. The corrective action is plasma replacement.
[Rodak 2020, p191]



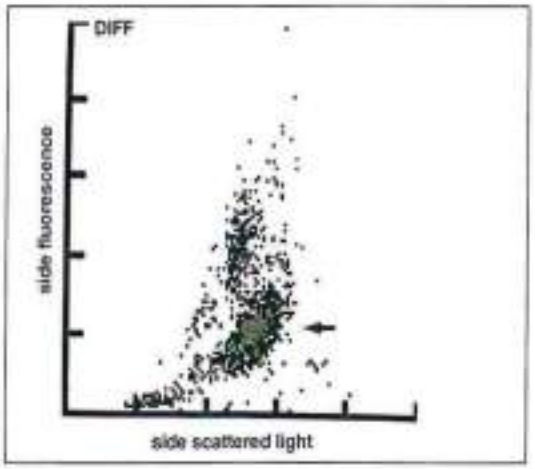
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232. **c** Average the number of cells counted on both sides of the chamber, then use the formula: total count = number of cells counted (average of both sides) × dilution factor divided by the area counted ($mm^2$) × depth.
[QCMLS 2021 4.10.1] [Rodak 2020, p158]

233. **a** The platelet count matches the estimate well. The average number of platelets per 100x oil immersion field (17.8 in this case) × 20,000 approximates the platelet count.
[QCMLS 2021: 4.10.4.3] [Rodak 2020, p207]



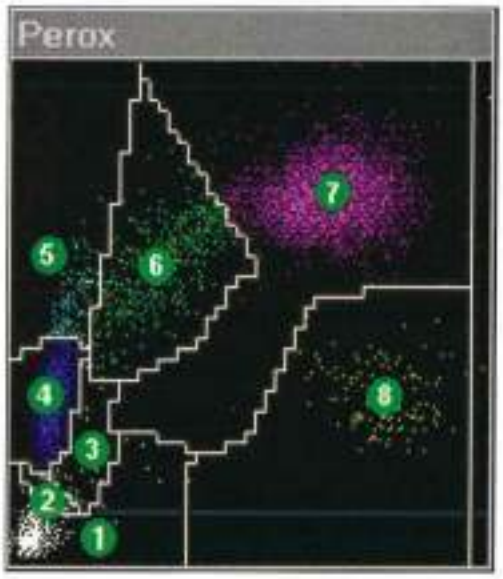

234. **b** Automated hematology analyzers generate scattergrams that show normal and abnormal populations. Each instrument has its own scatter plot to depict various cell types. This scattergram is from a Sysmex instrument and indicates an elevation in eosinophils, which should be seen on the stained peripheral blood smear.
[McKenzie 2020, p972]




235. **d** Automated hematology analyzers generate scattergrams that show normal and abnormal populations. Each instrument has its own scatter plot to depict various cell types. This scattergram is from a Sysmex instrument, and the arrow identifies the normal location of neutrophils.
[Rodak 2020, p183; McKenzie 2020, p972]

236. **b** Whenever an automated analyzer generates an alert flag, the smear should be reviewed to evaluate the presence of abnormal cells.
[Rodak 2020, p193; McKenzie 2020, p1070]

237. **c** Abbott instruments combine several technologies to count and identify blood cells. MAPPS is the primary, patented method that generates data used to classify leukocyte subpopulations.
[Rodak 2020, p178; McKenzie 2020, p980-981]




238. **a** This scatter plot represents a normal display of cell populations as generated by a Siemens Advia instrument. The channel shown identifies white blood cell populations based on light scatter and peroxidase staining. The position of the numeral 6 indicates the location of monocytes.
[McKenzie 2020, p986]

*B. Differentials and Morphology Evaluation*

239. **d** Primary myelofibrosis presents with dacryocytes (teardrop cells) nucleated RBCs and megakaryocytic fragments. A clonal myeloproliferative neoplasm characterized by ineffective hematopoiesis associated with bone marrow hypercellularity, extra-medullary hematopoiesis, fibrosis and increased megakaryocytes.
[QCMLS 2021: 4.9.3.1.3, I4.37] [Rodak 2020, p574-576]

240. **b** Cabot rings are believed to be remnants of spindle fibers formed during mitosis. They can be in ring or figure-eight forms. They are associated with megaloblastic anemia and myelodysplasic syndromes.
[QCMLS 2021: 4.10.4.2, I4.18] [Rodak 2020 p256]

241. **d** RBCs in sideroblastic anemia can contain iron deposits known as Pappenheimer bodies. Pappenheimer bodies are irregular clusters of small light to dark granules found near the periphery of the red blood cell (with Wright stain). They can also be stained with Prussian blue for iron. They can be seen in hemoglobinopathies, thalassemias, megaloblastic anemia, myelodysplasia, hyposplenism, and postsplenectomy.
[QCMLS 2021: 4.10.4.2, I4.18] [Rodak 2020, p256]

242. **c** The round nucleus, pyknotic or condensed chromatin and abundant gray-pink cytoplasm with nearly complete hemoglobin production are characteristic of the orthochromatic normoblast (metarubricyte).
[Gulati 2021, p23; Rodak 2020, p68-69]



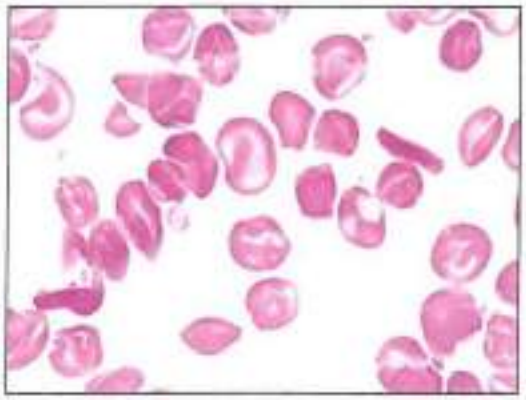
 ISBN 978-089189-6845 ©ASCP 2022

243. **d** Malarial parasites, especially *Falciparum* species, commonly appear on a Wright-stained peripheral blood slide as small blue rings with a red nucleus or chromatin dot.
[QCMLS 2021: 4.10.4.2, t4.18]

244. **a** Spherocytes are small, round, dense RBCs with no central pallor. Sometimes they occur with a high MCHC. They are associated with hereditary spherocytosis, immune hemolytic anemia, and extensive burns.
[QCMLS 2021: 4.10.4.2, t4.18] [Rodak 2020, p255]

245. **c** A calculation of the RBC indices can be used to classify anemia. The MCV determines the volume or size of the cell. If the MCV is <80 fL the RBCs are referred to as microcytic; if >100 fL, macrocytic; and if 80-100 fL, normocytic. The MCHC refers to the concentration of hemoglobin in the RBC. If the MCHC is <32 g/dL the RBCs are referred to as hypochromic; if 32-36 g/dL, normochromic
[QCMLS 2021: 4.10.5] [Rodak 2020, p163-164]

246. **a** Nucleated RBCs and polychromasia can be associated with accelerated RBC production. The polychromatic RBC has no nucleus and is the last stage before becoming a mature erythrocyte. Other immature (nucleated) RBCs may be seen in the peripheral blood as an indicator that the bone marrow is producing and releasing erythrocytes in an effort to regenerate cells that are being lost as a result of many possible mechanisms.
[McKenzie 2020, p210]



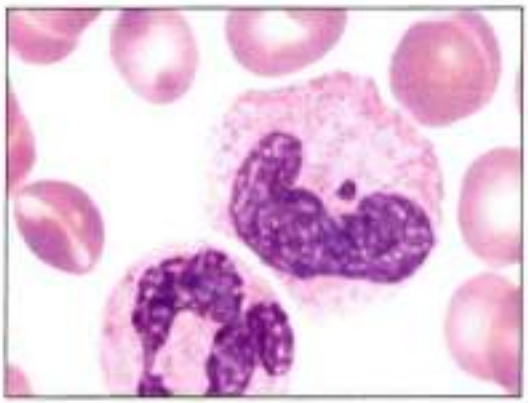
247. **b** HbC crystals can be seen in patients with HbC disease. Hemoglobin C crystals are rigid and hexagonal in shape. They can be seen as intracellular inclusions when RBCs are dehydrated. Crystals are densely stained and vary in size. Hemoglobin C disease presents with a mild to moderate normochromic, normocytic anemia.
[QCMLS 2021: 4.7.1.3.7.3] [Rodak 2020, p411-412]

248. **b** Rouleaux formation occurs when plasma proteins are increased and results in an elevated ESR. Rouleaux represents aggregation of stacked RBCs caused by elevated plasma proteins and abnormal monoclonal proteins as seen in multiple myeloma.
[QCMLS 2021: 4.10.11.1, 4.9.6.2.4] [Rodak 2020, p169, 619]

249. **b** The peripheral smear allows microscopic examination of the blood cells. The most characteristic finding in multiple myeloma is rouleaux formation of the red cells. An abnormal and excess number of plasma cells secrete monoclonial immunoglobins that allow red blood cells to form rouleaux.
[QCMLS 2021: 4.9.6.2.4] [Rodak 2020, p619]

250. **d** Patients who have suffered severe burns to >15% of their body generally show evidence of intravascular hemolysis. RBCs show changes including fragmentation, budding and microspherocyte formation.
[McKenzie 2020, p188; Rodak 2020, p373]

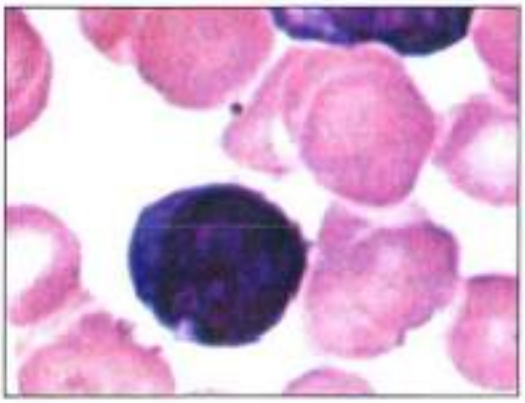
**251. d** Plasma cells are cells with eccentric dark nuclei, blue cytoplasm, and a prominent, pale, central Golgi apparatus. The chromatin in the nucleus may have a clock face appearance.
[QCMLS 2021 4.9.6.2.4, I4.47] [Rodak 2020, p232]



**252. d** Döhle bodies are light blue, usually single inclusions of varying sizes. They are seen in the cytoplasm of neutrophils and bands and are remnants of rough endoplasmic reticulum. Döhle bodies are often associated with toxic granulation and toxic vacuoles.
[QCMLS 2021: 4.10.4.1] [Rodak 2020, p457-458]

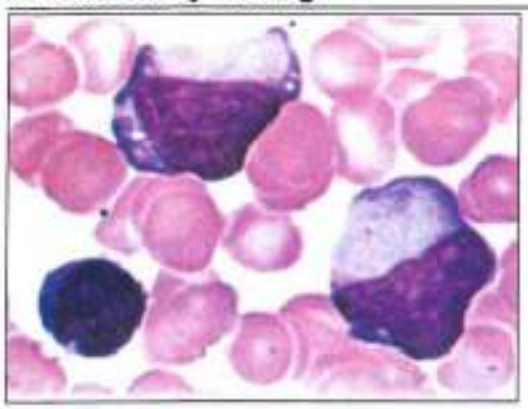
**253. b** The term "left shift" is based on the use of old-style cell counters, which had keys for immature neutrophils to the left of those for mature neutrophils. A left shift is defined as an increase in the production of neutrophils in the bone marrow or as a transfer of neutrophils from the bone marrow storage pool to the circulation pool. Immature cells will increase, and will include bands, metamyelocytes, and myelocytes.
[QCMLS 2021: 4.8.1.1] [Rodak 2020, p454-455]

**254. c** Leukoerythroblastosis is characterized by an increase in the WBC count and the presence of immature red blood cells and granulocytes in the peripheral blood and bone marrow. Leukoerythroblastosis is seen in primary myelofibrosis.
[QCMLS 2021: 4.9.3.1.3] [Rodak 2020, p454, 576, 858]

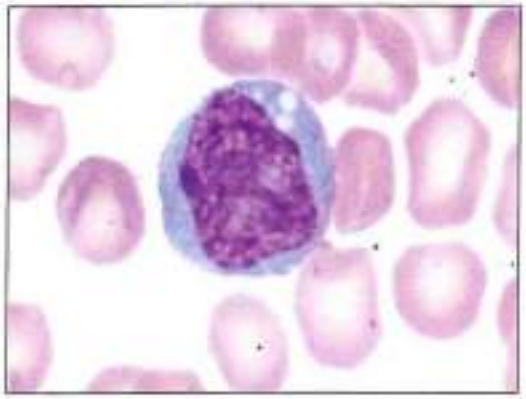


**255. b** Mature (resting) lymphocytes are small with a round nucleus and scant blue cytoplasm. The chromatin pattern is arranged in blocks, and the nucleolus is rarely seen.
[QCMLS 2021: 4.10.4.1, I4.17] [Rodak 2020, p130-131]

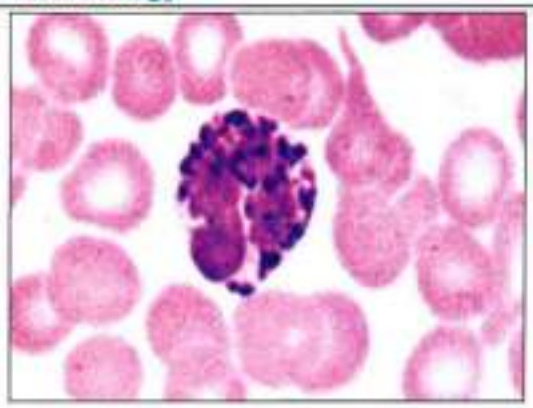
 ISBN 978-089189-6845 ©ASCP 2022



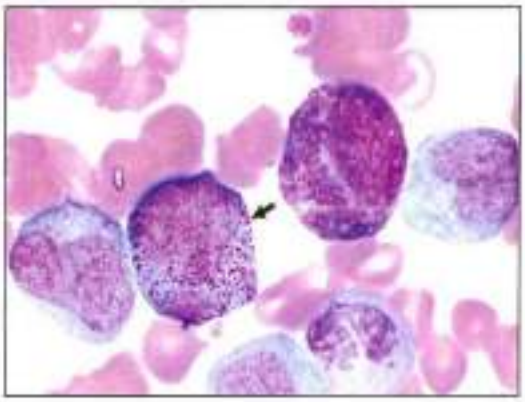
**256. b** Reactive changes in lymphocyte morphology occur when lymphocytes are stimulated, often in response to viral infection, as in infectious mononucleosis. Reative lymphocytes vary in size and shape. The nucleus is irregular with less condensed chromatin. Nucleoli may be seen. The cytoplasm is abundant and basophilic with darker edges, and they may be indented by surrounding RBCs.
[QCMLS 2021 4.8.3.1, I4.17, I4.8] [Rodak 2020, p460-461]



**257. d** Monocytes appear larger than neutrophils (15-20 µm). The nucleus may be round, oval, or kidney or horseshoe shapes. The nucleus tends to be folded on itself with a chromatin pattern that is loose, often described as lacy. Nucleoli are usually not seen. The cytoplasm is blue-gray and may appear like ground glass. Sometimes fine azure granules and vacuoles may be seen. Small cytoplastic pseudopods or blebs may also be seen.
[QCMLS 2021: 4.10.4.1, I4.17] [Rodak 2020, p128]



**258.** **b** Mature basophils contain a lobulated (typically 2 lobes) nucleus. The chromatin pattern if visible is usually clumped. The cytoplasm is colorless to pinkish and contains numerous, large, blue-black granules which often obscure the nucleus.
[QCMLS 2021: 4.10.4.1, I4.16] [Rodak 2020, p126]



**259.** **c** Promyelocytes are large cells with a high N:C ratio. The nucleus is round or oval and 2 or more nucleoli may be visible. The cytoplasm is basophilic and characteristically has few to numerous prominent, dark red or purple primary granules.
[QCMLS 2021: 4.10.4.1, I4.16]

**260.** **a** A granulocytic leukemoid reaction is also known as a reactive neutrophilia in which the WBC count is very elevated with a shift to the left. The condition occurs in response to a nonmalignant condition such as a bacterial infection or inflammation. Neutrophils and immature neutrophil forms may be seen.
[QCMLS 2021: 4.8.11] [Rodak 2020, p454, 863]

**261.** **d** Reactive changes in lymphocyte morphology occur when lymphocytes are stimulated. Reactive lymphocytes vary in size and shape. The nucleus is irregular with less condensed chromatin. Nucleoli may be seen. The cytoplasm is abundant and basophilic with darker edges that may be indented by surrounding RBCs.
[QCMLS 2021: 4.8.3.1, I4.25, I4.8] [Rodak 2020 p460-461]

**262.** **b** Blood smears should be dried quickly to avoid drying artifacts.
[Rodak 2020, p204]

**263.** **a** High-quality blood smears can be made from an EDTA tube within 4 hours of drawing the specimen. After 5 hours cell artifacts develop. EDTA generally prevents platelets from clumping on glass slides.
[Rodak 2020, p202]



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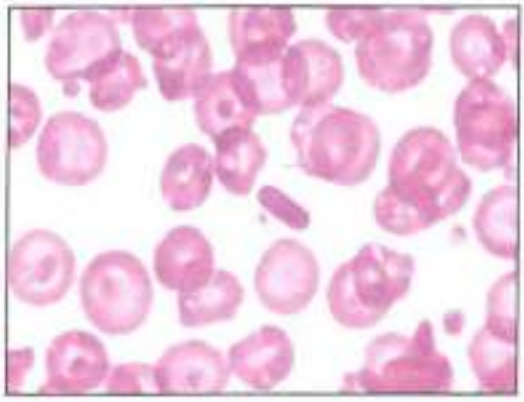
264. **a** Platelet clumping is expected if smears are made directly from the finger stick draw. Smears must be made promptly before any clotting begins and should be prepared at the patient's bedside. It is important to examine the edges of the slide for platelet clumping.
[Rodak 2020, p202-203]

265. **b** In a hemolytic event, the blood smear may reveal polychromasia as evidence of bone marrow compensation for anemia.
[Gulati 2021, p24]

*C. Hemoglobin*

266. **b** A very high WBC count (>20,000) or a high platelet count (>700,000) in cyanmethemoglobin reagent-treated specimens causes turbidity, which will result in falsely-elevated hemoglobin values.
[QCMLS 2021: 4.10.1.2] [Rodak 2020, p161]

267. **a** In beta-thalassemia minor the $HbA_2$ level is 3-7% and HbF is <5%. In beta-thalassemia major HbA is absent or decreased, $HbA_2$ is variable but HbF is at 70%. Thalassemia presents with microcytic RBCs and usually a normal iron.
[QCMLS 2021: 4.7.1.1.2] [Rodak 2020, p431-432]



268. **d** HbC crystals are hexagonal in shape, stain densely, and vary in size and shape. They appear oblong and have rounded ends. Definitive diagnosis is via electrophoresis. HbC crystals can be seen in patients with HbC disease, but the condition must be confirmed with hemoglobin electrophoresis.
[QCMLS 2021: 4.7.1.3.7.3] [Rodak 2020, p412]

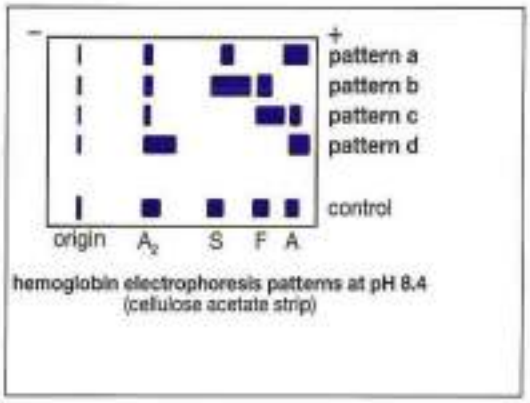
269. **b** The most common screening test for HbS is called the hemoglobin solubility test, which capitalizes on the decreased solubility of deoxygenated hemoglobin when in solution, producing a turbidity. The confirmation test is hemoglobin electrophoresis.
[QCMLS 2021: 4.10.6.1] [Rodak 2020, p405]

270. **c** Deletion of 3 alpha globins results in HbH. It is characterized by the accumulation of excess beta chains.HbH is a fast-migrating hemoglobin.
[QCMLS 2021: 4.10.6.2] [Rodak 2020, p435-436]

271. **d** HPFH results when there is either a deletion or inactivation of the beta and gamma gene complex. There is a delayed switch from gamma to beta or delta production with no HbA or $HbA_2$ produced. HbF persists into adulthood. Both gamma and alpha chains are produced in approximately equal amounts.
[QCMLS 2021: 4.7.1.3.7.9] [McKenzie 2020, p309-310]

272. **d** Anemia, as indicated by a decreased hematocrit, may result in a falsely negative result because the amount of HbS is too low to be accurately detected.
[McKenzie 2020, p277]

***Please consult the diagram below for questions 273-274 regarding electrophoresis patterns.***




hemoglobin electrophoresis patterns at pH 8.4 (cellulose acetate strip)

**273. c** In homozygous beta-thalassemia, reduced synthesis of beta chains affects the production of HbA. $HbA_2$ and HbF are increased and HbA decreased.
[QCMLS 2021: 4.7.1.1.2] [Rodak 2020 431-432]

**274. a** In sickle cell trait electrophoresis reveals ~40% of HbS and ~60% of HbA and $HbA_2$
[QCMLS 2021: 4.7.1.3.7.2, f4.3] [Rodak 2020, p411]

**275. b** HbE has a high incidence in Southeast Asia and migrates with $HbA_2$, HbC and $HbO_{Arab}$ on cellulose acetate electrophoresis.
[QCMLS 2021: 4.10.6.2] [Rodak 2020 412-413]

**276. d** In the cyanmethhemoglobin method, hemoglobin is oxidized to methemoglobin ($Fe^{3+}$) by potassium ferricyanide, and is then converted to cyanmethoglobin by potassium cyanide. The cyanmethemoglobin concentration is directly proportional to hemoglobin concentration. Sulfhemoglobin is not converted to cyanmethemoglobin and cannot be measured by this method.
[Rodak 2020, p160]

**277. c** The solubility test is a common procedure used to screen for the presence of HbS. The principle of the procedure involves the insolubility of HbS when RBCs are added to a solution of sodium hydrosulfite (sodium dithionite). The reagent lyses RBCs to release hemoglobin and then reduces the hemoglobin, which then forms tactoids when HbS is present. The solution becomes cloudy (turbid), indicating a positive reaction.
[QCMLS 2021: 4.10.6.2] [Rodak 2020, p406]

**278. d** HbH inclusions are seen after incubation of blood with brilliant cresyl blue. The inclusions give the RBC the appearance of a golf ball.
[QCMLS 2021: 4.7.1.1.2] [Rodak 2020, p438]

**279. c** On alkaline electrophoresis, HbE migrates with $HbA_2$, HbC and $HbO_{Arab}$. HbE is seen commonly in Southeast Asia and presents with a microcytic,hypochromic anemia.
[QCMLS 2021: 4.7.1.3.7.5] [Rodak 2020, p412-413]

### *D. Hematocrit*

**280. a** Most automated instruments determine the hematocrit using the directly measured MCV and RBC count with the formula: MCV (fL) × RBC ($\times 10^{12}$/L)/1000. Sysmex uses the cumulative detection of RBC pulse heights.
[QCMLS 2021: 4.10.7.2] [Rodak 2020, p178]



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**281.** **c** The trapping of plasma causes the microhematocrit to be 1-3% higher than the value obtained using automation. Sickle cell anemia, macrocytic anemias, hypochromic anemias, spherocytes and thalassemia may cause plasma to be trapped in the RBC layer. Other errors may result through improper sample collection and handling or when the procedure is not followed correctly.
[QCMLS 4.10.7.1] [Rodak 2020, p162]

*E. Indices*

**282.** **c** Spherocytes are RBCs without central pallor (due to decreased RBC membrane) and often have a high MCHC. Spherocytes may be referred to as hyperchromic because of increased MCHC.
[QCMLS 2021: 4.10.4.2, t4.18] [Rodak 2020, p255]

**283.** **d** The MCH is the average weight of hemoglobin in a RBC, expressed in picograms (pg). The formula to calculate the MCH is: hemoglobin × 10 divided by the RBC count.
[QCMLS 2021: 4.10.8.1] [Rodak 2020, p164]

**284.** **a** The RDW is the coefficient of variation of RBC volume expressed as a percent. It indicates the variation in the RBC volume (size) and correlates with anisocytosis.
[QCMLS 2021: 4.10.8.1] [Rodak 2020, p254]

**285.** **a** In determining the appearance of RBCs for classification of anemia, the MCV and the MCHC are usually used. When the MCV <80 fL the cell is microcytic; when >100 fL the cell is macrocytic. When the MCHC <32 g/dL the cell is hypochromic. The MCH may also sometimes be used to classify anemia. When the MCH is <28 pg, the cell is hypochromic.
[QCMLS 2021: 4.10.8.2] [Rodak 2020, p163]

**286.** **c** The MCV is calculated using the formula: hematocrit × 10 divided by the RBC count. It is commonly expressed in femtoliters (fL).
[QCMLS 2021: 4.10.8.1] [Rodak 2020, p164]

**287.** **d** The MCH is calculated using the formula: hemoglobin × 10 divided by the RBC count and is measured in picograms (pg).
[QCMLS 2021: 4.10.8.1] [Rodak 2020, p163]

**288.** **a** The MCH is calculated using the formula: hemoglobin × 10 divided by the RBC count and is measured in picograms (pg).
[QCMLS 2021: 4.10.8.1] [Rodak 2020, p163]

**289.** **a** The MCHC is calculated using the formula HBG × 100 / HCT. It is expressed in grams per deciliter (g/dL).
[QCMLS 2021: 4.10.8.1] [Rodak 2020, p163]

**290.** **b** The MCHC is calculated using the formula HBG × 100 / HCT. It is expressed in grams per deciliter (g/dL).
[QCMLS 2021: 4.10.8.1] [Rodak 2020, p163]

**291.** **c** The MCV is calculated using the formula HCT × 10 / RBC count. It is expressed in femtoliters (fL).
[QCMLS 2021: 4.10.8.1] [Rodak 2020, p163]

**292.** **c** An MCHC >38 g/dL should be investigated for an error in determining the hemoglobin value. False increases in MCHC can occur when hemoglobin and hematocrit values are spuriously affected as these parameters are used to calculate the MCHC.
[QCMLS 2021: 4.10.8.1] [Rodak 2020, p163]

*F. Hemolytic Indicators*

**293.** **b** Bilirubin results from increased heme catabolism and will be increased in extravascular hemolysis. Microcytes are associated with anemias related to defects in hemoglobin synthesis, and hemoglobinuria is associated with intravascular hemolysis. RBC membrane structural defects result in rigid cells, like spherocytes, which are less able to negotiate the small passages of the spleen and result in extravascular hemolysis.
[QCMLS 2021: 4.6.3, 4.7.1.3.1.1, t4.18] [Rodak 2020, p329]

**294.** **a** Extravascular hemolysis accounts for 90% of RBC destruction, occurs primarily in the spleen, and involves the key breakdown products of heme, iron, and globin. Intravascular hemolysis is responsible for 10% of RBC destruction; however, the lysis is visible in the peripheral blood (circulation) as free hemoglobin.
[QCMLS 2021: 4.6.3]

*G. Special Stains*

**295.** **a** Laboratory findings in iron deficiency anemia include decreased serum iron, serum ferritin, % transferrin saturation and increased TIBC.
[QCMLS 2021: 4.7.1.1.1] [Rodak 2020, p268-269]

**296.** **c** Hemochromatosis is iron overload. It can be primary, as in hereditary hemochromatosis, or secondary, due to chronic anemia or treatments such as multiple transfusions. Secondary hemochromatosis is associated with anemia and abnormally high serum iron studies. Prussian blue is the stain for iron.
[QCMLS 2021: 2.10.2.1.3] [Rodak 2020, p274-275]

**297.** **c** The Kleihauer-Betke procedure is commonly used as a screening test to determine the amount of fetal blood that has mixed with maternal blood. Blood smears are immersed in an acid buffer, which causes HgA to be eluted form the RBCs. The smear is stained and the RBCs that contain fetal hemoglobin take up the stain. This stain can be used in assessing hereditary persistence of fetal hemoglobin (HPFH), fetal maternal hemorrhage, and beta thalassemia.
[QCMLS 2021: 4.10.10.3] [Rodak 2020, p858]

**298.** **b** Siderocytes (and sideroblasts) may be identified with the Prussian blue iron stain, which stains iron deposits. Siderocytes are RBCs that have no nuclei but contain iron. When the iron surrounds the nucleus of immature RBCs, the cells are called ringed sideroblasts.
[Rodak 2020, p273]

**299.** **d** The Prussian blue stain is used for assessing iron stores in bone marrow.Prussian blue is considered the gold standard for the assessment of iron stores.
[QCMLS 2021 4.10.5.2] [Rodak 2020, p111]

**300.** **b** The Prussian blue stain is used to differentiate siderotic (iron) granules (Pappenheimer bodies) from basophilic stippling. The term "Pappenheimer" describes these granules when viewed in RBCs with a Wright stain. The term "siderocyte" describes the granules in RBCs when Prussian blue is used. Pappenheimer bodies are composed of iron; basophilic stipping is composed of precipitated RNA.
[QCMLS 2021: t4.18] [Rodak 2020, p256 t18.3]

**301.** **a** Myeloperoxidase (MPO) is an enzyme found in the primary granules of granulocytic cells. Lymphocytes do not exhibit MPO. The stain is used to differentiate AML from ALL.
[QCMLS 2021: 4.10.10.1] [Rodak 2020, p549]

**302.** **b** Esterases are used to differentiate myeloblasts from cells of monocytic origin. Alpha-naphthyl acetate and alpha-naphthyl butyrate are nonspecific esterases. Naphthol AS-D chloroacetate is a specific esterase, which means that only granulocytic cells show staining. The nonspecific esterase stains monocytic cells, while granulocytes are negative.
[QCMLS 2021: 4.10.10.1] [Rodak 2020, p550]

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303. **c** Myeloperoxidase is positive, along with chloroacetate, both of which are granulocytic markers. Alpha-naphthyl acetate is also positive and indicates cells of monocytic origin are present. Blasts that are positive for all the stains suggest the presence of both myeloblasts and monoblasts, or the diagnosis of a myelomonocytic leukemia.
[QCMLS 2021: 4.10.10.1] [Rodak 2020, p549-551]

304. **a** Myeloperoxidase (MPO) is an enzyme found in the primary granules of granulocytic cells. Lymphocytes do not exhibit MPO. The stain is used to differentiate AML from ALL.
[QCMLS 2021: 4.10.10.1] [Rodak 2020, p549]

*H. Other Studies*

305. **d** The composition of Heinz bodies is denatured hemoglobin; these inclusions and are not seen with Wright stain. They are associated with a G6PD deficiency.
[QCMLS 2021: 4.10.10.4] [Rodak 2020, p256]

306. **b** Specimen handling and technical factors can affect the ESR. A clotted specimen and delay in testing will decrease the ESR.
[Rodak 2020, p168-169]

307. **a** A falsely-elevated ESR can be caused by high room temperature, tilting of the tube, and vibration.
[Rodak 2020, p168-169]

*I. Flow Cytometry Immunophenotyping*

308. **c** PNH is an acquired clonal hematopoietic stem cell disorder; it results in a rare, chronic, intravascular hemolytic anemia. Blood cell membranes are missing GPI-anchored proteins designated CD55 and CD59. Absence of these proteins causes the RBCs to be lysed by complement. Inmmunophenotyping is the confirmatory testing method and typically uses fluorescent monoclonal antibodies to demonstrate the GPI-anchored proteins.
[QCMLS 2021: 4.7.1.3.1.1] [Rodak 2020, p347-354]

309. **d** Blood cell membranes are missing GPI-anchored proteins, ie, CD55 and CD59. Immunophenotyping by flow cytometry shows diminished staining for CD55.
[QCMLS 2021: 4.7.1.3.1.1] [Rodak 2020, p347-354]

310. **d** Terminal deoxynucleotidyl transferase (TdT) is a marker for precursor B and T-cell ALL.
[QCMLS 2021: 4.10.12] [Rodak 2020, p483]

311. **a** CD surface markers that identify mature (post-thymic) T cells include CD5.
[QCMLS 2021: 4.10.12] [Rodak 2020, p483]

312. **c** T-cell markers include CD34, CD2, CD3, CD4, CD5, CD7, CD8 and TDT. B-cell markers include CD34, CD10, CD19, HLA-DR, TDT, and surface immunoglobulin.
[QCMLS 2021: 4.10.12] [Rodak 2020, p347-354]

313. **d** CD138, or kappa and lambda clonal excess, is useful in delineating malignant plasma cells from a reactive process. Multiple myeloma is a bone marrow based disease in which plasma cells are found in the marrow. This disorder also presents with a monoclonial protein or M spike, representing the immunoglobin portion of protein electrophoresis.
[QCMLS 2021: 4.9.6.2.4] [Rodak 2020, p619-620]

314. **a** Hairy cells are mature B lymphocytes that are positive for CD markers CD19, CD20, CD22, CD25, CD11c, CD103, surface immunoglobulin and TRAP. CD5 is a T-cell marker.
[QCMLS 2021: 4.9.6.1.3] [Rodak 2020, p611-612]

315. **d** CD117 is one of several markers that will be positive in most cases of AML. Others include CD13, CD15, CD33 and CD64.
[QCMLS 2021: 4.9.1.2, (4.9)] [Rodak 2020, p483]

316. **d** Clusters of differentiation (CD) markers are cellular antigens. Most are determined by flow cytometry and are instrumental in determining the particular cell's lineage in a blood disorder.
[QCMLS 2021: 4.10.12] [McKenzie 2020, p998]

*J. Molecular and Cytogenetic Testing*

317. **d** The specific *JAK2* mutation is detected in 95% of patients with polycythemia vera (PV), which is a myeloproliferative neoplasm with elevations seen in all 3 cell lines.
[QCMLS 2021: t4.11] [Rodak 2020, p565-572]

318. **c** The t(8;14) typically found in Burkitt leukemia brings the master cell cycle control factor *c-myc* on chromosome 8 under the influence of the strong immunoglobulin heavy chain promoter on chromosome 14, driving a leukemogenic process.
[QCMLS 2021: 4.9.7.2]

319. **a** The *BCR/ABL* gene rearrangement is seen in CML. Use of tyrosine kinase inhibitors results in prolonged survival in most patients with this leukemia and is the first treatment option, except in pregnant patients.
[QCMLS 4.9.3.1.2.1; t4.11] [McKenzie 2020, p528]

320. **d** Disseminated intravascular coagulation (DIC) is often the presenting condition in patients eventually diagnosed with the t(15;17) translocation. This AML is also known as acute promyelocytic leukemia. Most patients respond well to targeted therapy with all-trans-retinoic acid (ATRA).
[QCMLS 4.9.1.2; t4.9]



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Chapter 5

# Hemostasis

The following items have been identified generally as appropriate for those preparing for both the MLS and MLT examinations. Items that are appropriate for the MLS examination **only** are marked with MLS ONLY.


**365 Questions**
365 I. Physiology
369 II. Disease States
381 III. Laboratory Testing

**394 Answers with Explanations & Citations**
394 I. Physiology
396 II. Disease States
403 III. Laboratory Testing


## I. Physiology

1. Vasoconstriction is caused by several regulatory molecules, which include:

   **a** fibrinogen and vWF
   **b** ADP and EPI
   **c** thromboxane A2 and serotonin
   **d** collagen and actomyosin

2. Warfarin is classified as a vitamin K antagonist. The factors that are impacted by warfarin therapy are:

   **a** VIII, IX and X
   **b** I, II, V and VII
   **c** II, VII, IX and X
   **d** II, V and VII

3. When a patient is placed on warfarin therapy, the first factor that will be decreased is:

   **a** factor II
   **b** factor V
   **c** factor VII
   **d** factor VIII

4. A patient with a positive family history of bleeding presents to the ED with the results shown in this table:

| test | result | reference range |
|---|---|---|
| aPTT | 29.5 seconds | 25-35 seconds |
| PT | 19.2 seconds | 10.5-12.5 seconds |

   The patient is not on any medication and so is likely to be deficient in factor:

   **a** II
   **b** V
   **c** VII
   **d** VIII

5. Hageman factor (XII) is involved in each of the following reactions **except**:

   **a** activation of C1 to C1 esterase
   **b** activation of plasminogen
   **c** activation of factor XI
   **d** transformation of fibrinogen to fibrin

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6. The most potent plasminogen activator in the contact phase of coagulation is:

**a** kallikrein
**b** streptokinase
**c** HMWK
**d** fibrinogen

7. The activation of plasminogen to plasmin resulting in the degradation of fibrin occurs by:

MLS ONLY

**a** PAI-1
**b** alpha-2 antiplasmin
**c** tPA
**d** alpha-2 macroglobulin

8. How does tissue factor pathway inhibitor inhibit coagulation?

**a** inhibition of tissue factor-factor VIIa-factor Xa complex
**b** conversion of thrombin to prothrombin
**c** uncoupling factor XIII dependent crosslinking of fibrin
**d** binding and hiding tissue factor on the endothelial surface

9. An inhibitor of plasmin activity is:

**a** tPA
**b** PAI-1
**c** alpha-2 antiplasmin
**d** plasminogen

10. Antithrombin inhibits factors:

**a** IIa and Xa
**b** Va and VIIIa
**c** VIIa and XIIa
**d** IXa and Va

11. The propagation phase of the *in vivo* coagulation model includes:

MLS ONLY

**a** a final burst of thrombin
**b** activation of platelets
**c** feedback mechanism to thrombin
**d** activation of factor Va and factor VIIIa

12. In the cell-based model of coagulation the intrinsic pathway operates on the:

MLS ONLY

**a** activated platelet surface to produce the burst of thrombin
**b** tissue factor bearing cell to initiate and amplify coagulation
**c** proteolytic digestion of fibrinogen
**d** activation of the contact pathway

13. A new oral anticoagulant apixaban has been given to a patient who was previously on warfarin. This drug directly inhibits:

MLS ONLY

**a** vitamin-K-dependent factors
**b** factor IIa
**c** factor Xa
**d** both factor IIa and Xa

14. The anticoagulant that directly inhibits thrombin is:

MLS ONLY

**a** LMWH
**b** argatroban
**c** warfarin
**d** rivorxaban


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15. MLS ONLY A patient who presents with renal impairment is being started on oral anticoagulant therapy. The DOAC that should be avoided would be:

a dabigatran
b apixaban
c rivaroxaban
d warfarin

16. In secondary hemostasis, coagulation proteins become activated to form a fibrin clot. Prior to these proteins being activated, their inactive proenzymes are known as:

a serine proteases
b cofactors
c zymogens
d substrates

17. Alpha granules are found on the platelet in the:

a peripheral zone
b sol gel zone
c organelle zone
d membranes

18. Which of the following best represents the 3 steps of normal hemostasis (in order)?

a decreased heart rate, adhesion of platelets, plug formation
b platelet aggregation, formation of FXIII, fibrin plug
c vasoconstriction, platelet aggregation, fibrin formation
d vascular damage, stasis, endothelial injury

19. Which of the following platelet antigens is the receptor for collagen?

a GPIb/V/IX complex
b GPIIb/IIIa complex
c GPIa/IIa complex
d GPIV/X complex

20. Which platelet surface antigen acts as the receptor for fibrinogen?

a GPIb/V/IX
b GPIIb/IIIa
c GPIa/IIa
d GPIc/IIa

21. How does GPIb become activated *in vivo* and *in vitro*, respectively?

a shear force, ristocetin
b ristocetin, compression
c activation of ADP receptor, ristocetin
d binding vWF, epinephrine

22. A patient is on aspirin 100 mg/day to prevent the formation of clots caused by platelets. The mechanism in which aspirin impairs platelet function is by:

a inactivating cyclooxygenase which blocks thromboxane A2
b impairs vWF by via GPIb/IX/V receptor
c decreased amounts of arachidonic acid
d inactivation of ADP and phospholipase A2

23. A patient is diagnosed with a factor V Leiden mutation. The factor V activity level should be:

a shortened
b prolonged
c undetectable
d within reference range

24. The International Sensitivity Index (ISI) is determined by comparing the PT results of the manufacturer's reagent against:

a rabbit thromboplastin
b micronized silica
c tissue factor
d human brain thromboplastin

25. A coagulation reagent should be sensitive to factor deficiencies. A reagent is considered sensitive to a factor when the level of factor activity is approximately:

a 30%
b 40%
c 50%
d 60%

26. What factors are considered heat labile?

a II and IX
b V and VIII
c VII and XI
d X and XII

27. What factor serves as a carrier for factor VIII?

a factor V
b factor IX
c tissue factor
d von Willebrand factor

28. Patients may present with either a bleeding or a thrombotic event in:

a dysfibrinogenemia
b hypofibrinogenemia
c afibrinogenemia
d hyperfibrinogenemia

29. The key enzyme of the fibrinolytic system is:

a plasmin
b thrombin
c urokinase
d streptokinase

30. The major serine protease responsible for clot breakdown is:

a TPA
b alpha 2 antiplasmin
c streptokinase
d PAI-1

31. What subendothelial structural protein triggers coagulation through activation of FVII?

a thrombomodulin
b nitric oxide
c tissue factor
d silica

32. Protein C and its cofactor protein S proteolytically inactivate factors:

a IIa and Xa
b Va and VIIIa
c VIIIa and IXa
d XIa and XIIa


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## II. Disease States

33. Arterial thrombosis is caused by:

a RBC & platelets
b fibrin & WBC
c thrombin & FXIII
d WBC & platelets

34. The 2 factors that differentiate liver disease from vitamin K deficiency are:

a II and VII
b IX and VII
c VIII and IX
d V and VII

35. In a patient diagnosed with liver disease, which one of the following factors typically shows an increase?

a factor VII
b factor VIII
c factor IX
d factor X

36. A 4-year-old boy presents with chronic ear infections and is on prophylactic antibiotics. He also presents with a bleeding diathesis. Factor assay results are shown in this table:

| test | patient result | reference range |
|---|---|---|
| factor VIII | 100% | 50-150% |
| factor V | 75% | 50-150% |
| factor IX | 38% | 50-150% |
| factor II | 22% | 50-150% |

Possible causes are:

a factor II deficiency
b lupus anticoagulant
c hemophilia
d vitamin K deficiency

37. A hemophiliac male and a normal female can produce a:

a female carrier
b male carrier
c male hemophiliac
d normal female

38. Hemophilia B is a sex-linked recessive disorder that presents with a decrease in factor:

MLS ONLY

a VIII
b IX
c X
d XI

39. MLS ONLY To distinguish between hemophilia and von Willebrand disease, a patient with von Willebrand will present with which of the following test results?

| results | aPTT | platelet screen | ristocetin cofactor |
|---|---|---|---|
| a | abnormal | normal | normal |
| b | normal | abnormal | normal |
| c | abnormal | abnormal | abnormal |
| d | normal | normal | abnormal |

**a** result a
**b** result b
**c** result c
**d** result d

40. A patient presents with bleeding 48 hours after tooth extraction. Results are shown in this table:

| test | patient result | reference range |
|---|---|---|
| PT | 11.5 seconds | 10-13 seconds |
| aPTT | 32.5 seconds | 23-35 seconds |
| fibrinogen | 345 mg/dL (3.45 g/L) | 200-400 mg/dL (2.0-4.0 g/L) |
| platelets | 324 × $10^3$/µL (324 × $10^9$/L) | 150-450 × $10^3$/µL (150-450 × $10^9$/L) |

The cause of bleeding is most likely a deficiency in:

**a** plasminogen
**b** factor XIII
**c** alpha-2 antiplasmin
**d** factor XII

41. Acute disseminated intravascular coagulation is characterized by:

**a** hypofibrinogenemia
**b** thrombocytosis
**c** negative D-dimer
**d** shortened thrombin time

42. MLS ONLY A patient develops unexpected bleeding and the test results shown in this table are obtained:

| test | result |
|---|---|
| PT and aPTT | prolonged |
| fibrinogen | decreased |
| D-dimer | increased |
| platelets | decreased |

What is the most probable cause of these results?

**a** familial afibrinogenemia
**b** primary fibrinolysis
**c** DIC
**d** liver disease

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43. A patient presents with the results shown in this table:

| test | result | reference range |
|---|---|---|
| thrombin time | 48 seconds | 12-21 seconds |
| reptilase time | 38 seconds | 14-22 seconds |

These results are characteristic of:

**a** dysfibrinogenemia
**b** increased D-dimer
**c** fibrin monomer-split product complexes
**d** therapeutic heparinization

44. Only an abnormal aPTT would be seen in the following disorder:

MLS ONLY

**a** deficiencies of factors X, V, and II
**b** disseminated intravascular coagulation
**c** liver disease
**d** anti-FVIII antibodies

45. What is usually the factor VIII level in a hemophiliac patient with spontaneous bleeding?

**a** <1%
**b** 5-10%
**c** 20-30%
**d** 50-60%

46. What is the most common presentation of factor XIII deficiency?

**a** clinically inapparent
**b** delayed bleeding tendency
**c** severe bleeding responsive to DDAVP
**d** severe bleeding not responsive to DDAVP

47. A patient is diagnosed with amyloidosis, they will be deficient in which of the following factors?

**a** factor II
**b** factor V
**c** factor VII
**d** factor X

48. Which of the following tests are the most beneficial tool to aid in the diagnosis of DIC?

**a** fibrinogen, aPTT, FDP
**b** FDP, aPTT, reptilase
**c** thrombin time, fibrinogen, aPTT
**d** thrombin time, D-dimer, fibrinogen

49. Results of a factor X assay are shown in this table:

MLS ONLY

| dilution | result |
|---|---|
| 1:10 | 50% |
| 1:20 | 77% |
| 1:40 | 127% |

These results indicate the presence of:

**a** factor deficiency
**b** clotted sample
**c** inhibitor
**d** bad draw

50. A patient that has a lupus anticoagulant may bleed due to:

MLS ONLY

a factor VIII deficiency
b drugs
c antibodies to prothrombin
d infection

51. Patients with factor XIII deficiency have:

a an abnormal aPTT
b delayed bleeding
c tissue damage
d joint bleeds

52. In hemolytic disease of the newborn, babies present with:

a hyperbilirubinemia
b thrombocytosis
c hyperalbuminemia
d leukocytosis

53. Occasional spontaneous bleeding may occur in a hemophiliac who is classified as:

a acquired
b mild
c moderate
d severe

54. In liver disease patients present with decreased synthesis of:

a common pathway factors
b intrinsic pathway factors
c fibrinogen pathway factors
d vitamin-K-dependent factors

55. Alloantibodies are found in:

MLS ONLY

a congenital hemophilia
b acquired hemophilia
c congenital von Willebrand disease
d acquired von Willebrand disease

56. When there is a disparity between the results of the fibrinogen antigen and the activity, the most likely diagnosis is:

a dysfibrinogenemia
b hypofibrinogenemia
c hyperfibrinogenemia
d afibrinogenemia

57. Bleeding doesn't correlate well with factor levels in a deficiency of:

MLS ONLY

a factor VIII
b factor IX
c factor XI
d factor VII

58. A newborn baby boy is known to have a homozygous protein C deficiency. This puts him at a risk for:

a DVT
b warfarin induced skin necrosis
c increased risk of thrombosis
d purpura fulminans


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59. Protein S forms a reversible complex with:

a C4b binding protein
b protein C
c total protein S
d Fc receptors

60. MLS ONLY A 48-year-old male is screened pre-operatively. He has a positive family history for bleeding. The patient is of Ashkenazi Jewish descent. His results are as follows:

| test | result | reference range |
|---|---|---|
| PT | 11.5 seconds | 10.5-13.5 seconds |
| aPTT | 45.1 seconds | 25-35 seconds |
| 1:1 mixing study | patient = 28.1 seconds | pooled normal plasma = 29 seconds |

Based on this history and the results of these tests, this patient's most likely diagnosis is a deficiency in factor:

a VIII
b IX
c XI
d XII

61. Unregulated and excessive formation of thrombin and plasmin is seen in:

a liver disease
b vitamin K deficiency
c hemolytic disease of the newborn
d disseminated intravascular coagulation

62. MLS ONLY A patient presents with a factor VIII level of 2%. The vWF activity (ristocetin cofactor) is <1% with a vWF antigen of 3%. The most likely diagnosis is:

a hemophilia A
b hemophilia B
c type II vWD
d type III vWD

63. A patient with a clot presents with the results shown in this table:

| test | result | reference range |
|---|---|---|
| PT | 15.5 seconds | 10.5-12.5 seconds |
| aPTT | 50.3 seconds | 24-34.0 seconds |
| D-dimer | 1.62 mg/L | <1.10 mg/L |
| fibrinogen | 35 mg/dL | 180-400 mg/dL |
| fibrinogen antigen | 268 mg/dL | 150-350 mg/dL |

The most likely diagnosis is:

a DIC
b hypofibrinogenemia
c dysfibrinogenemia
d afibrinogenemia

64. The following results are obtained on a pregnant woman who is short of breath.

| test | result | reference range |
|---|---|---|
| protein C activity | 62% | 70-120% |
| protein S activity | 50% | 62-145% |
| vWF activity | 199% | >50% |
| vWF antigen | 192% | 50-150% |
| factor VIII | 189% | 50-150% |

The patient's obstetrician calls because of concern about the patient's clotting risk. Based on the pregnancy status of this patient, the results appear:

**a** elevated
**b** that the patient is at risk for bleeding
**c** that the patient is at risk for clotting
**d** normal

65. A 65-year-old male with metastatic pancreatic carcinoma shows elevated PT and aPTT, platelet count 15 × $10^3$/μL (15 × $10^9$/L) and elevated D-dimer. A blood smear would show:

**a** Howell-Jolly bodies
**b** macro-ovalocytes
**c** schistocytes
**d** target cells

66. A 25-year-old male with celiac disease presents with occult positive stools. What vitamin deficiency should be considered?

**a** vitamin A deficiency
**b** vitamin D deficiency
**c** vitamin E deficiency
**d** vitamin K deficiency

67. vWF antigen can be found in which of the following?

**a** myeloblasts
**b** monoblasts
**c** lymphoblasts
**d** megakaryocytes

68. The life span of a platelet is:

**a** 5 days
**b** 10 days
**c** 20 days
**d** 30 days

69. Aspirin affects platelet function by interfering with platelet metabolism of:

MLS ONLY

**a** cyclooxygenase
**b** lipids
**c** carbohydrates
**d** nucleic acids

70. In patients who present with bleeding disorders caused by platelets, the most common type of bleeding is:

**a** mucosal bleeding
**b** hemarthrosis
**c** delayed bleeding
**d** deep hematomas


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71. The anticoagulant that is best directed against platelets is:

MLS ONLY

**a** argatroban
**b** hirudin
**c** tirofiban
**d** dabigatran

72. How do ticlopidine and clopidogrel inhibit platelets?

MLS ONLY

**a** binding von Willebrand factor
**b** ADP mediated platelet aggregation
**c** inhibit GPIIb/IIIa
**d** depletion of platelet alpha granule content

73. von Willebrand factor mediates platelet adhesion by binding to platelet receptor:

**a** GPIb/IIa
**b** GPIb/GPIX/GPV
**c** GPIIb/IIa
**d** GPIb/GPIIIa/GP X

74. The disease state that presents with a quantitative platelet disorder is:

MLS ONLY

**a** von Willebrand disease
**b** hemophilia A
**c** Glanzmann thrombasthenia
**d** May-Hegglin anomaly

75. A patient presents with a very low platelet count and is diagnosed with acute idiopathic thrombocytopenic purpura (ITP). Which statement is associated with acute ITP?

MLS ONLY

**a** it is found primarily in adults
**b** spontaneous remission usually occurs within several weeks
**c** women are more commonly affected
**d** peripheral destruction of platelets is decreased

76. The most common cause of bleeding in patients is:

**a** qualitative platelet defect
**b** qualitative abnormality of fibrinogen
**c** quantitative abnormality of fibrinogen
**d** quantitative abnormality of platelets

77. A 53-year-old man is in recovery following a triple bypass operation. Oozing is noted from his surgical wound. The laboratory data shown in this table are obtained:

MLS ONLY

| test | result |
|---|---|
| hemoglobin | 12.5 g/dL (125 g/L) |
| hematocrit | 37% |
| prothrombin time | 12.3 seconds |
| aPTT | 34 seconds |
| platelet count | 40.0 × $10^3$/µL (40.0 × $10^9$/L) |
| fibrinogen | 250 mg/dL (2.5 g/L) |

The most likely cause of bleeding would be:

**a** dilution of coagulation factors due to massive transfusion
**b** intravascular coagulation secondary to microaggregates
**c** hypofibrinogenemia
**d** dilutional thrombocytopenia

78. ADAMTS13 deficiency is responsible for thrombocytopenia found in:
MLS ONLY

**a** TTP
**b** DIC
**c** HUS
**d** ITP

79. Heparin induced thrombocytopenia (HIT) is an immune mediated complication associated with heparin therapy. Antibodies are produced against:
MLS ONLY

**a** ACLA
**b** PF4
**c** AT
**d** B2GP1

80. In polycythemia vera, the platelet count is:

**a** elevated
**b** normal
**c** decreased
**d** variable

81. A 60-year-old man has a painful right knee and a slightly enlarged spleen. Hematology results include:
MLS ONLY

| test | result |
|---|---|
| hemoglobin | 15 g/dL (150 g/L) |
| absolute neutrophil count | $10.0 \times 10^3/\mu L$ ($10.0 \times 10^9/L$) |
| platelet count | $900 \times 10^3/\mu L$ ($900 \times 10^9/L$) |
| uncorrected retic count | 1% |
| morphology | normal red cell morphology and indices<br>a slight increase in bands<br>rare metamyelocyte and myelocyte<br>giant and bizarre-shaped platelets |

This is most compatible with:

**a** congenital spherocytosis
**b** rheumatoid arthritis with reactive thrombocytosis
**c** myelofibrosis
**d** idiopathic thrombocythemia (essential or primary)

82. The type of bleeding that is the most characteristic in patients with platelet disorders is:

**a** deep muscle hemorrhages
**b** retroperitoneal hemorrhages
**c** mucous membrane hemorrhages
**d** severely prolonged clotting times

83. When reviewing platelet morphology on a blood smear, the presence of giant platelets is noted. This may indicate:

**a** Bernard-Soulier
**b** von Willebrand Disease
**c** Glanzmann thrombasthemia
**d** Ehler-Danlos


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84. A patient with multimer pattern that has all bands present but in decreased concentrations would be diagnosed with: MLS ONLY

a hemophilia
b type 1 vWD
c type 2 vWD
d type 3 vWD

85. Which characteristic of Bernard-Soulier syndrome helps distinguish it from von Willebrand disease?

a concomitant storage pool defects in platelets
b giant platelets
c thrombocytosis
d thrombocytopenia

86. Which of the following characteristics are common between Hĕrmanský-Pudlák and Chédiak-Higashi syndromes?

a giant inclusion granules in granulocytes
b alpha granule storage pool defects
c inclusions in macrophages
d oculocutaneous albinism

87. Patients with Wiscott-Aldrich syndrome present with:

a thrombocytosis, giant platelets
b thrombocythemia, eczema
c thrombocytopenia, giant platelets
d thrombocytopenia, eczema

88. When evaluating a patient for von Willebrand disorder, the ABO blood type that has the lowest level of vWF is:

a O
b A
c B
d AB

89. Which subtype of von Willebrand disease is the most common? MLS ONLY

a type 1
b type 2A
c type 2B
d type 3

90. Which of the following types of von Willebrand disease should not be treated with DDAVP? MLS ONLY

a type 1
b type 2A
c type 2B
d type 3

91. TTP presents with a pentad of symptoms that does not include:

a fever
b anemia
c thrombocytopenia
d liver failure

92. A 40-year-old female has pinpoint hemorrhages on her legs, with an Hct of 43% and a platelet count of 19 × $10^3$/μL (19 × $10^9$/L). Steroids fail to increase platelet count. The patient's diagnosis is:

**a** decreased platelet production
**b** suppressed pluripotent stem cells
**c** immune thrombocytopenic purpura
**d** defective platelet endothelial interaction

93. A 60-year-old female presents with blurred vision, confusion, and petechiae with a platelet count of 29 × $10^3$/μL (29 × $10^9$/L).

| test | result |
|---|---|
| PT | 12.1 seconds (11.4-13.5 seconds)) |
| aPTT | 32.6 seconds (25-35 seconds) |

The CBC RBC morphology shows schistocytes. The differential diagnosis should include:

**a** ITP
**b** TTP
**c** HUC
**d** DIC

94. One of the main difference between TTP and HUS is:

**a** neurological involvement
**b** kidney dysfunction
**c** thrombocytopenia
**d** microangiopathic hemolytic anemia

95. Several hours after birth an infant develops petechiae, purpuric hemorrhages and a platelet count of 21 × $10^3$/μL (21 × $10^9$/L). The most likely diagnosis is:

**a** drug induced immune thrombocytopenia
**b** thrombotic thrombocytopenic purpura
**c** autoimmune neonatal thrombocytopenia
**d** neonatal idiopathic thrombocytopenia

96. MLS ONLY In TTP, a deficiency of ADAMTS13 causes clotting because of:

**a** increased vW factor
**b** inability to cleave ULVWM
**c** increased production of thrombin
**d** inability to produce antibodies to prevent platelet aggregation

97. MLS ONLY An orthopedic patient who is on heparin has a platelet count of 50 × $10^3$/μL (50 × $10^9$/L); his platelet count the previous day was 120 × $10^3$/μL (120 × $10^9$/L). The patient is tested for HIT and has a positive result. The first step in the treatment of HIT is:

**a** start LMWH
**b** stop heparin
**c** switch to warfarin
**d** give platelets

98. In HELLP syndrome patients present with:

**a** normal liver enzymes
**b** elevated platelets
**c** high blood pressure
**d** hemochromatosis

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99. von Willebrand factor serves as a stabilizer for:

MLS ONLY

a platelets
b factor IX
c ristocetin
d factor VIII

100. The platelet disorder in which the abnormality is due to a defect in platelet aggregation is:

MLS ONLY

a Glanzmann thrombasthenia
b von Willebrand disease
c storage pool disease
d Bernard-Soulier syndrome

101. Large blue inclusions bodies in WBC are found in patients with:

a Wiskott-Aldrich syndrome
b May-Hegglin Anomaly
c Ehlers-Danlos syndrome
d Hĕrmanský-Pudlák syndrome

102. An initial prenatal evaluation at 10 weeks is conducted on a healthy female. PT, aPTT and fibrinogen are normal with the exception of her platelet count, which is 42,000/mL. The peripheral smear shows platelet clumps. What is the next step?

a conduct a workup for an autoimmune disease
b evaluate for risk of thromboembolism
c recollect the sample in EDTA
d recollect the sample in sodium citrate

103. The initial platelet adhesion to subendothelial matrix proteins is abnormal in:

a von Willebrand disease and Bernard-Soulier syndrome
b hemophilia A or B
c GPIIb/IIIa inhibitors
d factor XIII deficiency

104. A 2-year-old child presents with recurrent epistaxis, gum bleeding and occasional hematuria, upon physical exam showed petechiae and ecchymoses. The patient is not on any medication. PT and aPTT and platelet count are normal. Platelet function screen tests are abnormal, and a platelet aggregation test is ordered. Results are:

absent aggregation with ADP, collagen, epinephrine
ristocetin inducted platelet aggregation: within reference range

The differential diagnosis should include:

a Bernard-Soulier
b Glanzmann thrombasthenia
c von Willebrand disease
d grey platelet syndrome

105. Quantitative platelet disorders can be due to:

a deficient platelet production
b abnormal platelet distribution
c increased destruction
d abnormal structure or function

106. Post-transfusion purpura is a rare form of:

a autoimmune disorder
b qualitative platelet disorder
c alloimmune thrombocytopenia
d antibody platelet destruction

107. Disorders of platelet aggregation are found in:

a von Willebrand disease
b Bernard-Soulier
c Glanzmann thrombasthenia
d storage pool deficiency

108. Spontaneous bleeding occurs when the platelet count is:

a $<10 \times 10^3/\mu L$
b $<20 \times 10^3/\mu L$
c $<50 \times 10^3/\mu L$
d $<100 \times 10^3/\mu L$

109. In hemolytic uremic syndrome, patients present with:

a severe CNS symptoms
b fever
c acute renal failure
d decreased ADAMTS 13

110. A 65-year-old male has a valve replacement, and 5 days postoperatively his platelet count had fallen to 80,000 after an initial count of 320,000. Also noted is swelling of his right calf. The following day the platelet count is 30,000. What should be investigated, and what therapeutic step is appropriate?

a platelet disorder; administer platelets
b postoperative bleeding; administer plasma
c HIT; administer direct thrombin inhibitor
d HIT; administer LMWH

111. In renal dysfunction uremia can cause:

a thrombocytosis
b thrombocytopenia
c neutropenia
d nucleocytosis

112. A 70-year-old female with metastatic ovarian carcinoma shows elevated PT and aPTT, platelets $15,000/mm^3$ and an elevated D-dimer. The differential diagnosis should include:

a TTP
b APS
c DIC
d HUS

113. A patient presents with a platelet count of $223 \times 10^9/L$ with an abnormal PFA most possibly suggest:

a decreased platelet production
b defective platelet function
c increased platelet production
d increased platelet destruction

114. Which of the following is not an inherited vascular defect?

a Ehlers Danlos syndrome
b Marfan syndrome
c May-Hegglin disorder
d Osler-Weber-Rendu disease


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## *III. Laboratory Determinations*

**115.** MLS ONLY A patient is placed on clopidogrel. The clinician wants to determine if the dose is sufficient to impair platelet function. A platelet aggregation test is ordered. The agonist which would result in a decreased aggregation pattern would be:

**a** collagen
**b** epinephrine
**c** ristocetin
**d** adenosine disphosphate

**116.** MLS ONLY A platelet aggregation tracing appears to confirm the diagnosis of Glanzmann thrombasthenia in a patient presenting with a platelet disorder. How would these tracings look if they were performed using light transmittance optical density aggregation?

**a** decreased platelet aggregation to ristocetin
**b** increased platelet aggregation to ristocetin
**c** normal response for all agonists
**d** markedly decreased aggregation to epinephrine, ADP and collagen

**117.** A patient has been taking aspirin regularly for arthritic pain. Which one of the following tests is most likely to be abnormal in this patient?

**a** platelet count
**b** PFA-100
**c** prothrombin time
**d** activated partial thromboplastin time

**118.** MLS ONLY Which of the following platelet responses is most likely associated with type IIb von Willebrand disease?

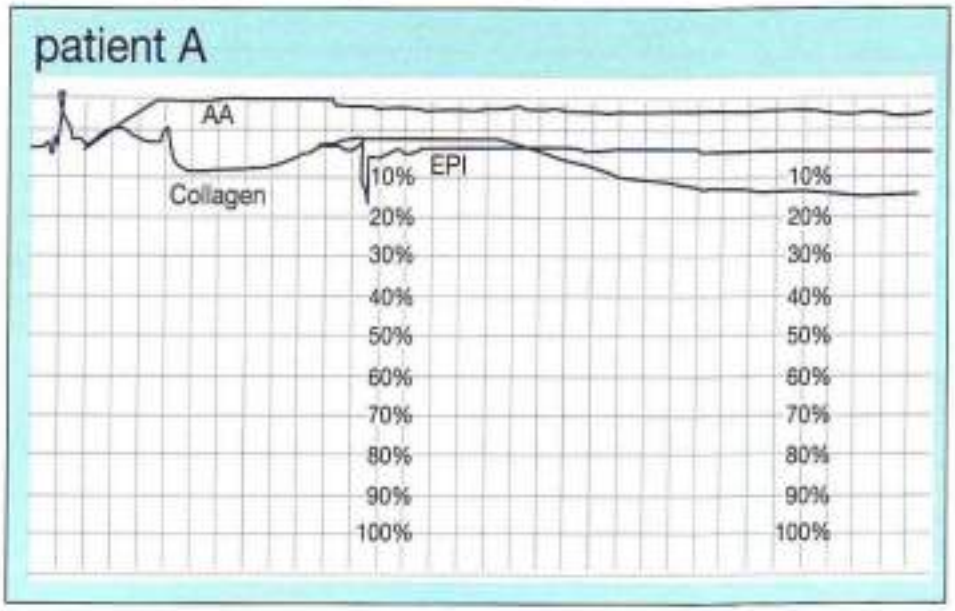
**a** decreased platelet aggregation to low-dose ristocetin
**b** normal platelet aggregation to low-dose ristocetin
**c** increased aggregation to low-dose ristocetin
**d** decreased aggregation to high-dose ristocetin

**119.** MLS ONLY Platelet aggregation will occur with the end production of:

**a** cyclooxygenase
**b** arachidonic acid
**c** prostacyclin
**d** thromboxane $A_2$

120. The platelet aggregation tracing shown in this figure represents:

MLS ONLY




a von Willebrand disease
b storage pool disease
c Glanzmann thrombastheniaww
d aspirin

121. Which of the following will not affect the PFA-100 closure time?

a aspirin
b thrombocytopenia
c anemia
d leukopenia

122. What does the secondary wave of platelet aggregation seen with the biphasic low-dose ADP and epinephrine response represent?

MLS ONLY

a increased binding to collagen
b release of platelet granules
c increased activation by collagen
d formation of fibrin dimers

123. Which of the following tests provides an appropriate laboratory confirmation of immune mediated heparin induced thrombocytopenia (type 2)?

MLS ONLY

a anti-PF3 antibody
b serotonin release assay
c ristocetin induced platelet aggregation assay
d reptilase assay

124. The principle of platelet aggregation is based on:

MLS ONLY

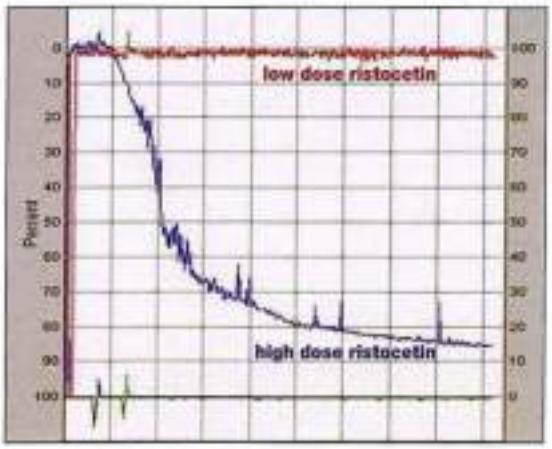
a decreased light transmission
b increased light transmission
c decreased light absorbance
d increased light absorbance


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125. Whole blood evaluation of blood clotting can be performed by using thromboelastography. MLS ONLY This methodology looks at:

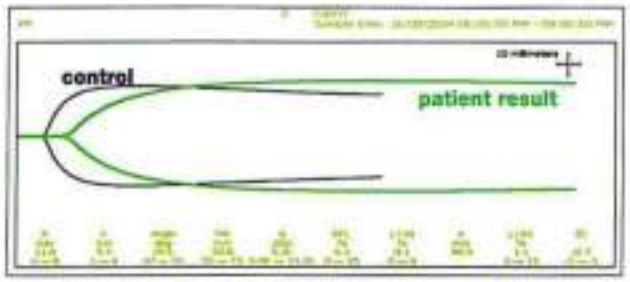
a clot onset, agglutination, fibrinolysis
b clot onset, aggregation, clot strength
c aggregation, clot onset, clot strength
d clot onset, clot strength, fibrinolysis

126. The tracing shown in this figure represents a patient with: MLS ONLY




a type 1 vW disease
b type 2 vW disease
c type 3 vW disease
d no vW disease

127. The tracing in this figure is from a cardiac surgery patient using thromboelastography. MLS ONLY This tracing shows the patient during cardiac surgery is:




a under-anticoagulated
b sufficiently anticoagulated
c over-anticoagulated
d normal

128. A patient sample has been tested in the routine laboratory with a PT and an aPTT. Six hours later factor assays are ordered on the same sample. Which factor is the most labile?

a II
b VII
c VIII
d X

129. A patient presents with an aPTT of 62.5 seconds (25-35 seconds) and the only factor that is decreased is factor XII. What is the clinical picture for this patient?

a negative bleeding history
b prolonged PFA
c decreased risk of thrombosis
d epistaxis

**130.** The results from a patient are shown in this table:

| test | patient result | reference range |
|---|---|---|
| PT | 18.5 seconds | 11.0-13.5 seconds |
| aPTT | 47.5 seconds | 24-35 seconds |
| thrombin time | 14.0 seconds | 12-19 seconds |
| ATIII | 82% | 70-130% |
| protein C | 54% | 77-167% |
| protein S | 48% | 65-140% |
| activated protein C resistance | 2.6 | >2.1 |

These results reflect:

**a** thrombophilia
**b** factor IX deficiency
**c** heparin
**d** warfarin

**131.** A deficiency of protein C is associated with which of the following?

MLS ONLY

**a** prolonged aPTT
**b** decreased fibrinogen level (<100 mg/dL [<1.0 g/L])
**c** increased risk of thrombosis
**d** spontaneous hemorrhage

**132.** Biological assays for antithrombin (AT) are based on the inhibition of:

MLS ONLY

**a** factor VIII
**b** heparin
**c** serine proteases
**d** anti-AT globulin

**133.** A patient presents with a low protein S activity and low total and free protein S antigens. The C4b binding protein is normal. This is classified as:

MLS ONLY

**a** no deficiency
**b** type I
**c** type II
**d** type III

**134.** APC resistance is confirmed by the molecular test for:

MLS ONLY

**a** PAI 1 4G/5G
**b** MTHFR
**c** FVL
**d** G20210A

**135.** The aPTT is a coagulation screening test that is also used to:

**a** evaluate the extrinsic coagulation pathway
**b** monitor Coumadin® therapy
**c** require tissue thromboplastin
**d** monitor unfractionated heparin therapy


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136. A patient presents with an aPTT of 49 seconds (25-35 seconds). A mixing study is performed with results as follows:

1:1 mix = 39.8 seconds
pooled normal plasma = 32.0 seconds
Incubated mix 1 hour = 39.0 seconds

The results indicate:

**a** circulating anticoagulant
**b** factor deficiency
**c** decreased platelets
**d** fibrinolysis

137. A patient is taking 10 mg per day of Coumadin® (warfarin). The results of which of the following laboratory tests will be most impacted?

**a** protein C
**b** antithrombin
**c** factor V Leiden
**d** factor VIII

138. The best test to determine if a sample contains residual heparin:

**a** fibrinogen
**b** thrombin time
**c** prothrombin time
**d** D-dimer

139. In the Clauss fibrinogen method, the time to clot formation in plasma is measured after the addition of:

**a** calcium
**b** thrombin
**c** phospholipids
**d** kaolin

140. An orthopedic patient is placed on low molecular weight heparin after back surgery, in order to determine if the level of LMWH is therapeutic the test to monitored levels would be:

**a** anti-Xa assay
**b** aPTT
**c** PT
**d** anti-IIa assay

141. When a patient presents with an elevated factor VIII level, what test will be impacted?

**a** prolonged PT
**b** shortened PT
**c** prolonged PTT
**d** shortened PTT

142. Activated protein C (APC) resistance is associated with a mutation in:

**a** thrombin
**b** factor V
**c** factor VIII
**d** protein C

143. What is the most common cause of inherited thrombophilia?

**a** antiphospholipid syndrome
**b** prothrombin G20210A mutation
**c** factor V Leiden
**d** hyperhomocysteinemia

144. Which of the following causes of thrombophilia most often presents as thrombotic episodes with resistance to heparin?

a protein C deficiency
b antithrombin deficiency
c prothrombin G20210A deficiency
d factor V Leiden

145. What are patients with homozygous protein C deficiency particularly at an increased risk of developing?

a posttransfusion purpura
b warfarin skin necrosis
c purpura fulminans
d thrombocytophilia

146. Which of the following proteins provides the binding site for heparin?

a protein C
b protein S
c factor X
d antithrombin

147. The DRVV screen test will be prolonged in a patient with lupus due to the reagent containing:

MLS ONLY
a decreased concentration of phospholipid
b increased concentration of phospholipid
c hexagonal phase phospholipids
d bilayer phospholipids

148. The measurement principle that uses a fixed concentration of factor Xa added to patient plasma and applied to a specific substrate is a:

a factor assay
b chromogenic assay
c ELISA assay
d immunologic assay

149. The test used to quantitate a factor inhibitor is the:

MLS ONLY
a factor assay
b multimer test
c ristocetin cofactor
d Bethesda assay

150. Excess D-dimers indicate that clots have been:

a converted to fibrin monomers
b released into the circulation
c stimulated to activate platelets
d formed and are being excessively lysed

151. In the Clauss fibrinogen assay the time for clot formation in plasma is measured after the addition of:

a calcium
b thrombin
c phospholipids
d fibrin

152. D-dimers are produced from:

a crosslinked and stabilized fibrin clots
b decreased fibrinogen and platelets
c plasminogen being converted to plasmin
d generation of thrombin from endothelial cells

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153. A patient presents with an aPTT of 24 seconds (25-35 seconds) which is below the normal range. A possible explanation for this result is the presence of a(n):

**a** DVT
**b** lupus anticoagulant
**c** acute phase reactant
**d** inhibitor

154. IMLS ONLY A patient has their coagulation blood sample drawn from a line. This is the only option for obtaining the sample. The PT is normal at 11.5 seconds, the aPTT is prolonged at 67 seconds, and the thrombin time is prolonged at 30 seconds. Based on this information how should testing proceed?

**a** factor assays testing should be performed
**b** a mixing study should be done
**c** a heparin neutralization should be done
**d** an inhibitor assay should be performed

155. A patient presents to the coumadin clinic with an INR of 3.1. He has a mechanical heart valve. The level of anticoagulant should be:

**a** decreased
**b** increased
**c** stopped
**d** not adjusted

156. A new PT reagent is being set up in the coagulation laboratory. The ISI of the new reagent is 1.0; the previous reagent had an ISI of 2.1. The new reagent is said to be:

**a** more sensitive
**b** less sensitive
**c** insensitive
**d** no change

157. A 42-year-old male presents with fatigue, difficulty breathing, and tingling in the hands and feet. The laboratory results shown in this table are obtained:

| test | result | reference range |
|---|---|---|
| PT | 25.6 seconds | 11.5-13.9 seconds |
| aPTT | 77.5 seconds | 25.0-35.7 seconds |
| Hgb | 23.5 g/dL | |
| Hct | 76.2% | |

The clinician questioned the results, as the patient had no bleeding symptoms. This result could be caused by:

**a** aplastic anemia
**b** polycythemia vera
**c** chronic lymphocytic leukemia
**d** sickle cell disorder

158. A sample is sent to the laboratory for an anti-Xa assay. The result of the PTT is 65.7 seconds. The result of the anti-Xa assay is 0.9 U/mL of heparin. The patient is on Lovenox®. Their anti-Xa level is:

**a** subtherapeutic
**b** therapeutic
**c** supratherapeutic
**d** prophylactic

**159.** MLS ONLY An aPTT result from the main laboratory is 67 seconds, and is frozen to be saved for additional lupus testing in the special coagulation laboratory. The aPTT repeated the next day is 37 seconds. This can be caused by:

**a** wrong sample saved
**b** factors were activated
**c** platelets contaminated sample
**d** different reagent sensitivity used

**160.** MLS ONLY A pediatric cardiac patient presents with a prolonged PT= 18.1 sec (11.5-13.5 sec) and an abnormal aPTT = 51 sec (25-35 sec) with a normal platelet count. Symptoms include bruising and nosebleeds. This is the patients third cardiac surgery, and previously all coagulation results have been normal. A mixing study does not correct for both the PT and aPTT. What should be considered next?

**a** lupus anticoagulant
**b** acquired factor inhibitor
**c** dysfibrinogenemia
**d** DIC evaluation

**161.** An 11-year-old female presents to the ED with hemarthrosis in her knee after an injury. This is the third incidence of hemarthrosis. Patient presents with a pronged PT of 27 seconds and a normal aPTT of 30 seconds, with a normal platelet count. The PT corrects to 13 seconds upon a mixing study. Possible diagnosis may be a deficiency of:

**a** factor II
**b** factor VII
**c** factor VIII
**d** factor IX

**162.** A 65-year-old multiple myeloma patient presents with bruising and hematuria. PT is normal and aPTT and thrombin time are prolonged and do not correct with a mixing study. Fibrinogen and reptilase times are normal. This may be caused by:

**a** heparin-like anticoagulant
**b** acquired vitamin K deficiency
**c** factor deficiency
**d** factor Inhibitor

**163.** MLS ONLY A 68-year-old male presents with tingling in hands and feet, an irregular heart beat, severe fatigue and difficulty in swallowing. Coagulation results show an abnormal PT and aPTT, which completely correct to normal with a mixing study. The factor most likely that contributes to this is a deficiency in:

**a** factor VII
**b** factor X
**c** factor VIII
**d** factor XIII


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**164.** A patient presents to his physician with excessive bleeding. He was previously on aspirin, but it was discontinued 3 weeks ago, but he was placed on antibiotics for a streptococcal infection. Coagulation results are as follows:

| test | result |
|---|---|
| PT | 16.7 |
| aPTT | 45.2 |
| mixing study | corrects to normal |
| platelet count | normal |
| platelet function screen | normal |

| factor assay results | (50-150%) |
|---|---|
| factor II | 32% |
| factor V | 102% |
| factor VII | 21% |
| factor VIII | 123% |
| factor IX | 41% |
| factor XI | 89% |

Bleeding may be due to:

**a** aspirin
**b** platelet dysfunction
**c** vitamin K deficiency
**d** liver disease

**165.** A 74-year-old male over time has had persistent prolonged aPTTs of 89, 85 and 95 seconds, which correct on a mixing study, suggesting a factor deficiency. He has had no history of bleeding despite several surgeries. He is most likely deficient in:

**a** factor VIII
**b** factor IX
**c** factor XI
**d** factor XII

**166.** A 58-year-old male presents with acute painful swelling in his right leg after a long airplane flight. He is diagnosed with proximal venous thrombosis and placed on low molecular heparin (LMWH). Three days post treatment his aPTT (reference range 25-35 seconds) is performed. The expected result would be:

**a** 22.1 seconds
**b** 30.0 seconds
**c** 41.0 seconds
**d** 68.0 seconds

**167.** A 38-year-old male has been referred for a consult due to siblings who have presented with multiple deev venous thromboses (DVTs) and pulmonary emboli (PEs). He is worked up with a hypercoagulable panel. Results are as follows:

| test | result | reference range |
|---|---|---|
| protein C activity | 123% | 80-120% |
| protein S activity | 99% | 55-145% |
| AT III | 110% | 70-120% |
| dilute Russel viper venom | negative | negative |

What test should be considered next:

**a** MTHFR
**b** prothrombin mutation
**c** factor V Leiden
**d** PAI-1 4G/5G

**168.** MLS ONLY When trying to determine if a patient has either hemophilia A or hemophilia B disease, a patient with hemophilia B will present with:

**a** normal aPTT, normal FVIII activity, normal factor IX activity
**b** prolonged aPTT abnormal FVIII activity, abnormal factor IX
**c** normal aPTT, abnormal FVIII activity, normal factor IX
**d** abnormal aPTT, normal FVIII activity, abnormal factor IX

**169.** A 67-year-old female is admitted to the ICU, with positive blood cultures. Screening tests are as follows:

| test | result | reference range |
|---|---|---|
| PT | 17.2 | 11.5-13.5 |
| aPTT | 38.1 | 25.5-35.5 |
| fibrinogen | 175 | 180-340 |
| D-dimer | 12.2 | <0.500 FEU |

The differential diagnosis should include:

**a** DVT
**b** DIC
**c** TTP
**d** APA

**170.** A pregnant women has routine coagulation testing performed. PT and aPTT are normal. Her fibrinogen level is elevated at 450 (reference range 150-350) due to acute phase reactants during pregnancy. The other factor that may be elevated due to this is:

**a** FVII
**b** FVIII
**c** FXI
**d** FXIII

**171.** If a patient is diagnosed as a severe hemophiliac, their FVIII level will be in the range of:

**a** <1%
**b** 1-5%
**c** 6-30%
**d** > 30%


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172. If a patient presents with a FVIII level of 2% and a ristocetin cofactor activity of <1% with a vW antigen of 3%, the most likely diagnosis is:

MLS ONLY

a type 1 vWD
b type 2 vWD
c type 2B vWD
d type 3 vWD

173. A patient is being worked up for a lupus anticoagulant and the physician requests a dilute Russel viper venom test (DRVV). The DRVV screen is 62 seconds and the confirm is 49 seconds. The final ratio is:

MLS ONLY

a 0.79
b 1.27
c 3.04
d 13

174. A patient is being worked up for a lupus anticoagulant. The DRVV screen is prolonged at 75 seconds with the pooled normal plasma at 35 seconds. The DRVV confirm is 38 seconds with the pooled normal plasma at 36 seconds. What is the normalized ratio?

MLS ONLY

a 1.97
b 0.51
c 0.97
d 2.03

175. A mixing study will remain uncorrected in a patient with:

a hemophilia A
b heparin
c HIT
d hemophilia B

176. The type of vWD that has enhanced ristocetin activity is:

MLS ONLY

a type 2A
b type 2B
c type 2M
d type 2N

177. Features of rare bleeding disorders include:

a autosomal dominant inheritance
b minimal bleeding outcomes
c commonly seen in consanguineous marriage
d occur in males only

178. When comparing acute DIC vs chronic DIC, acute DIC presents with:

a normal PT
b normal aPTT
c decreased D-dimer
d decreased platelets

179. A patient who has been on antibiotics presents with bleeding. This can be caused by:

a factor VIII deficiency
b vitamin E deficiency
c factor XIII deficiency
d vitamin K deficiency

180. Inhibitor development in hemophiliacs can occur after:

a exposure to factor concentrates
b joint bleed
c tissue damage
d exposure to platelet concentrates

181. Acquired hemophilia is distinguished from inherited hemophilia in that acquired hemophilia is:
MLS ONLY

a found in young people
b alloantibody
c autoimmune disease
d occurs in males

182. A 10-year-old male with history of bruising, abnormal joints, deep muscle bleeds. Platelets
MLS ONLY are normal, aPTT elevated. A mixing study is uncorrected. The differential diagnosis should include:

a DIC
b hemophilia B
c hemophilia A with factor inhibitor
d von Willebrand disease

183. An 80-year-old patient has been on long term warfarin therapy and presents with hematuria and a PT of 63 seconds. The quickest way to correct the PT is:

a vitamin K injection
b protamine sulfate injection
c fresh frozen plasma
d cryprecipitate

184. When screening for a lupus anticoagulant in the routine coagulation laboratory the best
MLS ONLY reagent would be:

a aPTT reagent with a high concentration of phospholipids
b aPTT reagent with a low concentration of phospholipids
c PT reagent with a high concentration of phospholipids
d PT reagent with a low concentration of phospholipids

185. Which of the following are characteristic of a factor XII deficiency?

a negative bleeding history
b negative screening test
c decreased risk of thrombosis
d decreased platelet count

186. A FXI assay gives different values at different dilutions. The results are as follows: 1:10 = 23%; 1:20= 42%, 1:140= 80%. This can be caused by a(n):

a clot
b factor deficiency
c inhibitor
d bad draw

187. A patient presents to the emergency department (ED) for a laceration to his arm that requires
MLS ONLY stitches. Procedure is uncomplicated and leaves the ED. He returns 15 hours later to the ED with excess bleeding. All PT, aPTT, fibrinogen and platelet counts are normal. The differential diagnosis should include:

a chronic DIC
b lupus
c factor XII deficiency
d factor XIII deficiency


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188. A patient that has a lupus anticoagulant may bleed due to:

a antibodies to prothrombin
b factor VII deficiency
c infection
d factor VIII deficiency

189. A type 1 AT deficiency patient will present with:

a normal antigen, decreased activity
b decreased antigen, normal activity
c decreased antigen, decreased activity
d normal antigen, normal activity

190. A patient is on unfractionated heparin after surgery and has an aPTT of 65 seconds, however the patient has developed a DVT. The best test to monitor this patient with is:

a factor X assay
b factor II assay
c anti-Xa assay
d antithrombin assay

191. An additional test to perform in a patient having a lupus workup after the DRVV would be:

a thrombin time
b hexagonal phase
c fibrinogen
d reptilase time

192. Patients diagnosed with hemophilia can now be prescribed:

a cryoprecipate
b direct thrombin inhibitors
c extended half-life products
d LMWH

193. MLS ONLY One of the coagulation test results that correlates with mortality in patients with the SARS-CoV-2 virus was:

a PT
b aPTT
c D-dimer
d thrombin time

194. MLS ONLY In the cellular based model of coagulation, what is true of the initiation phase?

a It occurs on a tissue factor bearing cell
b platelets and cofactors are activated
c large amounts of thrombin are generated
d fibrinogen is converted to fibrin

## I. Physiology

1. **c** Thromboxane A2 promotes platelet aggregation and vasoconstriction. Serotonin released from the dense bodies of platelets is a vasoconstrictor that binds endothelial cells and platelet membranes.
[QCMLS 2021: 5.2.1] [Rodak 2020, p143, 631]

2. **c** Warfarin interferes with the carboxylation of vitamin K factors by interrupting the enzymatic phase of the reaction. Factors are inhibited according to their half life, with VII having the shortest (6 hours) and II the longest (2-3 days).
[QCMLS 2021: 5.7.5.1.1] [Rodak 2020, p748-749]

3. **c** Factor VII (proconvertin) is a single-chain glycoprotein that is vitamin-K-dependent and remains stable for 6 hours in blood. Produced in the liver, it has the shortest half-life; therefore, it is the first factor affected when a vitamin K antagonist such as warfarin is administered. FII has the longest half-life (2-3) days and FVIII is not impacted by warfarin.
[QCMLS 2021: 5.7.5.1.1] [Rodak 2020, p748-749]

4. **c** The extrinsic pathway is initiated by the release of tissue thromboplastin that has been expressed after damage to a vessel. During this process factor VII forms a complex with tissue thromboplastin and calcium. The PT will be abnormal with deficiencies in the factors found in both the extrinsic (FVII) and common pathways (I, II, V and X) and the aPTT is abnormal when there are deficiencies in the intrinsic pathway (FVIII, FIX, FXI and FXII). Deficiencies in the intrinsic pathway prevent the activation of factor X in the common pathway resulting in a prolongation of the aPTT. Factors II and V are found in the common pathway, and factor VIII is in the intrinsic pathway. Since the aPTT is normal in this case, the likely deficiency is in factor VII.
[QCMLS 2021: 5.6.1.2] [Kottke-Marchant 2016, p9, 176]

5. **d** Factor XII is a contact factor, which is activated to factor XIIa. Factor XIIa is responsible for the activation of factor XI to factor XIa and the conversion of plasminogen to plasmin. Fibrinogen is converted to fibrin by the action of thrombin.
[QCMLS 2021: 5.4.3, t5.3] [Kottke-Marchant 2016, p10, 35]

6. **a** A clot is degraded by plasmin in the fibrinolytic system. Plasminogen is the zymogen produced when factor XIIa and kallikrein are produced by contact activation.
[QCMLS 2021: 5.4.4] [Kottke-Marchant 2016, p48]

7. **c** The dissolution of clots occurs several hours after the clot is formed and the key component is plasminogen. This is converted to plasmin through TPA. This substance is released as a result of the activity of endothelial damage and the production of thrombin.
[QCMLS 2021: 5.4.4, f5.4]

8. **a** TFP1, also known as extrinsic pathway inhibitor, functions predominantly by inhibiting the critical first step of coagulation, the formation of the tissue factor-factor VIIa-factor Xa complex.
[QCMLS 2021: 5.4.2, f5.3]

9. **c** Alpha-2 antiplasmin is synthesized in the liver and is the primary inhibitor of free plasmin.
[QCMLS 2021: 5.4.4, f5.4] [Rodak 2020, p644]

10. **a** AT is a serine protease inhibitor that binds and neutralizes serine proteases, including thrombin (FIIa) and factors IXa, Xa, XIa, and XIIa.
[QCMLS 2021: 5.7.2.1] [Rodak 2020, p642]

11. **a** During propagation, 95% of thrombin generation occurs. In the initiation phase, platelets are activated by the low levels of thrombin that are generated, and adhere to exposed collagen. The cofactors Va and VIIIa are activated by thrombin during the initiation phase.
[QCMLS 2021: 5.4.3] [Rodak 2020, p640]

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12. **a** In the cell based model of coagulation, the extrinsic or tissue factor pathway operates on the tissue factor-bearing cell and the intrinsic pathway as occurring on the platelet surface.
[QCMLS 2021: 5.4.3] [Rodak 2020, p640]

13. **c** Apixaban is an oral direct anti-FXa inhibitor used to prevent VTE. Direct thrombin inhibitors will inhibit factor IIa, vitamin K antagonists inhibit the vitamin-K-dependent factors, while LMWH can inhibit both IIa and Xa.
[QCMLS 2021: 5.7.5.1.2] [Rodak 2020, p747-754]

14. **b** Argatroban is a direct thrombin inhibitor that reversibly binds and inactivated free and clot-bound thrombin. LMWH will inhibit both factor IIa and Xa, whereas warfarin inhibits vitamin-K-dependent factors. Rivorxaban is an oral direct anti-Xa inhibitor.
[QCMLS 2021: 5.7.5.2.2] [Rodak 2020, p757-758]

15. **a** The drug dabigatran is cleared by the kidneys with a half-life of 12-17 hours; however, in renal disease the half-life may be prolonged up to 60 hours and cause an overdose. Apixaban and rivaroxaban is only partially eliminated by the kidney, warfarin is eliminated hepatically.
[QCMLS 2021: 5.7.5.1.2.2] [Rodak 2020, p756-757]

16. **c** In secondary hemostasis, serine proteases are activated to form a fibrin clot. These proenzymes circulate as zymogens. Zymogens are an inactive protein precursor of an enzyme converted into an active form.
[QCMLS 2021: 5.4.1, t5.2] [Rodak 2020, p631-632]

17. **c** The peripheral zone is associated with platelet adhesion and aggregation. The sol-gel zone provides a cytoskeletal system. The organelle zone contains alpha, dense, and lysosome granules. Membranes contain the dense tubular system.
[QCMLS 2021: 5.3.1, f5.1]

18. **c** Initially vascular damage is met with the body's reaction of "clamping down" to avoid further blood loss. This usually takes the form of vasoconstriction. In response platelets adhere to vessels and then aggregate with each other. The primitive platelet plug is then replaced by the fibrin clot produced through the action of the clotting cascade.
[QCMLS 2021: 5.1, 5.3.2]

19. **c** All of the glycoprotein complexes are receptors for extracellular antigens. GPIb/V/IX is the vWF receptor, GPIIb/IIIa is the fibrinogen receptor, and GPIa/IIa is the collagen receptor.
[QCMLS 2021: 5.3.3, f5.2, t5.1] [Kottke-Marchant 2016, p20-21]

20. **b** The GPIb/V/IX complex (CD42) serves as the receptor for platelet adhesion by binding with von Willebrand factor (vWF). This is followed by GPIIb/IIIa which assists with platelet aggregation through the binding of fibrinogen. Deficiencies in GPIb lead to Bernard-Soulier syndrome while a mutation of GPIIb/IIIa accounts for the weakened ("thrombasthenia") platelet binding in Glanzmann thrombasthenia.
[QCMLS 2021: 5.3.3, f5.2, t5.1] [Kottke-Marchant 2016, p22-23, 213-218]

21. **a** GPIb along with GPV/IX acts as a receptor for vWF on the exposed basement membrane of the endothelium. Shear forces from the circulation activate GPIb *in vivo*. Ristocetin is an antibiotic with the side effect of promoting platelet adhesion and will activate GP1b *in vitro*.
[QCMLS 2021: 5.3.3] [Kottke-Marchant 2016, p20]

22. **a** The acetylation of aspirin inactivates cyclooxygenase, which blocks thromboxane A2 production, which causes impaired platelet function.
[QCMLS 2021: 5.7.5.2.1, t5.13]

23. **d** The Factor V Leiden mutation has no effect on the procoagulant activity of Factor V. The activity of FV should be within the reference range.
[QCMLS 2021: 5.7.2.4] [Rodak 2020, p728]

24. **d** Human brain thromboplastin is the most sensitive of reagents and assigned an ISI of 1.0. Manufacturers compare their reagents to the most sensitive reagent to determine their reagents ISI. Micronized silica is an activator in aPTT reagents, rabbit thromboplastin is a type of PT reagent.
[QCMLS 2021: 5.8.2.2] [Rodak 2020, p750]

25. **a** A coagulation reagent is considered sensitive to a factor deficiency if it is prolonged when a patient's factor level is less than 30%, this is the level that will cause most patients to bleed.
[QCMLS 2021: 5.8.2.3] [Rodak 2020, p749]

26. **b** Factors V and VIII are considered heat labile and are not stable.
[QCMLS 2021: I.5.2]

27. **d** vWF serves as a carrier for factor VIII. It protects FVIII in circulation and it co-localizes FVIII at the cite of vascular injury.
[QCMLS 2021: 5.5.2.3] [Rodak 2020, p659]

28. **a** In a dysfibrinogenemia, the fibrinogen molecule is dysfunctional and can cause either clotting or bleeding. In hypofibrinogenemia patients present with low fibrinogen, and in a fibrinogenemia patients have absent fibrinogen, both are at a risk for bleeding. In hyperfibrinogenemia, there are elevated levels of fibrinogen, if these levels are persistently high it could put a patient at a risk for thrombosis.
[QCMLS 2021: 5.6.1.3]

29. **a** Plasmin if the key enzyme responsible for the breakdown of clots. Thrombin is the key enzyme in the formation of clots. Both urokinase and streptokinase are activators of fibrinolysis.
[QCMLS 2021: 5.4.4]

30. **a** Tissue plasminogen activator is a serine protease that catalyzes the conversion of plasminogen to plasmin and is responsible for clot breakdown. Alpha-2 antiplasmin inactivates plasmin and plasminogen activator inhibitor inhibits the plasminogen activators. Streptokinase is a fibrinolytic used to breakdown clots.
[QCMLS 2021: 5.4.4]

31. **c** Factor VII is activated in the coagulation cascade by tissue factor. Silica is an activator for the aPTT.
[QCMLS 2021: 5.4.2]

32. **b** Factor Va and VIIIa inactivation leads to a thrombotic condition. Factors IIa and Xa are inactivated by AT.
[QCMLS 2021: 5.7.2.2]

## *II. Disease States*

33. **d** Activated platelets, monocytes, and macrophages become embedded within fatty plaque combining with vWF to form arterial platelet plugs resulting in "white" thrombi seen in arterial thrombosis, while venous thrombosis is caused by fibrin and RBC.
[QCMLS 2021: 5.7.1.2] [Rodak 2020, p721]

34. **d** Factors V and VII are helpful in distinguishing between liver disease and vitamin K deficiency. FVII is a vitamin-K-dependent factor; however, FV is not and will not be decreased with a vitamin K deficiency. Both factors will be decreased in liver disease.
[QCMLS 2021: 5.6.2.4] [Rodak 2020, p654]

35. **b** Liver disease causes a decrease in all factors produced in the liver (factors II, V, VII, IX, and X); since FVIII is predominantly produced in endothelial cells, it is not decreased. It also is an acute phase reactant that may be elevated in liver disease.
[QCMLS 2021: 5.6.2.2] [Rodak 2020, p654]

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36. **d** Long-term antibiotic therapy disrupts normal flora, which provide a source of vitamin K synthesis. This results in a vitamin K deficiency (II, VII, IX and X). Factors V and VIII are vitamin K independent factors and, as such, would be within the reference range.
[QCMLS 2021: 5.6.2.4] [Rodak 2020, p656]

37. **a** All daughters of hemophiliac men are carriers of the disease; all sons are normal. The gene for factor VIII lies on the X chromosome.
[QCMLS 2021: 5.6.1.1.1] [Rodak 2020, p664]

38. **b** Individuals with hemophilia B lack factor IX clotting factor. Symptoms mimic hemophilia A which presents with a deficiency in FVIII. A factor XI deficiency is defined as hemophilia C. Treatment for hemophilia B includes factor IX concentrate.
[QCMLS 2021: 5.6.1.1.2] [Rodak 2020, p667]

39. **c** Laboratory testing of hemophilia will result in a prolonged aPTT, reflecting a decreased factor VIII. A patient with vWD will also present with an abnormal platelet function screen (bleeding time or PFA-100), and an abnormal ristocetin cofactor assay. These tests will both be normal in hemophilia.
[QCMLS 2021: 5.6.1.1.1] [Rodak 2020, p660-661]

40. **b** Factor XIII activity is <5% in congenital or acquired disorders. In adults, bleeding is slow and delayed. The PT, aPTT, fibrinogen and platelets will be normal. FXIII activity is not measured in the PT or aPTT.
[QCMLS 2021: 5.6.1.2] [Rodak 2020, p668, 784]

41. **a** The laboratory profile for a patient with acute DIC is: increased PT, aPTT and D-dimer, a decrease in platelets and hypofibrinogenemia.
[QCMLS 2021: 5.6.2.3.1]

42. **c** The laboratory profile for a patient with acute DIC is: increased PT, aPTT, and D-dimer, a decrease in platelets and hypofibrinogenemia.
[QCMLS 2021: 5.6.2.3.1] [Rodak 2020, p735]

43. **a** A prolonged thrombin time can indicate diminished or abnormal fibrinogen or the presence of FDPs, paraproteins, and heparin. Reptilase is insensitive to the effects of heparin and sensitive to dysfibrinogenemia. Therefore, when the TT is prolonged and the reptilase test is prolonged, this suggests a possible fibrinogen deficiency.
[QCMLS 2021: 5.8.2.5, 5.8.2.6, t5.14] [Rodak 2020, p780-781]

44. **d** An antibody against factor VIII presents as a prolongation of the aPTT with a relatively normal PT due to the fact that factor VIII is required for the activation of factor X in the intrinsic pathway but is not involved in the extrinsic pathway of coagulation.
[QCMLS 2021: 5.6.3.3.2] [Rodak 2020, p633, 637-638]

45. **a** For spontaneous bleeding to occur, there must be a severe deficiency of FVIII. Even small amounts of FVIII are protective. Patients with FVIII levels of 5-10% will only present with excessive bleeding following trauma or surgery. The aPTT as a screen is relatively insensitive in detecting FVIII deficiency. Most reagents demonstrate normal aPTT values until there is <30% of factor VIII levels.
[QCMLS 2021: 5.6.1.1.2, t5.9] [Rodak 2020, p664]

46. **b** Factor XIII deficiency causes a mild bleeding disorder where patients initially clot but later start to bleed. This makes sense in respect to the role of FXIII in the stabilization of the formed clot.
[QCMLS 2021: 5.8.2.1] [Rodak 2020, p637, 668]

47. **d** Amyloid fibrils in the vasculature can selectively adsorb factor X leading to an acquired factor X deficiency. It presents with a prolonged PT and PTT since FX is in the common pathway, which corrects with mixing studies.
[QCMLS 2021: 5.6.2, t5.10] [Rodak 2020, p666]



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48. **d** No single test is specific for DIC, but rather a combination of tests is used to aid in establishing a diagnosis. However, all of these tests will be impacted in DIC.
[QCMLS 2021: 5.6.2.3.1] [Rodak 2020, p735]

49. **c** Dilutions performed in factor assay should agree within 10% of each other. If the results do not match within 10%, they are considered to be nonparallel and suggestive of the presence of an inhibitor. As the patient sample with the inhibitor is diluted, it dilutes out the inhibitor and the factor activity increases.
[QCMLS 2021: 5.6.2.9] [Rodak 2020, p783]

50. **c** Although lupus anticoagulants are most often associated with thrombosis, some patients (approximately 30%) may experience bleeding due to the presence of anti-prothrombin (FII) antibodies.
[QCMLS 2021: 5.7.3.2]

51. **b** Adults with factor XIII (fibrin stabilizing factor) deficiency experience slow but progressive bleeding accompanied by poor would healing and slowly resolving hematomas. A congenital deficiency in infants involves seepage at the umbilical stump.
[QCMLS 2021: 5.6.1.2] [Rodak 2020, p668]

52. **a** In HDNB neonates present with decreased hemoglobin, increased reticulocyte count, and increased level of serum indirect bilirubin.
[QCMLS 2021: 5.6.2.5] [Rodak 2020, p388]

53. **c** Occasional spontaneous bleeding is seen in moderate hemophilia (1-5% FVIII), while frequent spontaneous bleeding is seen in severe hemophilia (<1% FVIII).
[QCMLS 2021: 5.6.1.1, t5.9] [Rodak 2020, p664]

54. **d** Coagulation factors are produced in the liver, (some of factor VIII is made in endothelial cells) therefore liver disease alters the production of the vitamin-K-dependent factors (II, VII, IX, X, proteins C, S, and Z).
[QCMLS 2021: 5.6.2.2] [Rodak 2020, p654]

55. **a** Alloantibodies arise in response to treatment, 30% of patients with severe hemophilia and 3% of those with moderate hemophilia. Autoantibodies are found in acquired hemophilia.
[QCMLS 2021: 5.6.2.1] [Rodak 2020, p657]

56. **a** Dysfibrinogenemia is caused by a structurally abnormal fibrinogen. The diagnosis is confirmed by demonstrating a low fibrinogen activity to fibrinogen antigen ratio as well as an elevated thrombin time and a prolonged reptilase time.
[QCMLS 2021: 5.6.1.3] [Kottke-Marchant 2016, p190]

57. **c** Factor XI supplements or boosts the activation of Factor IX, so deficiencies of factor XI are less severe clinically than deficiencies of factors IX or VIII.
[QCMLS 2021: 5.6.1.1.3] [Rodak 2016, p687]

58. **d** Inherited homozygous protein C deficiency is a severe condition that is associated with fetal central nervous system and retinal thrombotic events as well as neonatal purpura fulminans. In contrast, heterozygous protein C deficiency puts a patient at moderate risk for venous thromboembolism (VTE).
[QCMLS 2021: 5.7.2.2] [Kottke-Marchant 2016, p133]

59. **a** Protein S is carried through the circulation by C4b binding protein (C4bBP). An elevation in C4bBP occurs in response to inflammatory conditions. When C4bBP is elevated, there is an increase in the amount of bound protein S with a concomitant decrease in free and functional protein S levels.
[QCMLS 2021: 5.6.2.3] [Kottke-Marchant 2016, p 303]

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60. **c** There is an unusually high prevalence of factor XI deficiency in Ashkenazi Jews. Deficiency of factor XI causes an isolated prolongation of the aPTT and must be diagnosed and monitored by performing a factor XI assay. It may cause bleeding during surgery although the bleeding risk does not always correlate well with the factor level.
[QCMLS 2021: 5.6.1.1.3] [Kottke-Marchant 2016, p163, 364]

61. **d** In DIC, there is a generalized activation of hemostasis as a result of a systemic disease. Circulating thrombin activates platelets and coagulation proteins and catalyzes fibrin formation. As a result, the fibrinolytic system, at the level of plasminogen, becomes activated leading to the presence of soluble fibrin monomers, fibrin polymer, and cross-linked fibrin in the plasma, all of which serve to further activate plasminogen to plasmin. As platelets become enmeshed in the fibrin polymer or are exposed to thrombin, platelet activation continues, which continues to drive the coagulation system. At the same time protein C, protein S, and antithrombin are consumed, resulting in a loss of control in the coagulation system. The hemorrhagic picture in DIC is a result of thrombin activation, circulating plasmin, loss of control, and thrombocytopenia.
[QCMLS 2021: 5.6.2.3] [Rodak 2020, p667]

62. **d** Type III von Willebrand disease is a rare, autosomal recessive disorder in which all vWF multimers are absent or nearly absent from plasma. Factor VIII is proportionately diminished or absent and so primary and secondary hemostasis is impaired. Type II vWD is comprised of several qualitative abnormalities. Hemophilia A is a deficiency of FVIII with normal vWF activity and antigen. Hemophilia B is a deficiency of factor IX.
[QCMLS 2021: 5.5.2.3.1, t5.7] [Rodak 2020, p660-661]

63. **c** In dysfibrinogenemia, fibrinogen values using immunologic assays will be normal. Common pathway testing will be prolonged, and D-dimer will be elevated.
[QCMLS 2021: 5.6.1.3] [Rodak 2020, p780, 782]

64. **d** vWF and FVIII are acute phase reactants and will be increased in pregnancy, as well as trauma, infection and stress. Protein C and protein S levels are diminished in pregnancy and when using oral contraceptive, as well as in liver disease, renal disease, vitamin K deficiency and DIC. A deficiency of protein C or S cannot be determined during these conditions.
[QCMLS 2021: 5.7.2.2, 5.7.2.3] [Rodak 2020, p636, 730-731]

65. **c** In DIC the PT and aPTT are prolonged, the fibrinogen is decreased and the D-dimer in increased. The presence of schistocytes may be seen on the peripheral smear.
[QCMLS 2021: 5.6.2.3.1] [Rodak 2020, p735]

66. **d** Conditions such as liver disease, kidney failure, chronic infections, autoimmune disorders and dietary deficiencies such as vitamin K deficiency may all be associated with bleeding.
[QCMLS 2021: 5.6.2.4] [Rodak 2020, p654-655]

67. **d** vWF is a constituent of platelet alpha granules and is synthesized in the megakaryocyte as it develops.
[QCMLS 2021: 5.3.1] [Rodak 2020, p658]

68. **b** Average life span of platelets in peripheral blood is 9.5 days.
[QCMLS 2021: 5.3.1] [Rodak 2016, p720]

69. **a** Aspirin interferes with prostaglandin metabolism in the platelet by inhibiting cyclooxygenase, which participates in the conversion of arachidonic acid to protein G2; protein G2 is necessary to produce thromboxane, which stimulates secretion from the platelet granules.
[QCMLS 2021: 5.5.2.2] [Rodak 2020, p684]

70. **a** Certain characteristics of platelet-type bleeding disorders include mucosal bleeding, petechiae, and a female predominance. Other types of bleeding are due to defects in coagulation factors. Hemarthrosis and deep hematomas occur in factor VIII deficiency (hemophilia), delayed bleeding occurs in factor XIII deficiency, and deep hematomas presents with the other choices in the question.
[QCMLS 2021: 5.5] [Rodak 2020, p650-651]

71. **c** Hirudin, argatroban and dabigatran are direct thrombin inhibitors. They are either isolated from the leech, *Hirudo medicinalis* (hirudin), or are recombinant forms of the same protein (lepirudin, argatroban, dabigatran). Tirofiban is not related and is a platelet GPIIb/IIIa inhibitor and is an anti-platelet drug.
[QCMLS 2021: 5.7.5.2.4]

72. **b** Ticlopidine and clopidogrel both inhibit ADP mediated aggregation and present with blunted ADP response with aggregometry studies, but normal results with other agonists. Another class of inhibitors, like abciximab, block platelet function through GPIIb/IIIa.
[QCMLS 2021: 5.7.5.2.4] [Rodak 2020, p757-758]

73. **b** vWF binds platelets through their GP Ib/IX/V membrane receptor. The importance of platelet adhesion is demonstrated by bleeding disorders such as von Willebrand disease, in which vWF is missing or defective, and Bernard-Soulier disease, in which the GP Ib/IX/V receptor is absent.
[QCMLS 2021: 5.3.3, t5.1] [Rodak 2020, p658]

74. **d** May-Hegglin anomaly is characterized by decreased platelet counts; vWD and hemophilia patients do not present with low platelet counts. Glanzmann patients present with normal platelet counts that do not function.
[QCMLS 2021: 5.5.2] [Rodak 2020, p678, 696]

75. **b** Acute ITP typically resolves within weeks and is more frequently seen in children; it is not gender-dependent; platelet destruction is increased.
[QCMLS 2021: 5.5.1.2.1] [Rodak 2020, p700]

76. **d** A quantitative abnormality of platelets (thrombocytopenia) is the most common cause of excessive or abnormal bleeding. Fibrinogen deficiencies are rare.
[QCMLS 2021: 5.5] [Rodak 2020, p696]

77. **d** Abnormalities in coagulation factors and DIC are ruled out by the normal PT/aPTT and fibrinogen. The low platelet count is due to the dilutional effect of the bypass surgery.
[QCMLS 2021: 5.5.2.2] [Rodak 2020, p711, 735-736]

78. **a** Patients with thrombotic thrombocytopenic purpura (TTP) present with platelet counts $<20 \times 10^3/\mu L$ ($<20 \times 10^9/L$). Platelet thrombi are dispersed throughout the arterioles and capillaries subsequent to the accumulation of large vWF multimers made by endothelial cells and platelets. This is related to a deficiency of ADAMTS-13. DIC is caused by inflammation, infections and cancer. HUS is related to a toxin producing organism. ITP is a result of antibodies to platelets.
[QCMLS 2021: 5.5.1.2.2] [Rodak 2020, p707-709]

79. **b** The pathogenesis of HIT is that antibodies are produced against heparin- platelet factor 4 complex. This complex binds to FC receptors causing platelet activation and the formation of platelet microparticles, thrombocytopenia and hypercoagulability. ACLA and B2GP1 are seen in anti-phospholipid syndrome.
[QCMLS 2021: 5.7.4] [Rodak 2020, p737, 724]

80. **a** Polycythemia vera is characterized by increased WBC, RBC, and platelet counts.
[QCMLS 2021: 4.9.3.1.2.2, t4.11] [Rodak 2020, p571]

81. **d** Congenital spherocytosis is characterized by an increased MCHC and an increased reticulocyte count; reactive thrombocytosis is not usually accompanied by abnormal platelets; myelofibrosis is characterized by abnormal RBC morphology and decreased platelets and reticulocytes.
[QCMLS 2021: 5.5.2.2]

82. **c** Mucous membrane hemorrhage is typical of platelet disorders; remaining choices of other types of hemorrhage and prolonged clotting times are typical of coagulation factor disorders.
[QCMLS 2021: 5.5] [Rodak 2020, p651, 676]

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83. **a** Giant platelets, abnormal PFA100, normal aggregation with ADP, and decreased platelet count are characteristic of Bernard-Soulier.
[QCMLS 2021: 5.5.1] [Rodak 2020, p678-679]

84. **b** In type 1 von Willebrand disease (vWD) the plasma concentration of all vWF multimers and FVIII are variable, although proportionally reduced. Hemophilia presents with a normal multimer pattern. Type 3 vWD has absent or greatly reduced multimers, with type 2 will have varying multimer patterns based on the subtype.
[QCMLS 2021: 5.5.2.3.1, t5.7] [Rodak 2020, p659]

85. **b** The platelets seen in the peripheral smear of patients with Bernard-Soulier syndrome are typically larger than normal. Thrombocytopenia is usually present, yet is nonspecific. Otherwise, the clinical presentation of von Willebrand disease (especially platelet-type, and types 2M and 3) and Bernard-Soulier are strikingly similar.
[QCMLS 2021: 5.5.1.1] [Rodak 2020, p661, 678-679]

86. **d** Both Hĕrmanský-Pudlák and Chédiak-Higashi are due to defects in dense granule storage, where platelets store small molecules in order to initiate the secondary wave of aggregation. Inclusion granules are more common in Chédiak-Higashi, while ceroidlike inclusions are more common in Hĕrmanský-Pudlák. Alpha granule defects are not usually seen in either.
[QCMLS 2021: 5.5.2.1] [Rodak 2020, p661]

87. **d** A similar syndrome of thrombocytopenia with absent radii is most commonly seen in patient without radii and thumbs. Wiskott-Aldrich also presents with platelet-type bleeding, like Hĕrmanský-Pudlák and Chédiak-Higashi.
[QCMLS 2021: 5.5.1.1] [Rodak 2020, p681-682]

88. **a** The choices are organized in order of the relative amount of von Willebrand factor. There can be an almost 50% difference in the amount of vWF between type O and types B or AB. The lower reference range for vWF for type O is 35% vs 64% in type AB.
[QCMLS 2021: 5.5.2.3.3] [Rodak 2020, p662]

89. **a** By far the majority of cases of von Willebrand disease (vWD) are type 1, up to 75% of patients. Type 1 is due to decreased levels of von Willebrand factor (vWF). Type 3 is an absence of vWF and is found in less than 5% of vWD patients, while type 2 (and its subclasses) are due to functional defects in vWF and makes up between 7-30% of vWD cases.
[QCMLS 2021: 5.5.2.3.1.1] [Rodak 2020, p659-660]

90. **c** DDAVP stimulates the release of vWF and FVIII from Weibel-Palade bodies. Type 2B vWD reacts to DDAVP with enhanced platelet aggregation leading to potentially dangerous thrombosis, while type 3 does not react at all.
[QCMLS 2021: 5.5.2.3.1.1] [Rodak 2020, p663]

91. **d** The classic pentad of symptoms for TTP can be remembered with the insensitive "FAT RN" mnemonic: Fever, anemia, thrombocytopenia, renal failure, neurological symptoms. Clinically TTP can overlap with HUS.
[QCMLS 2021: 5.5.1.2.2, t5.5] [Rodak 2020, p365]

92. **c** Most cases of chronic immune thrombocytopenic purpura occur in patients between the ages of 20 and 50 years. Females with the disorder generally outnumber males. Presenting features include petechiae and ecchymoses.
[QCMLS 2021: 5.5.1.2.1] [Rodak 2020, p702]

93. **b** TTP is characterized by the triad of macroangiopathic hemolytic anemia, thrombocytopenia, and neurologic abnormalities. In addition, fever and renal dysfunction (forming a pentad) are often present. The PT and aPTT, are usually normal and may be useful in differentiating this disorder from DIC.
[QCMLS 2021: 5.5.1.2.2] [Rodak 2020, p707-708]

94. **b** The lack of neurologic symptoms, the presence of renal dysfunction, and the absence of other organ involvement suggest HUS.
[QCMLS 2021: 5.5.1.2.2, t5.5] [Rodak 2020, p710]

95. **c** The causes of neonatal thrombocytopenia are numerous and present at or within 72 hours of birth.
[QCMLS 2021: 5.5.1.2.2] [Rodak 2020, p705]

96. **b** In TTP, inherited or acquired defective ADAMTS13 enzyme activity is associated with the presence of ultra-large vWF multimers in plasma resulting in platelet aggregations and microvascular thrombosis.
[QCMLS 2021: 5.5.1.2.2] [Rodak 2020, p708]

97. **b** HIT is a relatively common side effect of unfractionated heparin administration with about 1-5% of patients developing this complication. In patients who develop HIT heparin should be stopped as soon as the diagnosis is made. It is not recommended to switch to LMWH or warfarin, and giving a patient platelets will cause clotting since platelets are already aggregating *in vivo*. Patients are treated with antithrombins.
[QCMLS 2021: 5.7.4] [Rodak 2020, p737]

98. **c** HELLP syndrome affects an estimated 4-12% of pregnant women with severe pre-eclampsia (hypertension). It is named for its characteristic presentation of hemolysis, elevated liver enzymes, and low platelet count.
[QCMLS 2021: 5.5.2.2] [Rodak 2020, p367]

99. **d** von Willebrand factor (vWF) functions as a protective carrier protein for factor VIII, thus participating in the formation of the fibrin clot. FVIII levels may be decreased in von Willebrand disease because the half-life of FVIII shortens when it is not bound to vWF.
[QCMLS 2021: 5.5.2.3] [Kottke-Marchant 2016, p241]

100. **a** Glanzmann thrombasthenia is a disorder of platelet aggregation in which there is a deficiency or abnormality of the platelet membrane glycoprotein GP IIb/IIIa, a receptor that is capable of binding fibrinogen and von Willebrand factor. In contrast, von Willebrand disease and Bernard-Soulier syndrome are both disorders of platelet adhesion and storage pool diseases are disorders of platelet secretion.
[QCMLS 2021: 5.5.2.1, t5.6] [Rodak 2020, p677]

101. **b** May-Hegglin anomaly is a rare, autosomal dominant platelet disorder. Patients present with variable thrombocytopenia, giant platelets, and large Döhle body-like inclusions in the neutrophils, eosinophils, basophils, and monocytes. The majority of patients are asymptomatic, but a few have shown mild bleeding tendencies related to the degree of thrombocytopenia. Wiskott-Aldrich syndrome and Hĕrmanský-Pudlák syndrome are both disorders of platelet secretion, specifically storage pool disorders. Ehlers-Danlos syndrome is a vascular disorder in which a bleeding tendency is manifested primarily in subcutaneous hematoma formation.
[QCMLS 2021: 4.8.2]

102. **d** Low platelet counts can be caused by EDTA-dependent platelet agglutinins and occur in healthy people, they are not known to be associated with any pathological condition. There is no risk to the patient for thromboembolism or autoimmune disease. Recollecting the sample in sodium citrate should normalize the platelet count.
[QCMLS 2021: 5.5]

103. **a** In Bernard-Soulier the abnormality is on the platelet receptor for vWF, and in vWD is it in the structure of vWF resulting in an abnormality in the initial platelet adhesion to subendothelium. GP IIb/IIIa inhibitors effect the formation of platelet aggregation. Patients with hemophilia A or B have abnormality of thrombin generation and do not have any platelet abnormality, nor do patients with factor XIII deficiency. They present with delayed bleeding due to a decreased level of fibrin stabilizing factor.
[QCMLS 2021: 5.5.2.1]

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104. **b** Glanzmann thrombasthenia is characterized by an absent aggregation to all agonists, the normal ristocetin induced platelet aggregation distinguishes it from Bernard-Soulier. This would also be abnormal in von Willebrand disorder.
[QCMLS 2021: 5.5.2]

105. **d** Platelets can be decreased in number based on any of the disorders. A qualitative disorder of platelets can be caused by abnormal structure or function of platelets.
[QCMLS 2021: 5.5.1]

106. **d** Post-transfusion purpura results in severe thrombocytopenia 3-12 days after a blood transfusion, in which platelet destruction is caused by antibodies produced against platelet antigen PLA-1a (HPA-1a).
[QCMLS 2021: 5.5.1.3.2]

107. **c** vWD and Bernard-Soulier are disorders of adhesion, while storage pool deficiency is an abnormality of platelet granules.
[QCMLS 2021: 5.5.2]

108. **b** A spontaneous bleed is seen in patients with counts $<20 \times 10^3/\mu L$, while a life-threatening bleed may be seen at $<10 \times 10^3/\mu L$. Clinical symptoms do not usually appear until counts are $<100 \times 10^3/\mu L$, but are more likely at $<50 \times 10^3/\mu L$.
[QCMLS 2021: 5.5.1]

109. **c** Acute renal failure is a frequent presenting symptom of HUS. Severe CNS symptoms and fever occur in patients with TTP.
[QCMLS 2021: 5.5.2.1]

110. **c** The treatment for patients with HIT is to first stop heparin, and a direct thrombin inhibitor should be administered. LMWH should not be administered because it can cause cross-reactivity and platelet aggregation. Neither platelets nor plasma are indicated and would be more harmful to the patient.
[QCMLS 2021: 5.7.4]

111. **b** In acute and chronic renal diseases, often a bleeding tendency is associated with several hemostatic abnormalities. Thrombocytopenia frequently develops in uremia.
[QCMLS 2021: 5.6.2.5]

112. **c** The patient has all of the characteristics of a patient with disseminated intravascular coagulation (DIC), which is a consumptive coagulopathy and occurs in many patients with cancer. HUS occurs in children and TTP presents with neurological symptoms and organ involvement.
[QCMLS 2021: 5.6.2.3]

113. **b** The platelet count for this patient is normal, however when screen the functionality of the platelets the result is abnormal suggesting defective platelet function.
[QCMLS 2021: 5.8.1.1]

114. **c** May-Hegglin disorder is a rare inherited blood platelet disorder characterized by abnormally large platelets and WBC defects. All of the rest are inherited vascular disorders.
[QCMLS 2021: 5.2.2.1]

## *III. Laboratory Determinations*

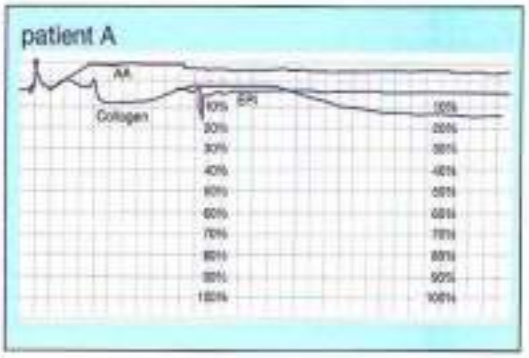
115. **d** Clopidogrel is a member of the thienopyridine drug family which occupy the platelet membrane adenosine disphosphate (ADP) P2Y12. As a result, you would expect a suppressed platelet aggregation response to ADP.
[QCMLS 2021: 5.7.5.2.2] [Rodak 2020, p685, 774]

116. **d** Glanzmann thrombasthenia is characterized by abnormal aggregation to ADP, epinephrine and collagen, and normal aggregation to ristocetin.
[QCMLS 2021: 5.5.2] [Kottke-Marchant 2016, p217]

117. **b** The PFA-100 assesses platelet function and is abnormal in the presence of aspirin. The platelet count only assesses platelet number. The PT and aPTT cannot be used to assess platelet number or function.
[QCMLS 2021: 5.7.5.2.1] [Kottke-Marchant 2016, p104-106]

118. **c** vWD is characterized by abnormal platelet aggregation to ristocetin and normal platelet aggregation to epinephrine, ADP and collagen. In type IIb vWD, as well as in platelet-type vWD, the response to low-dose ristocetin is increased, whereas in all other forms of vWD the ristocetin response is decreased or normal.
[QCMLS 2021: 5.5.2.3.1.2] [Kottke-Marchant 2016, p243-247]

119. **d** Thromboxane $A_2$ is necessary for normal platelet aggregation. Its formation initiates a series of events, which results in the secretion and aggregation of platelets.
[QCMLS 2021: 5.8.1.3]




120. **c** Glanzmann thrombasthenia is an autosomal recessive disorder. Patients will have a prolonged bleeding time and PFA-100, normal platelet count and morphology, and abnormal aggregation with all aggregating agents except ristocetin.
[QCMLS 2021: 5.5.1.2] [Kottke-Marchant 2016, p214-217]

121. **d** The PFA-100 assay measures platelet function by exposing a blood sample to high shear forces and causing the platelets to aggregate, thereby simulating platelet plug formation. The time for the aperture to close is directly proportional to platelet function. Aspirin interferes with platelet function and will cause an abnormal PFA. However, not only do platelet function disorders interfere with the PFA-100 closure times but thrombocytopenic and anemic samples can also give artifactually lengthened closure times.
[QCMLS 2021: 5.8.1.1] [Kottke-Marchant 2016, p104-106]

122. **b** Platelet-dense granules store molecules such as ATP, $Ca^{2+}$, and serotonin, which cause recruitment of additional platelets when released leading to further stimulation of platelet aggregation. Initial stimulation (primary wave) is due to the direct action of low-dose ADP or low-dose epinephrine on the GPIIb/IIIa platelet membrane receptor.
[QCMLS 2021: 5.8.1.3.1] [Kottke-Marchant 2016, p18, 107]

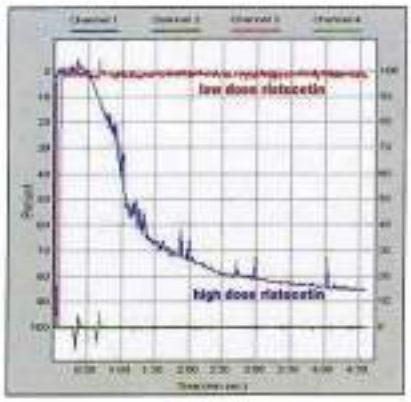
123. **b** While the assay for anti-PF4 antibody is the most commonly performed confirmatory test, it is less specific than either the gold standard serotonin release assay or the heparin induced platelet aggregation assay.
[QCMLS 2021: 5.7.4]



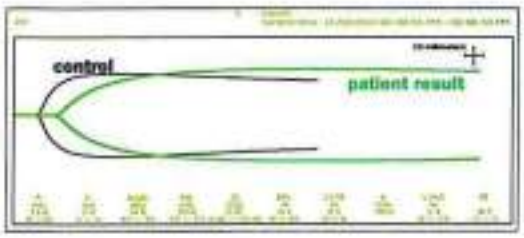
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124. **b** The optical turbidimetric platelet aggregation assay measures the increase in light transmission that occurs in direct proportion to platelet aggregation when various agonists such as ADP, collagen, epinephrine, arachidonic acid, and ristocetin are used to induce platelet aggregation. The test sample is platelet-rich plasma. As aggregation proceeds, there is a gradual clearing of the plasma as platelet aggregates form. This clearing results in an increase in light transmission measured spectrophotometrically and tracings of the reactions are displayed on a graph.
[QCMLS 2021: 5.8.1.3.1] [Rodak 2020, p771-772]

125. **d** Thromboelastography tracings can detect and report clot onset, clot strength, and fibrinolysis within 15 minutes of blood collection.
[QCMLS 2021: 5.8.1.4] [Rodak 2020, p803]




126. **d** The agonist ristocetin supports platelet agglutination in the presence of vWF. Additional tests are needed to confirm type 2 vWD and to differentiate subtype 2A from subtype 2B. Low dose ristocetin-induced platelet aggregometry (RIPA), also called the ristocetin response curve, identifies subtype 2B. The low-dose RIPA test is performed on platelet-rich plasma. In subtype 2B the patient's platelets, because they are coated with abnormal vWF multimers, agglutinate in response to <0.5 mg/mL (low-dose ristocetin). In comparison, normal platelets or platelets from a patient with subtype 2A agglutinate only at ristocetin concentrations >0.5 mg/mL (high-dose ristocetin).
[QCMLS 2021: 5.5.2.3.1.2] [Rodak 2020, p661]




127. **b** This is a global whole-blood analyzer that measures clotting time and dynamics, clot strength, antithrombotic effects, platelet effects on clot dynamics and strength, and fibrinolysis. It is used mainly in liver and cardiac surgeries. Analysis of the tracing determine the speed, strength, and stability of clot formation as well as the effect of fibrinolysis.
[QCMLS 2021: 5.8.1.4] [Rodak 2020, p803]

128. **c** Factors V and VIII are considered labile factors because their activity diminishes quickly at room temperature.
[QCMLS 2021: 5.4.1, t5.2] [Kottke-Marchant 2016, p48]

129. **a** Patients with a deficiency of factor XII tend to have thrombotic complications. They do not have bleeding problems, most likely due to the lack of activation of fibrin lysis as well as the lack of both the activation of factor IX by the factor VIIa/TF complex and the activation of factor XI by thrombin.
[QCMLS 2021: 5.6.1.2] [Rodak 2020, p636]

130. **d** Warfarin is a vitamin K antagonist which causes coagulation factors II, VII, IX, X, protein C, and protein S to be reduced as nonfunctional molecules are produced. The rate of reduction is based on the half-life of the factors.
[QCMLS 2021: 5.7.5.1.1] [Rodak 2020, p748-749]

131. **c** Protein C is inherited as an autosomal dominant trait. Venous thrombi (DVT) and pulmonary emboli occur in these patients. Patients with heterozygous deficiency have a 1.6-11.5 fold increased risk of thrombosis.
[QCMLS 2021: 5.7.2.2] [Rodak 2020, p730]

132. **c** AT inhibits the serine proteases thrombin and factors IXa, Xa and XIa. AT function is enhanced by heparin. Factor VIII does not play a role in assaying AT.
[QCMLS 2021: 5.7.2.1] [Rodak 2020, p642]

133. **b** In a type I deficiency of protein S, there is a decreased activity. To determine if the deficiency is the result of a dysfunctional molecule or a quantitative disorder, an ELISA assay should be performed to determine a type II deficiency.
[QCMLS 2021: 5.7.2.3]

134. **c** The molecular test available to confirm the specific point mutation of activated protein C resistance, which is factor V Leiden (FVL) This substitutes glutamine for arginine at position 506.
[QCMLS 2021: 5.7.2.4] [Rodak 2020, p728-729]

135. **d** The aPTT is performed to monitor the effects of unfractionated heparin, to detect factor deficiencies, and to detect the presence of inhibitors. The PT tests the extrinsic pathway and requires tissue thromboplastin. It is also used to monitor warfarin therapy.
[QCMLS 2021: 5.7.2.3] [Rodak 2020, p752-753]

136. **a** A mixing study, when corrected, indicates a factor deficiency. Lack of correction indicates an inhibitor. Some inhibitors may only become evident after the patient's plasma is allowed to interact with the normal plasma during an incubation period. In this case, correction did not occur initially, as well as after incubation, confirming a circulating inhibitor.
[QCMLS 2021: 5.8.2.6] [Rodak 2020, p779]

137. **a** Warfarin is a vitamin K antagonist. Protein C is a vitamin-K-dependent protein; hence warfarin therapy will lower levels of this protein. The other tests should not be impacted.
[QCMLS 2021: 5.7.5.1.1] [Rodak 2020, p748]

138. **b** A prolonged thrombin time may be considered evidence of diminished or abnormal fibrinogen; however, the presence of AT activity, such as heparin, must be ruled out. A prolonged thrombin time can indicate a sample may be contaminated with heparin.
[QCMLS 2021: 5.8.2.5] [Rodak 2020, p780]

139. **b** In a Clauss fibrinogen, a standard amount of thrombin is added to diluted plasma, and the time required for clot formation is recorded. The seconds are read off a standard curve to measure the final concentration.
[QCMLS 2021: 5.8.2.4] [Rodak 2020, p781]

140. **a** The chromogenic anti-Xa assay is the only assay available to monitor LMWH; it may also be used to measure UFH. The aPTT cannot be used to monitor low molecular weight heparin therapy.
[QCMLS 2021: 5.7.5.1.4] [Rodak 2020, p754]

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141. **d** Factor VIII, a constituent of the intrinsic pathway, is most sensitively monitored by aPTT. The aPTT is responsive to changes in factors involved in either the intrinsic or common pathways, showing a prolongation when factors are reduced. Conversely, an elevated FVIII causes a shortening of the aPTT, which could indicate an increased risk of thromboembolic disease.
[QCMLS 2021: 5.8.2.3] [Kottke-Marchant 2016, p80]

142. **b** Inhibition of coagulation by protein C involves the cleavage of factor V at a conserved sequence. The mutation of that site on factor V (Leiden) results in a protein that is resistant to degradation and cleavage by activated protein C. Hence, the patient is at risk for thrombosis.
[QCMLS 2021: 5.7.2.4] [Rodak 2020, p641]

143. **c** Activated protein C resistance is almost always due to a mutation in factor V that prevents protein C dependent cleavage and inhibition of factor V. This is the so-called factor V Leiden mutation due to a K506Q substitution. It has a prevalence of 10-15% in the general population. The second most common inherited thrombophilia is due to prothrombin G20210A mutation and occurs in 2% of Caucasians.
[QCMLS 2021: 5.7.2.4, t5.12] [Rodak 2020, p728]

144. **b** Antithrombin deficiency most often presents in this way, and can be treated in counterintuitive fashion by administering fresh frozen plasma to the patient with thromboses. Plasma contains antithrombin in addition to all the requisite clotting and patients subsequently respond to heparin. It is also possible to use antithrombin concentrate to treat antithrombin deficiency.
[QCMLS 2021: 5.7.2.1] [Rodak 2020, p729, 737]

145. **c** Purpura fulminans is a very severe intravascular coagulopathy with a significant risk for poor outcomes unless rapidly treated with plasma and heparin. Warfarin skin necrosis occurs due to an acquired protein C deficiency following warfarin treatment. Post-transfusion purpura occurs due to alloimmunization against platelet antigens, resulting in severe thrombocytopenia following transfusion of RBC or platelets.
[QCMLS 2021: 5.7.2.2] [Rodak 2020, p711]

146. **d** The anticoagulative properties of heparin are mediated through its binding to antithrombin, which facilitates the inhibition of thrombin and factor Xa. In addition, factors IXa, XIa, and XIIa are inhibited in a similar fashion. Protein C is bound by endothelial cell protein C receptor (EPCR); protein S binds to negatively charged phospholipids.
[QCMLS 2021: 5.7.2.1] [Rodak 2020, p751]

147. **a** When LA is suspected, perform a dRVVT screen, which contains a low concentration of phospholipid that if positive will prolong the clotting time in seconds. This compares the patient DRVVT screen result to the control normal plasma. If the patient-to-control normal plasma DRVVT in seconds is greater than a defined cut-off (seconds or ratio), repeat the test using the high phospholipid DRVVT confirm reagent.
[QCMLS 2021: 5.8.3.1.2] [Rodak 2020, p727]

148. **b** The chromogenic anti-FXa consists of a reagent that provides a fixed concentration of FXa and substrate specific to FXa. Excess FXa is added. A factor assay is a clot based assay using deficient plasma and routine coagulation reagents, an ELISA assay using a plate coated with antibodies and uses a sandwich method of detection.
[QCMLS 2021: 5.8.3.3.1]

149. **d** The Bethesda assay is used to detect and quantitate specific factor inhibitors. It is most commonly performed to quantitate inhibitors to factor VIII, but can also be used for inhibitors to factors V, IX, X, XI, and XII. The assay is based on the ability of the patient's plasma containing the inhibitor to inactivate the factor VIII present in pooled normal plasma at various dilutions. Factor VIII levels at each of these dilutions are compared to a normal control and calculations are performed to determine the Bethesda units (BU). Factor assays alone can only demonstrate the presence of an inhibitor but cannot quantitate the inhibitor. The ristocetin cofactor assay is used to measure von Willebrand activity, and the multimer assay separates vWF multimers by size in order to diagnose the various forms of von Willebrand disease.
[QCMLS 2021: 5.8.3.3.2] [Kottke-Marchant, p90-91, 246-247]

150. **d** D-dimer is a marker of thrombosis and fibrinolysis and is used to identify chronic and acute DIC and to rule out venous thromboembolism.
[QCMLS 2021: 5.8.2.7] [Rodak 2020, p733]

151. **b** The Clauss fibrinogen assay is a modified thrombin time test. A high concentration of thrombin is added to dilute citrated plasma, catalyzing the conversion of fibrinogen to fibrin polymer. There is an inverse relationship between the clotting time and the concentration of functional fibrinogen in patient plasma. The concentration is derived from a calibration curve prepared with standards of known assay values.
[QCMLS 2021: 5.8.2.4] [Rodak 2020, p781]

152. **a** D-dimer is made up of two D domains from separate fibrin molecules cross-linked by the action of factor XIIIa. Fragments X, Y, D, and E are produced as a result of the digestion of either fibrin or fibrinogen by plasmin. D-dimer is a specific product of the digestion of cross-linked fibrin.
[QCMLS 2021: 5.8.2.7] [Rodak 2020, p645]

153. **c** Several plasma proteins become elevated in acute phase reactions, including factor VIII, vWF, and fibrinogen. Elevations in any of these could result in a shortened aPTT. An elevation of a low value for vWF into the normal range may occur in an acute phase reaction, thus masking a diagnosis of von Willebrand disease. In this case, it would be helpful to measure fibrinogen to determine whether or not the patient is in an acute phase reaction at the time of testing. A DVT should not impact the aPTT; an LA and an inhibitor should prolong the aPTT
[QCMLS 2021: 5.8.2.3] [Kottke-Marchant 2016, p244]

154. **c** Although blood may be drawn from vascular access devices (VADs), management of these devices requires strict adherence to protocol in order to ensure the patient's safety as well as to ensure that an optimal specimen is collected. Before collecting blood for hemostasis testing, the line must be flushed with 5 mL of saline, and the first 5 mL of blood must be discarded. The line must never be flushed with heparin. When heparin contamination occurs, the thrombin time will be prolonged and a reptilase time will be normal. Plasma containing heparin may produce false-positive LA screening tests, inhibitor effects on factor assays, and a variety of other interferences. A heparin neutralization procedure should be performed, which would enable testing to proceed.
[QCMLS 2021: 5.8.2.5]

155. **d** The desired dosage of coumadin should result in an INR of 2-3. However, in the case of a mechanical heart valve, the desired INR range is 2.5-3.5, making the dose of anticoagulant acceptable. An INR greater than 4 can result in an increased risk of bleeding.
[QCMLS 2021: 5.7.5.1.1] [Kottke-Marchant 2016, p329]

156. **a** The most responsive PT thromboplastin reagents have ISI (International Sensitivity Index) near 1, they are said to be the most sensitive, as they are nearest to the assigned an ISI of 1 found in the WHO (World Health Organization) human brain thromboplastin. The higher the ISI, the less responsive the thromboplastin reagent.
[QCMLS 2021: 5.8.2.2] [Rodak 2020, p750]

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157. **b** The 9:1 blood-to-anticoagulant ratio is effective, provided the patient's hematocrit is 55% or less. In polycythemia, the decrease in plasma volume relative to whole blood unacceptably raises the anticoagulant-to-plasma ratio, which causes falsely prolonged results, due to the increased level of anticoagulant in the tube, for clot-based coagulation tests.
[Rodak 2020, p767]

158. **b** The range is therapeutic. Lovenox is a LMWH. The prophylactic range for LMWH is 0.2 to 0.5 U/mL, and the therapeutic range is 0.5 to 1.2 U/mL. Levels above 1.2 U/mL are supratherapeutic and can result in bleeding.
[QCMLS 2021: 5.7.5.1.4] [Rodak 2020, p755]

159. **c** Using platelet-poor plasma avoids neutralization of the lupus anticoagulant. If a sample is not platelet poor (platelet count <10,000) during freezing and thawing, platelet membrane fragments, which are phospholipids can neutralize lupus antibodies and lead to a false-negative result.
[QCMLS 2021: 5.8.3.1.1] [Rodak 2020, p727]

160. **b** Cardiac patients are exposed to fibrin glue which can cause the formation of antibodies to factor V, resulting in bleeding and an uncorrected mix. Patients with a lupus anticoagulant are more prone to thrombosis. A dysfibrinogenemia and DIC, both the mixing study would correct.
[QCMLS 2021: 5.6.2.1] [Rodak 2020, p657]

161. **b** A prolongation of just PT, with a corrected mixing study indicates a factor VII deficiency. Factor II is a common pathway factor, so both the PT and aPTT would be prolonged, while a factor VIII or IX deficiency would result in a normal PT and a prolonged aPTT.
[QCMLS 2021: t5.8] [Rodak 2016, p684]

162. **a** A heparin-like anticoagulant occurs when the monoclonal protein from multiple myeloma acts like a glycosaminoglycan. The normal reptilase time, which is not prolonged with heparin therapy, supports this diagnosis.
[QCMLS 2021: 5.8.2.6] [Rodak 2020, p687]

163. **c** Clinical symptoms suggest amyloidosis. This is an acquired factor deficiency related to the adsorption of factor X to the extracellular amyloid fibrils. A factor VII deficiency would only prolong the PT, and a FVIII deficiency would only prolong the aPTT. FXIII would not prolong either the PT or the aPTT.
[QCMLS 2021: t5.10] [Rodak 2020, p687, 774-775]

164. **c** Vitamin K deficiency can be caused by antibiotics, which alter the bacterial flora within the gut and impact the vitamin-K-dependent factors (II, VII, IX and X), which are all decreased. The effect of aspirin on platelets lasts 7-10 days should platelet function should be normal, as reflected in the platelet function screen. Patients with liver disease would demonstrate a decrease in all factors, except FVIII which is normally elevated.
[QCMLS 2021: 5.6.2.4] [Rodak 2020, p656]

165. **d** A deficiency of factor XII is not associated with a risk of bleeding, while factor VIII, XI, and XI deficient patients will all present with a bleeding diathesis. All deficiencies will correct in a mixing study.
[QCMLS 2021: 5.6.1.2] [Rodak 2020, p636]

166. **b** Patients who are receiving therapeutic doses of LMWH do not present with a prolonged aPTT and cannot be used to monitor dosage. The aPTT is prolonged on patients that receive standard unfractionated heparin and can be used to monitor dosage.
[QCMLS 2021: 5.7.5.1.4] [Rodak 2020, p754]

167. **c** Factor V Leiden patients have an increased risk of venous thrombosis due to the resistance to the activity of protein C. This is one of the most commonly inherited thrombophilia disorders. The incidence of the others are variable and not independently associated with thrombosis.
[QCMLS 2021: 5.7.2.4] [Rodak 2020, p728-729]

168. **d** Patient with hemophilia A will present with an abnormal FVIII level, but normal factor IX level; whereas a patient with hemophilia B will have a normal FVIII, but an abnormal FIX level. Both patients will have a prolonged aPTT.
[QCMLS 2021: 5.6.1.1] [Rodak 2020, p663-664]

169. **b** DIC is seen in patients with infections and present with a prolonged PT, aPTT and a decreased fibrinogen. An increased D-dimer can mean either a blood clot or DIC, but the patient would have normal PT and aPTT with a DVT.
[QCMLS 2021: 5.6.2.3.1] [Rodak 2020, p735-736]

170. **b** Factor VIII, fibrinogen and vWF are all acute phase reactants that are elevated during inflammation, stress and during pregnancy.
[QCMLS 2021: 5.6.1.3] [Rodak 2020, p636]

171. **a** A severe hemophiliac would be in the range of <1%, a moderate hemophilic would be in the range of 1-5% and a mild hemophiliac would be in the range of 6-30%.
[QCMLS 2021: 5.6.1.1.2] [Rodak 2020, p665]

172. **d** Type 1 vWD will have a decreased level of both antigen and activity. Type 2 will have different combinations of decreased activity and or antigen based on the subtype, however type 3 will present with severe deficiency of all parameters.
[QCMLS 2021: 5.5.2.3.1.2] [Rodak 2020, p661]

173. **b** To determine the final ratio for the DRVV the screen/confirm = 62/49 = 1.27.
[QCMLS 2021: 5.6.3.1.2] [Rodak 2020, p727]

174. **d** The normalized ratio for the DRVV test is determined by the ratio for the screen/PNP = 2.14 and the confirm/PNP = 1.05. The ratio of the normalized screen/normalized confirm = 2.14/1.05=2.01.
[QCMLS 2021: 5.6.3.1.2] [Rodak 2020, p727]

175. **b** Heparin acts as an inhibitor of factors II and X and will therefore not correct a mixing study. Both hemophilia A and B are due to factor deficiencies and will correct during a mixing study.
[QCMLS 2021: 5.7.5.1.3] [Rodak 2020, p751]

176. **b** Type 2B has enhanced response to ristocetin; all of the other types have a decreased response to ristocetin activity.
[QCMLS 2021: 5.5.2.3.1.2] [Rodak 2020, p660]

177. **c** Rare bleeding disorders are recessive and have more aggressive bleeding outcomes. Since they are recessive, they are frequently transmitted in consanguineous marriages.
[QCMLS 2021: 5.6.1.3] [Rodak 2020, p667]

178. **d** In acute DIC, patients present with a prolonged PT aPTT, increased d-dimer and decreased platelets.
[QCMLS 2021: 5.6.2.3] [Rodak 2020, p735-736]

179. **d** Vitamin K deficiency can be seen in patients on antibiotics, in malabsorption syndrome and in the absence of bile salts.
[QCMLS 2021: 5.6.2.4] [Rodak 2020, p656]

180. **a** When hemophiliacs are repeatedly given factor concentrates, a complication of them is the formation of inhibitors. They are not given platelet concentrates, for hemophilia is a disorder of secondary hemostasis.
[QCMLS 2021: 5.6.1.1] [Rodak 2020, p665-667]

181. **c** Acquired hemophilia is caused by autoimmune disease and by autoantibodies. It occurs in elderly and pregnant patients. Inherited hemophila is caused by alloantibodies and is X-linked recessive and occurs in young males.
[QCMLS 2021: 5.6.2.1] [Rodak 2020, p656]



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182. **c** All diagnoses will present with a prolonged aPTT, but will correct in a mixing study. A factor inhibitor to factor VIII will cause bleeding, but will not correct in a mixing study.
[QCMLS 2021: 5.8.2.8] [Rodak 2020, p779]

183. **a** Due to the impact of warfarin on the vitamin-K-dependent factors (II, VII, IX and X) the patients PT is prolonged. An injection of vitamin K would help to correct the PT, protamine sulfate is used to neutralize heparin. When there is a large bleed, FFP is given to replace the vitamin K-dependent factors.
[QCMLS 2021: 5.7.5.1.1] [Rodak 2020, p748]

184. **b** *In vitro*, the LA presents in the laboratory as an antibody to phospholipids. The best way to detect for an LA is to screen patients using an aPTT reagent with a low concentration of phospholipids to allow the LA to manifest, resulting in a prolonged aPTT. An aPTT with a high concentration of phospholipids will mask the antibody and will not prolong the aPTT. PT reagents are not used to screen for LA.
[QCMLS 2021: 5.8.3.1.1] [Rodak 2020, p726-727]

185. **a** Patients with a FXII deficiency with present with a prolonged aPTT, a normal platelet count and an increased risk of thrombosis. They do not bleed.
[QCMLS 2021: 5.6.1.2] [Rodak 2020, p536]

186. **c** An inhibitor that is caused by an interfering substance (like heparin) will cause an increase in factor activity levels as the interfering substance is diluted.
[QCMLS 2021: 5.8.2.9] [Rodak 2020, p782-783]

187. **d** The delayed bleeding and the normal screening tests suggest a FXIII deficiency. Both lupus and FXII would result in a thrombotic event and chronic DIC would be the result of an underlying disorder.
[QCMLS 2021: 5.6.1.2] [Rodak 2020, p784]

188. **a** Due to antibodies to prothrombin, generally an LA patient is prone to thrombosis and does not exhibit abnormal bleeding. They do not present with factor deficiencies. An LA patient may bleed if they form antibodies to prothrombin. It is a rare event.
[QCMLS 2021: 5.7.3.2]

189. **c** A type 1 AT deficiency is a quantitative deficiency resulting in a decreased antigen and activity, while a type 2 AT deficiency is qualitative and will demonstrate a normal antigen and a decreased activity level.
[QCMLS 2021: 5.7.2.1]

190. **c** The anti-Xa assay is used to monitor unfractionated heparin the most accurately. This should be used when the aPTT may not reflect the therapeutic range of UFH due to either a wrong dose or an incorrect therapeutic range. The factor X and II assay are used to determine factor deficiencies, and the AT assay is used to detect levels of AT.
[QCMLS 2021: 5.8.3.3.1] [Rodak 2020, p753]

191. **b** When investigating a lupus anticoagulant, the DRVV looks at the alteration of the phospholipids and the hexagonal phase test looks at the clot based function of an LA. It is based on distinguishing an anti-phospholipid antibody which configures in a bilayer configuration versus an LA, which configures in a hexagonal phase configuration.
[QCMLS 2021: 5.8.3.1.3] [Rodak 2020, p727]

192. **c** Extended half-life products have improved the quality of life of hemophilia patients. By using a product that lasts longer, they have fewer bleeding events and more time between infusions. Extended half life products have been developed for both hemophilia A and B patients.
[QCMLS 2021: 5.6.1.1.3] [Rodak 2020, p665-667]

193. **c** Patients with COVID-19 present with prolonged coagulation screening tests, but the higher and more persistently high D-dimer levels are associated with poorer outcomes and an increased risk of mortality.
[QCMLS 2021: 5.9.4]

194. **a** The initiation phase occurs on a tissue factor bearing cell. Amplification occurs when platelets and cofactors are activated to set the stage for large scale thrombin generation. Propagation occurs when large amounts of thrombin are generated. Fibrinogen is converted into fibrin by thrombin in the last step of the coagulation cascade.
[QCMLS 2021: 5.4.3] [Rodak 2020, p639]



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Chapter 6

# Microbiology

*The following items have been identified generally as appropriate for those preparing for both the MLS and MLT examinations. Items that are appropriate for the MLS examination **only** are marked with MLS ONLY.*


**413 Questions**
- 413 I. Preanalytic Procedures
- 419 II. Analytic Procedures for Bacteriology
- 458 III. Analytic Procedures for Mycology, Mycobacteriology, Parasitology & Virology
- 483 IV. Postanalytic Procedures

**484 Answers with Explanations**
- 484 I. Preanalytic Procedures
- 488 II. Analytic Procedures for Bacteriology
- 509 III. Analytic Procedures For Mycology, Mycobacteriology, Parasitology & Virology
- 525 IV. Postanalytic Procedures


## I. Preanalytic Procedures

### A. Specimen Collection & Transport

1. The proper blood-to-broth ratio for blood cultures to reduce the antibacterial effect of serum in adults is:

   a 1:2
   b 1:3
   c 1:10
   d 1:30

2. The most appropriate method for collecting a urine specimen from a patient with an indwelling catheter is:

   a remove the catheter, cut the tip, and submit it for culture
   b disconnect the catheter from the bag, and aseptically collect urine from the terminal end of the catheter
   c aseptically collect urine directly from the drainage bag
   d aspirate urine aseptically from the catheter tubing

3. Which of the following groups of specimens would be acceptable for anaerobic culture?

   a vaginal swab, eye swab
   b intraoral surface swab, leg tissue
   c pleural fluid, brain abscess fluid
   d urine, sputum

4. Sodium polyanetholsulfonate (SPS) is used as an anticoagulant for blood cultures because it:

   a inactivates penicillin and cephalosporins
   b prevents clumping of red cells
   c inactivates neutrophils and components of serum complement
   d facilitates growth of anaerobes

5. The optimal collection of a wound specimen for culture of anaerobic organisms is a:

   a swab of lesion obtained before administration of antibiotics
   b swab of lesion obtained after administration of antibiotics
   c syringe filled with pus, obtained before administration of antibiotics
   d syringe filled with pus, obtained after administration of antibiotics

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6. The most important variable in the recovery of organisms in adult patients with bacteremia (bacterial sepsis) is:

   **a** subculture of all bottles at day 5 of incubation
   **b** the recommended volume of blood cultured
   **c** collection of daily blood culture sets for 3 consecutive days
   **d** collection of multiple blood culture sets from a single venipuncture

7. Virus transport medium should contain agents that:

   **a** enable rapid viral growth during the transport time
   **b** inhibit bacterial and fungal growth
   **c** destroy nonpathogenic viruses
   **d** inhibit complement-fixing antibodies

*B. Specimen Processing*

8. A bronchoscopy sample with the request for culture of *Legionella* is sent to the laboratory. The correct plating protocol is:

   **a** culture on thiosulfate citrate bile salt media
   **b** incubate the culture media anaerobically
   **c** reject the specimen and request a sputum sample
   **d** culture on buffered charcoal yeast extract agar with antibiotics

9. A community hospital microbiology laboratory is processing significant numbers of stool cultures because of an outbreak of diarrhea following heavy rains and flooding in the county. A media that should be incorporated in the plating protocol is:

   **a** colistin nalidixic acid for *Listeria*
   **b** MacConkey agar with sorbitol for *Campylobacter*
   **c** mannitol salt agar for *Enterococcus* species
   **d** thiosulfate citrate bile salts sucrose for *Vibrio* species

10. A male urethral discharge specimen submitted for culture should be inoculated to:

    **a** sheep blood and phenylethyl alcohol agars
    **b** eosin-methylene blue and sheep blood agars
    **c** thioglycolate broth and chocolate agar
    **d** chocolate and modified Thayer-Martin agars

11. Which selective medium is used for the isolation of gram-positive microorganisms?

    **a** Columbia CNA with 5% sheep blood
    **b** trypticase soy agar with 5% sheep blood
    **c** eosin methylene blue
    **d** modified Thayer-Martin

12. *Campylobacter jejuni* isolation requires the fecal specimen be:

    **a** inoculated onto selective plating media and incubated in reduced oxygen with added $CO_2$ at 42°C
    **b** stored in tryptic soy broth before plating to ensure growth of the organism
    **c** inoculated onto selective plating media and incubated at both 35°C and at room temperature
    **d** incubated at 35°C for 2 hours in Cary-Blair media before inoculating onto selective plating media

13. Which of the following specimen requests is acceptable?

    **a** feces submitted for anaerobic culture
    **b** Foley catheter tip submitted for aerobic culture
    **c** rectal swab submitted for direct smear for gonococci
    **d** urine for culture of acid-fast bacilli


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14. A cerebrospinal fluid specimen containing only 2 drops of CSF is collected by a lumbar puncture from a febrile 25-year-old male and is submitted for a stat Gram stain and culture. The direct specimen Gram stain is reported as many neutrophils and no microorganisms seen. The remaining drop of CSF should be inoculated to:

**a** blood agar
**b** CNA agar
**c** chocolate agar
**d** Thayer-Martin agar

15. A diabetic foot swab from an 82-year-old woman with recurrent infections is submitted for culture. The Gram stain reveals:

many neutrophils, no squamous epithelial cells
many gram-negative bacilli
many gram-positive cocci in chains

The physician requests that all pathogens be worked up. In addition to the sheep blood, chocolate and MacConkey agar plates routinely used for wound cultures, the technologist might also process a(n):

**a** anaerobic blood agar plate
**b** BCYE agar plate
**c** CNA agar plate
**d** XLD agar plate

16. Which of the following is the most appropriate specimen source and primary media selection?

**a** CSF: Columbia CNA, MacConkey
**b** endocervical: chocolate, Martin Lewis
**c** sputum: sheep blood, Thayer-Martin, KV-laked blood
**d** urine: sheep blood, chocolate, Columbia CNA

17. Which of the following is the most appropriate organism and media combination?

**a** *Vibrio* species—Skirrow
**b** Enterohemorrhagic *E. coli*—phenylethyl alcohol (PEA)
**c** *Campylobacter* species—charcoal yeast extract
**d** *Yersinia enterocolitica*—cefsulodin-irgasan-novobiocin (CIN)

18. A Gram stain from a swab of a hand wound reveals:

moderate neutrophils
no squamous epithelial cells
moderate gram-positive cocci in clusters
moderate large gram-negative bacilli

Select the appropriate media that will selectively isolate each organism.

**a** KV-laked agar, Thayer-Martin
**b** sheep blood, MacConkey
**c** Columbia CNA, chocolate
**d** Columbia CNA, MacConkey

19. An organism that must be incubated in a microaerophilic environment for optimal recovery is:

**a** *Campylobacter jejuni*
**b** *Escherichia coli*
**c** *Pseudomonas aeruginosa*
**d** *Proteus mirabilis*

20. *Vibrio parahaemolyticus* is best isolated from feces on:

**a** eosin methylene blue (EMB) agar
**b** Hektoen enteric (HE) agar
**c** Salmonella-Shigella (SS) agar
**d** thiosulfate citrate bile salts (TCBS) agar

21. Media used to support growth of *Legionella pneumophila* should contain the additives:

**a** X and V factors
**b** hemin and vitamin K
**c** charcoal and yeast extract
**d** dextrose and laked blood

22. The best medium for culture of *Bordetella pertussis* is:

**a** phenylethyl alcohol agar
**b** potassium tellurite blood agar
**c** Regan-Lowe agar
**d** Tinsdale agar

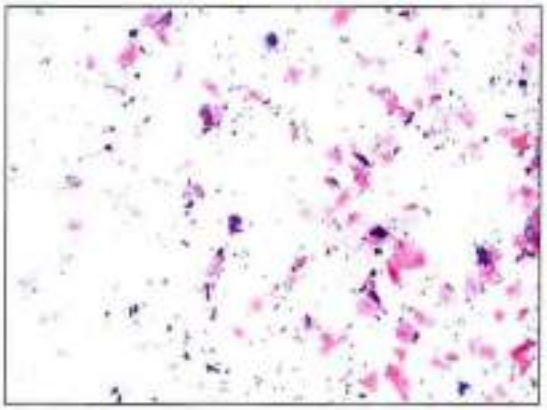
23. A 21-year-old patient presents with pharyngitis. A throat swab is collected and submitted for anaerobic culture. This specimen should be:

**a** set up immediately
**b** rejected as unacceptable
**c** inoculated into thioglycolate broth
**d** sent to a reference laboratory

24. An antibiotic used to suppress or kill contaminating fungi in media is:

**a** amphotericin B and penicillin
**b** chloramphenicol
**c** cycloheximide
**d** streptomycin

25. MLS ONLY A sputum specimen is received for culture and Gram stain. The Gram-stained smear from this specimen is seen in the image (total magnification 100x):



The technologist's best course of action would be to:

**a** inoculate appropriate media and incubate anaerobically
**b** inoculate appropriate media and incubate aerobically
**c** call the physician and notify him of this "life-threatening" situation
**d** call the patient care area and request a new specimen

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26. A vaginal/rectal swab is collected from a pregnant patient to screen for group B *Streptococcus* colonization. The best medium to inoculate the specimen to is:

**a** CNA agar
**b** LIM broth
**c** sheep blood agar
**d** thioglycolate broth

27. When processing a patient specimen for Gram stain and culture, the proper use of a biological safety cabinet includes:

**a** bringing into the cabinet all required media and equipment just prior to setting up each individual specimen
**b** keeping the ultraviolet light on for the first 30 minutes of working in the cabinet
**c** not using any heat generating equipment such as open flames or microburner/incinerators
**d** not disrupting the air curtain barrier by keeping air flow and exhaust grills unobstructed

28. A wound specimen grows 2 colony types on sheep blood agar and 1 clear colony type on MacConkey agar. Sheep blood agar growth is documented as:

colony type #1 swarming over entire plate, Gram stain: gram-negative bacilli
colony type #2 white colony, Gram stain: gram-positive cocci in clusters

The best way to isolate colony type #2 from colony type #1 is to subculture:

**a** colony #1 to sheep blood and chocolate agars
**b** colony #1 to sheep blood and/or MacConkey agar
**c** colony #2 to sheep blood and chocolate agars
**d** colony #2 to CNA and/or PEA agar

29. A differential medium that can be used as a primary isolation agar producing predictable colored colonies that can be distinguished from other organism colony types describes:

**a** buffered charcoal yeast extract agar
**b** blood phenylethyl alcohol agar
**c** campylobacter blood agar
**d** chromagar

30. Anticoagulants acceptable for use with blood, bone marrow and synovial fluid specimens that are to be cultured include:

**a** EDTA and sodium citrate
**b** heparin and sodium citrate
**c** sodium polyanethol sulfonate (SPS) and heparin
**d** sodium polyanethol sulfonate (SPS) and EDTA

31. Appropriate culture requirements for a specimen from a patient suspected of having tularemia include:

**a** a media with cysteine such as buffered charcoal yeast extract agar
**b** colistin nalidixic acid agar
**c** Mueller-Hinton agar with 5% sheep blood agar
**d** Regan-Lowe media

32. The primary isolation of *Neisseria gonorrhoeae* requires:

**a** anaerobic conditions
**b** starch media
**c** carbon dioxide
**d** blood agar

33. Acceptable specimen sources for culture of anaerobic bacteria includes:

**a** sputum
**b** stool
**c** suprapubic bladder aspiration
**d** vaginal

34. In general, anaerobic infections differ from aerobic infections in which one of the following?

**a** they usually respond favorably to aminoglycoside therapy
**b** they usually arise from exogenous sources
**c** they are usually polymicrobic
**d** Gram stains of specimens are less helpful in diagnosis

### *C. Stains: Procedure, Principle, and Interpretation*

35. An expectorated sputum is sent to the laboratory for culture from a patient with respiratory distress. The direct specimen Gram stain shows many squamous epithelial cells (>25/lpf) and rare neutrophils. The microscopic appearance of the organisms present include:

moderate gram-positive cocci in chains and diplococci
moderate gram-negative diplococci
moderate palisading gram-positive bacilli all in moderate amounts

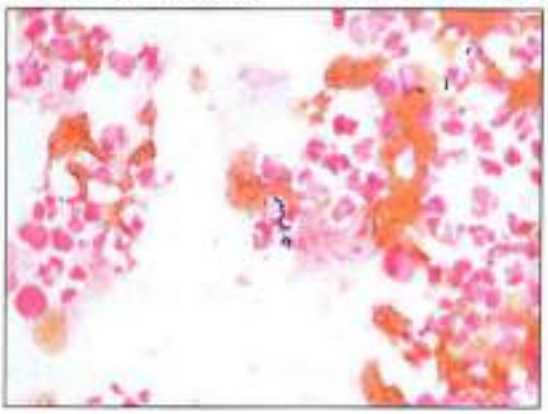
This Gram stain is most indicative of:

**a** a pneumococcal pneumonia
**b** an anaerobic infection
**c** a *Haemophilus* pneumonia
**d** oropharyngeal flora

36. Upon review of a sputum Gram stain, the technician notes that all the neutrophil nuclei in the smear stained dark blue. The best explanation for this finding is the:

**a** iodine was omitted from the staining procedure
**b** slide was inadequately decolorized with acetone/alcohol
**c** sputum smear was prepared too thin
**d** cellular components have stained as expected

37. MLS ONLY The image depicts a Gram stain (final magnification 1000x) of a knee fluid from a patient who has recently undergone knee replacement surgery:



The best interpretation of this Gram stain is:

**a** gram-positive cocci suggestive of *Staphylococcus*
**b** gram-positive bacilli suggestive of *Corynebacterium*
**c** gram-positive bacilli suggestive of *Listeria*
**d** gram-positive cocci suggestive of *Streptococcus*

38. The principle difference between the Ziehl-Neelsen acid-fast stain technique and the Kinyoun acid-fast stain technique is the:

**a** type of dyes used
**b** type of microscope used to interpret stained smears
**c** strength of acid decolorizer
**d** use of heat to allow the dye to penetrate organism


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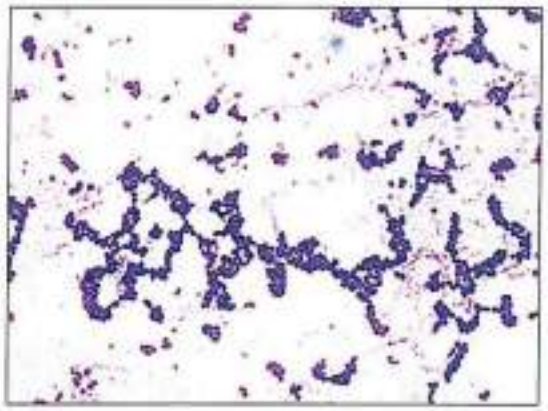
## *II. Analytic Procedures for Bacteriology*

### *A. Blood & Bone Marrow*

**39.** Relapsing fever in humans is caused by:

**a** *Borrelia recurrentis*
**b** *Brucella abortus*
**c** *Leptospira interrogans*
**d** *Spirillum minus*

**40.** Three sets of blood cultures are obtained from an adult patient with fever and suspected endocarditis. Growth in one aerobic bottle is seen after 5 days incubation, and the Gram stain is seen in the image. This indicates that:



**a** there is low-grade bacteremia
**b** the organism is most likely a contaminant
**c** the patient has a line infection
**d** the blood culture bottles are defective

**41.** The Gram stain from a blood culture shows gram-positive cocci in chains. The subcultured plates from the blood culture bottle show no growth. Additional testing should be done to detect the presence of:

**a** *Staphylococcus saprophyticus*
**b** *Aerococcus urinae*
**c** *Abiotrophia defectiva*
**d** *Streptococcus pneumoniae*

**42.** Gram stain examination from a blood culture bottle shows dark blue, spherical organisms in clusters. Growth on sheep blood agar shows small, round, pale yellow colonies. Further tests should include:

**a** catalase production and agglutination test for Protein A
**b** bacitracin susceptibility and latex grouping
**c** oxidase and indole reactions
**d** Voges-Proskauer and methyl red reactions

**43.** Gram-positive cocci in chains are seen on a Gram stain from a blood culture. The organism grows as a beta-hemolytic colony. Further tests that could be performed include:

**a** PYR, bacitracin, and hippurate
**b** catalase and agglutination test for Protein A
**c** oxidase and mass spectrometry
**d** Voges-Proskauer and methyl red

44. "Nutritionally variant" streptococci are:

**a** enterococci
**b** group D enterococci
**c** beta-hemolytic streptococci
**d** in the genera *Granulicatella* and *Abiotrophia*

45. The most frequent cause of prosthetic heart valve infections occurring within 2-3 months after surgery is:

**a** *Streptococcus pneumoniae*
**b** *Streptococcus pyogenes*
**c** *Staphylococcus aureus*
**d** *Staphylococcus epidermidis*

46. Gram-positive cocci isolated from a blood culture have the characteristics shown in this table:

| test | result |
|---|---|
| optochin susceptibility | negative |
| bacitracin (0.04 U) susceptibility | negative |
| bile esculin hydrolysis | negative |
| hippurate hydrolysis | positive |
| catalase | negative |

This organism is most likely:

**a** *Staphylococcus aureus*
**b** *Streptococcus pneumoniae*
**c** *Streptococcus pyogenes*
**d** *Streptococcus agalactiae*

47. During the previous month, *Staphylococcus epidermidis* has been isolated from blood cultures at 2-3 times the rate from the previous year. The most logical explanation for the increase in these isolates is that:

**a** the blood culture media are contaminated with this organism
**b** the hospital ventilation system is contaminated with *Staphylococcus epidermidis*
**c** there has been a break in proper skin preparation before drawing blood for culture
**d** a relatively virulent isolate is being spread from patient to patient

48. A 55-year-old man presents to the emergency room with chest pain and is found to have suffered a heart attack. He has a past history of hypertension, and high cholesterol. The patient is admitted and scheduled for a triple bypass procedure. During recovery, he becomes septic, developing a high grade fever and pneumonia. Gram-positive cocci in clusters, isolated from both his lungs via an induced sputum specimen and the surgical incision, produce beta-hemolytic, catalase-positive colonies on sheep blood agar.

Identify the organism most likely isolated and the biochemical test performed to confirm the identification.

**a** *Staphylococcus aureus* and latex agglutination
**b** *Streptococcus pyogenes* and PYR
**c** *Streptococcus agalactiae* and latex agglutination
**d** *Enterococcus faecium* and PYR


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**49.** MLS ONLY A patient with a prosthetic heart valve visits the dentist for her yearly checkup. She presents to her primary care physician 2 weeks later with a high fever, chills, and shortness of breath and receives a diagnosis of subacute endocarditis. Multiple blood culture sets are drawn on the patient and sent to the lab. The bottles are positive for bacterial growth 24 hours later. Which organism would be expected to grow?

**a** *Staphylococcus lugdunensis*
**b** *Staphylococcus saprophyticus*
**c** *Streptococcus viridans*
**d** *Streptococcus agalactiae*

**50.** A blood culture from a 64-year-old male with lymphoma is positive blood culture at 18 hours incubation. The organisms are nonlactose fermenting gram-negative bacilli on MacConkey agar. Further testing gives the reactions shown in this table:

| test | result |
|---|---|
| oxidase | negative |
| TSI | alkaline/acid, no hydrogen sulfide |
| motility | positive |
| indole | positive |
| citrate | positive |
| ornithine decarboxylase | negative |
| urea | positive |
| phenylalanine deaminase | positive |
| VP | negative |

The genus is:

**a** *Morganella*
**b** *Proteus*
**c** *Providencia*
**d** *Serratia*

**51.** MLS ONLY A blood culture bottle with macroscopic signs of growth is Gram stained and the technician notes small, curved gram-negative bacilli resembling "gull wings." It is subcultured to blood and chocolate agar, and incubated aerobically and anaerobically. After 24 hours, no growth is apparent. The next step should be to:

**a** subculture the bottle, and incubate in microaerophilic conditions
**b** assume the organism is nonviable, and ask for repeat specimen
**c** utilize the oxidase and indole test to detect *Aeromonas*
**d** subculture the bottle to a medium containing X and V factors

**52.** MLS ONLY Which of thes specimen types is considered to be the most sensitive for the recovery of *Brucella* in cases of chronic infection?

**a** blood
**b** urine
**c** bone marrow
**d** lymph node

53. A college student attends a beach party where raw oysters and other shellfish are consumed. The next day, he has symptoms of septicemia. The blood cultures grow gram-negative bacilli with the characteristics shown in the table:

| test | result |
|---|---|
| oxidase | positive |
| MacConkey agar | pink colonies |
| O/129 (150 µg) | susceptible |

The most likely organism is:

**a** *Aeromonas hydrophila*
**b** *Pseudomonas putida*
**c** *Serratia marcescens*
**d** *Vibrio vulnificus*

54. MLS ONLY The laboratory receives a blood culture from a veterinarian who has been ill for many weeks with fevers in the afternoon and evenings, arthritis, and fatigue. The blood culture is positive after 5 days, and the organism has the characteristics shown in this table:

| test | result |
|---|---|
| Gram stain | small, gram-negative coccobacilli |
| sheep blood agar | growth after 48 hours with small, smooth, raised colonies |

What should the microbiologist do next?

**a** consider the growth contamination and perform another Gram stain
**b** perform biochemical identification for HACEK organisms
**c** perform identification and susceptibility testing using an automated system
**d** take extra safety precautions for possible *Brucella*

55. *Cutibacterium* (formerly *Propionibacterium) acnes* is most often associated with:

**a** food poisoning
**b** post-antibiotic diarrhea
**c** tooth decay
**d** blood culture contamination

56. MLS ONLY Which one of the following anaerobes is inhibited by sodium polyanethol sulfonate (SPS)?

**a** *Bacteroides fragilis*
**b** *Cutibacterium acnes*
**c** *Peptostreptococcus anaerobius*
**d** *Veillonella parvula*

57. MLS ONLY A patient has a suspected diagnosis of subacute bacterial endocarditis. His blood cultures grow non-spore-forming pleomorphic gram-positive bacilli only in the anaerobic bottle. What test(s) will give a presumptive identification of this microorganism?

**a** beta-hemolysis and oxidase
**b** catalase and spot indole
**c** esculin hydrolysis
**d** hydrolysis of gelatin

58. MLS ONLY Microorganisms resembling *Mycoplasma pneumoniae* have been isolated from the blood of patients treated with antibiotics that:

**a** complex with flagellar protein
**b** interfere with cell membrane function
**c** inhibit protein synthesis
**d** interfere with cell wall synthesis

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*B. Cerebrospinal Fluid*

**59.** Cerebrospinal fluid test results that are most consistent with viral meningitis include:

MLS ONLY

**a** decreased protein level
**b** increased glucose level
**c** increased lactate level
**d** lymphocytes predominant

**60.** The organism most commonly associated with neonatal purulent meningitis is:

**a** *Neisseria meningitidis*
**b** *Streptococcus pneumoniae*
**c** group B streptococci
**d** *Haemophilus influenzae*

**61.** MLS ONLY Beta-hemolytic gram-positive cocci are isolated from the cerebrospinal fluid of a 2-day-old infant with signs of meningitis. The isolate grows on sheep blood agar under aerobic conditions and is resistant to a bacitracin disc. Which of the following should be performed for the identification of the organism?

**a** oxidase production
**b** catalase formation
**c** latex antigen grouping
**d** esculin hydrolysis

**62.** MLS ONLY A 4-year-old is admitted with symptoms of meningitis, and a Gram stain of the cerebrospinal fluid reveals small, pleomorphic, gram-negative coccobacilli. After 24 hours incubation at 35°C, small, moist, gray colonies, which are oxidase variable, are found on the chocolate agar plate only. Which of the following biochemical data would be consistent with this isolate?

| | | |
|---|---|---|
| **a** | CTA dextrose | positive |
| | CTA maltose-positive | |
| | ONPG-negative | |
| **b** | sodium hippurate hydrolysis-positive | |
| | A disc | negative |
| | CAMP test | positive |
| **c** | X factor | no growth |
| | V factor | no growth |
| | XV factor | growth |
| | horse blood | no hemolysis |
| **d** | catalase-positive | |
| | esculin hydrolysis-positive | |
| | methyl red | positive |
| | "umbrella" motility at 22°C | positive |

63. A technologist is reading a Gram stain from a CSF and observes many intracellular gram-negative diplococci. Which set of chemistry and hematology CSF results would most likely be seen in someone with this type of infection?

| CSF results | WBC | glucose | protein |
|---|---|---|---|
| a | increased | increased | increased |
| b | decreased | decreased | decreased |
| c | increased | decreased | increased |
| d | decreased | increased | decreased |

**a** result a
**b** result b
**c** result c
**d** result d

64. An 18-year-old boy is admitted to the hospital with suspected meningitis. He is lethargic and presents with a rigid neck. He has not had most of the recommended vaccines from childhood to now. Gram stain of his spinal fluid shows many PMNs with intra- and extracellular gram-negative diplococci. The suspected pathogen is:

**a** *Listeria monocytogenes*
**b** *Haemophilus influenzae*
**c** *Streptococcus agalactiae*
**d** *Neisseria meningitidis*

### *C. Body Fluids from Normally Sterile Sites*

65. A 25-year-old man who had recently worked as a steward on a transoceanic grain ship presented to the emergency room with high fever, diarrhea and prostration. Axillary lymph nodes are hemorrhagic and enlarged. A Wright-Giemsa stain of the aspirate shows bacilli that are bipolar, resembling safety pins. The most likely identification of this organism is:

**a** *Brucella melitensis*
**b** *Streptobacillus moniliformis*
**c** *Spirillum minus*
**d** *Yersinia pestis*

66. Anaerobic gram-positive bacilli with subterminal spores are isolated from a peritoneal abscess. The colony has a swarming appearance. The most likely identification of this organism is:

**a** *Bacillus cereus*
**b** *Clostridium septicum*
**c** *Eggerthella lenta*
**d** *Bifidobacterium dentium*

67. A Gram stain of a peritoneal fluid shows large gram-positive bacilli. There is 3+ growth on anaerobic media only, with colonies producing a double zone of hemolysis. To assist with the classic identification of the organism, the microbiologist could:

**a** determine if the organism ferments glucose
**b** perform the oxidase test
**c** set up egg yolk agar plate
**d** test for bile tolerance

68. An organism from a peritoneal abscess is isolated on kanamycin-vancomycin laked blood agar and grows black colonies on BBE agar. It is nonpigmented, catalase-positive, and indole-negative. The genus of this organism is:

**a** *Acidaminococcus*
**b** *Bacteroides*
**c** *Porphyromonas*
**d** *Prevotella*

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69. MLS ONLY Thin, gram-negative bacilli with tapered ends isolated from an empyema specimen grow only on anaerobic sheep blood agar. They are found to be indole-positive, lipase-negative, and are inhibited by 20% bile. The most probable identification of this isolate would be:

**a** *Bacteroides distasonis*
**b** *Prevotella melaninogenica*
**c** *Fusobacterium nucleatum*
**d** *Clostridium septicum*

### *D. Lower Respiratory*

70. MLS ONLY A 10-year-old child with cystic fibrosis presents with cough and shortness of breath. Her sputum Gram stain is seen in the image:



Based on the Gram stain, the best medium and incubation condition to optimize recovery of the organism seen is:

**a** MacConkey agar incubated in $CO_2$
**b** Tinsdale agar incubated in ambient air
**c** chocolate agar incubated in $CO_2$
**d** CNA agar incubated in ambient air

71. A sputum culture from an alcoholic seen in the ER grows gray, mucoid, stringy colonies on sheep blood agar. The isolate grows readily on MacConkey agar and forms mucoid, dark pink colonies. The colonies yield the test results shown in this table:

| test | result |
|---|---|
| ONPG | positive |
| indole | negative |
| glucose | positive |
| oxidase | negative |
| citrate | positive |
| VP | positive |

The organism is most likely:

**a** *Edwardsiella tarda*
**b** *Klebsiella pneumoniae*
**c** *Escherichia coli*
**d** *Proteus vulgaris*

72. MLS ONLY A patient with a nosocomial pneumonia has a sputum Gram stain that shows many neutrophils and numerous small gram-negative coccobacilli. The organism grows in 24 hours as a mucoid, hemolytic colony on blood agar and as a colorless colony on a MacConkey agar. The organism has the characteristics shown in this table:

| test | result |
|---|---|
| oxidase | negative |
| catalase | positive |
| nitrate | negative |
| ONPG | negative |
| ornithine decarboxylase | negative |
| lysine decarboxylase | negative |

The organism is:

**a** *Stenotrophomonas maltophilia*
**b** *Alcaligenes faecalis*
**c** *Moraxella lacunata*
**d** *Acinetobacter baumannii*

73. MLS ONLY Serum samples collected from a patient with pneumonia demonstrate a rising antibody titer to *Legionella*. A bronchoalveolar lavage (BAL) specimen from this patient has a positive antigen test for *Legionella*, but no organisms are recovered on buffered charcoal yeast extract medium after 2 days of incubation. The best explanation is that the:

**a** antibody titer represents an earlier infection
**b** positive antigen test is a false-positive
**c** specimen was cultured on the wrong media
**d** culture was not incubated long enough

74. MLS ONLY A 17-year-old female with cystic fibrosis is diagnosed with pneumonia. A sputum sample grows gram-negative bacilli with yellow, smooth colonies that have the biochemical reactions shown in this table:

| test | result |
|---|---|
| oxidase | positive |
| TSI | alk/alk |
| glucose | oxidized |
| fluorescence | negative |
| lysine decarboxylase | positive |

The most likely organism is:

**a** *Burkholderia cepacia*
**b** *Klebsiella pneumoniae*
**c** *Shewanella putrefaciens*
**d** *Stenotrophomonas maltophilia*

75. MLS ONLY A college student who recently studied a semester abroad in Southeast Asia is admitted to the hospital with a diagnosis of "glanders-like" infection (melioidosis). A sputum specimen grows small, gram-negative bacilli that are positive for oxidase, reduce nitrate to gas, and oxidize glucose, lactose, and mannitol. What is the most likely organism?

**a** *Stenotrophomans maltophilia*
**b** *Burkholderia pseudomallei*
**c** *Pseudomonas aeruginosa*
**d** *Acinetobacter baumannii*


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76. An organism previously thought to be nonpathogenic, *Moraxella catarrhalis*, is now known to be associated with opportunistic respiratory infection and nosocomial transmission. Characteristic identification criteria include:

**a** beta-lactamase-negative
**b** butyrate esterase-positive
**c** gram-negative bacilli
**d** oxidase-negative

77. An organism recovered from a sputum has the following characteristics:

| test | result |
|---|---|
| culture | growth at 6 days on buffered charcoal yeast extract (BCYE) agar, incubated under aerobic conditions with $CO_2$ at 35°C |
| Gram stain | delicate branching gram-positive bacilli |
| modified acid-fast stain | branching, filamentous, "partially" acid-fast bacterium |

These results are consistent with which of the following genera?

**a** *Nocardia*
**b** *Mycobacterium*
**c** *Actinomyces*
**d** *Streptomyces*

78. The Gram stain of drainage from a pulmonary sinus tract shows many WBCs and 3+ branching gram-positive bacilli. Colonies grow only on anaerobic media after 3 days incubation. They are yellow-tan and have a molar tooth appearance. The most likely genus is:

**a** *Actinomyces*
**b** *Bacteroides*
**c** *Fusobacterium*
**d** *Nocardia*

79. MLS ONLY A 1-2 mm translucent, nonpigmented colony, isolated from an anaerobic culture of a lung abscess after 72 hours, is found to fluoresce brick-red under ultraviolet light. A Gram stain of the organism reveals coccobacilli that have the characteristics shown in this table:

| test | result |
|---|---|
| growth in bile | inhibited |
| vancomycin | resistant |
| kanamycin | resistant |
| colistin | susceptible |
| catalase | negative |
| esculin hydrolysis | negative |
| indole | negative |

The identification of this isolate is:

**a** *Bacteroides ovatus*
**b** *Cutibacterium acnes*
**c** *Prevotella melaninogenica*
**d** *Porphyromonas asaccharolytica*

*E. Upper Respiratory*

**80.** Psittacosis is transmissible to man via contact with:

**a** insects
**b** birds
**c** cattle
**d** dogs

**81.** MLS ONLY Which 2 diseases are usually preceded by infection with the organism seen in the image below?

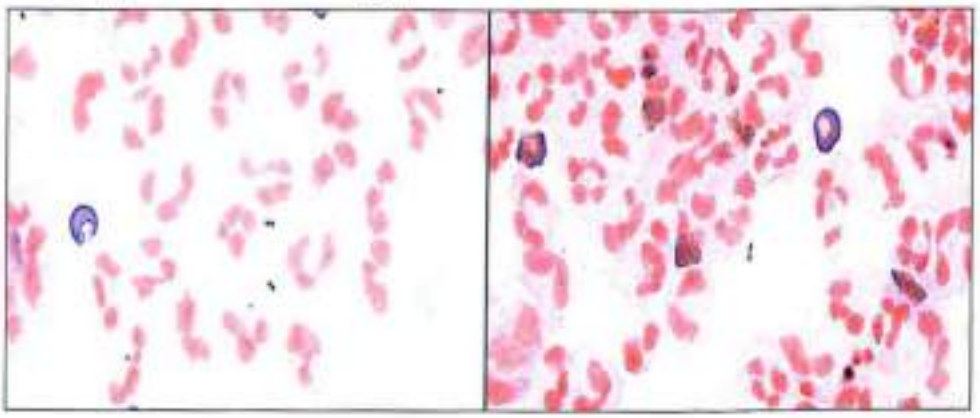


**a** rheumatic fever, undulant fever
**b** glomerulonephritis, rheumatic fever
**c** rheumatic fever, tularemia
**d** glomerulonephritis, undulant fever

**82.** A common cause of acute exudative pharyngitis is:

**a** *Staphylococcus aureus* (beta-hemolytic]
**b** *Streptococcus pneumoniae*
**c** *Streptococcus agalactiae*
**d** *Streptococcus pyogenes*

**83.** A 15-year-old is admitted to the ER with severe sinusitis. Aspiration specimens from the nasal passage reveal a pure culture of alpha-hemolytic, depressed center colonies with a distinctive mucoid appearance on a blood agar plate. Gram stains of the colonies are shown below:



Which of the following could aid in the identification of the organism recovered?

**a** bile solubility, optochin sensitivity
**b** hippurate hydrolysis, bile esculin
**c** bacitracin sensitivity, Lancefield grouping
**d** PYR positivity, catalase negativity


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84. A 6-year-old male presents to his pediatrician with a severe case of tonsillitis. The physician collects a throat swab specimen and orders a GAS (group A *Streptococcus*) probe test. The following day, the probe comes back negative. A culture is requested. The laboratory results are shown in this table:

| test | result |
|---|---|
| catalase | negative |
| bacitracin disk | sensitive |
| hippurate hydrolysis | negative |
| CAMP test | negative |
| PYR | negative |
| Gram stain | gram-positive cocci in chains |

Which of the following organisms is most likely causing the tonsillitis?

**a** group A beta-hemolytic streptococci
**b** group B beta-hemolytic streptococci
**c** group C beta-hemolytic streptococci
**d** group D beta-hemolytic streptococci

85. Small, pleomorphic gram-negative bacilli are isolated from an eye culture. They grow only on chocolate agar and are oxidase-variable. The most likely organism is:

**a** *Acinetobacter lwoffii*
**b** *Haemophilus influenzae*
**c** *Stenotrophomonas maltophilia*
**d** *Pseudomonas aeruginosa*

86. An isolate on chocolate agar from a patient with epiglottitis is suggestive of *Haemophilus* species. Additional testing shows that the isolate required NAD for growth and is nonhemolytic. The organism is most likely *Haemophilus*:

**a** *haemolyticus*
**b** *ducreyi*
**c** *influenzae*
**d** *parainfluenzae*

87. MLS ONLY A 3-year-old unimmunized female presents to the ER with a severe cough, fever, and flu-like symptoms. The parents report that the child had vomited a few times due to the severe coughing. A nasopharyngeal swab is used to collect the specimen and planted on a chocolate, Bordet-Gengou, and Regan Lowe media. After 5 days of incubation, colonies grow on all of the media, with the growth on Bordet-Gengou described as "drops of mercury." The Gram stain shows minute coccobacilli that are catalase-positive and oxidase-positive. The most likely identification of this isolate is:

**a** *Bordetella parapertussis*
**b** *Yersinia pestis*
**c** *Francisella tularensis*
**d** *Bordetella pertussis*

88. A Gram stain performed on a sinus aspirate reveals gram-negative diplococci and PMNs. Oxidase testing is positive and carbohydrate degradation tests are inert. The organism most likely is:

**a** *Neisseria lactamica*
**b** *Moraxella catarrhalis*
**c** *Neisseria meningitidis*
**d** *Neisseria sicca*

89. Establishing the pathogenicity of a microorganism isolated from a child's throat and identified as *Corynebacterium diphtheriae* would depend upon: (MLS ONLY)

**a** the morphological appearance as revealed by Gram stain
**b** the type of hemolysis on blood agar
**c** a positive toxigenicity test
**d** the appearance of growth on Tinsdale tellurite agar

90. *Chlamydia trachomatis* infections have been implicated in:

**a** LGV and conjunctivitis
**b** gastroenteritis and urethritis
**c** neonatal pneumonia and gastroenteritis
**d** neonatal meningitis and conjunctivitis

### *F. Gastrointestinal*

91. A liquid fecal specimen from a 3-month-old infant is submitted for culture. The stool culture should detect *Salmonella*, *Shigella* and:

**a** *Campylobacter* species
**b** *Clostridium botulinum*
**c** *Entamoeba hartmanni*
**d** enterotoxigenic *Escherichia coli*

92. When performing a stool culture, a colony type typical of an enteric pathogen is subcultured on a blood agar plate. The resulting pure culture is screened with several tests to obtain the results shown in this table:

| test | result |
|---|---|
| TSI | acid butt, alkaline slant, no gas, no $H_2S$ |
| phenylalanine deaminase | negative |
| motility | nonmotile |
| serological typing | *Shigella flexneri* (*Shigella* subgroup B) |

The serological typing is verified with new kit and controls. The best course of action would be to:

**a** report the organism as *Shigella flexneri* without further testing
**b** verify reactivity of motility medium with positive and negative controls
**c** verify reactivity of the TSI slants with positive and negative controls for $H_2S$ production
**d** verify reactivity of phenylalanine deaminase with positive and negative controls

93. MacConkey media for screening suspected cases of hemorrhagic *E. coli* O157:H7 must contain:

**a** indole
**b** citrate
**c** sorbitol
**d** lactose


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94. An isolate from a stool culture gives the growth characteristics and biochemical reactions shown in this table:

| test | result |
|---|---|
| MacConkey agar | colorless colonies |
| Hektoen agar | yellow-orange colonies |
| TSI | acid slant/acid butt, no gas, no $H_2S$ |
| urea | positive |

These screening reactions are consistent with which of these enteric pathogens?

**a** *Yersinia enterocolitica*
**b** *Shigella sonnei*
**c** *Vibrio parahaemolyticus*
**d** *Campylobacter jejuni*

95. Which of the following organisms can grow in the small bowel and cause diarrhea in children, traveler's diarrhea, or a severe cholera-like syndrome through the production of enterotoxins?

**a** *Yersinia enterocolitica*
**b** *Escherichia coli*
**c** *Salmonella typhi*
**d** *Shigella dysenteriae*

96. *Shigella* species characteristically are:

**a** urease-positive
**b** nonmotile
**c** oxidase-positive
**d** lactose fermenters

97. Gram-negative bacilli have been isolated from feces, and the confirmed biochemical reactions fit those of *Shigella*. The organism does not agglutinate in *Shigella* antisera. What should be done next?

**a** test the organism with a new lot of antisera
**b** test with Vi antigen
**c** repeat the biochemical tests
**d** boil the organism and retest with the antisera

98. Biochemical reactions of an organism are consistent with *Shigella*. A suspension is tested in antiserum without resulting agglutination. However, after 15 minutes of boiling, agglutination occurs in group-D antisera. The *Shigella* species is:

**a** *dysenteriae*
**b** *flexneri*
**c** *boydii*
**d** *sonnei*

**99.** MLS ONLY An 8-year-old girl is admitted to the hospital with a 3-day history of fever, abdominal pain, diarrhea, and vomiting. A stool culture grows many lactose-negative colonies that yielded the laboratory results shown in this table:

| test | result |
|---|---|
| oxidase | negative |
| TSI | acid slant/acid butt |
| indole | negative |
| urease | positive |
| ornithine decarboxylase | positive |
| sucrose | positive |
| $H_2S$ | negative |
| motility at 25°C | positive |

The most probable identification of this organism is:

**a** *Escherichia coli*
**b** *Providencia stuartii*
**c** *Yersinia enterocolitica*
**d** *Edwardsiella tarda*

**100.** A fecal specimen, inoculated to xylose lysine deoxycholate (XLD) and Hektoen enteric (HE) produced colonies with black centers. Additional testing results are shown in this table:

| biochemical screen | result | serological test | result |
|---|---|---|---|
| Glucose fermentation | positive | polyvalent | no agglutination |
| $H_2S$ | positive | group A | no agglutination |
| lysine decarboxylase | positive | group $B_1$ | no agglutination |
| urea | negative | group C | no agglutination |
| ONPG | negative | group D | no agglutination |
| indole | positive | group $V_1$ | no agglutination |

The most probable identification is:

**a** *Salmonella enterica*
**b** *Edwardsiella tarda*
**c** *Proteus mirabilis*
**d** *Shigella sonnei*

**101.** MLS ONLY A 10-year-old boy is admitted to the emergency room with lower right quadrant pain and tenderness that mimicks appendicitis. The laboratory results shown in this table are obtained:

| | patient value | normal range |
|---|---|---|
| % segmented neutrophils | 75% | 16-60% |
| WBC count | $200 \times 10^3/\mu L$ ($200 \times 10^9/L$) | $13.0 \times 10^3/\mu L$ ($13.0 \times 10^9/L$) |

The admitting diagnosis is appendicitis. During surgery, the appendix appears normal; an enlarged node is removed and cultured. Small gram-negative bacilli are isolated from the room temperature plate. The organism most likely is:

**a** *Prevotella melaninogenica*
**b** *Shigella sonnei*
**c** *Listeria monocytogenes*
**d** *Yersinia enterocolitica*


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102. Biochemical reactions of an organism are consistent with *Salmonella*. A suspension is tested in polyvalent antiserum A through G and in Vi antiserum. There is agglutination in the Vi antiserum only. What should be done next?

**a** boil suspension of the organism for 10 minutes to inactivate the Vi antigen
**b** test organism with individual antisera for agglutination
**c** report "no *Salmonella* isolated"
**d** repeat biochemical identification of the organism

103. The optimal incubator temperature for isolation of the *Campylobacter jejuni/coli* group is:

**a** 4°C
**b** 20°C
**c** 25°C
**d** 42°C

104. Which test can be used to diagnose infection and confirm eradication of *Helicobacter pylori*?

MLS ONLY

**a** DNase
**b** hippurate hydrolysis
**c** string test
**d** urea breath test

105. A Gram stain of a touch prep from a gastric biopsy shows gram-negative bacilli that are slender and curved. The most likely pathogen is:

MLS ONLY

**a** *Burkholderia cepacia*
**b** *Corynebacterium urealyticum*
**c** *Helicobacter pylori*
**d** *Pasteurella multocida*

106. A very bloody stool is received by the laboratory. The following day a pathogenic strain of *E. coli* is isolated. Which sugar should this isolate be tested against to begin the identification process?

**a** mannitol
**b** sorbitol
**c** lactose
**d** arabinose

107. Optimum growth of *Campylobacter jejuni* is obtained on suitable media incubated at 42°C in an atmosphere containing:

**a** 6% $O_2$, 10-15% $CO_2$, 85-90% nitrogen
**b** 10% $H_2$, 5% $CO_2$, 85% nitrogen
**c** 10% $H_2$, 10% $CO_2$, 80% nitrogen
**d** 25% $O_2$, 5% $CO_2$, 70% nitrogen

108. A medical technologist is working on a stool culture from a patient with severe, bloody diarrhea. She wants to set up biochemicals to differentiate *Shigella* and *E. coli*. Which of the following tests would be the most appropriate?

**a** hydrogen sulfide, ONPG, motility, urease
**b** lactose, indole, ONPG, motility
**c** urease, citrate, VP, hydrogen sulfide
**d** gas, MR, urease, citrate

109. Which organism commonly causes food poisoning by consumption of foods containing excessive populations of organisms and/or preformed enterotoxin?

**a** *Salmonella enteritidis*
**b** *Shigella sonnei*
**c** *Bacillus cereus*
**d** *Aeromonas hydrophila*

110. The laboratory is considering adoption of a rapid and sensitive "stand alone" method that detects *Clostridioides difficile* toxins A and B. Which one of the following testing methodologies will provide this? (ILS ONLY)

**a** cell culture cytotoxin assay
**b** latex agglutination
**c** lecithinase production
**d** NAAT

111. *Clostridioides difficile* can be detected by:

**a** fluorescent staining
**b** glutamate dehydrogenase
**c** growth on LKV media
**d** high pressure liquid chromatography

112. If a stool sample is sent to the laboratory to rule out *Clostridioides difficile*, what medium should the microbiologist use, and what is the appearance of this organism on this medium?

**a** BBE: colonies turn black
**b** *Brucella* agar: red pigmented colonies
**c** CCFA: yellow, ground glass colonies
**d** CNA: double zone hemolytic colonies

113. The enterotoxin produced by certain strains of hemolytic, coagulase-positive *Staphylococcus aureus*:

**a** is the primary cause of subacute endocarditis
**b** creates a biofilm on indwelling catheters
**c** causes a rapidly occurring (2-6 hours after ingestion) food poisoning
**d** is of extremely low virulence

### *G. Skin, Soft Tissue, and Bone*

114. After being admitted to the hospital with an oozing leg wound post-operation, A 56-year-old male has 2 sets of blood cultures that grow gram-positive cocci. The infected area is red, swollen, and warm to the touch. A red line has appeared at the sight of the wound and is beginning to travel up the patient's leg. Biochemicals performed from the beta-hemolytic colonies on the sheep blood agar plate reveal the results shown in this table:

| test | result |
|---|---|
| CAMP test | negative |
| hippurate hydrolysis | negative |
| PYR | positive |
| bacitracin | sensitive |
| 65% NaCl | no growth |
| bile esculin | negative |

The most likely identification is:

**a** *Streptococcus pyogenes*
**b** *Streptococcus agalactiae*
**c** *Staphylococcus aureus*
**d** *Enterococcus faecalis*

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115. A young boy who routinely bites is finger nails develops a wound on his right pointer finger. A culture reveals alpha-hemolytic, dry colonies on the blood and CNA plates that are catalase-negative, resistant to optochin, and 6.5% NaCl negative. Gram stain of the colony is gram-positive cocci in chains. The organism most likely isolated is:

**a** *Enterococcus faecium*
**b** *Enterococcus faecalis*
**c** *Streptococcus viridans*
**d** *Streptococcus agalactiae*

116. An organism isolated from the surface of a skin burn is found to produce a diffusible green pigment on a blood agar plate. Further studies of the organism would most likely show the organism to be:

**a** *Staphylococcus aureus*
**b** *Serratia marcescens*
**c** *Elizabethkingia meningoseptica*
**d** *Pseudomonas aeruginosa*

117. Nonfermenting gram-negative bacilli are isolated from a wound. The nitrate and oxidase are strongly positive. The growth on Mueller-Hinton agar produces pyoverdin. The organism is:

**a** *Burkholderia cepacia*
**b** *Moraxella lacunata*
**c** *Elizabethkingia meningoseptica*
**d** *Pseudomonas aeruginosa*

118. MLS ONLY Gram-negative bacilli with bipolar staining are isolated from a wound infection caused by a bite from a pet cat. The characteristic reactions shown in this table are seen:

| test | result |
|---|---|
| oxidase | positive |
| glucose OF | fermentative |
| motility | negative |
| MacConkey agar | no growth |

Which of the following is the most likely organism?

**a** *Pseudomonas aeruginosa*
**b** *Pasteurella multocida*
**c** *Aeromonas hydrophila*
**d** *Vibrio cholerae*

119. While swimming in a lake near his home, a young boy cut his foot, and an infection developed. The culture grew a nonfastidious gram-negative, oxidase-positive, beta-hemolytic, motile bacilli that produced deoxyribonuclease (DNase). The most likely identification is:

**a** *Enterobacter cloacae*
**b** *Serratia marcescens*
**c** *Aeromonas hydrophila*
**d** *Escherichia coli*

**120.** MLS ONLY A child is bitten on the arm by her sibling and the resulting wound grows a slender gram-negative bacilli that has the characteristics shown in this table:

| test | result |
|---|---|
| growth on SBA | colonies that "pit" the agar |
| colonies odor | like bleach |
| catalase | negative |
| oxidase | positive |
| TSI | no growth |

The identification of this organism is:

**a** *Moraxella catarrhalis*
**b** *Eikenella corrodens*
**c** *Kingella kingae*
**d** *Legionella pneumophila*

**121.** A specimen from a foot ulcer of a 52-year-old male diabetic patient is sent to the microbiology laboratory for culture. The results shown in this table are obtained from a clear colony growing on the MacConkey agar:

| test | result |
|---|---|
| oxidase | positive |
| catalase | positive |
| OF tubes | oxidation positive |
| pigment production | blue/green |
| growth at 42°C | positive |

The results indicate which of the following organisms has been isolated from the culture?

**a** *Acinetobacter baumannii*
**b** *Serratia marcescens*
**c** *Stenotrophomonas maltophilia*
**d** *Pseudomonas aeruginosa*

**122.** MLS ONLY A 26-year-old female goes to her doctor with several lacerations on her right hand and a swollen knuckle. The patient tells the clinician she is a boxer and received the lacerations 2 days prior while sparring without gloves when she accidentally struck her partner in the mouth. At 48 hours a culture reveals

colonies on the blood plate and chocolate plate with no growth on the MacConkey agar plate.
Gram stain of the colonies reveals small, slender, gram-negative bacilli
oxidase reaction is positive
indole reaction is negative
reduces nitrate to nitrite
does not require X and V factors
catalase is negative

What is the most likely identification of this organism?

**a** *Pasteurella multicoda*
**b** *Eikenella corrodens*
**c** *Pseudomonas aeruginosa*
**d** *Escherichia coli*

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123. A young girl cuts her foot on a rock while swimming in the ocean. Her foot begins to show signs of infection, and her parents take her to the ER. A culture grows non-lactose fermenting gram-negative bacilli that produce copious amounts of hydrogen sulfide and gas, is indole-positive and motile. The organism most likely isolated is:

**a** *Shigella* species
**b** *Escherchia* species
**c** *Edwardsiella* species
**d** *Klebsiella* species

124. Fluid from a cutaneous black lesion is submitted for routine bacterial culture. After 18 hours of incubation at 35°C there is no growth on MacConkey agar, but 3+ growth on sheep blood agar. The colonies are nonhemolytic, nonmotile, 4-5 mm in diameter and off-white with a ground glass appearance. Each colony has an irregular edge with comma-shaped outgrowths that stand up like "beaten egg whites" when gently lifted with an inoculating needle. A Gram stain of a typical colony shows large, gram-positive rectangular bacilli. The organism is most likely:

**a** *Clostridium perfringens*
**b** *Aeromonas hydrophila*
**c** *Bacillus anthracis*
**d** *Mycobacterium marinum*

125. An aspirate of a deep wound is plated on blood agar plates and incubated aerobically and anaerobically. At 24 hours there is growth on both plates. This indicates that the organism is a(n):

**a** nonfermenter
**b** obligate anaerobe
**c** aerobe
**d** facultative anaerobe

126. The characteristic that is most commonly associated with the presence of strict anaerobic bacteria and can be taken as presumptive evidence of their presence in a clinical specimen is the:

**a** presence of a single bacterial species
**b** production of gas in a thioglycolate broth culture
**c** growth on a blood agar plate incubated in an anaerobic jar
**d** presence of a foul, putrid odor from tissue specimens and cultures

127. Gram stain of a thigh wound shows many gram-positive spore-forming bacilli. The specimen is placed on brain-heart infusion blood agar and incubated aerobically at 35°C for 3 days. At the end of that time, the plates show no growth. The most likely explanation is that some of the specimen should have been incubated:

**a** on chocolate agar
**b** for 5 days
**c** under 5% $CO_2$
**d** anaerobically

128. An aspirate of a deep wound is plated on blood agar plates aerobically and anaerobically. At 48 hours there is growth on the anaerobic plate only. The next step in the evaluation of this culture is to:

**a** reincubate both plates for another 24 hours
**b** Gram stain and begin organism identification
**c** call physician and request blood culture
**d** set up a Bauer-Kirby susceptibility test

129. MLS ONLY The growth results shown in this table are observed on media inoculated with a foot abscess aspirate and incubated in 3-5% $CO_2$.

| test | result |
|---|---|
| SBA | 2+ large gray colonies |
| PEA | no growth |
| chocolate | 3+ large gray colonies |
| MacConkey | 3+ lactose fermenters |
| thioglycolate broth | gram-negative bacilli and gram-positive bacilli |

Biochemicals are set up on the colonies from the MacConkey agar plate. What should the microbiologist do next?

**a** set up biochemicals on the colonies from SBA
**b** Gram stain colonies on SBA
**c** subculture thioglycolate broth to SBA aerobic and SBA anaerobic
**d** test colonies on chocolate agar with hemin and NAD

130. Anaerobic, box-car shaped, beta-hemolytic gram-positive bacilli isolated from a foot wound are most likely:

**a** *Actinomyces israelii*
**b** *Clostridium perfringens*
**c** *Bacillus subtilis*
**d** *Eggerthella lenta*

131. Which organism is the most common anaerobic bacteria isolated from infectious processes of soft tissue and anaerobic bacteremia?

**a** *Bacteroides fragilis*
**b** *Fusobacterium nucleatum*
**c** *Porphyromonas asaccharolytica*
**d** *Clostridium perfringens*

### *H. Genital Tract*

132. An organism that may be mistaken for *Neisseria gonorrhoeae* in Gram-stained smears of uterine cervix exudates is:

**a** *Lactobacillus* species
**b** *Streptococcus agalactiae*
**c** *Pseudomonas aeruginosa*
**d** *Moraxella osloensis*


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133. MLS ONLY A 24-year-old man presents with pain on urination and urethral discharge. A Gram stain of the discharge is seen in the image:



What is the most likely identification of this organism?

**a** *Acinetobacter baumannii*
**b** *Neisseria gonorrhoeae*
**c** *Haemophilus ducreyi*
**d** *Escherichia coli*

134. MLS ONLY A pregnant patient is screened at 36 weeks' gestation for group B *Streptococcus* (GBS). A vaginal swab is collected and cultured in Todd-Hewitt broth with 8 µg/mL gentamicin and 15 µg/mL nalidixic acid. The broth is subcultured onto sheep blood agar after 24 hours of incubation. No GBS are seen on the subculture and the results are reported as negative. The patient later goes on to deliver an infant with early onset GBS disease. What is the most likely reason for the negative GBS culture?

**a** the patient was screened too early since screening is recommended after 38 weeks' gestation
**b** a vaginal swab was collected instead of a vaginal/rectal swab
**c** the Todd-Hewitt broth used was inhibitory to the organism
**d** the selective broth was incubated only 24 hours before subculture

135. A urethral swab obtained from a man with a urethral exudate is plated directly on chocolate agar and modified Thayer-Martin agar, and a Gram stain is made. The Gram stain shows gram-negative diplococci. The culture plates are incubated at 35°C in ambient air, but have no growth at 48 hours. The most likely failure for organism growth is that the:

**a** wrong media are used
**b** anaerobic chocolate agar plate not set up
**c** organism only grows at room temperature
**d** organism requires $CO_2$ for growth

136. MLS ONLY Which nonculture method is best for the diagnosis of *Neisseria gonorrhoeae* in an adult female?

**a** clinical history
**b** Gram stain of cervical secretions
**c** MALDI-TOF MS
**d** NAAT

137. The laboratory aid prepared and performed a Gram stain of a vaginal smear for *Neisseria gonorrhoeae*, as requested by a resident. The findings on the stain were:

MLS ONLY

many white blood cells
few epithelial cells
many gram-positive bacilli
few gram-negative diplococci
few gram-positive cocci in chains

The technologist should:

**a** report out smear positive for gonorrhea
**b** report out smear negative for gonorrhea
**c** request a new specimen due to number of white blood cells
**d** not read or report a Gram stain on a vaginal specimen

138. A wet mount of vaginal fluid is examined microscopically and large squamous epithelial cells are seen with gram-variable bacilli clustered on the cell edges. The pH of the fluid is 5.0. The most likely pathogen is:

**a** *Escherichia coli*
**b** *Arcanobacterium haemolyticum*
**c** *Gardnerella vaginalis*
**d** *Lactobacillus* species

139. Which organism fails to grow on artificial media or in cell cultures?

**a** *Chlamydia trachomatis*
**b** *Neisseria gonorrhoeae*
**c** *Treponema pallidum*
**d** herpes simplex virus

140. Darkfield microscopy is used to visualize:

MLS ONLY

**a** *Borrelia recurrentis*
**b** *Mycoplasma pneumoniae*
**c** *Treponema pallidum*
**d** *Legionella pneumophila*

141. A 29-year-old man is seen for recurrence of a purulent urethral discharge 10 days after the successful treatment of culture proven gonorrhea. The most likely etiology of his urethritis is:

MLS ONLY

**a** *Mycoplasma hominis*
**b** *Chlamydia trachomatis*
**c** *Trichomonas vaginalis*
**d** *Neisseria gonorrhoeae*

142. *Ureaplasma urealyticum* is difficult to grow in the laboratory on routine media because of its requirement for:

**a** sterols
**b** horse blood
**c** ferric pyrophosphate
**d** surfactant such as Tween® 80

### *I. Urine*

143. The colony count from a suprapubic urine culture growing 10 colonies of *Staphylococcus saprophyticus* is:

**a** 0 CFU/mL
**b** 100 CFU/mL
**c** 1,000 CFU/mL
**d** 100,000 CFU/mL


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144. A catheterized urine is inoculated onto blood and MacConkey agar using a 0.01 mL loop. After 24 hours, 68 colonies of a small translucent nonhemolytic organism grew on blood agar, but not MacConkey. Testing reveals small gram-positive, catalase-negative cocci. The preliminary report and follow up testing would be:

**a** growth of 680 CFU/mL of gram-positive cocci, optochin and bacitracin susceptibility tests to follow
**b** growth of 6800 CFU/mL of a *Staphylococcus* species, latex agglutination test to follow
**c** growth of 6800 CFU/mL of a *Streptococcus* species, esculin hydrolysis and 6.5% NaCl growth test to follow
**d** growth of 6800 CFU/mL of a *Streptococcus* species, no further testing

145. Children who have infections with beta-hemolytic streptococci can develop:

**a** acute pyelonephritis
**b** acute glomerulonephritis
**c** chronic glomerulonephritis
**d** nephrosis

146. A clean-catch urine culture (obtained with a 0.01 mL calibrated loop) grows 60 colonies of *Escherichia coli*. Which of the following represents the final colony count in CFU/mL?

**a** 60 CFU/mL
**b** 600 CFU/mL
**c** 6000 CFU/mL
**d** 60,000 CFU/mL

147. Infection of the urinary tract is most frequently associated with:

**a** *Staphylococcus aureus*
**b** *Escherichia coli*
**c** *Enterococcus faecalis*
**d** *Serratia marcescens*

148. >100,000 CFU/mL of a gram-negative bacilli are isolated on MacConkey from a urine specimen. Biochemical results are shown in this table:

| test | result |
|---|---|
| glucose | acid, gas produced |
| indole | negative |
| urea | positive |
| TDA | positive |
| $H_2S$ | positive |

The organism is most likely:

**a** *Morganella morganii*
**b** *Proteus mirabilis*
**c** *Proteus vulgaris*
**d** *Providencia stuartii*

**149.** A urine culture has the results shown in this table:

MLS ONLY

| test | result |
|---|---|
| sheep blood | swarming |
| Columbia CNA | no growth |
| MacConkey | >100,000 CFU/mL non-lactose-fermenter |
| | >100,000 CFU/mL non-lactose-fermenter with red pigment |

The isolates from MacConkey agar have the biochemical reactions shown in this second table:

| test | isolate 1 | isolate 2 |
|---|---|---|
| TSI | alk/acid | alk/acid |
| urea | positive | negative |
| TDA | positive | negative |
| $H_2S$ | positive | negative |

The organisms are most likely:

**a** *Proteus vulgaris* and *Enterobacter cloacae*
**b** *Proteus mirabilis* and *Serratia marcescens*
**c** *Morganella morganii* and *Klebsiella pneumoniae*
**d** *Providencia stuartii* and *Serratia liquefaciens*

**150.** A clean-catch urine sample from a nursing home patient is cultured using a 0.001 mL loop. It grows 67 colonies of a lactose fermenter that has the biochemical reactions shown in this table:

| test | result |
|---|---|
| TSI | acid/acid |
| oxidase | negative |
| motility | positive |
| indole | negative |
| citrate | positive |
| VP | positive |
| lysine decarboxylase | negative |
| ornithine decarboxylase | positive |
| urea | negative |

What should the microbiologist report?

**a** 670 CFU/mL *Serratia marsecens*
**b** 6,700 CFU/mL *Providencia stuartii*
**c** 67,000 CFU/mL *Enterobacter cloacae*
**d** 67,000 CFU/mL *Klebsiella oxytoca*

**151.** MLS ONLY A Foley catheter urine specimen from an 88-year-old male patient is received by the microbiology laboratory for culture. At 24 hours, the culture is growing 100,000 CFU/mL colonies of non-lactose-fermenting gram-negative bacilli. The isolate also tested positive for indole, ornithine decarboxylase, urease, motility, and phenylalanine deaminase, and negative for hydrogen sulfide production. The organism isolated is most likely:

**a** *Edwardsiella* species
**b** *Morganella* species
**c** *Hafnia* species
**d** *Shigella* species


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152. A jaundiced 7-year-old boy, with a history of playing in a pond in a rat-infested area, has a urine specimen submitted for a direct darkfield examination. Several spiral organisms are seen. Which of the following organisms would most likely be responsible for the patient's condition?

MLS ONLY

**a** *Cardiobacterium hominis*
**b** *Streptobacillus moniliformis*
**c** *Listeria monocytogenes*
**d** *Leptospira interrogans*

*J. Identification Methods (Theory, Interpretation, and Application)*

153. What organism combination is appropriate to serve as quality control for the listed test or organism characteristic ?

**a** beta-hemolysis: *Staphylococcus aureus* and *Streptococcus pyogenes*
**b** catalase: *Staphylococcus aureus* and *Staphylococcus epidermidis*
**c** $H_2S$ production: *Proteus mirabilis* and *Salmonella* species
**d** indole: *Escherichia coli* and *Proteus mirabilis*

154. A urine isolate Gram stain shows gram-positive cocci in clusters. The organism tested catalase-positive. To identify this organism from culture, the technician should perform a coagulase test and a/an:

**a** polymyxin B susceptibility
**b** novobiocin susceptibility
**c** oxidase
**d** beta-lactamase

155. Viridans streptococci can be differentiated from *Streptococcus pneumoniae* by:

**a** alpha hemolysis
**b** colony morphology
**c** catalase reaction result
**d** bile solubility

156. A reliable test for distinguishing *Staphylococcus aureus* from other staphylococci is:

**a** oxidase
**b** coagulase
**c** catalase
**d** optochin susceptibility

157. The optochin disk is used for the identification of:

**a** *Haemophilus influenzae*
**b** group A beta-hemolytic streptococci
**c** *Streptococcus pneumoniae*
**d** *Enterococcus*

158. Interpret the test shown in the image. The most likely organism is:



**a** *Staphylococcus aureus*
**b** group A *Streptococcus*
**c** *Streptococcus pneumoniae*
**d** group B *Streptococcus*

159. Gamma-hemolytic streptococci that blacken bile esculin agar but do not grow in 6.5% NaCl broth are most likely:

**a** group B *Streptococcus*
**b** *Enterococcus*
**c** group D *Streptococcus (Streptococcus bovis* group)
**d** *Streptococcus pneumoniae*

160. After 24 hours a blood culture from a newborn grows catalase-negative, gram-positive cocci. The bacterial colonies are small, translucent and beta-hemolytic on a blood agar plate. Biochemical test results of a pure culture are:

| test | result |
|---|---|
| bacitracin | resistant |
| CAMP reaction | positive |
| bile esculin | not hydrolyzed |
| 6.5% NaCl broth | no growth |

Assuming that all controls react properly and reactions are verified, the next step would be to:

**a** perform a *Streptococcus* group typing
**b** report the organism as *Streptococcus pneumoniae*
**c** report the organism as *Staphylococcus aureus*
**d** report the organism as *Staphylococcus epidermidis*

161. Nonhemolytic streptococci that have been isolated from an ear culture grow up to the edge of a 0.04 unit bacitracin disk. Which of the following tests would help to determine if the organism is *Enterococcus*?

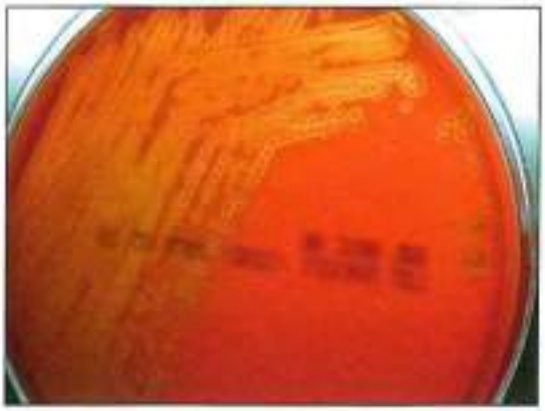
**a** hydrolysis of PYR
**b** growth in the presence of penicillin
**c** optochin susceptibility
**d** fermentation of mannitol

162. A beta-hemolytic *Streptococcus* that is bacitracin-sensitive and CAMP-negative is:

**a** group B
**b** group A
**c** beta-hemolytic, not group A, B, or D
**d** group D


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163. The organism seen in the image below is bacitracin-resistant and CAMP-positive. The identification is:



a *Staphylococcus saprophyticus*
b *Streptococcus agalactiae*
c *Moraxella catarrhalis* group B
d *Streptococcus pneumoniae*

164. Group B, beta-hemolytic streptococci may be distinguished from other hemolytic streptococci by which of the following procedures?

a latex antigen grouping
b growth in 6.5% NaCl broth
c growth on bile esculin medium
d bacitracin susceptibility

165. It is important to differentiate between *Enterococcus* and group D streptococci because:

a viridans streptococci are often confused with enterococci
b several enterococci cause severe puerperal sepsis
c group D streptococci are avirulent
d enterococci often show more antibiotic resistance than group D streptococci

166. *Streptococcus pneumoniae* can be differentiated best from the viridans group of streptococci by:

a Gram stain
b the type of hemolysis
c colonial morphology
d bile solubility

167. Characteristically, enterococci are:

a unable to grow in 6.5% NaCl
b relatively resistant to penicillin
c sodium hippurate positive
d bile esculin negative

168. Which of the following would best differentiate *Streptococcus agalactiae* from *Streptococcus pyogenes*?

a ability to grow in sodium azide broth
b a positive bile-esculin reaction
c hydrolysis of sodium hippurate
d beta-hemolysis on sheep blood agar

169. A yellow colony from a wound culture tested catalase-positive and coagulase-negative. The organism stained as gram-positive cocci in clusters. Which of the following tests would differentiate between a coagulase-negative *Staphylococcus* and *Micrococcus*?

**a** novobiocin susceptibility
**b** leucine aminopeptidase (LAP) production
**c** furazolidone (100 µg/disk) susceptibility
**d** hydrolysis of bile esculin

170. NLS ONLY A light-yellow colony from a skin lesion grows aerobically and tests as catalase-positive and coagulase-negative. The organism stains as gram-positive cocci in clusters. The organism is modified oxidase-positive, bacitracin (0.04U) susceptible and resistant to lysostaphin. What is the identification of this organism?

**a** *Staphylococcus aureus*
**b** *Micrococcus luteus*
**c** *Staphylococcus epidermidis*
**d** *Peptostreptococcus anaerobius*

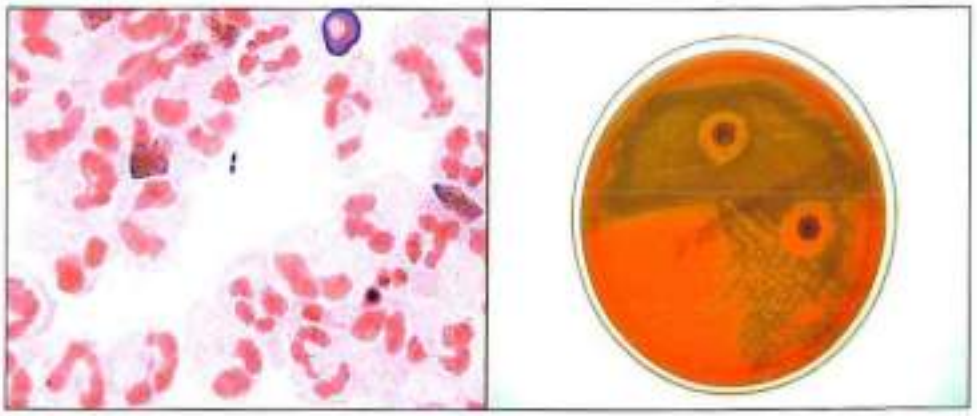
171. An isolate of an unknown beta-hemolytic *Streptococcus* is streaked perpendicular to a streak of beta-lysin-producing *Staphylococcus aureus*. After incubation a zone of arrowhead hemolysis is noted at the interface of the 2 streaks. What is the name of the test and the presumptive identification of the unknown *Streptococcus*?

**a** hippurate hydrolysis and *S. agalactiae*
**b** CAMP test and *S. pyogenes*
**c** hippurate hydrolysis and *S. pyogenes*
**d** CAMP test and *S. agalactiae*

172. Which of the following may be used as a positive quality control organism for the bile esculin test?

**a** *Staphylococcus epidermidis*
**b** *Staphylococcus aureus*
**c** *Streptococcus pyogenes*
**d** *Enterococcus faecalis*

173. Interpret the Gram stain and test shown in the images. This organism is:



**a** *Streptococcus pyogenes*
**b** *Streptococcus pneumoniae*
**c** *Enterococcus faecalis*
**d** Viridans group *Streptococcus*


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174. A gray, nonhemolytic, catalase-negative colony grows on a CNA plate. The biochemical results shown in this table are obtained:

| test | result |
|---|---|
| 65% NaCl | negative |
| bile esculin | positive |
| PYR | negative |
| bacitracin | resistant |
| hippurate hydrolysis | negative |
| CAMP test | negative |

The most likely identification is:

**a** *Enterococcus faecalis*
**b** *Streptococcus bovis*
**c** *Streptococcus viridans*
**d** *Streptococcus pneumoniae*

175. The ONPG test allows organisms to be classified as a lactose fermenter by testing for which of the following?

**a** permease
**b** beta-galactosidase
**c** beta-lactamase
**d** phosphatase

176. The most rapid method for detection of *Francisella tularensis* is:

MLS ONLY

**a** serological slide agglutination utilizing specific antiserum
**b** dye-stained clinical specimens
**c** fluorescent antibody staining techniques on clinical specimens
**d** polymerase chain reaction

177. Members of the family *Enterobacteriaceae* share which one of the following characteristics?

**a** produce cytochrome oxidase
**b** ferment lactose
**c** produce beta-hemolysis
**d** reduce nitrate to nitrite

178. Which one of these genera is among the least biochemically reactive members of the *Enterobacteriaceae*?

**a** *Proteus*
**b** *Pseudomonas*
**c** *Citrobacter*
**d** *Shigella*

179. Which one of these gram-negative bacilli ferments glucose?

**a** *Alcaligenes faecalis*
**b** *Burkholderia cepacia*
**c** *Acinetobacter lwoffii*
**d** *Yersinia enterocolitica*

**180.** An organism is inoculated to a TSI tube and gives the reactions shown:

alkaline slant/acid butt, $H_2S$, gas produced

This organism most likely is:

**a** *Klebsiella pneumoniae*
**b** *Shigella dysenteriae*
**c** *Salmonella typhimurium*
**d** *Escherichia coli*

**181.** A TSI tube is inoculated with an organism and gives the reactions shown:

alkaline slant, acid butt; no $H_2S$, no gas produced

This organism is most likely:

**a** *Yersinia enterocolitica*
**b** *Salmonella typhi*
**c** *Salmonella enteritidis*
**d** *Shigella dysenteriae*

**182.** An organism gave the reactions shown in this table:

| test | result |
|---|---|
| TSI | acid slant, acid butt; no $H_2S$ gas produced |
| indole | positive |
| motility | positive |
| citrate | negative |
| lysine decarboxylase | positive |
| urea | negative |
| VP | negative |

This organism most likely is:

**a** *Klebsiella pneumoniae*
**b** *Shigella dysenteriae*
**c** *Escherichia coli*
**d** *Enterobacter cloacae*

**183.** *Plesiomonas shigelloides* is a member of the family *Enterobacteriaceae*. The characteristic that differentiates *Pleisiomonas* from other *Enterobacteria*eae is:

**a** positive oxidase
**b** glucose fermentation
**c** reduction of nitrates to nitrites
**d** growth on MacConkey agar

**184.** The stock cultures needed for quality control testing of motility are:

**a** *Salmonella typhimurium—Escherichia coli*
**b** *Escherichia coli—Pseudomonas aeruginosa*
**c** *Serratia marcescens—Escherichia coli*
**d** *Klebsiella pneumoniae—Escherichia coli*

**185.** The stock cultures needed for quality control testing of oxidase production are:

**a** *Escherichia coli—Klebsiella pneumoniae*
**b** *Salmonella typhimurium—Escherichia coli*
**c** *Escherichia coli—Pseudomonas aeruginosa*
**d** *Proteus mirabilis—Escherichia coli*


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186. The stock cultures needed for quality control testing of deamination activity are:

- **a** *Escherichia coli—Klebsiella pneumoniae*
- **b** *Salmonella typhimurium—Escherichia coli*
- **c** *Escherichia coli—Pseudomonas aeruginosa*
- **d** *Proteus mirabilis—Escherichia coli*

187. The stock cultures needed for quality control testing of deoxyribonuclease (DNase) production are:

- **a** *Salmonella typhimurium—Escherichia coli*
- **b** *Escherichia coli—Pseudomonas aeruginosa*
- **c** *Proteus mirabilis—Escherichia coli*
- **d** *Serratia marcescens—Escherichia coli*

188. Quality control of the spot indole test requires the use of ATCC cultures of:

- **a** *Pseudomonas aeruginosa—Proteus mirabilis*
- **b** *Salmonella typhi—Shigella sonnei*
- **c** *Escherichia coli—Proteus vulgaris*
- **d** *Escherichia coli—Enterobacter cloacae*

189. An organism that exhibits the satellite phenomenon around colonies of *Staphylococcus* aureus is:

- **a** *Haemophilus influenzae*
- **b** *Neisseria meningitidis*
- **c** *Neisseria gonorrhoeae*
- **d** *Klebsiella pneumoniae*

190. *Acinetobacter lwoffii* differs from *Neisseria gonorrhoeae* in that *Acinetobacter*:

MLS ONLY

- **a** exhibits a gram-negative staining reaction
- **b** will grow on MacConkey and EMB media
- **c** is oxidase-positive
- **d** produces hydrogen sulfide on a TSI slant

191. A characteristic that is helpful in separating *Pseudomonas aeruginosa* from other members of the *Pseudomonas* family is:

- **a** a positive test for cytochrome oxidase
- **b** oxidative metabolism in the OF test
- **c** production of fluorescein pigment
- **d** growth at 42°C

192. The porphyrin test is devised to detect strains of *Haemophilus* capable of:

- **a** ampicillin degradation
- **b** capsule production
- **c** hemin synthesis
- **d** chloramphenicol resistance

**193.** MLS ONLY A genus that is found in soil and water and causes infections in immunocompromised patients has the characteristics shown in the table:

| test | result |
|---|---|
| sheep blood agar | violet pigment |
| MacConkey agar | growth |
| 42°C incubation | growth |
| oxidase | positive |
| OF glucose | fermenter |
| indole | negative |

The genus is:

**a** *Campylobacter*
**b** *Chromobacterium*
**c** *Aeromonas*
**d** *Serratia*

**194.** Which characteristic best differentiates *Acinetobacter* species from *Moraxella* species?

**a** production of oxidase
**b** growth on MacConkey agar
**c** motility
**d** susceptibility to penicillin

**195.** An organism has been identified as a member of the fluorescent group of *Pseudomonas*. Which of the following sets of tests should be used to determine the species of the organism?

**a** growth at 42°C, pyocyanin production, gelatinase production
**b** pyocyanin production, gelatinase production, OF glucose
**c** growth at 37°C, pyocyanin production, OF glucose
**d** gelatinase production, growth at 52°C, $H_2S$

**196.** Characteristics of the genus *Capnocytophaga* include:

MLS ONLY
**a** grows in ambient air
**b** colonies are large and spreading after 2-4 days
**c** considered "nonfermenter"
**d** gram-positive bacilli

**197.** Differentiating tests that will separate *Burkholderia* from *Stenotrophomonas* include:

MLS ONLY
**a** Gram stain reaction
**b** growth on MacConkey agar
**c** glucose fermentation
**d** oxidase

**198.** Characteristics of the HACEK group of bacteria include:

MLS ONLY
**a** association with urinary tract infections
**b** Gram stain of pleomorphic gram-positive bacilli
**c** requirement of 5-10% $CO_2$ for growth
**d** requirement of 42°C for growth

**199.** What are the most appropriate screening tests to presumptively differentiate and identify the nonfermentative gram-negative bacilli from the *Enterobacteriaceae*?

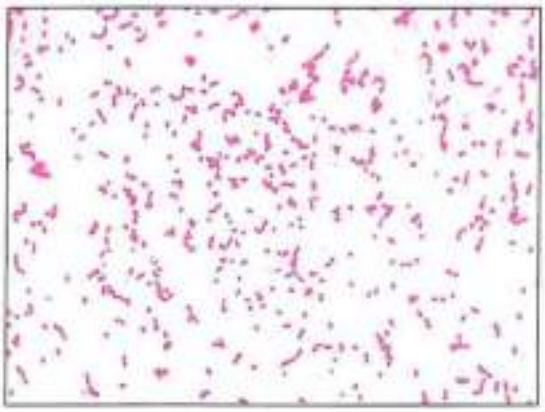
**a** catalase, decarboxylation of arginine, growth on blood agar
**b** motility, urease, morphology on blood agar
**c** oxidase, nitrate reduction, growth on MacConkey agar
**d** oxidase, indole, and growth on blood agar


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**200.** Which genera are positive for phenylalanine deaminase (PAD) production?

**a** *Klebsiella, Serratia, Enterobacter*
**b** *Proteus, Providencia, Morganella*
**c** *Escherichia, Edwardsiella, Salmonella*
**d** *Citrobacter, Klebsiella, Pantoea*

**201.** MLS ONLY A nonfermenting organism that grows on MacConkey, is nonmotile, oxidase-negative, and has the characteristic coccobacilli Gram stain seen in the image is:



**a** *Pseudomonas aeruginosa*
**b** *Stenotrophomonas maltophilia*
**c** *Proteus mirabilis*
**d** *Acinetobacter baumannii*

**202.** All species of the genus *Neisseria* have the enzyme to oxidize:

**a** naphthylamine
**b** dimethylaminobenzaldehyde
**c** glucopyranoside
**d** tetramethyl-phenylenediamine

**203.** Which of the following is the most reliable test to differentiate *Neisseria lactamica* from *Neisseria meningitidis*?

**a** growth on a modified Thayer-Martin agar
**b** nitrite reduction to nitrogen gas
**c** rapid ONPG
**d** utilization of maltose

**204.** A method for the definitive identification of *Neisseria gonorrhoeae* is:

**a** degradation of amino acids
**b** EIA
**c** utilization of carbohydrates
**d** resistance to penicillins and cephalosporins

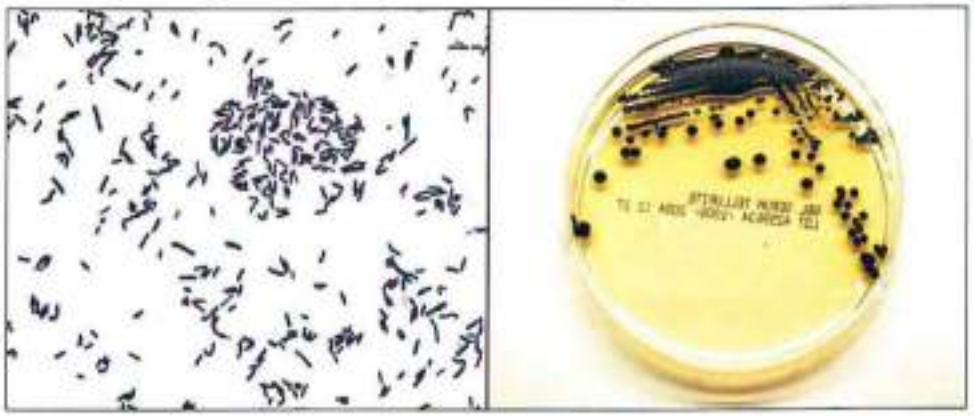
**205.** Gram-negative diplococci that grow on modified Thayer-Martin medium can be further confirmed as *Neisseria gonorrhoeae* if they are:

**a** oxidase-positive, glucose-positive, and maltose-positive
**b** oxidase-positive and glucose-positive, maltose-negative
**c** oxidase-positive, maltose-positive, and glucose-negative
**d** glucose-positive, oxidase-negative and maltose-negative

206. Which DNase-positive organism's colony is not easily broken up and therefore displays the "hockey puck" characteristic, where it can be pushed across the plate of medium?

a *Moraxella catarrhalis*
b *Neisseria gonorrhoeae*
c *Neisseria meningitidis*
d *Vibrio cholerae*

207. MLS ONLY The left image shows the Gram stain of an organism growing on sheep blood agar, while the right image shows growth of this organism on Tinsdale agar. The most likely organism is:



a *Bacillus cereus*
b *Corynebacterium diphtheriae*
c *Listeria monocytogenes*
d *Nocardia asteroides*

208. The best test to differentiate *Listeria monocytogenes* from *Corynebacterium* species is:

a catalase
b motility at 25°C
c motility at 35°C
d Gram stain

209. Which feature distinguishes *Erysipelothrix rhusiopathiae* from other clinically significant non-spore-forming, gram-positive, facultatively anaerobic bacilli?

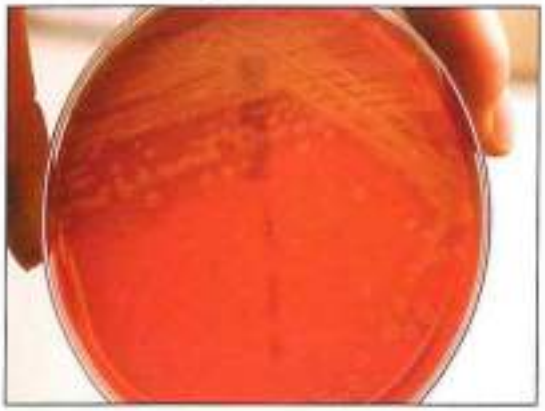
a "tumbling" motility
b beta-hemolysis
c more pronounced motility at 25°C than 37°C
d $H_2S$ production

210. *Listeria* can be confused with some Streptococcaceae because it is beta-hemolytic and:

a nonmotile
b catalase-negative
c oxidase-positive
d esculin positive


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211. Which of the following gram-positive bacilli is associated with infections in neonates, pregnant women, elderly adults, and outbreaks of food contamination? Its growth on sheep blood agar is shown in the image.



a *Clostridium perfringens*
b *Corynebacterium jeikeium*
c *Erysipelothrix rhusiopathiae*
d *Listeria monocytogenes*

212. Which of the following pairs of organisms usually grow on kanamycin, vancomycin, laked blood agar?

a *Bacteroides* and *Prevotella*
b *Mobiluncus* and *Gardnerella*
c *Porphyromonas* and *Enterococcus*
d *Veillonella* and *Capnocytophaga*

213. The reverse CAMP test, lecithinase production, double zone hemolysis, and Gram stain morphology are all useful criteria in the identification of:

a *Clostridium perfringens*
b *Streptococcus agalactiae*
c *Cutibacterium acnes*
d *Bacillus anthracis*

214. Which of the following sets of organisms may exhibit a brick red fluorescence?

MLS ONLY

a *Porphyromonas asaccharolytica* and *Clostridium ramosum*
b *Clostridioides difficile* and *Fusobacterium* species
c *Prevotella melaninogenica and Porphyromonas asaccharolytica*
d *Fusobacterium* species and *Bacteroides fragilis*

215. The presence of 20% bile in agar will allow growth of:

a *Fusobacterium necrophorum*
b *Bacteroides fragilis*
c *Prevotella melaninogenica*
d *Porphyromonas asaccharolytica*

216. Which testing platform meets this description? An isolated colony is irradiated by a laser, which ionizes the biomolecules and causes them to become accelerated in an electric field. The ionized biomolecules then enter a flight tube where they are separated by their mass-to-charge ratio.

a MALDI-TOF
b multiplex PCR
c pulsed-field gel electrophoresis
d sequencing

217. MLS ONLY Which one of the following organisms could be used as the positive quality control test for lecithinase on egg yolk agar?

a *Bacteroides fragilis*
b *Fusobacterium necrophorum*
c *Clostridium perfringens*
d *Clostridium sporogenes*

### *K. Antimicrobial Susceptibility Testing & Antibiotic Resistance*

218. MLS ONLY Which of the 2 different antimicrobial agents listed below are commonly used and may result in synergistic action in the treatment of endocarditis caused by *Enterococcus faecalis*?

a an aminoglycoside and a macrolide
b a penicillin derivative and an aminoglycoside
c a cell membrane active agent and nalidixic acid
d a macrolide and a penicillin derivative

219. The lowest concentration of antibiotic that inhibits growth of a test organism is the:

a minimum bactericidal concentration
b minimum inhibitory concentration
c serum bactericidal concentration
d serum inhibitory concentration

220. Penicillin resistance in *Neisseria gonorrhoeae* can be due to the organism producing:

a beta-D-galactosidase
b beta-lactamase
c butyrate esterase
d DNase

221. The most sensitive substrate for the detection of beta-lactamases is:

a penicillin
b ampicillin
c cefoxitin
d nitrocefin

222. An *Enterococcus* isolated from multiple blood cultures in a patient endocarditis should be:

a screened for high level aminoglycoside resistance
b checked for tolerance
c assayed for serum antimicrobial activity
d tested for beta-lactamase production

223. The procedure that assures the most accurate detection of mecA-mediated oxacillin resistance in routine broth microdilution susceptibility testing against *S. aureus* is:

a addition of 4% NaCl
b incubation at 30°C
c incubation for 48 hours
d testing with cefoxitin

224. MLS ONLY Susceptibility testing performed on quality control organisms using a new media lot number yields zone sizes that are too large for all antibiotics tested. The testing is repeated using media from a previously used lot number, and all zone sizes are acceptable. The unacceptable zone sizes are best explained by the:

a antibiotic disks are not stored with the proper desiccant
b depth of the media is too thick
c depth of the media is too thin
d antibiotic disks are not properly applied to the media

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225. In disk diffusion susceptibility testing, as an antimicrobial agent diffuses away from the disk, the concentration of antibiotic is:

**a** increased
**b** decreased
**c** unchanged
**d** inoculum dependent

226. When performing a disk diffusion susceptibility test, the antibiotic disks are placed on the agar 30 minutes after organism inoculation and then incubated within 15 minutes of the disk placement. This procedure will result in:

**a** the antibiotic not diffusing into the medium, resulting in no zone
**b** decreased zone diameters
**c** increased zone diameters
**d** no effect on the final zone diameter

227. An *Enterobacteriaceae* organism will appear to be more resistant on a disk diffusion susceptibility test if the

**a** depth of the agar is too thin
**b** inoculum is too concentrated
**c** the antimicrobial agent in disk is too concentrated
**d** test interpretation occurs after 12 hours of ambient air incubation

228. IILS ONLY When performing antimicrobial susceptibility testing on *Enterobacteriaceae*, first-generation cephalosporins can be adequately represented by:

**a** cefuroxime
**b** ceftriaxone
**c** cefazolin
**d** cefonicid

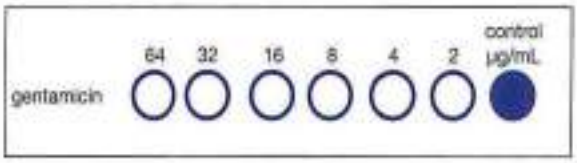
229. IILS ONLY An antibiotic that inhibits cell wall synthesis is:

**a** chloramphenicol
**b** colistin
**c** penicillin
**d** sulfamethoxazole

230. Tests for *Haemophilus influenzae* beta-lactamase production:

**a** are not commercially available
**b** include tests that measure a change to an alkaline pH
**c** should be performed on all blood and CSF isolates
**d** are not valid for any other bacterial species

231. IILS ONLY Interpret the gentamicin MIC shown in this broth microdilution susceptibility test:




**a** >64 µg/mL
**b** 32 µg/mL
**c** 16 µg/mL
**d** ≤2 µg/mL

**232.** MLS ONLY When using a control strain of *Staphylococcus aureus*, the technologist notices that the zone around the cefoxitin disk is too small. Which of the following is the most likely explanation?

- **a** inoculation of the plates 10 minutes after preparing the inoculum
- **b** incubation of the Mueller-Hinton plates at 35°C
- **c** use of a 0.25 McFarland standard to prepare inoculum
- **d** use of outdated cefoxitin disks

**233.** MLS ONLY In the disk diffusion method of determining antibiotic susceptibility, the size of the inhibition zone used to indicate susceptibility has been determined by:

- **a** testing 30 strains of 1 genus of bacteria
- **b** correlating the zone size with minimum inhibitory concentrations
- **c** correlating the zone size with minimum bactericidal concentrations
- **d** correlating the zone size with the antibiotic content of the disk

**234.** MLS ONLY A D test is performed on an isolate of *Staphylococcus aureus* to determine inducible clindamycin resistance:



Based on the result seen in the image how should the erythromycin and clindamycin be reported?

- **a** erythromycin: resistant; clindamycin: resistant
- **b** erythromycin: resistant; clindamycin: susceptible
- **c** erythromycin: susceptible; clindamycin: resistant
- **d** erythromycin: susceptible; clindamycin: susceptible

**235.** MLS ONLY An antimicrobial combination that is useful for confirming the presence of extended spectrum beta-lactamases in *E. coli* is:

- **a** ampicillin + cefepime
- **b** cefoxitin + penicillin
- **c** ceftazidime + clavulanic acid
- **d** cefpodoxime + cefotaxime

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**236.** MLS ONLY Susceptibility testing is performed on a *Staphylococcus aureus* isolate from a blood culture with the results shown in this table:

| antibiotic | susceptibility interpretation |
|---|---|
| oxacillin | resistant |
| cefazolin | susceptible |
| clindamycin | susceptible |
| erythromycin | susceptible |
| trimethoprim/sulfamethoxazole | susceptible |
| vancomycin | susceptible |

What should the technologist do next?

**a** ceftriaxone should be reported instead of cefazolin
**b** clindamycin should be tested for inducible resistance prior to reporting
**c** the trimethoprim/sulfamethoxazole result should be removed since all *S. aureus* are resistant
**d** the cefazolin result should be changed to resistant since the oxacillin result is resistant

**237.** A quality control procedure on a new batch of Mueller-Hinton agar plates using a *Staphylococcus aureus* ATCC strain and incubation in ambient air results in all the disk zone sizes measuring too small. The most likely reason for this is that the:

**a** Mueller-Hinton plates are poured too thin
**b** potency of the antibiotic disks is too high
**c** bacterial suspension is not diluted to the proper concentration
**d** incubation should have been in a 5-10% $CO_2$ atmosphere

**238.** MLS ONLY The antimicrobial susceptibility test for a *Klebsiella pneumoniae* isolated from a pleural fluid had the results shown in this table:

| antibiotic | susceptibility interpretation |
|---|---|
| amikacin | susceptible |
| ampicillin | susceptible |
| cefazolin (1st generation) | susceptible |
| cefoxitin (2nd generation) | susceptible |
| gentamicin | susceptible |
| tobramycin | susceptible |

The next best step is to:

**a** report out the antimicrobial susceptibility results without further investigation
**b** verify the results as it is unusual for *Klebsiella pneumoniae* to be susceptible to ampicillin
**c** verify the results as it is unusual for *Klebsiella pneumoniae* to be susceptible to cefazolin
**d** verify the results as it is unusual for *Klebsiella pneumoniae* to be susceptible to both ampicillin and cefazolin

**239.** MLS ONLY A *Klebsiella pneumoniae* isolate is known to produce carbapenemase; therefore, it therapeutically will not respond to the antibiotics:

**a** amoxicillin, cefazolin and imipenem
**b** colistin and rifampin
**c** sulfamethoxazole and trimethoprim-sulfamethoxazole
**d** tetracycline and doxycycline

240. Representative beta-lactam antibiotics and their mechanism of action include:

MLS ONLY

a ampicillin, cefazolin and imipenem; and inhibition of cell wall synthesis
b ampicillin, cefazolin and imipenem; and inhibition of DNA replication
c ciprofloxacin, levofloxacin and aztreonam; and inhibition of cell wall synthesis
d ciprofloxacin, levofloxacin and aztreonam; and inhibition of DNA replication

241. *Haemophilus influenzae* becomes resistant to ampicillin when the organism produces:

a a capsule of polysaccharide material
b NAD
c porphobilinogen
d the beta-lactamase enzyme

242. Clinical resistance to penicillin correlates most frequently with beta-lactamase production in:

a *Chlamydia trachomatis*
b *Neisseria gonorrhoeae*
c *Neisseria meningitidis*
d *Treponema pallidum*

### *L. MRSA/MSSA, VRE, ESBL/CRE Screening*

243. An antimicrobial susceptibility method recommended to detect vancomycin-intermediate *Staphylococcus aureus* is:

a MIC broth dilution
b agar dilution
c kinetic diffusion
d disk diffusion

### *M. BSL-3 Pathogens & Select Agents (Bioterrorism)*

244. Examples of Category A agents of bioterrorism include:

a *Bacillus anthracis*, MDR tuberculosis
b *Francisella tularensis, Yersinia pestis*
c Hanta virus, variola major virus
d VRE, *Bacillus cereus*

245. BSL-3 organisms are those that:

a cause human disease but are not readily transmitted among hosts
b do not ordinarily cause human disease; they require minimal safety procedures
c have very high risk of serious disease and no available treatment or vaccine
d produce serious disease and are transmitted by respiratory route

## *III. Analytic Procedures for Mycology, Mycobacteriology, Parasitology & Virology*

### *A. Mycobacteriology & Nocardia species*

246. A branching gram-positive, partially acid-fast organism is isolated from a bronchial washing on a 63-year-old woman receiving chemotherapy. The organism does **not** hydrolyze casein, tyrosine or xanthine. The most likely genus causing the infection is:

a *Actinomadura*
b *Erysipelothrix*
c *Nocardia*
d *Streptomyces*


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247. A first morning sputum is received for culture of mycobacteria. It is digested and concentrated by the N-acetyl-L-cysteine alkali method. Two Lowenstein-Jensen slants are incubated in the dark at 35°C with 5-10% $CO_2$. The smears reveal acid-fast bacilli, and after 7 days, no growth appears on the slants. The best explanation is:

**a** improper specimen submitted
**b** incorrect concentration procedure
**c** exposure to $CO_2$ prevents growth
**d** cultures held for insufficient length of time

248. A first morning sputum specimen is received for acid-fast culture. The specimen is centrifuged, and the sediment is inoculated on 2 Lowenstein-Jensen slants, which are incubated at 35°C in 5-10% $CO_2$. After 1 week, the slants show abundant growth over the entire surface. Stains reveal gram-negative bacilli. To avoid this problem:

**a** utilize a medium that inhibits bacterial growth
**b** add sodium hypochlorite to the sediment before inoculation
**c** incubate the tubes at room temperature to retard bacterial growth
**d** decontaminate the specimen with sodium hydroxide

249. A first morning sputum is received for acid-fast bacilli culture. It is digested and decontaminated by the N-acetyl-L-cysteine alkali method. Two Sabouraud dextrose slants are incubated in the dark at 35°C with 5-10% $CO_2$. The smears reveal acid-fast bacilli, but the slants show no growth after 8 weeks. The explanation is:

**a** improper media used
**b** incorrect decontamination procedure used
**c** improper specimen submitted
**d** improper incubation temperature and atmosphere

250. MLS ONLY In reviewing the number of *Mycobacterium* isolates for the current year, it is noted that there were 76% fewer isolates than the previous year (115 vs 28). The technologist in charge of the area has documented that the quality control of media, reagents and stains has been acceptable and there has been no gross contamination of the cultures noted. The most appropriate next course of action is to:

**a** stop the use of commercial media and produce in-house
**b** change to different formulations of egg and agar based media
**c** invest in an updated Bactec™ system for isolation of *Mycobacterium*
**d** review the digestion and decontamination procedure

251. A mucolytic, alkaline reagent for digestion and decontamination of a sputum for mycobacterial culture is:

**a** N-acetyl-L-cystine and NaOH
**b** NaOH alone
**c** zephiran-trisodium phosphate
**d** oxalic acid

252. The function of N-acetyl-L-cysteine in the reagent for acid-fast digestion-decontamination procedure is to:

**a** inhibit growth of normal respiratory flora
**b** inhibit growth of fungi
**c** neutralize the sodium hydroxide
**d** liquefy mucus

253. Middlebrook 7H10 and 7H11 media must be refrigerated in the dark, and incubated in the dark as well. If these conditions are not met, the media may prove toxic for mycobacteria because:

**a** carbon dioxide will be released
**b** growth factors will be broken down
**c** light destroys the ammonium sulfate
**d** formaldehyde may be produced

254. The best method to process specimens for mycobacterial culture contaminated with *Pseudomonas* is:

**a** N-acetyl-L-cystine and NaOH
**b** NaOH
**c** zephiran-trisodium phosphate
**d** oxalic acid

255. An AFB broth culture is positive for acid-fast bacilli at 1 week while the agar slant shows no growth. The most likely explanation for this is:

**a** the organism is a contaminant
**b** AFB grow more rapidly in liquid media
**c** PANTA is added to the broth
**d** the agar slant is incubated in 5% $CO_2$

256. What precaution should be taken to prevent infection of laboratory personnel when processing specimens for mycobacterial culture?

**a** add NALC in the ratio of 1 part NALC to 1 part specimen
**b** process all specimens under ultraviolet light
**c** centrifuge specimens only after the addition of preservative
**d** process all specimens in a biological safety hood

257. When staining mycobacteria, the primary stain used in the acid-fast staining process is:

**a** 1% acid fuchsin
**b** carbol fuchsin
**c** crystal violet
**d** methylene blue

258. A positive niacin test is characteristic of *Mycobacterium*:

MLS ONLY

**a** avium complex
**b** *fortuitum*
**c** *kansasii*
**d** *tuberculosis*

259. Characteristics necessary for the definitive identification of *Mycobacterium tuberculosis* are:

MLS ONLY

**a** buff color, slow growth at 37°C, niacin production-positive, nitrate reduction-negative
**b** rough colony, slow growth at 37°C, nonpigmented
**c** rough, nonpigmented colony, cording positive, niacin production-negative, catalase-negative at pH 7/68°C
**d** rough, nonpigmented colony, slow growth at 37°C, niacin production-positive, nitrate reduction-positive

260. The disease-producing capacity of *Mycobacterium tuberculosis* depends primarily upon:

MLS ONLY

**a** production of exotoxin
**b** production of endotoxin
**c** capacity to withstand intracellular digestion by macrophages
**d** lack of susceptibility to the myeloperoxidase system

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261. What *Mycobacterium* species includes a BCG strain used for vaccination against tuberculosis?

a *bovis*
b *fortuitum/chelonae* complex
c *kansasii*
d *tuberculosis*

262. MLS ONLY *Mycobacterium tuberculosis* complex can be identified directly in AFB smear positive respiratory specimens the same day the smear is read by:

a cording seen on the AFB smear
b molecular testing
c QuantiFERON®-TB test
d MALDI-TOF testing

263. MLS ONLY A primary drug used for the treatment of *Mycobacterium tuberculosis* is:

a ethionamide
b kanamycin
c rifabutin
d rifampin

264. When grown in the dark, yellow-to-orange pigmentation of the colonies is usually demonstrated by:

a *Mycobacterium tuberculosis*
b *Mycobacterium kansasii*
c *Mycobacterium fortuitum* group
d *Mycobacterium scrofulaceum*

265. The mycobacteria that produce a deep yellow or orange pigment both in the dark and in light are:

a nonchromogens
b photochromogens
c rapid growers
d scotochromogens

266. Mycobacteria that produce pigment only after exposure to light are classified as:

a nonchromogens
b photochromogens
c rapid growers
d scotochromogens

267. In a suspected case of Hansen disease (leprosy), a presumptive diagnosis is established by:

a isolation of organisms on Lowenstein-Jensen medium
b detection of weakly acid-fast bacilli in infected tissue
c isolation of organisms in a cell culture
d detection of niacin production by the isolated bacterium

268. Media used to culture *Mycobacterium tuberculosis* include:

a Bordet-Gengou agar and Middlebrook 7H10 agar
b Loeffler medium and PANTA medium
c Lowenstein-Jensen agar and Middlebrook 7H11 agar
d PANTA medium and cystine blood agar

269. A 27-year-old scuba diver has an abrasion on his left thigh. A culture of this wound grew an acid fast organism at 30°C. This isolate most likely is:

a *Mycobacterium chelonae*
b *Mycobacterium marinum*
c *Mycobacterium tuberculosis*
d *Mycobacterium xenopi*

270. MLS ONLY Differentiation of *Mycobacterium avium* from *Mycobacterium intracellulare* can be accomplished by:

a nitrate reduction test
b Tween® hydrolysis test
c resistance to 10 µg thiophene-2-carboxylic acid hydrazide (TCH)
d molecular testing

271. MLS ONLY A thoracic surgical wound specimen growa visible white colonies on sheep blood agar at 48 hours in 5-10% $CO_2$ incubation. When the isolated organism is Gram stained, faint staining gram-positive bacilli are observed. The most probable organism and appropriate stain to better view organism morphology is:

a *Corynebacterium jeikeium* and acridine orange stain
b *Mycobacterium fortuitum* and acid-fast stain
c *Mycobacterium tuberculosis* and auramine stain
d *Mycoplasma pneumoniae* and Giemsa stain

272. MLS ONLY When compared to direct specimen carbol fuchsin stains used for detecting mycobacteria, fluorochrome stains are less sensitive for detecting mycobacteria classified as:

a *Mycobacterium tuberculosis* complex
b nonphotochromogens
c photochromogens
d rapid growers

273. MLS ONLY What stain is most often used as a screening stain when mycobacteria is suspected in a respiratory specimen?

a acridine orange
b auramine O
c calcofluor white
d Gram stain with carbol fuchsin substituted as the counter stain

274. *Mycobacterium tuberculosis* is initially isolated from the sputum of a 35-year-old male. Antimicrobial susceptibility testing on this isolate is:

a routinely performed
b only performed if the isolate is from an immunosuppressed individual
c only performed if the isolate is recovered from a sterile body site
d not routinely performed as no standardized method is available

275. Organisms that are part of the *Mycobacterium tuberculosis* complex (MTBC) include *Mycobacterium tuberculosis* and:

a *M. africanum* and *M. bovis*
b *M. africanum* and *M. kansasii*
c *M. avium* and *M. bovis*
d *M. avium hominissuis*

276. Specimens for mycobacterial culture that routinely undergo the digestion and decontamination process include:

a blood and pleural fluid
b bone marrow and tissue biopsies
c bronchial washings and sputum
d cerebrospinal fluid and pleural fluid

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277. The recommended medium/media to inoculate for the primary isolation of mycobacteria include:

**a** a broth-based medium only
**b** a solid-based medium only
**c** a broth-based and a solid-based medium
**d** solid-based media

*B. Virology*

278. Which one of the following provides a presumptive identification of a viral infection?

**a** cytopathic effect on cell cultures
**b** intranuclear inclusions in RBCs
**c** cell lysis of sheep red blood cells
**d** presence of mononuclear inflammatory cells

279. The specimen of choice for detection of RSV is:

**a** nasopharyngeal aspirate
**b** cough plate
**c** expectorated sputum
**d** throat swab

280. The genus of virus associated with anogenital warts, cervical dysplasia and neoplasia is:

**a** herpes simplex virus
**b** human papillomavirus
**c** cytomegalovirus
**d** coxsackievirus

281. Encephalitis is most commonly associated with which of the following viruses?

**a** Epstein-Barr
**b** herpes simplex virus
**c** coxsackie B
**d** varicella zoster virus

282. Colds and other acute respiratory diseases are most often associated with:

**a** Epstein-Barr virus
**b** adenovirus
**c** coxsackie B
**d** reovirus

283. The Epstein-Barr virus is associated with which of the following?

**a** chickenpox
**b** Hodgkin lymphoma
**c** Burkitt lymphoma
**d** smallpox

284. Which of the following agents is the most common cause of pediatric viral gastroenteritis?

**a** adenovirus, serotypes 40 and 41
**b** Norwalk virus
**c** coronavirus
**d** rotavirus

285. Hanta or Sin Nombre virus is a Bunyavirus found in the 4 Corners area of the US (Arizona, New Mexico, Nevada, Colorado). What is the vector?

**a** deer mouse
**b** Norwegian rat
**c** domestic canine
**d** *Ixodes* tick

286. Which type of virus causes severe acute respiratory syndrome?

a paramyxovirus
b enterovirus
c rhinovirus
d coronavirus

287. What is the animal reservoir of West Nile virus?

a mice
b rats
c domestic cats
d birds

288. Which of the following organisms is the causative agent of hand, foot and mouth disease?

a Adenovirus
b coxsackie A
c coxsackie B
d human herpes virus 6

289. Which one of the following viruses is responsible for the most common congenital infection in the United States?

a VZV
b CMV
c EBV
d adenovirus

290. Which one of the following clinical syndromes is associated with VZV infection?

a infectious mononucleosis
b shingles
c primary CNS lymphoma
d Burkitt lymphoma

291. Which of the following clinical presentations is associated with HHV8?

a Kaposi sarcoma
b Duncan disease
c fifth disease
d exanthem infectiosum

292. In a person vaccinated against hepatitis B virus several years prior, which serological marker would be expected?

a HBsAg
b HBeAg
c anti-HBs
d anti-HBc

293. The persistence of which marker is the best evidence of chronic HBV infection?

a HBeAg
b HBsAg
c anti-HBe
d anti-HBs

294. Which of the following statements regarding antigenic shift or drift in influenza is correct?

MLS ONLY

a antigenic drift is due to point mutations in the H&N genes
b antigenic drift is responsible for pandemics of influenza
c antigenic shift is responsible for seasonable epidemics of influenza
d local annual outbreaks of influenza are often due to antigenic shift


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**295.** This virus is responsible for nearly all cases of infantile respiratory bronchiolitis:

- **a** parainfluenza virus
- **b** metapneumovirus
- **c** coxsackie A virus
- **d** respiratory syncytial virus

**296.** Which test is the primary screening test for HIV?

- **a** serum enzyme linked immunosorbent assay
- **b** western blot
- **c** quantitative HIV RNA
- **d** CD4 count

**297.** Which of the following HIV tests is the assay for determining response to anti-retrovirals?

MLS ONLY

- **a** serum ELISA
- **b** quantitative HIV RNA
- **c** CD4 count
- **d** p24 antigen detection

**298.** Which one of the following is a characteristic of hepatitis B virus?

- **a** hepatitis B infects CD4+T lymphocytes.
- **b** it can be reactivated, causing "shingles".
- **c** it is an enveloped DNA virus that is primarily a blood-borne pathogen.
- **d** this RNA virus is almost always transmitted by the fecal-oral route.

### *C. Parasitology*

**299.** Artifacts found in a stool specimen that can be confused with ova or cysts are:

- **a** partially digested meat fibers
- **b** degenerated cells from the gastrointestinal mucosa
- **c** dried chemical crystals
- **d** pollen grains

**300.** Polyvinyl alcohol used in the preparation of permanently stained smears of fecal material:

MLS ONLY

- **a** concentrates eggs
- **b** dissolves artifacts
- **c** serves as an adhesive
- **d** enhances stain penetration

**301.** A method to culture *Acanthamoeba* species from corneal ulcer scrapings is to inoculate

MLS ONLY

- **a** McCoy cells
- **b** Novy, MacNeal and Nicolle (NNN) medium
- **c** an agar plate overlaid with *Escherichia coli*
- **d** Regan-Lowe medium

**302.** Primary amoebic encephalitis is caused by:

- **a** *Entamoeba coli*
- **b** *Dientamoeba fragilis*
- **c** *Endolimax nana*
- **d** *Naegleria fowleri*

**303.** A formed stool is received in the laboratory at 10:30 PM for ova and parasite exam. The night shift technologist is certain that the workload will prevent examination of the specimen until 7 AM when the next shift arrives. The technologist should:

- **a** request that a new specimen be collected after 7 AM
- **b** hold the specimen at room temperature
- **c** examine a direct prep for trophozoites and freeze the remaining specimen
- **d** preserve the specimen in formalin until it can be examined

**304.** The advantage of thick blood smears for malarial parasites is to:

**a** improve staining of the organisms
**b** improve detection of the organisms
**c** remove RBC artifacts
**d** remove platelets

**305.** This parasite is transmitted by accidental ingestion of cat feces that contain oocysts and by ingestion of undercooked meat that has tissue cysts. The parasite is:

**a** *Cryptosporidium parvum*
**b** *Cyclospora cayetanensis*
**c** *Toxoplasma gondii*
**d** *Trypanosoma cruzi*

**306.** MLS ONLY A 44-year-old man is admitted to the hospital following a 2-week history of low-grade fever, malaise and anorexia. Examination of a Giemsa stain reveals many intraerythrocytic parasites. Further history reveals frequent camping trips near Martha's Vineyard and Nantucket Island, but no travel outside the continental United States. This parasite could easily be confused with:

**a** *Trypanosoma cruzi*
**b** *Trypanosoma rhodesiense/gambiense*
**c** *Plasmodium falciparum*
**d** *Leishmania donovani*

**307.** A Wright-stained peripheral smear reveals the following:

erythrocytes enlarged 1½x to 2x normal size
Schüffner dots
parasites with irregular "spread-out" trophozoites, golden-brown pigment
12-24 merozoites
wide range of stages

This is consistent with *Plasmodium*:

**a** *falciparum*
**b** *malariae*
**c** *ovale*
**d** *vivax*

**308.** A patient is suspected of having amoebic dysentery. Upon microscopic examination of a fresh fecal specimen for ova and parasites, the data shown are obtained:

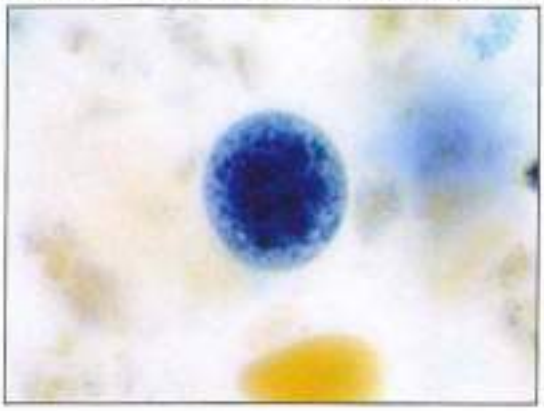
a trophozoite of 25 µm
progressive, unidirectional crawl
evenly distributed peripheral chromatin
finely granular cytoplasm

This information indicates:

**a** *Entamoeba coli*
**b** *Entamoeba histolytica*
**c** *Endolimax nana*
**d** *Iodamoeba bütschlii*

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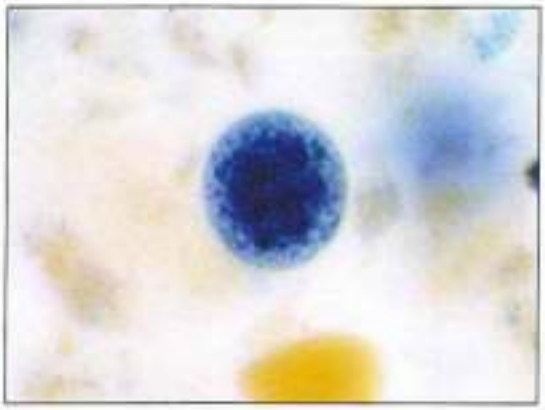
**309.** Trophozoites of the cyst shown in the image are likely to:



**a** contain red blood cells
**b** have clear, pointed pseudopodia
**c** contain few, if any, vacuoles
**d** have slow, undefined motility

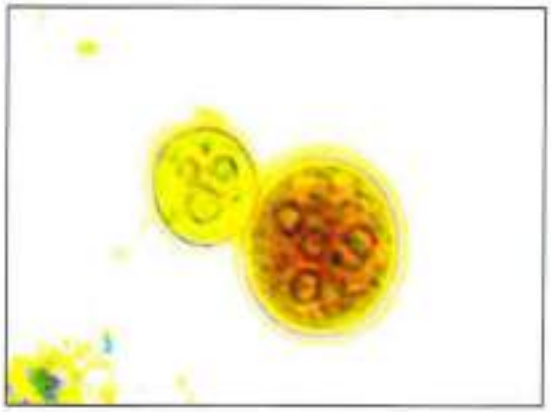
**310.** Upon finding the organism shown in a fecal concentrate, the technologist should:

MLS ONLY



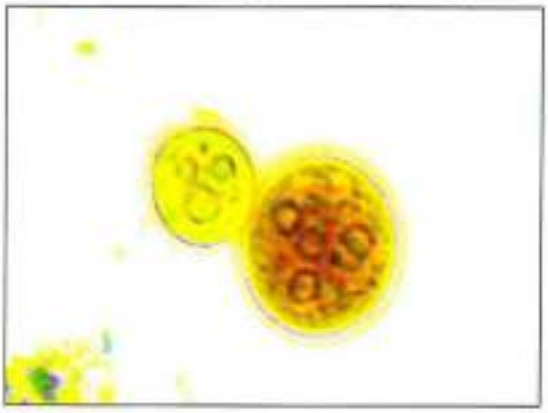
**a** immediately telephone the report of this pathogen to the physician
**b** review the fecal concentration carefully for the presence of other microorganisms that may be pathogenic
**c** look for motile trophozoites
**d** request a new specimen because of the presence of excessive pollen grains

311. Which organisms are shown together in the image?



**a** *Entamoeba histolytica* and *Entamoeba hartmanni*
**b** *Dientamoeba fragilis* and *Entamoeba histolytica*
**c** *Giardia lamblia* and *Entamoeba coli*
**d** *Entamoeba coli* and *Entamoeba histolytica*

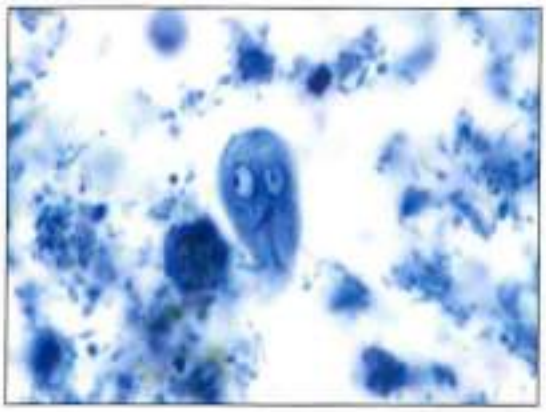
312. This structure on the right in the image depicts a:



**a** cyst of a nonpathogenic amoeba
**b** trophozoite of a nonpathogenic amoeba
**c** cyst of a pathogenic amoeba
**d** trophozoite of a pathogenic amoeba

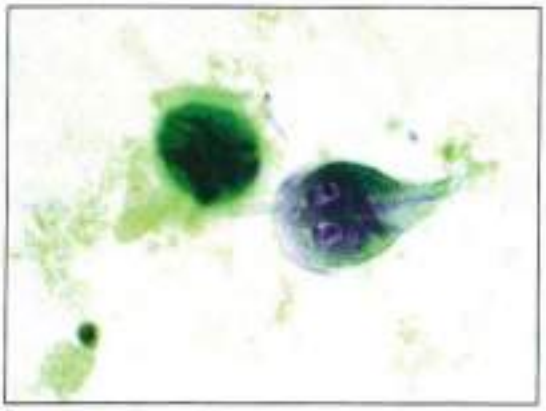

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313. The organism shown in the image is a(n):



a amoeba
b flagellate
c filaria
d sporozoan

314. A 24-year-old woman, who just returned from vacationing in Lebanon, becomes ill with diarrhea. The organisms shown in the image are found in her stool. The patient most likely is suffering from:



a giardiasis
b amebiasis
c ascariasis
d balantidiasis

315. A liquid stool specimen is collected at 10:00 PM and brought to the laboratory for culture and ova and parasite examination. It is refrigerated until 10:10 AM the next day, when the physician requests that the technologist look for amoebic trophozoites. The best course of action would be to:

a request a fresh specimen
b perform a concentration on the original specimen
c perform a trichrome stain on the original specimen
d perform a saline wet mount on the original specimen

316. Small protozoan cysts are found in a wet mount of sediment from ethyl-acetate concentrated material. Each cyst has 4 nuclei that do not have peripheral chromatin, and each nucleus has a large karyosome, which appears as a refractive dot. These oval cysts are most likely:

a *Endolimax nana*
b *Chilomastix mesnili*
c *Entamoeba histolytica*
d *Entamoeba hartmanni*

317. The term "internal autoinfection" is generally used in referring to infections with:

MLS ONLY

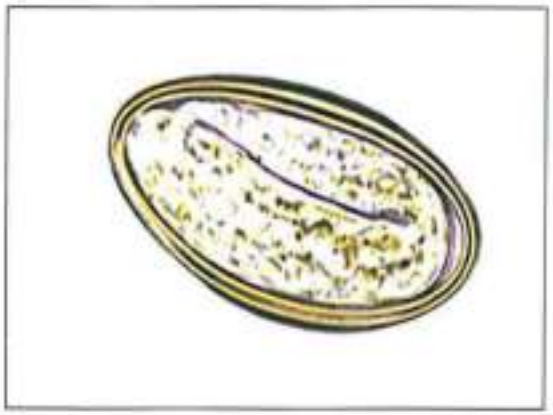
**a** *Ascaris lumbricoides*
**b** *Necator americanus*
**c** *Trichuris trichiura*
**d** *Strongyloides stercoralis*

318. A fibrous skin nodule is removed from the back of a patient from Central America. A microfilaria seen upon microscopic exam of the nodule is most likely:

MLS ONLY

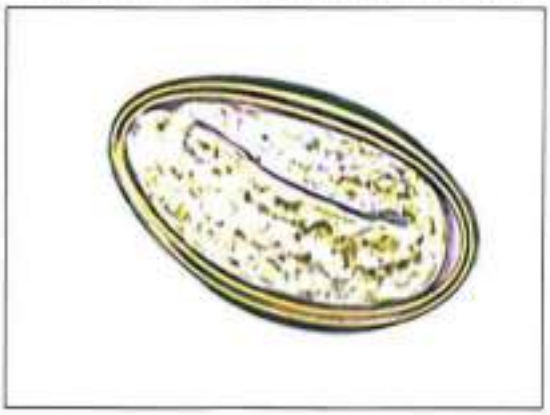
**a** *Wuchereria bancrofti*
**b** *Brugia malayi*
**c** *Onchocerca volvulus*
**d** *Loa loa*

319. The egg shown in the image is most likely to be found in children suffering from:



**a** diarrhea
**b** constipation
**c** perianal itching
**d** stomach pain

320. The specimen of choice for finding the parasite shown in the image is:



**a** stool
**b** duodenal washing
**c** rectal swab
**d** scotch tape preparation

321. Human feces is not a recommended specimen in the detection of:

**a** *Strongyloides stercoralis*
**b** *Entamoeba histolytica*
**c** *Echinococcus granulosus*
**d** *Ancylostoma duodenale*


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322. The causative agent of cysticercosis is:

a *Taenia solium*
b *Taenia saginata*
c *Ascaris lumbricoides*
d *Trichuris trichiura*

323. Organisms that can be easily identified to the species level from the ova in fecal specimens include: (MLS ONLY)

a *Metagonimus yokogawai, Heterophyes heterophyes*
b *Taenia solium, Taenia saginata*
c *Necator americanus, Ancylostoma duodenale*
d *Paragonimus westermani, Hymenolepis nana*

324. The preferred specimen for the diagnosis of paragonimiasis is:

a bile drainage
b blood smear
c skin snips
d sputum

325. A stool specimen for ova and parasite examination contained numerous rhabditiform larvae. Which factor(s) aid in the identification of larvae?

a larva tail nuclei and presence of sheath
b length of the buccal cavity and appearance of the genital primordium
c presence of hydatid cysts
d prominent kinetoplasts in trypomastigote

326. Which one of the following routine tests for *Entamoeba histolytica* has the highest sensitivity and specificity? (MLS ONLY)

a colonic ulcer biopsy
b stool microscopy
c stool EIA
d urine PCR

327. What is the principal means of distinguishing *Entamoeba histolytica* from *Entamoeba hartmanni* by light microscopy?

a size of trophozoite
b appearance of karyosome
c appearance of nuclear chromatin
d number of nuclei in cyst form

328. Which characteristic will identify *Iodamoeba bütschlii*?

a nuclei in mature cyst
b small (5-10 µm) size
c prominent vacuole in the cyst form
d presence of up to 8 nuclei in the cyst form

329. The only medically significant ciliate organism is:

a *Acanthamoeba*
b *Balantidium coli*
c *Cryptosporidium parvum*
d *Chilomastix mesnili*

330. Where do *Plasmodium* sporozoites proliferate?

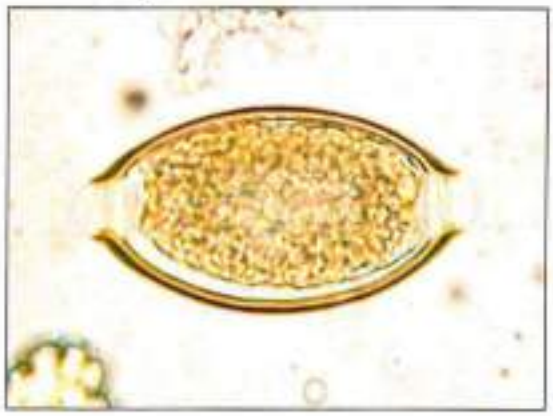
a bone marrow
b liver
c red blood cells
d nucleated erythrocyte precursors

331. Individuals who lack the Duffy antigen on the surface of their red blood cells are protected against which species of *Plasmodium*?

MLS ONLY

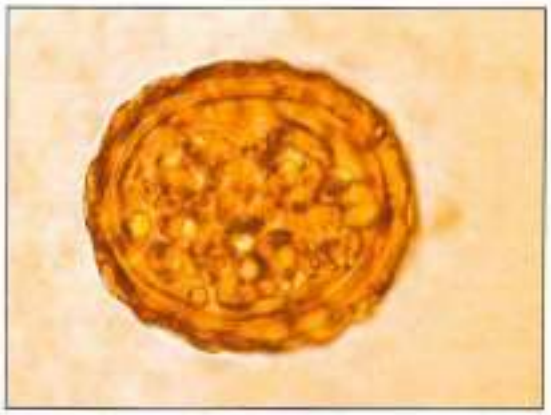
a *P. vivax*
b *P. falciparum*
c *P. malariae*
d *P. ovale*

332. Which nematode produces eggs with characteristic hyaline polar plugs at each end, as in the image?



a *Ascaris lumbricoides*
b *Necator americanus*
c *Strongyloides stercoralis*
d *Trichuris trichiura*

333. Which nematode has a characteristic mammillated bile stained egg, as in the image?



a *Ascaris*
b *Necator*
c *Strongyloides*
d *Trichuris*

334. Which organism is predominantly responsible for visceral larva migrans?

a *Ancylostoma braziliensis*
b *Onchocerca volvulus*
c *Toxocara canis*
d *Trypanosoma brucei*


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335. Which organism has the largest egg?

a *Clonorchis*
b *Diphyllobothrium*
c *Fasciola*
d *Paragonimus*

336. The eggs of which species of *Schistosoma* can be isolated from urine?

a *S. haematobium*
b *S. japonicum*
c *S. mansoni*
d *S. stercoralis*

337. Which one of the following features of *Taenia saginata* helps distinguish it from *T. solium*?

a egg with a radially striated wall
b pork tapeworm
c proglottid with <13 uterine branches
d unarmed rostellum

338. Infection by this organism can cause of $B_{12}$ deficiency:

a *Diphyllobothrium latum*
b *Echinococcus granulosus*
c *Taenia solium*
d *Schistosoma mansoni*

339. This organism is responsible for hydatid cysts of the liver:

a *Diphyllobothrium*
b *Echinococcus*
c *Hymenolepis*
d *Trichomonas*

340. How is *Trichomonas vaginalis* transmitted?

a ingestion of cyst stage
b ingestion of trophozoite stage
c larval stage burrows through the skin
d sexual contact

341. This parasite has the following characteristics:

banana shaped gametocytes in RBCs
multiple ring forms in RBCs
all RBCs are infected

The identification of the parasite is:

a *Plasmodium falciparum*
b *Plasmodium malariae*
c *Plasmodium ovale*
d *Plasmodium vivax*

**342.** This parasite has the following characteristics:

stains red with acid-fast stain
zoonotic transfer to humans
spread by fecal-oral route

The most likely organism is:

**a** *Cryptosporidium*
**b** *Giardia*
**c** *Naegleria*
**d** *Necator*

## *D. Mycology*

**343.** The major features by which molds are routinely categorized are:

**a** macroscopic growth characteristics and microscopic morphology
**b** biochemical reactions and microscopic morphology
**c** macroscopic characteristics and selective media
**d** specialized sexual reproductive cells and phialides

**344.** MLS ONLY A sputum specimen from a patient with a known *Klebsiella pneumoniae* infection is received in the laboratory for fungus culture. The proper procedure for handling this specimen is to:

**a** reject the current specimen and request a repeat culture when the bacterial organism is no longer present
**b** incubate culture tubes at room temperature in order to inhibit the bacterial organism
**c** include media that have cycloheximide and chloramphenicol added to inhibit bacterial organisms and saprophytic fungi
**d** perform a direct PAS stain; if no fungal organisms are seen, reject the specimen

**345.** MLS ONLY Many fungal infections are transmitted to man via inhalation of infectious structures. Which of the following is usually contracted in this manner?

**a** *Sporothrix schenckii*
**b** *Trichophyton rubrum*
**c** *Malassezia furfur*
**d** *Histoplasma capsulatum*

**346.** Using a fluorescent microscope, a wet preparation of skin tissue reveals fluorescent septate hyphae. The smear is prepared using:

**a** acridine orange
**b** calcofluor white
**c** Gomori methanamine silver
**d** periodic acid-Schiff

**347.** The formation of germ tubes presumptively identifies:

**a** *Candida tropicalis*
**b** *Candida parapsilosis*
**c** *Candida glabrata*
**d** *Candida albicans*

**348.** MLS ONLY An HIV-positive patient begins to show signs of meningitis. A spinal fluid specimen is collected and cultured for bacteria and fungus. A budding, encapsulated yeast is recovered. Which organism is consistent with this information?

**a** *Candida glabrata*
**b** *Cryptococcus neoformans*
**c** *Paracoccidioides brazilliensis*
**d** *Sporothrix schenckii*

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**349.** MLS ONLY Caffeic acid media inoculated with a yeast isolate show brown pigment production in a few hours. This is useful in the identification of:

- **a** *Candida albicans*
- **b** *Candida glabrata*
- **c** *Saccharomyces cerevisiae*
- **d** *Cryptococcus neoformans*

**350.** MLS ONLY The one characteristic by which an unknown *Cryptococcus* species can be identified as *Cryptococcus neoformans* is:

- **a** appearance of yellow colonies
- **b** positive urease test
- **c** presence of a capsule
- **d** positive phenol oxidase test

**351.** MLS ONLY A urine culture from a patient with a urinary tract infection yields a yeast with the following characteristics:

failure to produce germ tubes
hyphae not formed on cornmeal agar
urease-negative
assimilates trehalose

The most likely identification is:

- **a** *Geotrichum candidum*
- **b** *Cryptococcus laurentii*
- **c** *Candida tropicalis*
- **d** *Candida glabrata*

**352.** The recovery of some molds and yeasts may be compromised if the isolation media contains:

- **a** cycloheximide
- **b** gentamicin
- **c** chloramphenicol
- **d** penicillin

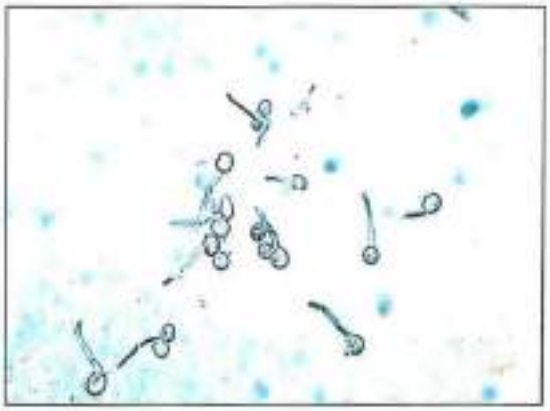
**353.** MLS ONLY A neonatal blood culture collected through a catheter grows a small yeast. Microscopically, the yeast appears round at one end, with a bud-like structure on a broad base at the other end. Growth is enhanced around olive oil-saturated discs. The organism isolated is:

- **a** *Candida tropicalis*
- **b** *Malassezia furfur*
- **c** *Candida lipolytica*
- **d** *Cryptococcus gattii*

**354.** Which medium could a technologist use to verify that a yeast isolate from a blood culture is pure?

- **a** Sabouraud dextrose agar
- **b** potato dextrose agar
- **c** cornmeal agar
- **d** CHROMagar™

355. The morphological characteristics of a yeast grown in rabbit plasma are shown in the image:



The most likely identification of this yeast is:

**a** *Candida tropicalis*
**b** *Candida krusei*
**c** *Candida albicans*
**d** *Candida glabrata*

356. The most sensitive test for the initial diagnosis of cryptococcal disease is:

MLS ONLY

**a** India ink
**b** Gram stain
**c** cryptococcal antigen
**d** Giemsa stain

357. Which of the following procedures is recommended to confirm that an unknown mold is one of the pathogenic dimorphic fungi?

**a** animal inoculation
**b** conversion from yeast to mold form
**c** demonstration of sexual and asexual reproduction
**d** molecular testing

358. Laboratory workers should always work under a biological safety hood when working with cultures of:

**a** *Streptococcus pyogenes*
**b** *Staphylococcus aureus*
**c** *Candida albicans*
**d** *Coccidioides immitis*

359. Structures important in the microscopic identification of *Coccidioides immitis* are:

MLS ONLY

**a** irregular staining, barrel-shaped arthrospores
**b** tuberculate, thick-walled macroconidia
**c** thick-walled sporangia containing sporangiospores
**d** small pyriform microconidia

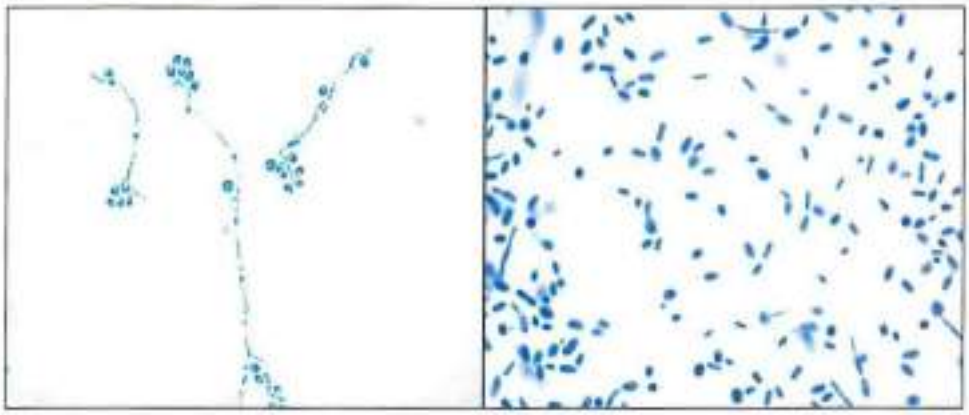
360. Which of the following is the most useful morphological feature in identifying the mycelial phase of *Histoplasma capsulatum*?

MLS ONLY

**a** arthrospores in every other cell
**b** 2-5 µm microspores
**c** 8-14 µm tuberculate macroconidia
**d** 5-7 µm nonseptate macroconidia


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**361.** MLS ONLY A mold growing at 25°C exhibits delicate, septate, hyaline hyphae and many conidiophores extending at right angles from the hyphae. Oval, 2-5 µm conidia are formed at the end of the conidiophores giving a flowerlike appearance. A 37°C culture of this organism produces small, cigar-shaped yeast cells as shown in the images. This organism is most likely:

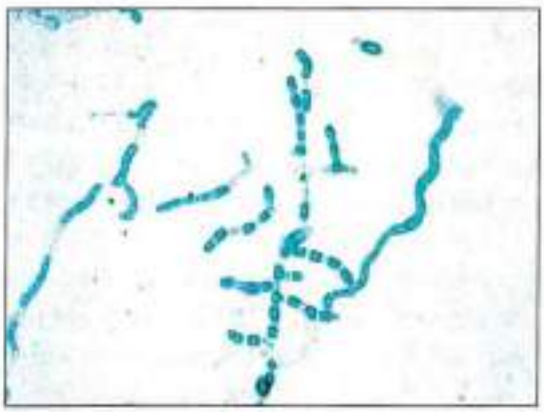


**a** *Histoplasma capsulatum*
**b** *Sporothrix schenckii*
**c** *Blastomyces dermatitidis*
**d** *Acremonium falciforme*

**362.** Which of the following is a dimorphic fungus?

**a** *Blastomyces dermatitidis*
**b** *Candida albicans*
**c** *Cryptococcus neoformans*
**d** *Aspergillus fumigatus*

**363.** MLS ONLY Examination of a fungal culture from a bronchial washing reveals white, cottony aerial mycelium. A tease preparation in lactophenol cotton blue shows the structures shown in the image:



The most rapid test for definitive identification is:

**a** MALDI-TOF mass spectrometry
**b** animal inoculation
**c** exoantigen test
**d** slide culture

**364.** The mold shown in the image is isolated from a bone marrow culture of a patient who travelled to southeast Asia. The isolate produced a red pigment that diffused into the medium. The technologist should suspect

MLS ONLY



**a** *Blastomyces dermatitidis*
**b** *Aspergillus niger*
**c** *Talaromyces* (formerly *Penicillium*) *marneffei*
**d** *Rhizopus* species

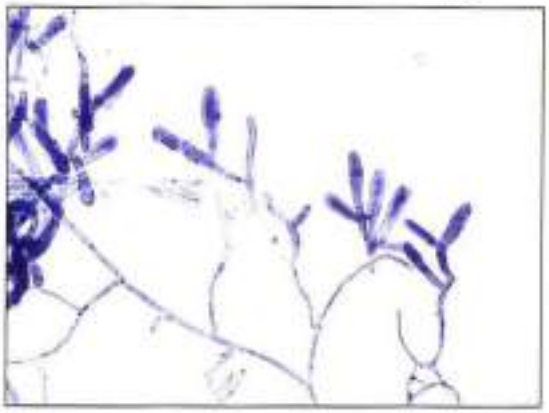
**365.** The microscopic structures that are most useful in the identification of dermatophytes are:

MLS ONLY

**a** septate and branching hyphae
**b** racquet and pectinate hyphae
**c** chlamydospores and microconidia
**d** macroconidia and microconidia

**366.** The correct identification of the dermatophyte seen in this image is:

MLS ONLY



**a** *Epidermophyton floccosum*
**b** *Fonsecaea pedrosoi*
**c** *Microsporum canis*
**d** *Trichophton tonsurans*

**367.** Culture of a strand of hair, that fluoresced yellow-green when examined with a Woods lamp, produced a slow-growing, flat gray colony with a salmon-pink reverse. Microscopic examination demonstrated racquet hyphae, pectinate bodies, chlamydospores, and a few abortive or bizarre-shaped macroconidia. The most probable identification of this isolate is:

**a** *Epidermophyton floccosum*
**b** *Microsporum canis*
**c** *Microsporum audouinii*
**d** *Trichophyton rubrum*

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368. On day 3 of a fungal culture, a dense grayish cottony growth is observed. It fills the container and looks like "cotton candy". The most likely mold isolated is a:

**a** dermatophyte
**b** dimorphic mold
**c** zygomycete
**d** dematiaceous mold

369. *Talaromyces* (formerly *Penicillium*) species can be separated from *Aspergillus* fumigatus by:

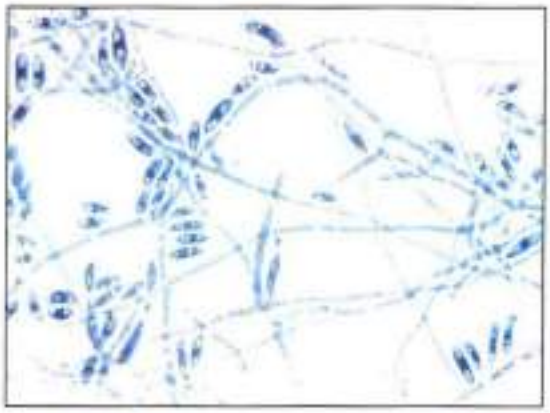
**a** production of conidia on phialids
**b** optimum growth temperature
**c** presence of rhizoids
**d** lack of vesicle

370. A fungus superficially resembles *Talaromyces* (formerly *Penicillium*) species but may be differentiated because its phialides are long and tapering and bend away from the central axis, as shown in the image. The most likely identification is:



**a** *Exophiala*
**b** *Acremonium*
**c** *Cladosporium*
**d** *Paecilomyces*

**371.** An isolate from a cornea infection had the Gram stain shown in the image, and the culture results shown in the table:



| medium | growth |
|---|---|
| Sabouraud dextrose | white & cottony at 2 days, violet at 6 days |
| slide culture | slender sickle shape macroconidia |

The most likely organism is:

**a** *Acremonium*
**b** *Aspergillus*
**c** *Fusarium*
**d** *Geotrichum*

**372.** In the USA, a common cause of eumycotic mycetoma is:

MLS ONLY

**a** *Pseudallescheria boydii*
**b** *Nocardia brasiliensis*
**c** *Coccidioides immitis*
**d** *Aspergillus fumigatus*

**373.** MLS ONLY Crust from a cauliflower-like lesion on the hand exhibited brown spherical bodies 6-12 µm in diameter when examined microscopically. A slow-growing black mold grew on Sabouraud dextrose agar. Microscopic examination revealed cladosporium, phialophora and fonsecaea types of sporulation. The probable identification of this organism is:

**a** *Fonsecaea pedrosoi*
**b** *Pseudallescheria boydii*
**c** *Phialophora verrucosa*
**d** *Cladosporium carrionii*

**374.** Which of the following scenarios presents the greatest risk for coccidioidomycosis?

**a** Missouri cattle rancher
**b** Wyoming rabbit rancher
**c** Central Valley California migrant worker
**d** pregnant woman with several house cats

**375.** Which one of the following media is most helpful in distinguishing the morphology of yeasts?

**a** commeal agar with Tween 80
**b** brain-heart infusion medium
**c** potato dextrose agar
**d** urea agar

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376. The most likely identification of the mold shown in the images is:



a *Aspergillus niger*
b *Gliocladium*
c *Aspergillus fumigatus*
d *Microsporum canis*

377. Which of the following dermatophytes can be identified with its clavate or peg-shaped microconidia that resemble "birds on a wire"?

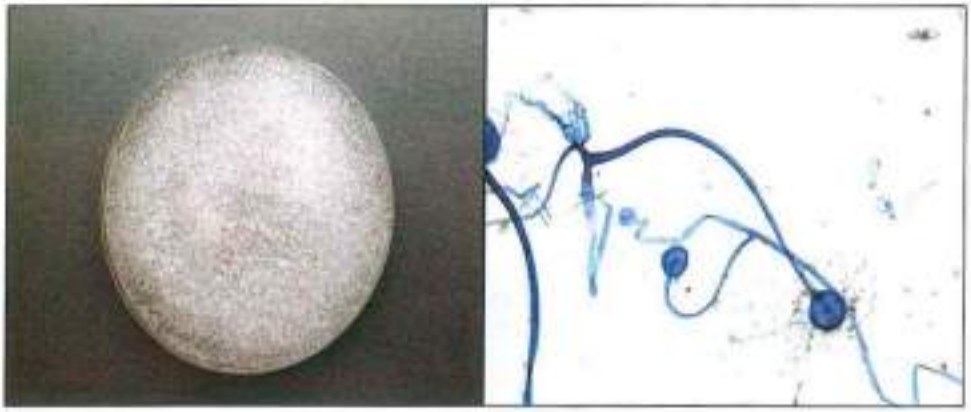
a *Microsporum canis*
b *Trichophyton rubrum*
c *Epidermophyton floccosum*
d *Trichophyton tonsurans*

378. Which fungal organism is urease+ and phenol oxidase+?

MLS ONLY

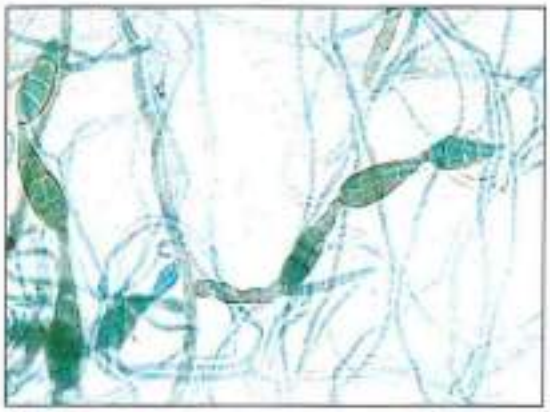
a *Cryptococcus neoformans*
b *Malassezia furfur*
c *Rhodotorula*
d *Trichophyton mentagrophytes*

379. A mold grows on SAB agar in 2 days, with "salt and pepper colonies as shown on the left image below. By day 3 the colony filled the petri dish. The LPCB stain of the colony is seen in the image on the right. The fungus identification is:



a *Aspergillus*
b *Cunninghamella*
c *Fusarium*
d *Rhizopus*

**380.** A dematiaceous mold grew in 4 days on Sabouraud agar, isolated from a foot wound. The image shows chaining macroconidia with longitudinal and horizontal septations.



The most likely identification is:

**a** *Alternaria*
**b** *Curvularia*
**c** *Paecilomyces*
**d** *Scopulariopsis*

**381.** MLS ONLY This fungus is encapsulated and produces by budding. It is associated with the eucalyptus tree and is usually acquired from the environment. The organism uses glycine as a sole carbon and nitrogen source in the presence of canavanine (canavanine glycine bromothymol blue agar). The organism is:

**a** *Candida krusei*
**b** *Cryptococcus gattii*
**c** *Cryptococcus neoformans*
**d** *Rhodotorula* species

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## IV. Postanalytic Procedures

### A. Documentation Practices

**382.** Microbiology culture results are made available within the hospital information system:

- **a** at regular intervals when information on bacterial growth is available, neither direct specimen nor "no growth" status information should be released
- **b** at regular intervals when information becomes available, such as direct specimen testing results, and all interim reports to include "no growth" updates
- **c** when all direct specimen and culture results are finalized, no interim results should be released
- **d** when the primary care practitioner requests updates on culture results, otherwise only final direct specimen and culture results are released

### B. Urgent & Critical Value Reporting

**383.** Microbiology-related urgent or critical results include positive:

- **a** blood cultures, cerebrospinal fluid cultures and acid-fast smears
- **b** blood cultures, urine cultures and ova & parasite examinations
- **c** cerebrospinal fluid cultures, fungal cultures and ova & parasite examinations
- **d** *Streptococcus pyogenes* throat screens, bronchial washing cultures and fungal cultures

### C. Result Review & Autoverification

**384.** Data on positive blood cultures due to contaminants are collected and distributed to outreach clinicians so they can review the:

- **a** laboratory's ability to identify contaminants growing from a blood culture
- **b** office phlebotomist's procedure for proper collection of a blood culture
- **c** rate at which the blood culture instrument identifies contaminated blood culture bottles
- **d** organism most commonly identified as a contaminant

### D. Issuing Corrected Reports

**385.** Incorrect results on a urine culture are recorded in the patient record in ICU. Therefore, a Problem Action Form is required and should contain:

- **a** date, problem, investigation, corrective action, outcome
- **b** employee certification number, problem, corrective action
- **c** employee years of experience, date, investigation of issue
- **d** signatures of attending physician and pathologist

### E. Reporting to Infection Control/Prevention & Public Health

**386.** A technologist on the respiratory bench notices an increased number of patients on the ICU growing the same, drug-resistant gram-negative bacilli from sputum specimens. Which department in the hospital will the microbiology laboratory work with to collect data on positive patients?

- **a** Infection control
- **b** Hospital administration
- **c** The Joint Commission
- **d** Food and Drug Administration

## I. Preanalytic Procedures

### A. Specimen Collection & Transport

1. **c** Human blood contains substances that may inhibit microbial growth such as complement, lysozyme and white blood cells. Diluting blood in culture broth reduces the concentration of these substances as well as any antibiotics that may be present. The recommended blood broth ratio is 1:5-1:10. Greater dilutions may increase the time to detection.
[QCMLS 2021: 6.1.2] [Mahon 2019, p869]

2. **d** Indwelling catheters are closed systems, and should not be disconnected for specimen collection. Urine samples should not be collected from catheter drainage bags, and Foley catheter tips are unsuitable for culture because they are contaminated with colonizing organisms. Urine from indwelling catheters should be collected by aseptically puncturing the tubing (collection port).
[QCMLS 2021: 6.1.2] [Mahon 2019, p689, 692]

3. **c** Materials collected from sites not harboring indigenous flora (sterile body fluids, abscess exudate and tissue) should be cultured for anaerobic bacteria. However, since anaerobes normally inhabit the skin and mucus membranes as part of the indigenous flora, specimens such as urine, sputum, vaginal, eye and ear swabs are not acceptable for culture.
[QCMLS 2021: 6.1.2] [Mahon 2019, p495]

4. **c** Most commercially available blood culture media contain sodium polyanetholsulfonate (SPS) in concentrations between 0.025 and 0.05%. SPS has anticoagulant activity, and inactivates neutrophils as well as some antibiotics including gentamicin and polymyxin. It also precipitates components of serum complement.
[QCMLS 2021: 6.1.2] [Mahon 2019, p871]

5. **c** The use of swabs for collection of specimens for anaerobic culture is discouraged. Aspiration with a needle and syringe is recommended. Whenever possible cultures should be obtained before the administration of antibiotics to optimize organism recovery.
[QCMLS 2021: 6.1.2] [Mahon 2019, p107]

6. **b** The volume of blood collected is the single most important variable in the recovery of organisms in patients with bloodstream infections. Since many cases of adult bacteremia are of low magnitude, there is a direct relationship between the yield of blood culture (positivity) and volume of blood collected. The collection of multiple blood culture sets from a single venipuncture is an unacceptable practice due to the potential for contamination. The practice of terminal subculture of blood culture bottles at 5 days is no longer recommended.
[QCMLS 2021: 6.1.2] [Procop 2017, p99-102]

7. **b** Antibiotics and antifungal agents are added to viral transport medium to inhibit the growth of bacteria and fungus.
[QCMLS 2021: 6.6.1] [Mahon 2019, p685]

### B. Specimen Processing

8. **d** Buffered charcoal yeast extract agar is recommended for culture of specimens for *Legionella*.
[QCMLS 2021: 6.1.6.4.4] [Tille 2016, p428]

9. **d** Thiosulfate citrate bile salt agar is a selective media for *Vibrio*, and it also differentiates sucrose-fermenting species, such as *V. cholerae* and *V. alginolyticus*.
[QCMLS 2021: 6.1.6.4.2] [Procop 2017, p455]

10. **d** Chocolate agar and chocolate agar-based selective media should be used for recovery of *Neisseria gonorrhoeae* from urethral discharge. Chocolate agar provides the nutrients required by *N. gonorrhoeae* and selective media contains antimicrobial agents that inhibits other organisms and permits recovery of pathogenic *Neisseria*.
[QCMLS 2021: 6.1.6.2] [Mahon 2019, p371]

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11. **a** Columbia CNA agar contains colistin and nalidixic acid, which inhibit most facultative gram-negative organisms. Eosin methylene blue is selective and inhibits gram-positive organisms, and modified Thayer Martin is selective and inhibits gram-positive organisms, gram-negative bacilli and yeast. Trypticase soy agar with 5% sheep blood is not a selective medium.
[QCMLS 2021: 6.1.5] [Mahon 2019, p13, 114]

12. **a** *Campylobacter jejuni* requires a microaerophilic and capnophilic atmosphere for optimal recovery. The use of selective media is recommended for recovery from fecal specimens. Selective media for *Campylobacter* contains antibiotics to inhibit the growth of enteric gram-negative flora. Unlike other enteric pathogens, *C. jejuni* grows well at 42°C.
[QCMLS 2021: 6.1.6.4.2] [Mahon 2019, p461]

13. **d** Urine is an appropriate specimen for the detection of renal tuberculosis. Since feces contain anaerobic organisms as part of the indigenous flora, it is an unacceptable specimen for anaerobic culture. Foley catheter tips are also not acceptable for culture, because they are contaminated with colonizing organisms. Gram stain smears of rectal swabs for *Neisseria gonorrhoeae* should also not be performed, since the presence of organisms with similar morphologies may lead to overinterpretation of smears.
[QCMLS 2021: 6.2.1.5] [Mahon 2019, p112, 371, 504, 567]

14. **c** Enriched media such as chocolate agar has no inhibitory effects on bacterial growth and contains additional nutrients that support the growth of fastidious organisms such as *Haemophilus influenzae* and *Neisseria*.
[QCMLS 2021: 6.1.5] [Mahon 2019, p371, 388]

15. **c** CNA agar is a selective medium commonly used in the isolation of gram-positive aerobic and anaerobic organisms. Since the Gram stain indicates a mixture of gram-positive and gram-negative organisms, use of CNA will aid in the recovery of the gram-positive cocci in culture. Surface swabs from foot ulcers are unacceptable specimens for anaerobic culture.
[QCMLS 2021: 6.1.5] [Mahon 2019, p114-115, 495]

16. **b** Chocolate agar and chocolate agar-based selective media (Martin Lewis) are routinely used for the recovery of *Neisseria gonorrhoeae* from genital specimens. Sputum and urine specimens are routinely processed using a general purpose media (blood agar) and a selective agar (EMB or MacConkey). In addition, respiratory specimen cultures routinely include chocolate agar to enhance recovery of fastidious organisms such as *Haemophilus influenzae*. CSF is routinely processed using blood and chocolate agars.
[QCMLS 2021: 6.1.5] [Mahon 2019, p115, 117]

17. **d** CIN agar is a selective and differential medium for the isolation and differentiation of *Yersinia enterocolitica*. This medium contains sodium desoxycholate, crystal violet, cefsulodin, irgason (triclosan), and novobiocin as selective agents, and mannitol as the carbohydrate. Additional stool culture pathogens specific isolation medium: *Vibrio* species—TCBS agar, *Campylobacter* species—Campy-BA, and enterohemorrhagic *Escherichia coli*—MacConkey sorbitol agar.
[QCMLS 2021: 6.1.5] [Mahon 2019, p844]

18. **d** Columbia CNA agar is a selective medium used for the isolation of gram-positive organisms. The medium contains colistin and nalidixic acid, which inhibits gram-negative organisms. MacConkey agar is a selective and differential medium used for the isolation of gram-negative organisms. The medium contains bile and crystal violet, which inhibits gram-positive organisms. Surface wound swabs are unacceptable for anaerobic culture.
[QCMLS 2021: 6.1.5] [Mahon 2019, p107, 114]

19. **a** Most *Campylobacter* species grow best under lower oxygen tension in an atmosphere of 5% oxygen, 10% carbon dioxide and 85% nitrogen. *Escherichia coli* and *Proteus mirabilis* are facultative anaerobes, and *Pseudomonas aeruginosa* is an aerobe.
[QCMLS 2021: 6.1.6.4.2] [Mahon 2019, p14, 469, 487, 844]

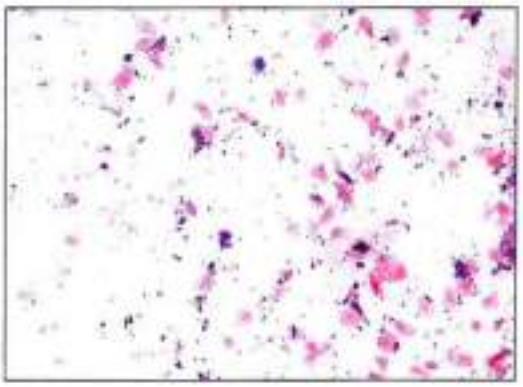
20. **d** TCBS is a highly selective and differential medium for the recovery of most *Vibrio* species, including *V. parahaemolyticus*. Hektoen and Salmonella-Shigella agars are selective and differential for the isolation and differentiation of enteric pathogens such as *Salmonella* and *Shigella*. EMB is a selective and differential medium for gram-negative enteric bacilli.
[QCMLS 2021: 6.1.6.4.2] [Mahon 2019, p434, 453]

21. **c** Buffered charcoal yeast extract (BCYE) medium is a specialized enrichment agar for the isolation of *Legionella*. The nutritive base includes yeast extract that includes L-cysteine. Charcoal is added to the medium as a detoxifying agent.
[QCMLS 2021: 6.1.6.4.4] [Mahon 2019, p404]

22. **c** Regan-Lowe agar is an enriched and selective medium formulated for the isolation of *Bordetella pertussis*. Cephalexin is added to inhibit nasopharyngeal flora. It provides better isolation of *B. pertussis* than Bordet-Gengou medium. Phenylethyl alcohol agar is selective for gram-positive cocci. Potassium tellurite agar and Tinsdale agar are selective/differential media useful in isolating *Corynebacterium diphtheriae*.
[QCMLS 2021: 6.1.6.4.4] [Mahon 2019, p347-348, 409]

23. **b** Throat swabs are unacceptable for anaerobic culture. Many anaerobic bacteria are commensal flora in the oropharynx. Anaerobic bacteria do not cause pharyngitis. The most common cause of pharyngitis is *Streptococcus pyogenes*.
[QCMLS 2021: 6.1.2] [Mahon 2019, p326, 488, 495]

24. **c** Cyclohexamide, which inhibits protein synthesis, is the common agent added to fungal isolation media to inhibit faster-growing saprophytic fungi. Chloramphenicol in fungal media suppresses bacterial growth, and streptomycin does not inhibit fungi. Amphotericin B is not routinely used as an additive in fungal media.
[QCMLS 2021: 6.5.1] [Mahon 2019, p609]



25. **d** There are several sputum-screening systems for assessing the quality of respiratory specimens. In general, neutrophils are a positive indicator of quality, and squamous epithelial cells are a negative indicator of quality, suggesting oropharyngeal contamination. The specimen shown in the image contains an abundance of squamous cells (>10/low power field), and would be unacceptable for culture.
[QCMLS 2021: 6.7.4] [Mahon 2019, p128]

26. **b** Detection of group B *Streptococcus* (GBS) in the genital and gastrointestinal tracts of pregnant women can identify infants at risk for GBS infection. The CDC currently recommends the collection of vaginal and rectal swabs or a single swab inserted first into the vagina and then the rectum at 35-37 weeks' gestation. The swab(s) should be inoculated into a selective broth medium such as LIM broth (Todd-Hewitt broth with colistin and nalidixic acid). The use of vaginal/rectal swabs and selective broth medium greatly increases the recovery of GBS. Trans Vag broth or StrepB Carrot broth can also be used.
[QCMLS 2021: 6.1.6.1.2.1] [Mahon 2019, p329]




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27. **d** The proper use of a biological safety cabinet includes keeping airflow unobstructed. An air curtain is a major barrier between the individual when the cabinet is not in use. Open flames are not to be used, but incinerators are acceptable. Moving hands in and out of the cabinet should be kept at a minimum. Clean and contaminated items should be segregated and organized to decrease hand movement. A biohazard discard receptacle should be placed in the rear of the cabinet without obstructing air flow.
[QCMLS 2021: 6.1.5] [Mahon 2019, p73-75]

28. **d** Colony #1 is a probable *Proteus* species To isolate the gram-positive cocci in clusters from the swarming *Proteus*, a medium that is selective for gram-positive cocci needs to be used. CNA and PEA agars are both selective for gram-positive cocci. The swarming GNR will not be able to grow on these media.
[QCMLS 2021: 6.1.5] [Mahon 2019, p114]

29. **d** Chromagars are various types of media that can be nonselective or selective, and are differential. Organism properties, such as the production of particular enzymes, will react with substrates in the agar to produce a predictable colored colony. The other agars listed are enriched and/or selective.
[QCMLS 2021: 6.5.1] [Mahon 2019, p114, 609]

30. **c** SPS and heparin are acceptable anticoagulants to use with specimens that are to be cultured. EDTA and citrate are not acceptable.
[QCMLS 2021: 6.7.5] [Mahon 2019, p110-111]

31. **a** *Francisella tularensis* is the causative agent of tularemia. It has a specific growth requirement for cysteine.
[QCMLS 2021: 6.1.6.4.4] [Procop 2017, p536]

32. **c** *Neisseria gonorrhoeae* requires an enhanced $CO_2$ atmosphere for optimal growth.
[QCMLS 2021: 6.1.6.2] [Mahon 2019, p372]

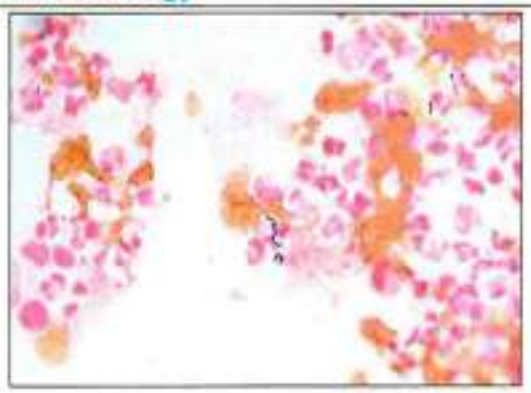
33. **c** Anaerobes normally inhabit skin and mucous membranes as part of the normal flora. Distractors **a**, **b** and **d** are virtually always unacceptable for anaerobic culture, because they normally contain anaerobic organisms. Only suprapubic bladder aspiration is an appropriate urine specimen for anaerobes. It is difficult to interpret culture results from these specimens and distinguish between pathogens and normal flora.
[QCMLS 2021: 6.1.2] [Mahon 2019, p495]

34. **c** Most infections involving anaerobes are polymicrobic and can include obligate aerobes, facultative anaerobes, microaerophilic bacteria in addition to anaerobic bacteria.
[QCMLS 2021: 6.1.6.5.1] [Mahon 2019, p490]

### *C. Stains: Procedure, Principle, and Interpretation*

35. **d** Sputum specimen quality is assessed to determine if the specimen is representative of the site of infection. The presence of white blood cells is an indicator of infection, and presence of squamous epithelial cells is an indicator of oropharyngeal contamination. In this specimen, >25 epithelial cells per low power field is an indicator of poor specimen quality, and the bacteria present are representative of oropharyngeal flora.
[QCMLS 2021: 6.7.4] [Mahon 2019, p128]

36. **b** Problems with analysis of Gram staining generally result from errors including interpretation of the slide (smear prepared too thick), excessive heat fixing, and improper decolorization. Inadequate decolorization with acetone/alcohol results in a smear in which host cells (neutrophils and squamous cells), as well as bacteria, all appear blue.
[QCMLS 2021: 6.1.4] [Mahon 2019, p11]



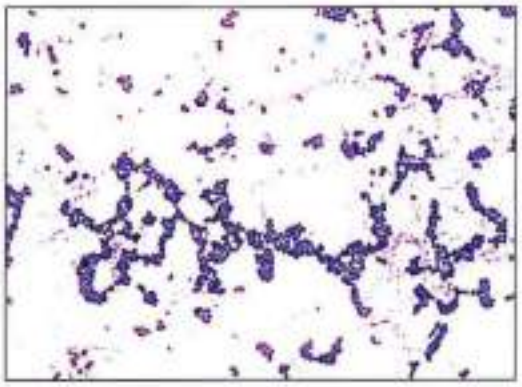
37. **d** The Gram stain depicts gram-positive cocci arranged in chains. Members of the genus *Streptococcus* characteristically grow in pairs and chains, and tend to chain more in fluid. Staphylococci are also gram-positive cocci that can appear singly, in pairs, short chains or, more typically, clusters.
[QCMLS 2021: 6.7.7] [Mahon 2019, p308, 313, 323]

38. **d** The Kinyoun acid-fast stain is also known as the "cold" technique. A surface-active agent is used to increase the stain's permeability. The Ziehl-Neelsen procedure uses heat to help the carbol fuchsin penetrate the organism's waxy cell wall.
[QCMLS 2021: 6.1.4] [Procop 2017, p569]

## *II. Analytic Procedures for Bacteriology*

### *A. Blood & Bone Marrow*

39. **a** Relapsing fever is caused by *Borrelia recurrentis* and is transmitted by the human body louse. Relapsing fever is characterized by the acute onset of high fever lasting 3-7 days, interspersed with periods of no fever lasting days to weeks.
[QCMLS 2021: 6.1.6.6.2] [Mahon 2019, p523]



40. **b** Coagulase-negative staphylococci are commonly associated with contaminated blood cultures; however, they are also increasing as a cause of true bacteremia. Significant bacteremia in a patient with endocarditis is usually continuous and low grade. In most cases, all blood cultures drawn will yield positive results. The facts that only 1 bottle of 1 set is positive, and that the bottle did not become positive until day 5 of incubation, indicate that this isolate is most likely a contaminant.
[QCMLS 2021: 6.7.5]



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41. **c** Nutritionally-deficient streptococci such as *Abiotrophia* do not grow on sheep blood agar without the addition a *Staphylococcus aureus* streak, the addition of a pyridoxal disc, or inclusion of a chocolate agar plate. Each will ensure the growth of nutritionally variant streptococci.
[QCMLS 2021: 6.7.5, 6.1.6.1.2.4] [Procop 2017, p808]

42. **a** The Gram stain and culture growth describe a *Staphylococcus* species. A positive catalase test result indicates that the organism is most likely a member of the genus *Staphylococcus*. Latex agglutination is used to differentiate *S. aureus* from coagulase-negative staphylococci. Latex agglutination tests for *S. aureus* detects clumping factor and the cell-wall protein, protein A, which binds IgG. The presence of clumps in the reagent after addition of a suspect colony indicates a positive reaction.
[QCMLS 2021: 6.1.6.1.1.1] [Procop 2017, p692, 693]

43. **a** PYR, bacitracin and hippurate are biochemicals/tests used in the presumptive or definitive identification of beta-hemolytic streptococci such as *Streptococcus pyogenes*, and *S. agalactiae*.
[QCMLS 2021: 6.1.6.1.2] [Procop 2017, p788, 790]

44. **d** Organisms that used to be categorized as nutritionally variant or deficient streptococci have been reclassified into the genera *Abiotrophia* and *Granulicatella*.
[QCMLS 2021: 6.1.6.1.2.4] [Procop 2017, p774]

45. **d** *Staphylococcus epidermidis* is the most common cause of prosthetic valve endocarditis.
[QCMLS 2021: 6.1.6.1.1.2] [Procop 2017, p687]

46. **d** *Streptococcus agalactiae* is catalase and bile esculin hydrolysis-negative and bacitracin and optochin resistant. *Streptococcus agalactiae* hydrolyzes hippurate.
[QCMLS 2021: 6.1.6.1.2] [Procop 2017, p788]

47. **c** Appropriate skin antisepsis is the most important factor in preventing contaminated blood cultures. *Staphylococcus epidermidis* is a common blood culture contaminant because it is a common inhabitant of the skin.
[QCMLS 2021: 6.7.5] [Procop 2017, p98]

48. **a** *Staphylococcus aureus* belongs to the normal flora of skin allowing the organism to be present in a variety of infections. The growth pattern on a sheep blood agar plate is characteristic for *S. aureus*, and the latex test is used as a rapid method of definitive identification.
[QCMLS 2021: 6.1.6.1.1.1] [Procop 2017, p672, 702]

49. **c** Subacute endocarditis caused by *Streptococcus viridans* typically occurs in patients with pre-existing heart valve disease, indwelling catheters, or other heart abnormalities. Poor oral hygiene or an invasive oral procedure can introduce the organism into the bloodstream, travel to the heart, and cause a vegetation.
[QCMLS 2021: 6.1.6.1.2.2] [Procop 2017, p763]

50. **c** These are typical biochemical reactions for *Providencia*. Key reactions that separate it from most other enteric organisms are lack of hydrogen sulfide production and phenylalanine deaminase positivity. Citrate and ornithine reactions differentiate *Providencia* and *Morganella*.
[QCMLS 2021: 6.1.6.4.1] [Procop 2017, p244]

51. **a** *Campylobacter* are gram-negative, curved bacilli that require microaerophilic conditions for growth.
[QCMLS 2021: 6.1.6.4.2] [Tille 2016, p415]

52. **c** Bone marrow is considered one of the most sensitive specimens for the recovery of *Brucella* along with blood, CSF, pleural, and synovial fluids, urine, abscesses, or other tissues.
[QCMLS 2021: 6.1.6.4.4] [Tille 2016, p432]

53. **d** The consumption of raw shell fish is a risk factor for *Vibrio vulnificus* and the biochemical reactions support a *Vibrio* species. *V. vulnificus* is one of the only vibrios that ferments lactose.
[QCMLS 2021: 6.1.6.4.2] [Procop 2017, p453, 454]

54. **d** *Brucella* causes undulant fever and is a cause of fever of unknown origin. It is slow growing and is associated with laboratory-acquired infections. It is also a potential agent of bioterrorism. Suspected *Brucella* isolates should not be tested in automated or manual identification systems.
[QCMLS 2021: 6.1.6.4.4] [Tille 2016, p431-434]

55. **d** *Cutibacterium acnes* is part of the normal flora of the skin, so it is frequently isolated from improperly collected blood cultures.
[QCMLS 2021: 6.1.6.5.4] [Mahon 2019, p25]

56. **c** SPS in blood culture media enhances recovery of most bacteria, including anaerobes. However, *Peptostreptococcus anaerobius* is inhibited by SPS.
[QCMLS 2021: 6.1.6.5.2] [Mahon 2019, p605, 869]

57. **b** *Cutibacterium acnes* is part of the normal flora on the skin and is a common blood culture contaminant. The Gram stain given is typical for *C. acnes*, and it is catalase and indole-positive.
[QCMLS 2021: 6.1.6.5] [Mahon 2019, p504]

58. **d** *Mycoplasma pneumoniae*, permanently lacks a cell wall so antibiotics that inhibit cell wall synthesis are not affective in its treatment.
[QCMLS 2021: 6.1.6.6.7] [Mahon 2019, p544, 557]

### *B. Cerebrospinal Fluid*

59. **d** Differential diagnosis of viral meningitis includes normal lactate level (increased in bacterial, tubercular and fungal), normal glucose level, and increased protein level. Lymphocytes are normally present.
[QCMLS 2021: 6.7.6] [Mahon 2019, p859]

60. **c** All of the organisms listed are potential causes of meningitis. Group B *Streptococcus* is associated with neonatal meningitis.
[QCMLS 2021: 6.7.6.4] [Mahon 2019, p850]

61. **c** The most likely organism isolated from this specimen is *Streptococcus agalactiae*. Polysaccharide latex agglutination will confirm the identification of *S. agalactiae* and differentiate it from other beta-hemolytic streptococci.
[QCMLS 2021: 6.7.6.4, 6.1.6.1.2.1] [Mahon 2019, p329]

62. **c** These are classic Gram stain, growth and biochemicals for *Haemophilus influenzae*.
[QCMLS 2021: 6.1.6.2] [Procop 2017, p475, 476]

63. **c** The classic CSF alterations associated with bacterial meningitis are a high WBC count with a neutrophil predominance as well as a low CSF glucose and a high CSF protein.
[QCMLS 2021: 6.7.6.5, 6.7.6.1] [Mahon 2019, p859]

64. **d** *Neisseria meningitidis* is the second leading cause of meningitis in young adults. A cytospin Gram stain of the cerebrospinal fluid can show gram-negative diplococci which may appear inside and outside of the PMNs.
[QCMLS 2021: 6.7.6] [Procop 2017, p624, 635]

### *C. Body Fluids from Normally Sterile Sites*

65. **d** *Yersinia pestis* is classically described as having a "safety pin" appearance on Wright-Giemsa stain. This patient's presentation is classic for bubonic plague.
[QCMLS 2021: 6.1.6.4.1] [Mahon 2019, p738-739]




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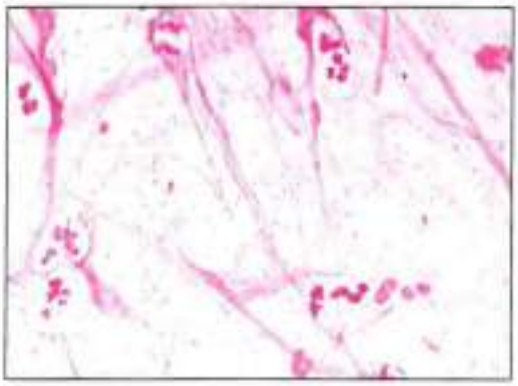
66. **b** *Bacillus cereus* is not an anaerobic organism; *Eggerthella* (formerly *Eubacterium*) and *Bifidobacterium* are anaerobic gram-positive bacilli that do not form spores. *Clostridium septicum* forms subterminal spores, and the colony can swarm.
[QCMLS 2021: 6.1.6.5.4] [Mahon 2019, p345, 363]

67. **c** The Gram stain and double zone of hemolysis are characteristics of *Clostridium perfringens*. *C. perfringens* is lecithinase-positive on the egg yolk agar test for lecithinase and lipase.
[QCMLS 2021: 6.1.6.5.4] [Mahon 2019, p506-507]

68. **b** *Bacteroides fragilis* grows on BBE agar and, because it can hydrolyze esculin, produces black colonies. *B. fragilis* is also catalase-positive and indole-negative.
[QCMLS 2021: 6.1.6.5.6] [Mahon 2019, p512-513]

69. **c** *Fusobacterium nucleatum* is classically described as long, slender, gram-negative bacilli with tapered ends. Inhibition of growth by 20% bile and a positive indole reaction narrow the selection process.
[QCMLS 2021: 6.1.6.5.5] [Mahon 2019, p513-514]

*D. Lower Respiratory*



70. **c** The sputum Gram stain shown demonstrates numerous neutrophils and small, pleomorphic gram-negative bacilli suggestive of *Haemophilus*. *H. influenzae* is an important cause of lower respiratory tract infections in patients with pre-existing lung disease, such as cystic fibrosis. *Haemophilus* are fastidious and require the use of an enriched medium such as chocolate agar and incubation at 35°-37°C in a moist environment supplemented with 5-10% $CO_2$.
[QCMLS 2021: 6.1.6.4.4] [Mahon 2019, p386, 388-389]

71. **b** *Klebsiella* is the only distractor that is VP-positive, and the other biochemical reactions are typical for *K. pneumoniae*.
[QCMLS 2021: 6.1.6.4.1] [Procop 2017, p234]

72. **d** *Alcaligenes* and *Moraxella* are oxidase-positive; *Stenotrophomonas* are gram-negative bacilli and are lysine and ONPG-positive; *Acinetobacter baumannii* are nitrate and ONPG-negative and are gram-negative coccobacilli.
[QCMLS 2021: 6.1.6.4.3] [Procop 2017, p370, 382, 390]

73. **d** Within 3 to 4 days colonies should be visible with plates held for a maximum of 2 weeks.
[QCMLS 2021: 6.1.6.4.4] [Tille 2016, p428]

74. **a** *Burkholderia cepacia* is associated with respiratory infections in cystic fibrosis patients, and the biochemicals are typical for this organism.
[QCMLS 2021: 6.1.6.4.3] [Procop 2017, p348, 351, 352]

75. **b** *Burkholderia pseudomallei* is an organism commonly found in the soil of Southeast Asian countries (Thailand, Vietnam) and parts of northern Australia. Infection with the organism causes melioidosis, which is also referred to as "glanders-like" disease. Key biochemicals for *B. pseudomallei* include oxidase, motility, reduction of nitrate to gas, and oxidation of glucose, lactose, and mannitol.
[QCMLS 2021: 6.1.6.4.3] [Procop 2017, p347]

76. **b** *Moraxella catarrhalis* are oxidase-positive gram-negative diplococci that are usually beta-lactamase-positive. It does not utilize carbohydrates. Unlike *Neisseria* species, it is butyrate-esterase-positive.
[QCMLS 2021: 6.1.6.2] [Mahon 2019, p378-379]

77. **a** *Nocardia* are capable of growing on BCYE agar within 3-6 days and demonstrate branching, beaded gram-positive bacilli on Gram stain. *Nocardia* are partially acid fast, and are stained best with the modified acid-fast stain. *Streptomyces* are not acid-fast; *Actinomyces* are anaerobic; *Mycobacterium* have a longer incubation period and grow on Lowenstein-Jensen or Middlebrook media
[QCMLS 2021: 6.1.6.3, 6.2.1.3] [Mahon 2019, p358-359]

78. **a** *Bacteroides* and *Fusobacterium* are anaerobic gram-negative bacilli and *Nocardia* are aerobic gram-positive bacilli. Gram stain and colony morphology described are classic for *Actinomyces*.
[QCMLS 2021: 6.1.6.5.5] [Mahon 2019, p511-512]

79. **c** Both *Prevotella* and *Porphyromonas* colonies fluoresce brick red. *Porphyromonas* is susceptible to vancomycin and can be catalase-positive.
[QCMLS 2021: 6.1.6.5.6] [Mahon 2019, p513-514]

### *E. Upper Respiratory*

80. **b** *Chlamydophila psittaci*, the agent of psittacosis, is transmitted to humans via inhalation, of dried excrement, urine, or respiratory secretions from specific birds (eg, parrots, parakeets, macaws, cockatiels).
[QCMLS 2021: 6.1.6.6.4]

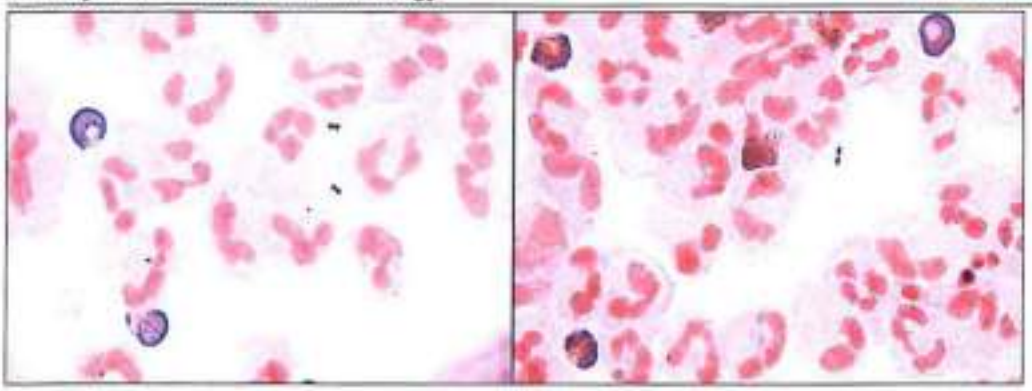


81. **b** Noninfectious sequelae associated with infection with *Streptococcus pyogenes* are glomerulonephritis and rheumatic fever.
[QCMLS 2021: 6.1.6.1.2.1] [Procop 2017, p743]

82. **d** *Streptococcus pyogenes* is the cause of streptococcal pharyngitis, which commonly afflicts school age children during the winter and spring months, with a lower percentage of adults contracting the infection.
[QCMLS 2021: 6.1.6.1.2.1] [Procop 2017, p743]




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83. **a** Colony morphology and Gram stain characteristics are 2 of the first indicators that an organism may be *Streptococcus pneumoniae*. Susceptibility to optochin and a positive bile solubility test indicating the organism's ability to be autolytic both definitively identify *S. pneumoniae*.
[QCMLS 2021: 6.1.6.1.2.2] [Procop 2017, p791]

84. **c** Group C beta-hemolytic streptococci are considered normal flora of the nasopharynx. This organism has been found to cause infections similar to *S. pyogenes* and *S. agalactiae*. Group C beta-hemolytic streptococci are also sensitive to bacitracin disk making this particular test no longer definitive for the identification of Group A beta-hemolytic streptococci.
[QCMLS 2021: 6.1.6.1.2] [Tille 2016, p249-250, 257]

85. **b** Growth only on chocolate agar is typical for *Haemophilus influenzae*, which are gram-negative coccobacilli that causes upper respiratory infections, including eye infections.
[QCMLS 2021: 6.1.6.4.4] [Tille 2016, p341]

86. **d** *Haemophilus parainfluenzae* requires NAD for growth but not hemin. This distinguishes it from *H. influenzae*. *H. haemolyticus* is hemolytic, and *H. ducreyi* does not cause epiglotittis.
[QCMLS 2021: 6.1.6.4.4] [Tille 2016, p408]

87. **d** Infection with *Bordetella pertussis* can be prevented through the use of a vaccine. Patients present with malaise, fever, and a characteristic cough which can resemble a sound of "whooping". Commonly used selective media include Bordet-Gengou and Regan Lowe. Colonies growing on Bordet-Gengou are described as "drops of mercury". Nasopharyngeal swabs are used to collect samples from patients.
[QCMLS 2021: 6.1.6.4.4] [Tille 2016, p438-439]

88. **b** The *Neisseria* species listed all ferment several carbohydrates, *Moraxella catarrhalis* is biochemically inert and does not ferment carbohydrates.
[QCMLS 2021: 6.1.6.2] [Mahon 2019, p372-373, 375]

89. **c** To determine if an isolate of *Corynebacterium diphtheriae* produces toxin, testing for the presence of diphtheria toxin must be performed using methods such as the Elek test or PCR.
[QCMLS 2021: 6.1.6.3] [Mahon 2019, p348]

90. **a** *Chlamydia trachomatis* is a well-known cause of sexually transmitted infections, including urethritis and cervicitis, as well as inclusion conjunctivitis and pneumonia in neonates. It also causes trachoma and lymphogranuloma venereum.
[QCMLS 2021: 6.1.6.6.4]

*F. Gastrointestinal*

91. **a** *Campylobacter* continues to be the most common enteric pathogen isolated from patients with gastroenteritis. Routinely fecal specimens should be cultured for *Salmonella*, *Shigella* and *Campylobacter*. Fecal specimens are not routinely cultured for enterotoxigenic *Escherichia coli* nor *Clostridium botulinum*. *Entamoeba hartmanni* is a nonpathogenic parasite and does not cause diarrhea.
[QCMLS 2021: 6.7.2]

92. **a** All of the biochemical and serological reactions listed are consistent with an identification of *Shigella flexneri*.
[QCMLS 2021: 6.1.6.4.1] [Mahon 2019, p429]

93. **c** D-sorbitol replaces lactose. *Escherichia coli* O157:H7 does not ferment sorbitol and grows clear on MacConkey with sorbitol. Other species of *E. coli* are positive for fermentation of sorbitol. This makes the medium a good screen for O157:H7.
[QCMLS 2021: 6.1.6.4.1] [Tille 2016, p316]

94. **a** *Shigella* has colorless colonies on both MacConkey and Hektoen agars. *Yersinia* is lactose-negative, but Hektoen agar contains both lactose and sucrose, causing *Yersinia* to produce yellow colonies from the fermentation of sucrose. *Vibrio parahaemolyticus* needs at least 1% NaCl to grow and *Campylobacter* does not grow on MacConkey or Hektoen agars.
[QCMLS 2021: 6.1.6.4.1] [Tille 2016, p958]

95. **b** *Escherichia coli* can produces several different types of toxins that result in different gastroenteritis manifestations.
[QCMLS 2021: 6.1.6.4.1] [Procop 2017, p254]

96. **b** The biochemical characteristic that best fits *Shigella* is that it is nonmotile. *Shigella* are urease-negative and oxidase-negative. *Shigella* are lactose nonfermenters.
[QCMLS 2021: 6.1.6.4.1] [Procop 2017, p258]

97. **d** Some *Shigella* produce capsular antigen that mask the cell wall and boiling removes the capsule.
[QCMLS 2021: 6.1.6.4.1] [Procop 2017, p322]

98. **d** Boiling removed the capsule so that the antiserum could react with cell wall antigen. Group D *Shigella* is *S. sonnei*.
[QCMLS 2021: 6.1.6.4.1] [Procop 2017, p322]

99. **c** *Edwardsiella* produces $H_2S$; *E. coli* is indole-positive; *Providencia* has a TSI reaction of alkaline/acid. *Yersinia* typically shows motility at 25°C and not 35°C.
[QCMLS 2021: 6.1.6.4.1] [Procop 2017, p234, 235]

100. **b** *Shigella* is $H_2S$-negative, while *Salmonella*, *Edwardsiella* and *Proteus* are $H_2S$-positive. *Proteus mirabilis* is indole-negative, so a lack of agglutination with *Salmonella* antisera indicates the presence of *Edwardsiella*.
[QCMLS 2021: 6.1.6.4.1] [Procop 2017, p234, 235]

101. **d** The history of the patient suggests an appendicitis-like syndrome, which is consistent with *Yersinia enterocolitica*. Also, *Y. enterocolitica* grows better at 25°C.
[QCMLS 2021: 6.1.6.4.1] [Procop 2017, p283, 284]

102. **a** If the Vi antigen is present, it will not permit agglutination of the polyvalent antisera. The Vi antigen is heat labile, so boiling will remove it and appropriate agglutination can take place.
[QCMLS 2021: 6.1.6.4.1] [Procop 2017, p322]

103. **d** *Campylobacter jejuni/coli* grow better at 42°C than 37°C and other organisms in the colon are inhibited at this high temperature.
[QCMLS 2021: 6.1.6.4.2] [Tille 2016, p419]



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104. **d** Presumptive identification of *Helicobacter pylori* can be made with positive oxidase, catalase, and rapidly positive urea (within minutes) reactions. The urea breath test has high sensitivity (96%) and specificity (100%) compared to patients who were *H. pylori* positive by culture and Gram stain. The breath test is also recommended for monitoring therapy.
[QCMLS 2021: 6.1.6.4.2] [Procop 2017, p443]

105. **c** *Helicobacter pylori* is known to cause gastritis and are gram-negative, curved bacilli.
[QCMLS 2021: 6.1.6.4.2] [Procop 2017, p443]

106. **b** *Escherichia coli* O157:H7 is unable to utilize sorbitol producing clear colonies on the agar, MacConkey with Sorbitol. Other strains of *E. coli* do utilize the sorbitol producing pink colonies on the same agar.
[QCMLS 2021: 6.1.6.4.1] [Tille 2016, p316]

107. **a** *Campylobacter* is microaerophilic, and requires a decreased oxygen and increased carbon dioxide atmosphere for growth.
[QCMLS 2021: 6.1.6.4.2] [Procop 2017, p434]

108. **b** These 2 genera both belong to the tribe, Escherichieae. While both are biochemically different and have differing growth patterns, both are closely related genetically, placing them in the same group. However, *Escherichia coli* characteristically ferments lactose, is motile, utilizes beta-galactosidase (ONPG), and is positive for indole production.
[QCMLS 2021: 6.1.6.4.1] [Procop 2017, p235, 240, 241, 250, 251]

109. **c** *Bacillus cereus* is the etiologic agent of 2 distinct types of food poisoning syndromes. Spores can survive cooking and germinate. Vegetative cells multiply and produce toxin.
[QCMLS 2021: 6.1.6.3] [Mahon 2019, p363-364]

110. **d** Lecithinase is not related to toxin production. Latex agglutination detects the organism in the stool specimen but not the toxins. Cell culture cytotoxicity assay (toxin B only) has been considered the gold standard, but it takes 2-3 days for results. EIAs are rapid and can detect both toxins A and B, and glutamate dehydrogenase is an enzyme often associated with *Clostridioides difficile*. If EIA and glutamate dehydrogenase are negative, they can be used as rapid screening tests to rule out *C. difficile*. However, not all strains of *C. difficile* produce toxins A and B. NAAT will determine the presence of both toxins A and B in the stool and is considered a stand alone test—the results do not need to be confirmed.
[QCMLS 2021: 6.1.6.5.4] [Mahon 2019, p510]

111. **b** Fluorescent staining, HPLC, and LKV media are not used for detection of *Clostridioides difficile*. Glutamate dehydrogenase is an enzyme associated with *C. difficile*.
[QCMLS 2021: 6.1.6.5.4] [Mahon 2019, p510]

112. **c** Selective media are needed to isolate *Clostridioides difficile* from stool and CCFA is also differential—fermentation of lactose produces classic colony morphology for this organism.
[QCMLS 2021: 6.1.6.5.4] [Mahon 2019, p509-510]

113. **c** Strains of *Staphylococcus aureus* produce a heat-stable enterotoxin that is associated with short-incubation food poisoning.
[QCMLS 2021: 6.1.6.1.1.1] [Procop 2017, p679]

### *G. Skin, Soft Tissue, and Bone*

114. **a** *Streptococcus pyogenes* belongs to the beta-hemolytic streptococci group and possesses the Lancefield group antigen, A. This organism is sensitive to bacitracin and definitively identified by being PYR-positive or agglutinating with a group A latex.
[QCMLS 2021: 6.1.6.1.2.1] [Procop 2017, p790]

115. **c** *Streptococcus viridans* makes up part of the flora found in the upper respiratory tract (which includes oral flora) and the genitourinary tract. The organism is alpha-hemolytic but, unlike *S. pneumoniae*, optochin resistant. Enterococci grow in 6.5% NaCl.
[QCMLS 2021: 6.1.6.1.2.2] [Procop 2017, p763, 794]

116. **d** *Pseudomonas aeruginosa* produces the blue-green pigment, pyocyanin.
[QCMLS 2021: 6.1.6.4.3] [Tille 2016, p341]

117. **d** *Pseudomonas aeruginosa* produces a blue-green pigment, reduces nitrate and is oxidase-positive.
[QCMLS 2021: 6.1.6.4.3] [Mahon 2019, p472-743]

118. **b** *Pasteurella multocida* does not grow on MacConkey agar, and is associated with wounds resulting from dog and cat bites. *Vibrio cholerae* is motile and *Pseudomonas* and *Aeromonas* grow on MacConkey.
[QCMLS 2021: 6.1.6.4.4] [Procop 2017, p459]

119. **c** *Enterobacteriaceae*, such as *E. coli*, *Serratia* and *Enterobacter*, are oxidase-negative. The only selection that is oxidase-positive is *Aeromonas*. It is associated with wounds contaminated with water.
[QCMLS 2021: 6.1.6.4.2] [Procop 2017, p462]

120. **b** "Pitting the agar" and the bleach smell of the colonies are hallmark characteristics of *Eikenella corrodens*.
[QCMLS 2021: 6.1.6.4.4] [Procop 2017, p495]

121. **d** *Pseudomonas aeruginosa* produces a characteristic appearance when grown on a BAP and streaked to a Mueller-Hinton agar plate. This commonly isolated opportunistic organism is known to cause infections of the respiratory tract, wounds (burn patients), eye, immunocompromised (diabetic), and urinary tract.
[QCMLS 2021: 6.1.6.4.3] [Procop 2017, p340]

122. **b** *Eikenella corrodens* is a residential organism of the mouth and upper respiratory tract. Introduction of mouth flora into an open wound or a human bite can cause infection usually presenting in the form of cellulitis.
[QCMLS 2021: 6.1.6.4.4] [Procop 2017, p493-495]

123. **c** *Edwardsiella* species are commonly isolated from reptiles and freshwater fish. Infections can occur during an aquatic accident resulting in an infected wound. Key biochemical characteristic is the production of hydrogen sulfide otherwise presents similarly to *E. coli*.
[QCMLS 2021: 6.1.6.4.1] [Procop 2017, p259]

124. **c** *Bacillus anthracis* are large, rectangular gram-positive bacilli that produce colonies with an irregular edge (often described as a "medusa-head" appearance) on blood agar. Colonies are nonhemolytic, catalase-positive and nonmotile.
[QCMLS 2021: 6.1.6.3] [Mahon 2019, p362-363]

125. **d** Facultative anaerobes are organisms that can grow under both aerobic and anaerobic conditions.
[QCMLS 2021: 6.1.5] [Procop 2017, p195]

126. **d** Anaerobic bacteria characteristically produce foul-smelling metabolic end products. Infections associated with anaerobes are typically polymicrobic. Gas production and growth on SBA can occur in anaerobic conditions with facultative anaerobes.
[QCMLS 2021: 6.1.6.5.1] [Mahon 2019, p490-491]

127. **d** The specimen Gram stain suggests the presence of *Bacillus* or *Clostridium*. Since no growth is observed aerobically, the specimen should be inoculated to media that are incubated anaerobically.
[QCMLS 2021: 6.1.6.5.1] [Mahon 2019, p502]

128. **b** A probable anaerobe is the only organism growing so the microbiologist can proceed with identification.
[QCMLS 2021: 6.1.6.5.1] [Mahon 2019, p502]



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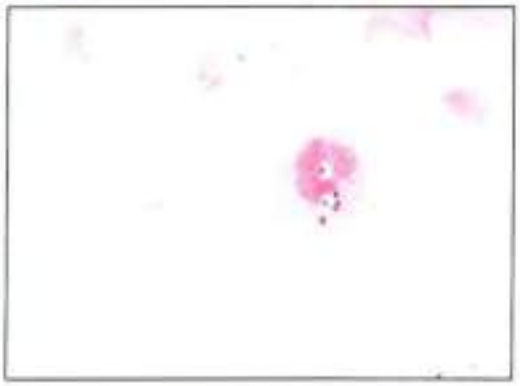
129. **c** In this culture there are aerobic or facultative gram-negative bacilli **and** a second organism growing in the thioglycolate broth only (gram-positive bacilli). This leads one to think it could be an anaerobe because it did not grow on any of the media incubated in 3-5% $CO_2$. The aerotolerance test is the first step in determining if an anaerobe is present.
[QCMLS 2021: 6.1.6.5.1] [Mahon 2019, p502]

130. **b** *Actinomyces* and *Eggerthella* are anaerobic gram-positive bacilli that do not form spores; *Bacillus* is not an anaerobic organism. *Clostridium perfringens* are spore-forming, box-car shaped, anaerobic gram-positive bacilli.
[QCMLS 2021: 6.1.6.5.4] [Mahon 2019, p506]

131. **a** Anaerobic gram-negative bacilli predominate among anaerobes in clinical infections. *Bacteroides fragilis* is the most commonly isolated anaerobic bacteria in clinical specimens.
[QCMLS 2021: 6.1.6.5.6] [Mahon 2019, p512]

### *H. Genital Tract*

132. **d** *Moraxella osloensis* are gram-negative coccobacilli that are often plump and occur in pairs, demonstrating a morphology similar to *Neisseria*. The presence of this organism in endocervical specimens contaminated with vaginal secretions can lead to overinterpretation of smears for *N. gonorrhoeae*.
[QCMLS 2021: 6.1.6.2] [Mahon 2019, p478]



133. **b** Gonococcal urethritis in adult males is often diagnosed by the observation of gram-negative intracellular diplococci within neutrophils in smears prepared from urethral discharge. The Gram stain in males has a sensitivity of 89% % and a specificity of >90% for diagnosing gonorrhea in symptomatic males.
[QCMLS 2021: 6.1.6.2, 6.7.9] [Mahon 2019, p901]

134. **b** Group B *Streptococcus* (GBS) colonizes both the genital and gastrointestinal tracts of pregnant women. Collection of a vaginal and rectal specimen is recommended by the CDC to maximize GBS detection in this population. Patients should be screened at 35-37 weeks' gestation. In addition selective broth culture is recommended (Todd-Hewitt broth with antibiotics) although other selective media are also available. When selective broth culture is used it should be incubated for 18-24 hours prior to subculture onto blood agar.
[QCMLS 2021: 6.1.6.1.2.1] [Mahon 2019, p329]

135. **d** *Neisseria gonorrhoeae* requires an enhanced carbon dioxide atmosphere for optimal growth.
[QCMLS 2021: 6.1.6.2] [Mahon 2019, p371-372]

136. **d** Clinical history does not distinguish *Neisseria gonorrhoeae* from *Chlamydia trachomatis*. Because of the presence of nonpathogenic *Neisseria* in the female genital tract, Gram stain does not differentiate these organisms from *N. gonorrhoeae*. MALDI-TOF requires colony to work from, while NAAT is direct detection.
[QCMLS 2021: 6.7.9] [Mahon 2019, p374-375]

137. **d** Because of the presence of nonpathogenic *Neisseria* in the female genital tract, Gram stain does not differentiate these organisms from *N. gonorrhoeae*. For this reason Gram stain results should not be reported on vaginal specimens.
[QCMLS 2021: 6.1.6.2] [Mahon 2019, p371]

138. **c** *Gardnerella vaginalis* are technically pleomorphic gram-positive bacilli, but they often stain gram-variable or even gram-negative. It causes, in concert with other organisms, bacterial vaginosis, where organisms cluster on large squamous epithelial cells—"clue cells"—and the pH of vaginal fluid is >4.5.
[QCMLS 2021: 6.1.6.3] [Mahon 2019, p356-357]

139. **c** *Chlamydia trachomatis*, *Neisseria gonorrhoeae* and herpes simplex virus can all be isolated in culture. Direct culture of *Treponema pallidum* on artificial media has not been achieved. Darkfield microscopy and serological techniques are used to diagnose *T. pallidum* infection.
[QCMLS 2021: 6.1.6.6.1] [Mahon 2019, p526]

140. **c** Darkfield microscopy can be performed to visualize *Treponema pallidum* in genital or skin lesions. Darkfield examination allows for the visualization of typical *T. pallidum* morphology and motility.
[QCMLS 2021: 6.1.6.6.1] [Mahon 2019, p526]

141. **b** Up to 1/3 of patients with *Neisseria gonorrhoeae* infection are also co-infected with *Chlamydia trachomatis*. Patients with identified infection with one organism are usually treated for both infections.
[QCMLS 2021: 6.1.6.6.4, 6.7.9] [Mahon 2019, p376]

142. **a** *Ureaplasma urealyticum*, like other *Mycoplasma*, lacks a cell wall and possesses an extremely small genome. As a result, this organism has limited biosynthetic capability and fastidious growth requirements. Culture medium should contain serum (provides sterols), growth factors such as yeast extract, and a metabolic substrate.
[QCMLS 2021: 6.1.6.6.7] [Mahon 2019, p550]

*I. Urine*

143. **c** Suprapubic aspiration removes urine directly from the bladder and yields a specimen free of urethral contamination. When a suprapubic culture grows >$10^2$ CFU/mL, the culture receives a full work-up and is considered a significant pathogen. A $10^2$ quantitative loop is used to culture the specimen, so 10 colonies × $10^2$ = 1000 CFU.
[QCMLS 2021: 6.7.1] [Tille 2016, p926]

144. **c** The number of colonies isolated is multiplied by 100 when a 0.01 mL loop is used for inoculation. Gram-positive, catalase-negative cocci are indicative of streptococci. In urine, *Enterococcus* is a major pathogen, so esculin hydrolosis followed by 6.5% NaCl growth test will test for it.
[QCMLS 2021: 6.7.1, 6.1.6.1.2.3] [Procop 2017, p30, 734]

145. **b** Noninfectious sequelae associated with infection with *Streptococcus pyogenes* are glomerulonephritis and rheumatic fever.
[QCMLS 2021: 6.1.6.1.2.1] [Procop 2017, p743]

146. **c** The criteria for a culture considered positive for infection now $10^2$ CFU/mL. The 60 CFU/mL is multiplied by the 0.01 mL delivered by the calibrated loop to get the organisms per mL.
[QCMLS 2021: 6.7.1]

147. **b** *Escherichia coli* is the most commonly isolated species in the clinical laboratory including in urine cultures.
[QCMLS 2021: 6.7.1.1] [Procop 2017, p254]

148. **b** *Morganella* and *Providencia* do not produce $H_2S$; the indole reaction differentiates *P. mirabilis* and *P. vulgaris*.
[QCMLS 2021: 6.1.6.4.1] [Procop 2017, p244]



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149. **b** *Serratia* can produce a red pigment; *Proteus mirabilis* swarms, is TDA-positive and produces $H_2S$.
[QCMLS 2021: 6.1.6.4.1] [Procop 2017, p276, 278.]

150. **c** The correct quantitation on a urine sample is obtained by counting the colonies and multiplying them by the dilution factor, which in this case is 1000 because a 0.001 µL loop was used for culture. The biochemicals are characteristic of *Enterobacter cloacae*.
[QCMLS 2021: 6.1.6.4.1] [Procop 2017, p273]

151. **b** *Morganella morganii* has been isolated in urinary tract infections, wound infections, and is known to also cause diarrhea in patients. Identifying biochemicals of the genus include motility, and ornithine decarboxylation, and phenylalanine deaminase.
[QCMLS 2021: 6.1.6.4.1] [Procop 2017, p235, 279]

152. **d** *Leptospira* are spiral-shaped organisms with hooked ends. They are ubiquitous in water (eg, lakes, ponds) and associated with renal infection in animals. Leptospirosis is a zoonosis, and humans are usually infected via direct or indirect contact with the urine of infected animals (including rats). Between 5-10% of patients with leptospirosis have the icteric form and develop jaundice, and may develop acute renal failure.
[QCMLS 2021: 6.1.6.6.1]

### *J. Identification Methods (Theory, Interpretation, and Application)*

153. **d** Of the combinations listed, the use of *Escherichia coli* and *Proteus mirabilis* will produce a positive and negative result for indole, respectively. The remainder of the organisms are all positive for the test or characteristic described.
[QCMLS 2021: 6.1.6.4.1] [Mahon 2019, p314, 327, 440, 442]

154. **b** The organism in this urine culture is a *Staphylococcus* species. Coagulase will differentiate *S. aureus* from coagulase-negative staphylococci (CNS) and novobiocin susceptibility will differentiate *S. saprophyticus* from other CNS. *S. saprophyticus* is a common cause of urinary tract infections in young, sexually active females of childbearing age.
[QCMLS 2021: 6.1.6.1.1.2] [Procop 2017, p691, 706]

155. **d** Bile solubility testing of alpha-hemolytic streptococci differentiates *Streptococcus pneumoniae* (soluble) from other alpha-hemolytic streptococci, such as viridans streptococci (insoluble).
[QCMLS 2021: 6.1.6.1.2] [Procop 2017, p791]

156. **b** Coagulase is the biochemical test used to distinguish *Staphylococcus aureus* (coagulase-positive) from other staphylococci (coagulase-negative).
[QCMLS 2021: 6.1.6.1.1.1] [Procop 2017, p702]

157. **c** Optochin susceptibility is used to differentiate *Streptococcus pneumoniae*, which are susceptible, from other alpha-hemolytic streptococci, which are resistant.
[QCMLS 2021: 6.1.6.1.2.2] [Procop 2017, p791]



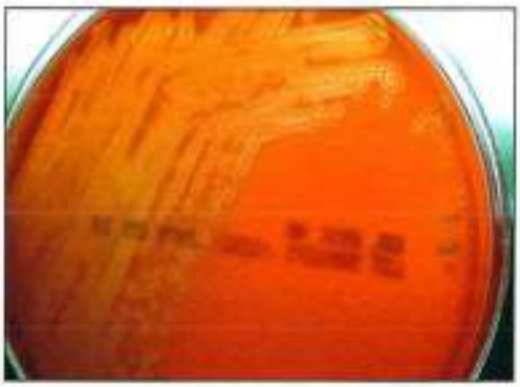
158. **b** Bacitracin susceptibility is used to presumptively identify *S. pyogenes* (Group A *Streptococcus*).
[QCMLS 2021: 6.1.6.1.2.1] [Procop 2017, p788]

159. **c** *Streptococcus bovis and Enterococcus* produce a positive bile esculin test; however, of these, only *Enterococcus* grows in the presence of 6.5% NaCl.
[QCMLS 2021: 6.1.6.1.2.3] [Procop 2017, p788]

160. **a** The colony description and biochemical results presented describe *Streptococcus agalactiae*. The identification of this organism is confirmed by streptococcus latex agglutination.
[QCMLS 2021: 6.1.6.1.2.1] [Procop 2017, p790]

161. **a** Of the biochemicals listed, only a positive PYR will aid in the identification of *Enterococcus*, which has the ability to hydrolyze the substrate L-pyrrolidonyl-B-naphthylamide to produce a B-naphthylamine. Combining the B-naphthylamine with a cinnamaldehyde reagent, a bright red color forms, indicating a positive reaction.
[QCMLS 2021: 6.1.6.1.2.3] [Procop 2017, p788; Tille 2016, p97]

162. **b** Group A streptococci (*Streptococcus pyogenes*) are susceptible to bacitracin and CAMP-test-negative.
[QCMLS 2021: 6.1.6.1.2.1] [Procop 2017, p788]



163. **b** *Streptococcus agalactiae* is beta-hemolytic, resistant to bacitracin and CAMP test positive.
[QCMLS 2021: 6.1.6.1.2.1] [Procop 2017, p788]

164. **a** Growth in 6.5% NaCl, growth in bile esculin medium and susceptibility are not used in the routine identification of *Streptococcus agalactiae*. Polysaccharide grouping for group B antigen is routinely used for identification of *S. agalactiae*.
[QCMLS 2021: 6.1.6.1.2.1] [Procop 2017, p790]

165. **d** *Enterococcus* species are more resistant to antimicrobial therapy than group D streptococci such as *S. bovis*.
[QCMLS 2021: 6.1.6.1.2.3] [Procop 2017, p767]

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166. **d** Bile solubility testing of alpha-hemolytic streptococci differentiates *S. pneumoniae* (soluble) from other alpha-hemolytic streptococci such as viridans streptococci (insoluble).
[QCMLS 2021: 6.1.6.1.2.2] [Procop 2017, p791]

167. **b** Enterococci are bile-esculin-positive, hippurate-negative and have the ability to grow in 6.5% NaCl. Enterococci are relatively resistant to penicillin and require combination therapy to treat serious infections.
[QCMLS 2021: 6.1.6.1.2.3] [Procop 2017, 773, 788]

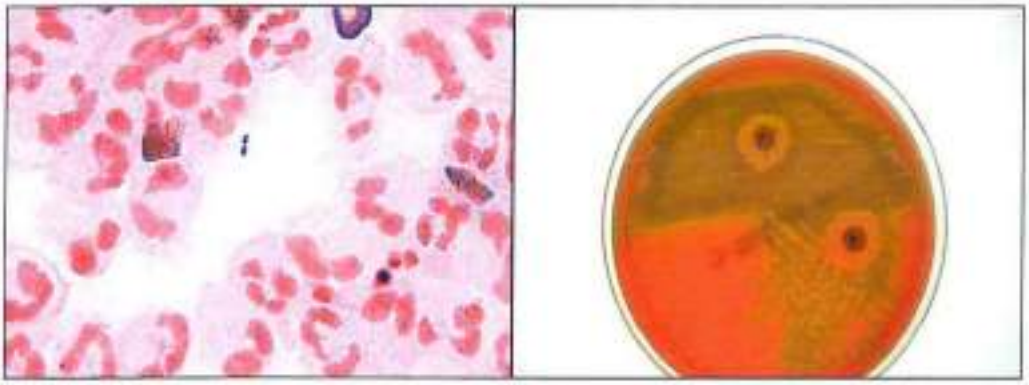
168. **c** Of the biochemicals listed, only hydrolysis of sodium hippurate will differentiate *Streptococcus agalactiae* (positive) from *S. pyogenes* (negative).
[QCMLS 2021: 6.1.6.1.2.1] [Procop 2017, p788]

169. **c** *Micrococcus* and *Staphylococcus* can be differentiated by susceptibility to furazolidone (100 µg/disk). *Staphylococcus* is susceptible; *Micrococcus* is resistant.
[QCMLS 2021: 6.1.6.1.1.1] [Procop 2017, p694]

170. **b** *Micrococcus* is modified oxidase-positive, bacitracin (0.04U) susceptible and resistant to lysostaphin.
[QCMLS 2021: 6.1.6.1.1.1] [Procop 2017, p693, 694]

171. **d** The question describes the CAMP test, which is positive for *Streptococcus agalactiae*.
[QCMLS 2021: 6.1.6.1.2.1] [Procop 2017, p788]

172. **d** *Enterococcus* is positive for the bile esculin test, while *Streptococcus pyogenes* and staphylococci are negative.
[QCMLS 2021: 6.1.6.1.2.3] [Procop 2017, p788]



173. **b** The viridans streptococci and enterococci are resistant to the optochin test (or P tab) differentiating the alpha-hemolytic colonies from *Streptococcus pneumoniae*. *S. pyogenes* is beta-hemolytic.
[QCMLS 2021: 6.1.6.1.2, 6.1.6.1.2.2] [Procop 2017, p791]

174. **b** Like the enterococci, *Streptococcus bovis* is a nonhemolytic streptococci that possesses the group D antigen. *Streptococcus bovis* can utilize bile esculin but does not hydrolyze PYR or grow in 6.5% NaCl.
[QCMLS 2021: 6.1.6.1.2.3] [Procop 2017, p796]

175. **b** Through the action of the enzyme beta-galactosidase, ONPG cleaves into galactose and o-nitrophenol (a yellow compound).
[QCMLS 2021: 6.1.6.4.1] [Mahon 2019, p178]

176. **d** Methods for rapid diagnosis of *Francisella tularensis* include fluorescent antibody staining of smears and tissues, antigen detection in urine detection of lipopolysaccharide using specific monoclonal antibodies and PCR. Only PCR has gained widespread use. PCR is appealing because smears and cultures are usually negative, and organism isolation may be hazardous. Serological diagnosis may take weeks to confirm.
[QCMLS 2021: 6.1.6.4.4]

177. **d** Members of *Enterobacteriaceae* are gram-negative bacilli or coccobacilli; oxidase-negative, capable of fermenting glucose, and reducing nitrates to nitrites.
[QCMLS 2021: 6.1.6.4.1] [Tille 2016, p308]

178. **d** *Shigella* is lactose-negative, most species do not produce gas, are VP, urea, lysine decarboxylase and citrate-negative, and they are nonmotile.
[QCMLS 2021: 6.1.6.4.1] [Procop 2017, p250]

179. **d** *Alcaligenes*, *Pseudomonas* and *Acinetobacter* are all nonfermenters; *Yersinia* is a member of the *Enterobacteriaceae* and, by definition, ferments glucose.
[QCMLS 2021: 6.1.6.4.1] [Procop 2017, p250]

180. **c** *Salmonella* is the only distractor that produces $H_2S$. Also, *Klebsiella* and *Escherichia coli* produce acid/acid reactions in TSI.
[QCMLS 2021: 6.1.6.4.1] [Procop 2017, p234]

181. **d** *Salmonella* produce $H_2S$ in TSI and *Yersinia* produces an acid slant and acid butt. *Yersinia* is negative for glucose fermentation but does ferment sucrose, so that turns the TSI acid/acid.*Shigella* fits this biochemical profile.
[QCMLS 2021: 6.4.6.4.1] [Procop 2017, p234]

182. **c** These biochemicals are characteristic for *Escherichia coli*. *Klebsiella pneumoniae* is indole-negative and nonmotile. *Shigella dysenteriae* is nonmotile and *Enterobacter cloacae* is indole-negative.
[QCMLS 2021: 6.4.6.4.1] [Procop 2017, p234]

183. **a** *Enterobacteriaceae* ferment glucose and are oxidase-negative. *Plesiomonas* was a member of the *Vibrio* family in part because it is oxidase-positive. However, it was moved to the *Enterobacteriaceae* family despite its positive oxidase reaction.
[QCMLS 2021: 6.1.6.4.2] [Procop 2017, p461]

184. **d** Of the organisms listed only *Klebsiella pneumoniae* is nonmotile.
[QCMLS 2021: 6.1.6.4.1] [Procop 2017, p240]

185. **c** Of the organisms listed only *Pseudomonas aeruginosa* is oxidase-positive.
[QCMLS 2021: 6.1.6.4.1, 6.1.6.4.3] [Tille 2016, p341]

186. **d** Of the organisms listed only *Proteus mirabilis* is phenylalanine-deaminase-positive.
[QCMLS 2021: 6.1.6.4.1] [Procop 2017, p237]

187. **d** Of the organisms listed only *Serratia marcescens* is DNase-positive.
[QCMLS 2021: 6.1.6.4.1] [Procop 2017, p276]

188. **d** Quality control of indole requires both a positive and a negative control. *E. coli* and *E. cloacae* respectively produce a positive and negative reaction with indole.
[QCMLS 2021: 6.1.6.4.1] [Procop 2017, p62]

189. **a** *Haemophilus influenzae* requires X (hemin) and V (NAD) factors. Sheep blood agar streaked with *Staphylococcus aureus* supplies X factor, and V factor, so colonies grow around the *S. aureus* colonies.
[QCMLS 2021: 6.1.6.4.4] [Procop 2017, p484]

190. **b** Both organisms are gram-negative. *Neisseria gonorrhoeae* is fastidious and does not grow on MacConkey or EMB agar, but *Acinetobacter* does. *Neisseria* is oxidase-positive and *Acinetobacter* is oxidase-negative.
[QCMLS 2021: 6.1.6.4.3] [Procop 2017, p373, 382]

191. **d** *Pseudomonas aeruginosa* grows at 42°C, but this temperature is inhibitory for other *Pseudomonas* species.
[QCMLS 2021: 6.1.6.4.3] [Tille 2016, p341]



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192. **c** The porphyrin test is an alternative method for detecting heme-producing species of *Haemophilus*. It detects whether or not the organism converts the substrate delta-amino levulinic acid into porphyrins or porphobilinogen, which are intermediates in synthesis of Factor X.
[QCMLS 2021: 6.1.6.4.4]

193. **b** The violet pigment on sheep blood agar is a characteristic for the genus *Chromobacterium*. *Serratia* is oxidase-negative, and *Campylobacter* does not produce a pigment and does not grow on MacConkey agar.
[QCMLS 2021: 6.1.6.4.3] [Procop 2017, p276, 434]

194. **a** Both *Acinetobacter* and *Moraxella* display resistance to penicillin, *Acinetobacter* grows on MacConkey agar and some species of Moraxella can grow on MacConkey agar (*M. osloensis* and *M. nonliquefaciens*.) *Acinetobacter* are oxidase-negative, and *Moraxella* are oxidase-positive.
[QCMLS 2021: 6.1.6.4.3] [Procop 2017, p381-382]

195. **a** Growth at 42°C and pyocyanin production are classic tests for the identification of *Pseudomonas aeruginosa*. Gelatin hydrolysis separates *Pseudomonas putida* (negative) from *Pseudomonas fluorescence* (positive).
[QCMLS 2021: 6.1.6.4.3]

196. **b** *Capnocytophaga* require $CO_2$ environment, ferment glucose, sucrose, and lactose, and are gram-negative bacilli. *Capnocytophaga* produce characteristic spreading colonies with finger-like projections.
[QCMLS 2021: 6.1.6.4.3] [Procop 2017, p499]

197. **d** Both organisms are gram-negative bacilli and grow on MacConkey agar. Neither ferments glucose. *Stenotrophomonas* is oxidase-negative, while most other nonfermenters are oxidase-positive.
[QCMLS 2021: 6.1.6.4.3] [Procop 2017, p390]

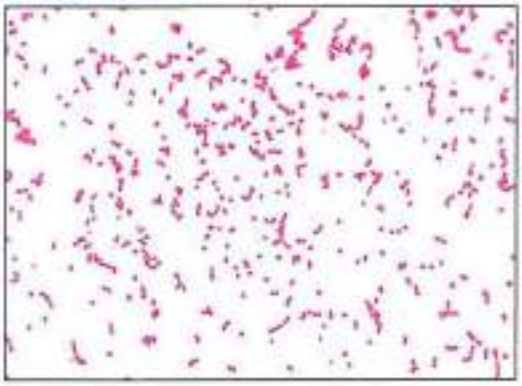
198. **c** The HACEK group of organisms are gram-negative bacilli that require increased $CO_2$ for growth. They are commonly associated with endocarditis, and include *Haemophilus* species (especially *H. aphrophilus*), *Aggregatibacter* (formerly *Actinobacillus*) *actinomycetemcomitans*, *Cardiobacterium hominis*, *Eikenella corrodens*, and *Kingella* species.
[QCMLS 2021: 6.1.6.4.4] [Tille 2016, p874]

199. **c** The organisms which belong to the *Enterobacteriaceae* group are oxidase-negative, reduce nitrate to nitrite, ferment glucose, and are capable of growing on a MacConkey agar.
[QCMLS 2021: 6.1.6.4.3, 6.1.6.4.1] [Procop 2017, p214]

200. **b** There are key reactions for *Enterobacteriaceae* that are important to remember. Phenylalanine deaminase is a biochemical that is helpful in initialing differentiation *Proteus*, *Providencia*, and *Morganella* from other *Enterobacteriaceae*.
[QCMLS 2021: 6.1.6.4.1] [Procop 2017, p235]



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201. **d** All species of *Acinetobacter* are oxidase-negative, nonmotile, and grow on MacConkey agar. The Gram stain of *Acinetobacter* is of coccobacilli that will elongate to gram-negative bacilli when placed in thioglycolate broth. *Pseudomonas aeruginosa* is oxidase-positive and motile while *Stenotrophomonas maltophilia* is motile. *Proteus mirabilis* is part of the *Enterobacteriaceae* family.
[QCMLS 2021: 6.1.6.4.3] [Tille 2016, p331-332]

202. **d** Since all *Neisseria* species are oxidase-positive, they possess the enzyme to oxidize tetramethyl-phenylenediamine.
[QCMLS 2021: 6.1.6.2] [Mahon 2019, p183, 372]

203. **c** Both *Neisseria meningitidis* and *N. lactamica* produce acid from maltose and grow on modified Thayer martin agar. *N. lactamica* utilizes lactose, *N. meningitidis* does not. Lactose positivity in *N. lactamica* can be delayed or absent so ONPG test will detect lactose utilization (30 minute test).
[QCMLS 2021: 6.1.6.2] [Mahon 2019, p183, 375]

204. **c** Of the choices provided, only utilization of carbohydrates provides definitive identification of *Neisseria gonorrhoeae*.
[QCMLS 2021: 6.1.6.2] [Mahon 2019, p375]

205. **b** *Neisseria gonorrhoeae* is oxidase-positive and utilizes glucose but not maltose.
[QCMLS 2021: 6.1.6.2] [Mahon 2019, p372, 375]

206. **a** *Moraxella catarrhalis* looks very similar to *Neisseria* species in the Gram stain, and it is also nonmotile, oxidase-positive and catalase-positive. Unlike *Neisseria*, it is DNase-positive and displays the "hockey puck" colonies that stay intact when pushed across the plate with a loop.
[QCMLS 2021: 6.1.6.2] [Mahon 2019, p375, 379]



207. **d** Media containing potassium tellurite (Tinsdale agar) is used for the recovery of *Corynebacterium diphtheriae*. The organism reduces potassium tellurite, giving black colonies on the media. The Gram stain morphology is consistent with *Corynebacteria*.
[QCMLS 2021: 6.1.6.3] [Mahon 2019, p347]

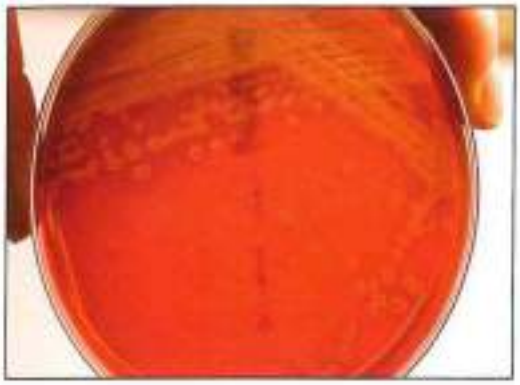



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208. **b** *Corynebacterium* and *Listeria* are catalase-positive and gram-positive bacilli. *Listeria* demonstrates "tumbling" motility that is best demonstrated following growth at 25°C. A few species of *Corynebacterium* species are motile when grown at 35°C.
[QCMLS 2021: 6.1.6.3] [Mahon 2019, p353-355]

209. **d** *Erysipelothrix rhusopathiae* are the only gram-positive bacilli that produce hydrogen sulfide when inoculated into triple sugar iron agar.
[QCMLS 2021: 6.1.6.3] [Mahon 2019, p354-355]

210. **d** *Listeria* may be confused with some streptococci because *Listeria* is beta-hemolytic and is capable of hydrolyzing esculin.
[QCMLS 2021: 6.1.6.3] [Mahon 2019, p355]



211. **d** *Listeria monocytogenes* is the only choice that causes infections in the noted groups. In pregnant women, it is usually seen during the 3rd trimester and causes spontaneous abortion and stillborn neonates. In newborns, it has fatality rates of up to 50%, where it causes sepsis and meningitis. *Listeria* is responsible for multistate outbreaks of contaminated foods, such as cheese or deli meats.
[QCMLS 2021: 6.1.6.3] [Mahon 2019, p351-352]

212. **a** Kanamycin-vancomycin laked blood agar is a selective medium used for the isolation of *Bacteroides* or *Prevotella*.
[QCMLS 2021: 6.1.6.5.1] [Mahon 2019, p498]

213. **a** Production of lecithinase and demonstration of double zone hemolysis are tests and characteristics used for the identification of *Clostridium perfringens*.
[QCMLS 2021: 6.1.6.5.4] [Mahon 2019, p506-507]

214. **c** *Prevotella* and *Porphyromonas* (as well as *Veillonella parvula*) produce colonies that fluoresce brick red.
[QCMLS 2021: 6.1.6.5.6] [Mahon 2019, p506-507]

215. **b** Bile tolerant *Bacteroides* fragilis will grow on agar with 20% bile (BBE agar), while *Fusobacterium*, *Prevotella*, and *Porphyromonas* are sensitive to 20% bile and will not grow on BBE agar. This reinforces the usefulness of bile esculin agar for differentiation of anaerobic gram-negative bacilli.
[QCMLS 2021: 6.1.6.5.6] [Mahon 2019, p508]

216. **a** MALDI-TOF is not a molecular application. It is a mass spectrometry platform. The other choices are molecular based, and and pulsed-field gel electrophoresis relies on a fluctuating electric field and not a laser pulse.
[QCMLS 2021: 6.1.6] [Mahon 2019, p243]

217. **c** *Clostridium perfringens* is lecithinase-positive. *Bacteroides*, *Fusobacterium* and *Clostridium sporogenes* are lecithinase-negative.
[QCMLS 2021: 6.1.6.5.4] [Mahon 2019, p509, 511]

218. **b** *Enterococcus* species are relatively resistant to beta-lactam agents and aminoglycosides. Combination therapy with a beta-lactam agent or vancomycin and an aminoglycoside provide a synergistic combination to effectively treat enterococcal infections.
[QCMLS 2021: 6.1.6.1.2.3]

219. **b** The MIC is a basic laboratory measurement of the activity of an antibiotic against an organism. It is the lowest concentration of antibiotic that inhibits visible growth of the organism. It does not represent the concentration of antibiotic that is lethal to the organism.
[QCMLS 2021: 6.3.3] [Mahon 2019, p270]

220. **b** Antimicrobial resistance in *Neisseria gonorrhoeae* is widespread. One mechanism of penicillin resistance is the production of beta-lactamase (penicillinase) that breaks open the beta lactam ring of penicillin, destroying its activity. Thus, *N. gonorrhoeae* that produce beta lactamase are resistant to penicillin. Detection of the other listed enzymes can be used to identify other gram-negative diplococci.
[QCMLS 2021: 6.3.3] [Mahon 2019, p282-283]

221. **d** The chromogenic cephalosporin test using nitrocefin is the most sensitive and specific test for detection of beta-lactamase. Acidimetric tests employing penicillin are less expensive, but not as sensitive, as the nitrocefin assay.
[QCMLS 2021: 6.3.3] [Procop 2017, p1119]

222. **a** Systemic enterococcal infections, such as endocarditis, are commonly treated with a cell-wall-active agent and an aminoglycoside. These agents act synergistically to kill the organism. If the organism is resistant to one or both, there is no synergy, and the combination will fail. It is important to detect aminoglycoside and beta-lactam resistance in these cases. Enterococci have intrinsic moderate level resistance to aminoglycosides. Acquired resistance corresponds to very high MICs (greater than 500 µg/mL) for gentamicin and is termed high level resistance.
[QCMLS 2021: 6.1.6.1.2.3] [Mahon 2019, p285, 286]

223. **d** Cefoxitin is used as a surrogate for mecA-mediated oxacillin resistance in *Staphylococcus aureus*. *S. aureus* with cefoxitin MICs >4 µg/mL are considered oxacillin resistant.
[QCMLS 2021: 6.3.3] [CLSI 2017, p56, 59, 132]

224. **c** All Mueller-Hinton agar used for disk diffusion susceptibility testing should be poured to a depth of 3-5 mm. If the depth of the media is <3 mm, this may be associated with excessively large zones and false-positive susceptibility results. Agar that is >5 mm in depth may cause excessively small zone sizes.
[QCMLS 2021: 6.3.3] [Mahon 2019, p278; CLSI 2017, p159]

225. **b** The amount of antibiotic used in disk diffusion susceptibility testing is standardized and constant. Once the disk is placed on the inoculated plate and makes contact with the agar, the antibiotic in the disk begins to diffuse out. As it diffuses into the media, the concentration of antibiotic gets lower the further it diffuses from the disk.
[QCMLS 2021: 6.3.3]

226. **b** Antibiotic disks should be placed on the agar within 15 minutes of organism inoculation. Extending the time to place the disks allows the organisms to get a head start on growing, which can cause decreased zone sizes.
[QCMLS 2021: 6.3.3]

227. **b** To ensure the reproducibility of disk diffusion testing, the inoculum must be standardized. If the inoculum is too concentrated (too many organisms), zone sizes would be smaller than expected and appear falsely resistant. If the antibiotic disk is too concentrated, the antibiotic inhibitory activity will be further from the disk, creating increased zone sizes. Incubation of less than the standard16-18 hours and using plates with agar that is too thin will cause zone sizes to be larger than expected.
[QCMLS 2021: 6.3.3] [CLSI 2017, p62]


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**228.** **c** Cefazolin is a first-generation cephalosporin; cefuroxime and cefonicid are second-generation cephalosporins; and ceftriaxone is a third generation cephalosporin. CLSI guidelines recommend cefazolin be tested against the *Enterobacteriaceae* to represent the first generation cephalosporins.
[QCMLS 2021: 6.3.1] [CLSI 2017, p18]

**229.** **c** Penicillin inhibits penicillin binding proteins that are essential to peptidoglycan (cell wall) synthesis. Penicillin is a beta-lactam antibiotic. All beta-lactam antibiotics inhibit cell wall synthesis. Chloramphenicol inhibits protein synthesis, colistin increases cell membrane permeability, and sulfamethoxazole inhibits folate metabolism.
[QCMLS 2021: 6.3.1] [Mahon 2019, p250-251]

**230.** **c** As many as 20-40% of *Haemophilus influenzae* produce beta-lactamases. Detection of these enzymes should be performed on any isolate considered to be a pathogen using the chromogenic cephalosporin (nitrocefin) test.
[QCMLS 2021: 6.1.6.4.4, 6.3.3] [Mahon 2019, p392]




**231.** **d** When reading a broth microdilution susceptibility test, growth in each well is determined by comparison with the growth control well and indicated by turbidity. The well with the lowest concentration of antibiotic displaying no growth is read as the minimum inhibitory concentration (MIC).
[QCMLS 2021: 6.3.3] [Mahon 2019, p275]

**232.** **d** Deterioration of the antimicrobial agent in the disk will cause the zone sizes to be too small (falsely resistant). Standardization of the inoculum turbidity to less than a 0.5 McFarland standard would result in an inoculum that is too light and resulting zone sizes that are too large. Incubation of the plates at 35°C and inoculating plates within 10 minutes of preparation would not have an adverse effect on zone sizes.
[QCMLS 2021: 6.3.3] [Procop 2017, p1108]

**233.** **b** The zone size observed has no meaning in and of itself. Interpretive standards are derived from a correlation between zone sizes and minimum inhibitory concentrations. Usually a large number of organisms from a given species or group (eg, *Enterobacteriaceae*) are tested.
[QCMLS 2021: 6.3.3] [Procop 2017, p1109]



**234.** **b** The image displays a negative D test result. There is no flattening of the zone of inhibition around the clindamycin disk adjacent to the erythromycin disk. Thus, there is no inducible clindamycin resistance, and the isolate is reported as clindamycin-susceptible, while the erythromycin is reported as resistant.
[QCMLS 2021: 6.3.3] [CLSI 2017, p136-137]

235. **c** Extended spectrum beta-lactamases (ESBL) are inhibited by clavulanic acid. Confirmatory tests of the presence of ESBL are based on the enhanced activity of a beta-lactam antibiotic, usually cefotaxime or ceftazidime, when it is tested with clavulanic acid compared to the activity of the beta-lactam tested alone.
[QCMLS 2021: 6.3.3] [CLSI 2017, p102-103]

236. **d** Oxacillin resistant staphylococci are resistant to all beta-lactam agents (with the exception of the cephalosporins with anti MRSA activity), beta-lactam/beta-lactamase inhibitor combinations and carbapenems. Results for these antibiotics should be reported as resistant or should not be reported.
[QCMLS 2021: 6.3.3] [CLSI 2017, p57]

237. **c** Quality control zone sizes that are too small could indicate that the organism inoculum is too high, plates were poured too thick, or that the potency of the antibiotic disks is too low.
[QCMLS 2021: 6.3.3] [CLSI 2017, p159]

238. **b** *Klebsiella pneumoniae* is usually resistant to ampicillin. This is an unusual antibiogram and should be verified.
[QCMLS 2021: 6.3.2] [Mahon 2019, p295]

239. **a** Organisms producing carbapenemase are typically resistant to carbapenems (eg, imipenem) and also penicillins (eg, amoxicillin), cephalosporins (eg, cefazolin), fluoroquinolones (eg, ciprofloxacin), and sometimes aminoglycosides (eg, gentamicin).
[QCMLS 2021: 6.3.3] [Procop 2017, p270-271]

240. **a** Beta-lactam antibiotics inhibit cell wall synthesis. Beta-lactam antibiotic examples include: penicillins (eg, ampicillin), cephalosporins (eg, cefazolin), and carbapenems (eg, imipenem).
[QCMLS 2021: 6.3.1] [Mahon 2019, p250]

241. **d** The beta-lactamase enzyme produced by *Haemophilus influenzae* inactivates the antibiotics that have a beta-lactam ring in their structure, such as penicillins and cephalosporins.
[QCMLS 2021: 6.3.3] [Tille 2016, p407, 408]

242. **b** *Neisseria meningitidis* is rarely resistant to penicillin, which can be beta-lactamase mediated in some isolates. The primary mechanism of penicillin resistance in *N. gonorrhoeae* is beta-lactamase. *Treponema pallidum* is sensitive to penicillin, while treatment for *Chlamydia* is doxycycline and azithromycin.
[QCMLS 2021: 6.3.3]

### *L. MRSA/MSSA, VRE, ESBL/CRE Screening*

243. **a** Agar dilution screens do not reliably detect all vancomycin-intermediate (VISA) *Staphylococcus aureus* isolates. MIC tests should be performed to determine the susceptibility of all isolates of staphylococci to vancomycin.
[QCMLS 2021: 6.3.3] [CLSI 2017, p61, 134]

### *M. BSL-3 Pathogens & Select Agents (Bioterrorism)*

244. **b** Category A agents of bioterrorism potentially have the greatest impact on public health for transmission, morbidity and mortality. *Francisella tularensis* and *Yersinia pestis* are Category A, as well as *Bacillus anthracis*, and variola major virus (smallpox). MDR tuberculosis and Hanta virus are Category C agents. VRE and *Bacillus cereus* are not considered agents of bioterrorism.
[QCMLS 2021: 6.1.5] [Mahon 2019, p725, 727]

245. **d** BSL-3 organisms produce serious disease and are transmitted by respiratory route. BSL-2 organisms are not readily transmitted; BSL-1 organisms don't cause human disease; BSL-4 organisms cause serious disease for which there is neither treatment nor vaccine.
[QCMLS 2021: 6.1.5] [Mahon 2019, p728]

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## III. Procedures For Mycology, Mycobacteriology, Parasitology & Virology

### A. Mycobacteriology & Nocardia species

**246. c** *Nocardia asteroides* are partially acid fast, and do not hydrolyze the substrates casein, tyrosine, or xanthine. The most reliable method for ID is 16S rRNA gene sequencing.
[QCMLS 2021: 6.1.6.3] [Mahon 2019, p358-359]

**247. d** Many species of mycobacteria are slow growing. The average time of recovery on egg-based media, such as Lowenstein-Jensen agar, is about 21 days but can be as long as 60 days.
[QCMLS 2021: 6.2.1.1, 6.2.1.3] [Procop 2017, p1232]

**248. d** Due to slow growth rate of most mycobacteria, and to prevent undesirable bacteria overgrowth, specimens contaminated by normal flora must be decontaminated prior to inoculation of media; NaOH is the most common decontaminant.
[QCMLS 2021: 6.2.1.1] [Procop 2017, p1223]

**249. a** Sabouraud dextrose is fungal media; mycobacterial media should have been inoculated.
[QCMLS 2021: 6.2.1.3] [Mahon 2019, p609]

**250. d** Since the quality control of media, reagents and stains is in range, the digestion and decontamination procedure should be reviewed. The concentration of sodium hydroxide may be too strong or the decontamination time too long, which can result in decreased recovery of AFB and lack of contaminated cultures. Exposure of specimens to strong NAOH or other decontamination agents needs to be carefully timed to prevent chemical injury to mycobacteria.
[QCMLS 2021: 6.2.1] [Procop 2017, p1223]

**251. a** A widely used digestion-decontamination method is the N-acetyl-L-cysteine-2% NaOH method. The N-acetyl-L-cystine (NALC) acts as a mucolytic agent in combination with the NaOH, which inhibits contaminant growth.
[QCMLS 2021: 6.2.1.1] [Mahon 2019, p568]

**252. d** N-acetyl-L-cysteine (NALC) is a mucolytic agent; the concentration of NALC may be increased to digest thick, mucoid specimens.
[QCMLS 2021: 6.2.1.1] [Mahon 2019, p568]

**253. d** If exposed to light, agar-based media such as Middlebrook 7H10 and 7H11 may release formaldehyde, which is toxic to mycobacteria.
[QCMLS 2021: 6.2.1.3] [Mahon 2019, p570]

**254. d** The oxalic acid method is superior to alkali methods for processing specimens contaminated with *Pseudomonas*.
[QCMLS 2021:] [Mahon 2019, p568]

**255. b** Liquid media is recommended to provide more rapid recovery of AFB than solid media. Isolation rates of mycobacteria is also higher with liquid media. PANTA is an antibiotic mixture added to some broth culture systems to inhibit the growth of contaminating bacteria. A 5-10% $CO_2$ environment is recommended for mycobacteria isolation.
[QCMLS 2021: 6.2.1.3] [Procop 2017, p1229-1230]

**256. d** Because inhalation is the primary route of mycobacterial infection, specimens for mycobacterial culture must be processed in a biological safety cabinet to minimize aerosol transmission of the organism. After specimen processing and cleaning surfaces with a disinfectant, an ultraviolet light is used to kill any remaining bacteria within the safety cabinet.
[QCMLS 2021: 6.1.5] [Mahon 2019, p566]

**257. b** The Ziehl-Neelsen and Kinyoun differential staining procedures use carbol fuchsin as the primary stain; mycobacteria cells pick up the red carbol fuchsin stain. Methylene blue is the counterstain.
[QCMLS 2021: 6.2.1.2] [Procop 2017, p1226]

258. **d** A positive niacin reaction differentiates *Mycobacterium tuberculosis* from *M. avium* complex, *M. kansasii* and *M. fortuitum*.
[QCMLS 2021: 6.2.1.5] [Procop 2017, p1234]

259. **d** Growth rate, colony morphology, niacin production, and nitrate reduction differentiate *Mycobacterium tuberculosis* from other mycobacteria.
[QCMLS 2021: 6.2.1.5] [Procop 2017, p1237-1239]

260. **c** *Mycobacterium tuberculosis* can remain viable but dormant in macrophages for many years.
[QCMLS 2021: 6.2.1.5] [Mahon 2019, p256]

261. **a** BCG strains of *Mycobacterium bovis* have been used as a vaccine in highly endemic areas of the world.
[QCMLS 2021: 6.2.1.5] [Procop 2017, p1240]

262. **b** Commercial molecular tests are available that identify *Mycobacterium tuberculosis* complex directly from acid-fast smear positive respiratory specimens.
[QCMLS 2021: 6.2.1.5] [Mahon 2019, p576]

263. **d** Primary drugs for *Mycobacterium tuberculosis* treatment include isoniazid, rifampin, pyrazinamide, streptomycin and ethambutol.
[QCMLS 2021: 6.2.1.5] [Mahon 2019, p558]

264. **d** *Mycobacterium scrofulaceum* is a scotochromogen, which produces carotene pigment when grown in the light or dark.
[QCMLS 2021: 6.2.1.5] [Mahon 2019, p560-561]

265. **d** Scotochromogens produce deep yellow or orange pigment when grown in either light or darkness.
[QCMLS 2021: 6.2.1.5] [Mahon 2019, p559]

266. **b** Photochromogens produce nonpigmented colonies only when grown in darkness, but pigmented colonies after exposure to light.
[QCMLS 2021: 6.2.1.5] [Mahon 2019, p559]

267. **b** *Mycobacterium leprae* cannot be cultured *in vitro*. A clinical diagnosis is made, supported by the presence of AFB in a biopsy specimen.
[QCMLS 2021: 6.2.1.7] [Mahon 2019, p652-563]

268. **c** Media commonly used to isolate *Mycobacterium tuberculosis* include Lowenstein-Jensen agar, and Middlebrook 7H10 and 7H11 agars. PANTA is an antibiotic mixture added to some broth culture systems to inhibit contaminant growth.
[QCMLS 2021: 6.2.1.5] [Procop 2017, p1228-1230]

269. **b** *Mycobacterium marinum* causes infections as a result of trauma to the skin and exposure to contaminated fish tanks or salt water. Its optimal growth temperature is 28-30°C.
[QCMLS 2021: 6.2.1.7] [Mahon 2019, p560]

270. **d** *Mycobacterium avium* and *M. intracellulare* have the same biochemical pattern, but can be differentiated by the use of molecular diagnostic tests.
[QCMLS 2021: 6.2.1.7] [Mahon 2019, p559]

271. **b** *Mycobacterium fortuitum* is a rapid growing mycobacteria that can grow on routine sheep blood agar within 48 hours and is acid-fast stain positive. Rapid growing mycobacteria are associated with skin and subcutaneous tissue infections. *Corynebacterium jeikeium* will stain well (with Gram stain) and appear as diphtheroids. *M. tuberculosis* is a slow growing mycobacteria and will not grow on SBA at 48 hours. *M. pneumoniae* does not have a cell wall and would not grow on SBA.
[QCMLS 2021: 6.2.1.5] [Procop 2017, p1249-1250]

272. **d** One potential drawback of using fluorochrome stains is the lower sensitivity in detecting rapid growing mycobacteria, such as *Mycobacterium fortuitum*, in direct specimen stains.
[QCMLS 2021: 6.2.1.2] [Procop 2017, p1225-1227]

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273. **b** Use of fluorescence stains, such as auramine O, facilitates the screening of smears for mycobacteria, particularly when a 25x objective is used to scan a direct specimen smear. This objective provides the ability to scan wide microscopic fields while able to view the yellow-fluorescing bacterial cells. One issue is the fluorescent stains are less sensitive in detecting rapidly growing mycobacteria when compared to a stain using carbol fuchsin.
[QCMLS 2021: 6.2.1.2] [Procop 2017, p24-25, 1225-1227]

274. **a** Antimicrobial susceptibility testing on MTB isolates is routinely preformed to detect multidrug-resistant strains and to select appropriate antimicrobial treatment.
[QCMLS 2021: 6.2.1.5] [Procop 2017, p1257]

275. **a** *Mycobacterium tuberculosis, M. africanum, M. bovis* and *M. microti* are genotypically related and are considered members of the MTBC. These organisms can be differentiated by phenotypic characteristics, including colony morphology, niacin and nitrate reactions.
[QCMLS 2021: 6.2.1.5] [Procop 2017, p1237]

276. **c** The digestion process liquefies mucus and the decontamination process kills rapidly growing bacteria that can overgrow slow growing mycobacteria. Specimens that contain mucin and rapidly growing bacteria, such as respiratory specimens, should undergo digestion and decontamination for optimal mycobacteria recovery.
[QCMLS 2021: 6.2.1.1] [Mahon 2019, p568]

277. **c** The use of a solid-based medium in combination with a liquid-based medium is recommended for routine mycobacteria recovery. Most mycobacteria grow more rapidly in liquid medium.
[QCMLS 2021: 6.2.1.3] [Mahon 2019, p569]

## *B. Virology*

278. **a** Many viruses produce changes in infected cells that is called cytopathic effect. Viruses cannot be visualized but CPE can be detected in specific cell lines. Most laboratories now use immunologic and molecular methods for ID of viruses, rather than the CPE.
[QCMLS 2021: 6.6.2] [Mahon 2019, p587-588]

279. **a** Respiratory syncytial virus (RSV) infects the ciliated respiratory epithelium of the upper respiratory tract. A nasopharyngeal swab or aspirate is the optimal specimen for RSV recovery.
[QCMLS 2021: 6.6.11.4] [Mahon 2019, p707]

280. **b** Human papilloma virus (HPV) infects epithelial tissues throughout the body, including skin, larynx, and anogenital tissue. Persistent infection with oncogenic types of HPV and integration of HPV DNA into the cellular genome is a pathway leading to HPV-induced neoplasia, such as cervical cancer.
[QCMLS 2021: 6.6.7]

281. **b** Herpes simplex virus is the most common cause of fatal sporadic encephalitis in the United States.
[QCMLS 2021: 6.6.4.1]

282. **b** Adenovirus infections are common. It causes up to 5% of all respiratory infections, and the prevalence of infection is higher (up to 14%).
[QCMLS 2021: 6.6.5]

283. **c** The classic clinical syndrome associated with Epstein-Barr virus (EBV) infection is infectious mononucleosis. However, in immunocompromised patients, EBV is associated with posttransplant lymphoproliferative disorders, and malignancies such as Burkitt lymphoma.
[QCMLS 2021: 6.6.4.4]

**284. d** Rotavirus is the most common cause of viral gastroenteritis in infants and children. It causes >600,000 deaths annually. Coronavirus causes cold-like infections in adults and a small percentage of pediatric diarrhea cases. Adenovirus is mostly asymptomatic, causing a small percentage of pneumonia and gastroenteritis in children. Norwalk virus is the most common cause of adult viral gastroenteritis and is often responsible for the diarrheal outbreaks on cruise ships, resorts, or nursing homes.
[QCMLS 2021: 6.7.2.2, 6.6.17] [Mahon 2019, p690, 698, 700]

**285. a** Deer mouse: *Peromyscus maniculatus*, or the deer mouse, spreads the virus in its infected feces. People living in close contact with animals are at a particularly high risk. Infected individuals develop a pulmonary syndrome.
[QCMLS 2021: 6.6.13] [Mahon 2019, p699-700]

**286. d** Coronavirus: SARS is a severe respiratory disease spread from human to human and caused by a coronavirus (SARS CoV).
[QCMLS 2021: 6.6.14] [Mahon 2019, p701]

**287. d** The host organism is the bird, which passes on the virus through a mosquito vector to humans.
[QCMLS 2021: 6.6.16] [Mahon 2019, p703]

**288. b** Coxsackie A & B are frequently confused for each other in terms of the diseases that they cause. Coxsackie A causes hand, foot, and mouth disease, so named for the most common locations of the lesions. Coxsackie B is responsible for the majority of cases of viral myocarditis.
[QCMLS 2021: 6.6.12.1] [Mahon 2019, p708]

**289. b** CMV: Passed transplacentally, the risk of *in utero* infection is greatest when the mother acquires a primary CMV infection while pregnant. The effects from *in utero* infection can range from severe to mild.
[QCMLS 2021: 6.6.4.3] [Mahon 2019, p692]

**290. b** All the other options are associated with EBV infection; shingles is the only option that is caused by VZV.
[QCMLS 2021: 6.6.4.2] [Mahon 2019, p815]

**291. a** Especially in the immune-compromised patient HHV8 or KSHV is associated with a number of disease processes, also including primary effusion lymphoma. The etiologic agent in these cases can be identified by PCR, IHC, or serological means.
[QCMLS 2021: 6.6.4.7] [Mahon 2019, p695-696]

**292. c** Anti HBs in serum indicates convalescence or immune status. All of the other choices, with perhaps the early appearance post vaccination of HBsAg, should not be seen in a vaccinated person. HBeAg is present during active viral replication. Anti HBc shows early in the course of disease and shows an acute infection.
[QCMLS 2021: 6.6.9.2] [Mahon 2019, p714-715]

**293. b** The persistence of surface antigen is an excellent indicator of chronicity. With acute infection and resolution surface antigen peaks 2-3 months post-infection and before symptoms, but is completely gone before 6 months, to be replaced by anti surface antibody (HBsAb) and the clinical symptoms of acute hepatitis. In chronic infection the surface antibody does not appear, while the surface antigen persists.
[QCMLS 2021: 6.6.9.2]

**294. a** Drift refers to the minor changes that occur due to point mutations. Drift is primarily responsible for the annual outbreaks and epidemics of influenza. Shift refers to a more major change due to recombination between different viral subtypes and is most often the cause of the periodic pandemic infections. It is due to antigenic drift and shift that lifelong immunity to influenza cannot currently be achieved.
[QCMLS 2021: 6.6.10.1]

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**295. d** RSV is the most common cause of severe lower respiratory tract disease in infants and young children. Persistence of immunity does not often occur so reinfection is common.
[QCMLS 2021: 6.6.11.4] [Mahon 2019, p707]

**296. a** The extremely high sensitivity of the serum ELISA test makes it the screening test of choice. There can be a window period between the infection and seroconversion, usually 6-8 weeks, leading to possible false-negative results. The confirmation of a positive result is usually done with western blot, which detects antibodies to specific antigens (eg, p24, p31, gp41 and gp120/160) are positive.
[QCMLS 2021: 6.6.1.9] [Mahon 2019, p709-711]

**297. b** Quantitative HIV RNA is an assessment of viral load. Quantitation of RNA levels is an excellent prognostic tool to predict progression, especially long term.
[QCMLS 2021: 6.6.20] [Mahon 2019, p711]

**298. c** It is an enveloped DNA virus that is transmitted through sexual, perinatal, and parenteral routes. The virus enters the host and travels from the blood to the liver and infects hepatocytes.
[QCMLS 2021: 6.6.9.2] [Mahon 2019, p713]

### *C. Parasitology*

**299. d** Pollen grains are common artifacts in stool specimens submitted for ova and parasite examination. Their appearance is similar, in particular, to protozoan cysts.
[QCMLS 2021: 6.4.1.1] [Ash & Orihel 2020, p588-589]

**300. c** Polyvinyl alcohol is an adhesive and is used in the preparation of smears for stains, such as trichrome.
[QCMLS 2021: 6.4.1.1] [Mahon 2019, p618]

**301. c** The recommended technique for culturing *Acanthamoeba* is the use of non-nutrient agar seeded with a lawn of enteric gram-negative bacilli (*Escherichia coli*). Specimens with suspected *Acanthamoeba* are inoculated onto a freshly inoculated lawn of *E. coli*, incubated and observed for 7 days. Identification is based on the characteristic patterns of locomotion and morphologic features of the trophic and cystic forms. The amoeba uses *E. coli* as a nutritional source, and in 1-2 days trails form on the agar surface as amoeba move.
[QCMLS 2021: 6.4.2.1] [Mahon 2019, p663]

**302. d** *Naegleria fowleri* is the etiologic agent of primary amoebic encephalitis.
[QCMLS 2021: 6.4.2.1] [Mahon 2019, p631]

**303. d** Formed stool is unlikely to contain trophozoites so direct examination of the stool is not necessary. The stool should be preserved as soon as possible to maintain any cysts, ova, or larvae that may be present in the specimen.
[QCMLS 2021: 6.4.1.1] [Mahon 2019, p619]

**304. b** The increased amount of blood placed on the slide of a thick smear for blood parasites increases likelihood to detect parasite.
[QCMLS 2021: 6.4.3.3] [Mahon 2019, p645]

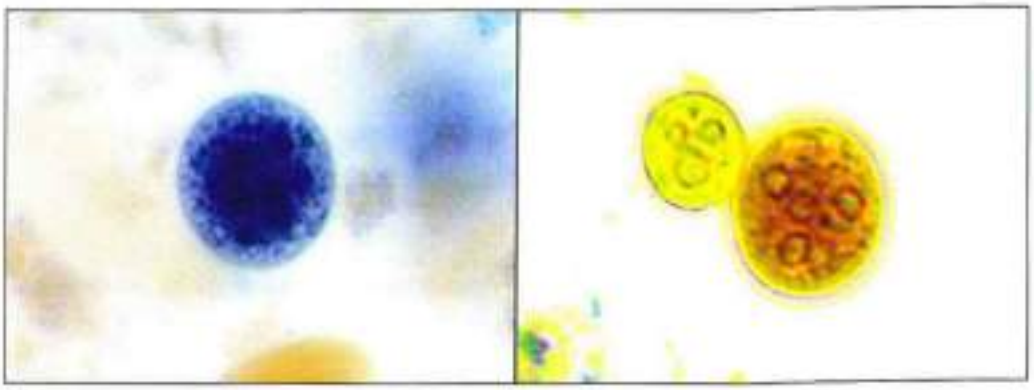
**305. c** *Cyclospora* and *Cryptosporidium* are found in feces and associated with diarrhea. *Trypanosoma cruzi* is a hemoflagellate introduced into the human host by the reduviid bug and causes Chagas disease or heart disease. *Toxoplasma* is the only choice that is found in cat feces, and infection often occurs when undercooked meat with tissue cysts is ingested.
[QCMLS 2021: 6.4.3.1] [Mahon 2019, p649-651]

**306. c** The patient's history is suggestive of *Babesia* infection. *Babesia microti* ring forms are similar to *Plasmodium falciparum*. A travel history is helpful in determining the cause of infection.
[QCMLS 2021: 6.4.3.4] [Mahon 2019, p647-648]

307. **d** Malarial parasites, especially *Plasmodium* species, commonly appear on a Wright-stained peripheral blood slide as small blue rings with a red nucleus or chromatin dot.
[QCMLS 2021: 4.10.4.2, t4.18] [Mahon 2019, p626]

308. **b** The trophozoite of *Entamoeba histolytica* ranges in size from 12-60 µm, which is significantly larger than *Endolimax nana*. The nucleus of *E. histolytica* displays evenly distributed peripheral chromatin unlike *Entamoeba coli*, which has coarse peripheral chromatin and *Iodamoeba bütschlii*, which has none.
[QCMLS 2021: 6.4.2.1] [Mahon 2019, p625-626]

***Please consult the images below for questions 309-310 (left) and questions 311-312 (right) regarding Entamoeba.***



309. **d** The image shows a cyst with more than 4 nuclei, an indication of *Entamoeba coli*. Trophozoites of *Entamoeba coli* demonstrate slow and undefined motility. They do not ingest RBCs.
[QCMLS 2021: 6.4.2.1] [Mahon 2019, p626, 628]

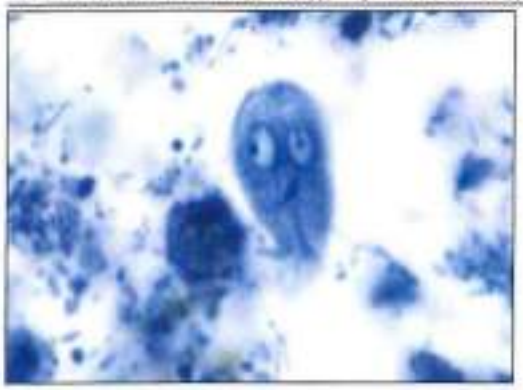
310. **b** *Entamoeba coli* is a nonpathogenic protozoan. Its presence indicates the ingestion of fecally contaminated food or water and should lead to a closer review of the specimen for pathogenic parasites or the collection of additional specimens.
[QCMLS 2021: 6.4.2.1] [Mahon 2019, p626]

311. **a** The cysts of *Entamoeba coli* and *E. histolytica* may appear similar to the unexperienced technologist. *E. coli* cysts are larger (right) and contain more than 4 nuclei in each cyst (up to 8 nuclei may be seen).
[QCMLS 2021: 6.4.2.1] [Mahon 2019, p628]

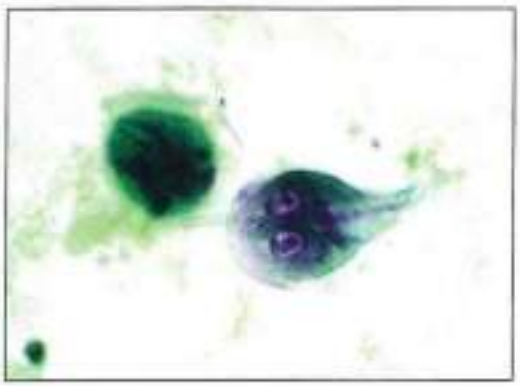
312. **a** The image displays the cyst form of *Entamoeba coli*, a nonpathogenic parasite, next to the smaller cyst of the pathogenic *E. histoytica*, the trophozoites of which may ingest RBCs.
[QCMLS 2021: 6.4.2.1] [Mahon 2019, p635]



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313. **b** The image displays the trophozoite form of *Giardia duodenalis* (syn *lamblia*), a protozoan flagellate parasite.
[QCMLS 2021: 6.4.2.2.1]



314. **a** Diarrhea associated with *Giardia duodenalis* (syn *lamblia*) is caused by the consumption of contaminated water and results in greasy, foul-smelling stools. The image shows both the trophozoite and cyst forms of *Giardia*.
[QCMLS 2021: 6.4.2.2.1] [Mahon 2019, p617]

315. **a** Protozoan trophozoites are fragile and begin to disintegrate as soon as they are passed. Liquid stool specimens should be preserved within 30 minutes of passage in order to adequately preserve parasite morphology. If a liquid specimen cannot be properly preserved, another specimen should be collected.
[QCMLS 2021: 6.4.1.1] [Mahon 2019, p626-627]

316. **a** While *Entamoeba hartmanni* and *Endolimax nana* are a similar size, *E. hartmanni* has similar peripheral chromatin to *E. histolytica*, while *E. nana* has none. *E. histolytica* cysts are larger and round when compared to *E. nana*. *Chilomastix* is a flagellated protozoan, and has only 1 nucleus.
[QCMLS 2021: 6.4.2.1] [Mahon 2019, p819]

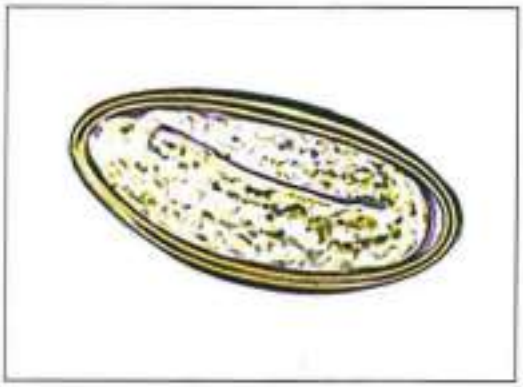
317. **d** *Strongyloides stercoralis* rhabditiform larvae are capable of transforming into filariform (infective) larvae in the intestines of immunocompromised patients. This establishes an autoinfective cycle.
[QCMLS 2021: 6.4.4.1] [Mahon 2019, p666-667]

318. **c** *Onchocerca volvulus* is the only microfilaria that is detected in the skin snips of patients with raised skin nodules. The microfilaria of *Wuchereria*, *Brugia* and *Loa loa* are found in the blood of infected patients.
[QCMLS 2021: 6.4.4.2]



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***Please consult the image below for questions 319-320 regarding Enterobius vermicularis.***



**319.** **c** The most common sign of *Enterobius vermicularis* infection is intense perianal itching.
[QCMLS 2021: 6.4.4.1] [Mahon 2019, p668]

**320.** **d** The ova of *Enterobius vermicularis* cannot be demonstrated in a routine ova and parasite examination. The adult female *Enterobius* worm migrates out of the anus and lays her eggs in the perianal folds. A scotch tape preparation of the skin of the perianal folds is used to collect ova.
[QCMLS 2021: 6.4.4.2] [Mahon 2019, p666]

**321.** **c** The diagnostic stages of *Strongyloides*, *Entamoeba* and *Ancylostoma* can be detected in the stool of infected patients. The diagnostic stage of *Echinococcus granulosus* is not detected in an infected patient's stool. Aspiration of the cyst in the liver or lung, for example, will show hydatid sand.
[QCMLS 2021: 6.4.4.5] [Mahon 2019, p661, 663]

**322.** **a** Humans may become infected with *Taenia solium* by either ingesting the larval form (in undercooked pork) or ova. If ova are ingested the parasite cannot complete the life cycle, and cysticerci encyst in various tissues including the brain.
[QCMLS 2021: 6.4.4.5] [Mahon 2019, p658-659-664]

**323.** **d** Of the organisms listed, only *Paragonimus* and *Hymenolepis* can be identified to the species level by the appearance of their ova in stool. The eggs of the other paired parasites are identical and offer no differentiation.
[QCMLS 2021: 6.4.4.5, 6.4.4.4]

**324.** **d** Adult *Paragonimus* worms live in cystic cavities in the lungs. Eggs are laid by the adult and transferred to the bronchial tree with sputum. Ova may be found in sputum or swallowed and passed in stool.
[QCMLS 2021: 6.4.4.4] [Murray 2019, p672]

**325.** **b** The rhabditiform larvae of hookworm and *Strongyloides stercoralis* can be differentiated by the size of their genital primordium. *Strongyloides* has a prominent genital primordium and a short buccal cavity. Hookworm larvae have a longer buccal cavity, and the genital primordium is not evident. *S. stercoralis* larvae have a notched tail whereas hookworm larvae have a tapering tail.
[QCMLS 2021: 6.4.4.1] [Murray 2019, p623-624]

**326.** **c** Biopsy and stool microscopy are limited by both the paucity of organisms and the possible presence of related nonpathogenic *Entamoeba* species, such as *E. dispar*. Urine is not routinely examined for this parasite, as it is an intestinal pathogen. Enzyme-linked immunoassay has the highest sensitivity and specificity.
[QCMLS 2021: 6.4.2.1] [Mahon 2019, p626]



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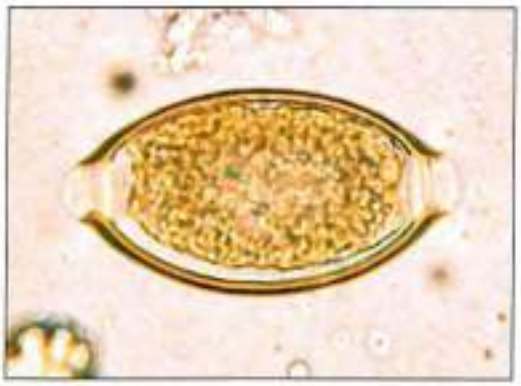
**327. a** The only morphological difference between the related organisms is the size of the trophozoite. *Entamoeba histolytica* is roughly twice the size at 20-30 µm diameter than *Entamoeba hartmanni*, which is only 5-10 µm in diameter. Also, though nonspecific, *E. histolytica* tends to be the most common trophozoite to contain ingested RBCs.
[QCMLS 2021: 6.4.2.1] [Mahon 2019, p626-628]

**328. c** *Iodamoeba bütschlii* has a large clear vacuole in the cyst form, with only one nucleus. *Entamoeba hartmanni* is small, and *Entamoeba coli* has up to 8 nuclei in the cyst form. Multiple amoeba have 4 nuclei in mature cyst.
[QCMLS 2021: 6.4.2.1] [Mahon 2019, p633]

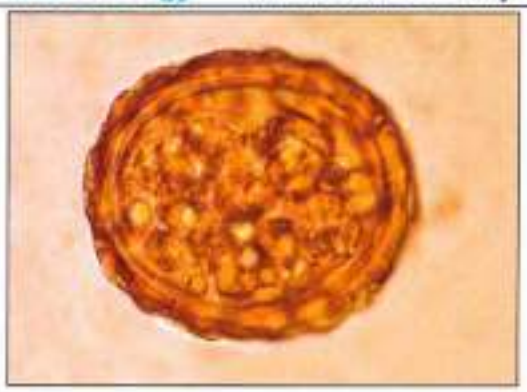
**329. b** Pigs are the natural host for *Balantidium coli*, and humans are "accidental" hosts. The intestinal infection can be asymptomatic, but it can also cause some diarrhea with nausea and vomiting. *B. coli* is the only ciliate among the choices.
[QCMLS 2021: 6.4.2.3.1] [Mahon 2019, p642-643]

**330. b** After sporozoites are introduced into the blood stream by an *Anopheles* mosquito, they travel to the liver to proliferate. The infected parenchymal cells rupture and release the merozoites, which infect RBCs.
[QCMLS 2021: 6.4.3.3] [Mahon 2019, p643, 645]

**331. a** The Duffy antigen is the receptor for *Plasmodium vivax* (as well as *Plasmodium knowlesi*); therefore, loss of the receptor affords some protection against infection. Sickle cell trait protects individuals predominantly against *P. falciparum*, which produces the most deadly form of malarial disease.
[QCMLS 2021: 6.4.3.3] [Mahon 2019, p668-669]



**332. d** *Trichuris*, the whipworm, produces barrel-shaped eggs that have "plugs" or thickenings on each end. It may co-infect with *Ascaris*.
[QCMLS 2021: 6.4.4.1] [Mahon 2019, p669-670]



**333.** **a** Fertile *Ascaris* eggs are oval with a thick hyaline and mammillated outer layer, though some lack the mammillated outer wall. Without the unique outer shell *Ascaris* eggs resemble those of hookworm or *Strongyloides*.
[QCMLS 2021: 6.4.4.1] [Mahon 2019, p674]

**334.** **c** Most cases of visceral larva migrans (and ocular larva migrans) are due to *Toxocara canis*. A small percentage of cases are due to *Toxocara cati*. *Onchocerca* causes corneal opacities, *Ancylostoma braziliensis* causes cutaneous larva migrans, and *Trypanosoma brucei* is the agent for sleeping sickness.
[QCMLS 2021: 6.4.4.3] [Mahon 2019, p656-658]

**335.** **c** *Fasciola hepatica* (& *Fasciolopsis buski*) have the largest human parasite eggs known (~150 µm in diameter). The eggs have a nonshouldered operculum and lack an abopercular knob.
[QCMLS 2021: 6.4.4.4] [Mahon 2019, p659-660]

**336.** **a** *Schistosoma haematobium* migrates to the veins of the bladder so eggs are seen in the urine. *S. mansoni* goes to the veins of the large intestine, while *S. japonicum* infects the veins of the small intestines, but both will shed eggs in the feces. *Strongyloides stercoralis* is not a *Schistosoma* species, but rather a nematode.
[QCMLS 2021: 6.4.4.4] [Mahon 2019, p663]

**337.** **d** The *Taenia* species have several features to distinguish them from each other. *T. saginata* is the beef tapeworm, *T. solium* is the pork tapeworm. *T. saginata* has an unarmed rostellum, unlike the armed rostellum of the pork tapeworm. The proglottid of the beef tapeworm has >13 uterine branches (as visualized by India ink injection), where the pork tapeworm has <13 branches. In addition the pork tapeworm can cause cysticercosis while the beef tapeworm does not. The eggs of both tapeworms are identical, with thick radially-striated walls.
[QCMLS 2021: 6.4.4.5] [Mahon 2019, p661]

**338.** **a** *Diphyllobothrium latum* is noted for its ability to cause vitamin $B_{12}$ deficiency by competing for binding the vitamin. It occurs primarily in people of northern European descent.
[QCMLS 2021: 6.4.4.5] [Mahon 2019, p666]

**339.** **b** The most common definitive host is the dog, where the eggs are found in the feces. Ingestion of eggs by intermediate hosts, such as sheep or cattle directly causes the disease. Humans accidentally ingest the eggs (sheepherders). It is seen mostly in sheep-raising countries or areas of the US. The primary manifestation is hepatic cysts.
[QCMLS 2021: 6.4.4.5] [Mahon 2019, p636-637]

**340.** **d** *Trichomonas vaginalis* only has a trophozoite stage, so it is sensitive to the environment. It is a pathogen of the urogenital tract of men and women. It is one of the most common STDs worldwide.
[QCMLS 2021: 6.4.2.2.3] [Mahon 2019, p646]



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341. **a** *Plasmodium vivax* and *P. ovale* are found in enlarged (young) RBCs only, and *P. malariae* usually invades older RBCs. *P. falciparum* invades RBCs of any age, and multiple ring forms in a single RBC are common. The banana-shaped gametocyte is produced by *P. falciparum*.
[QCMLS 2021: 6.4.3.3] [Mahon 2019, p652-653]

342. **a** *Cryptosporidium* is the only selection that is acid-fast-positive. While *Giardia* is spread by fecal-oral route, it is not limited to transfer by animals. Humans can transfer *Giardia* as well. *Naegleria* is transferred by water aspiration into the nasal passages, and *Necator* is transferred by larvae burrowing into the skin.
[QCMLS 2021: 6.4.3.1] [Mahon 2019, p610]

## *D. Mycology*

343. **a** Macroscopic characteristics and microscopic morphology are observations used for the routine identification of molds.
[QCMLS 2021: 6.5.1] [Mahon 2019, p609]

344. **c** Since fungi typically grow more slowly than bacteria, a medium with antimicrobials is included to assist in the recovery of fungi. Chloramphenicol is an antibacterial agent active against *Klebsiella* and many other bacteria. Cycloheximide inhibits rapidly growing saprophytic fungi.
[QCMLS 2021: 6.5.1] [Mahon 2019, p596]

345. **d** *Histoplasma capsulatum* is the only choice that is transmitted by inhalation.
[QCMLS 2021: 6.5.2] [Mahon 2019, p608]

346. **b** Calcofluor white binds to cellulose and chitin present in fungal cell walls. It flouresces when exposed to long-wave UV light.
[QCMLS 2021: 6.5.1] [Mahon 2019, p611]

347. **d** *Candida albicans* produces germ tubes.
[QCMLS 2021: 6.5.6.1] [Mahon 2019, p604]

348. **b** *Cryptococcus neoformans* is a cause of meningitis in immunocompromised patients, and produces a polysaccharide capsule.
[QCMLS 2021: 6.5.6.2] [Mahon 2019, p604]

349. **d** *Cryptococcus neoformans* produces phenoloxidase enzyme that breaks down caffeic acid to melanin, showing a dark brown color.
[QCMLS 2021: 6.5.6.1] [Mahon 2019, p604]

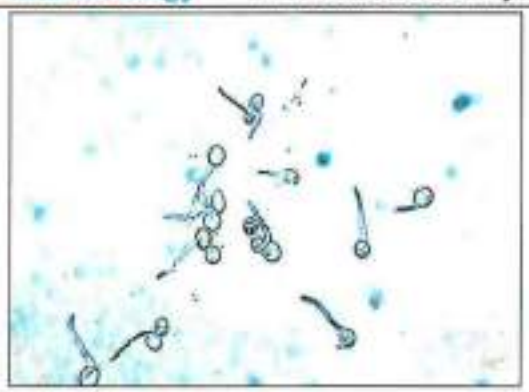
350. **d** *Cryptococcus neoformans* is the only clinically encountered yeast that is phenol-oxidase-positive.
[QCMLS 2021: 6.5.6.1] [Mahon 2019, p600, 604]

351. **d** *Candida glabrata* is urease-negative and does not produce pseudohyphae. *Candida tropicalis* is also urease-negative, but it produces pseudohyphae on cornmeal or rice agar. *Cryptococcus laurentii* does not produce pseudohyphae, but it is urease-positive. *Geotrichum* produces true hyphae.
[QCMLS 2021: 6.5.6.1] [Mahon 2019, p609]

352. **a** Cycloheximide is known to inhibit the growth of some fungal pathogens, including *Cryptococcus neoformans*.
[QCMLS 2021: 6.5.1] [Mahon 2019, p585-586]

353. **b** *Malassezia furfur* causes catheter-related sepsis, requires lipids for growth, and is a small yeast with a wide bud.
[QCMLS 2021: 6.5.6.2] [Mahon 2019, p612]

354. **d** Chromogenic agar is extremely effective in detecting mixed yeast populations in clinical specimens.
[QCMLS 2021: 6.5.6.1] [Mahon 2019, p611]



**355.** **c** *Candida albicans* produces tubular structures with no constrictions (germ tubes) when incubated in rabbit plasma.
[QCMLS 2021: 6.5.6.1] [Mahon 2019, p613]

**356.** **c** The cryptococcal latex antigen test has been proven to be significantly more sensitive than staining methods.
[QCMLS 2021: 6.5.6.2] [Mahon 2019, p595-596]

**357.** **d** Several monomorphic molds resemble the filamentous phase of dimorphic molds, so it is important to differentiate. Animal inoculation is not realistic in a routine microbiology laboratory, and conversion from mold to yeast form may take several days. Note: conversion is *not* from yeast to mold form (as stated in item **b**). Confirmatory identification is typically done by DNA probe.
[QCMLS 2021: 6.5.1, 6.5.2.1] [Mahon 2019, p595, 605]

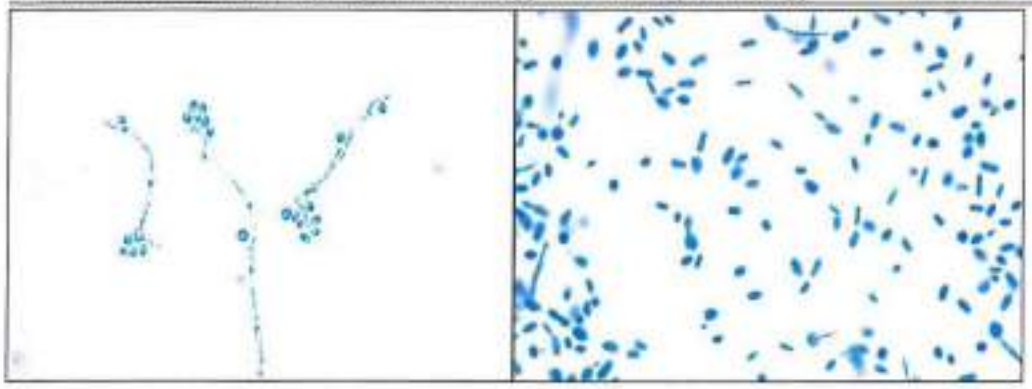
**358.** **d** The arthroconidia of *Coccidioides immitis* are highly infectious; cultures must be handled with care to minimize aerosols.
[QCMLS 2021: 6.5.2.3] [Mahon 2019, p595]

**359.** **a** The mycelial phase of *Coccidioides immitis* produces alternating, barrel-shaped arthrospores.
[QCMLS 2021: 6.5.2.3] [Mahon 2019, p596]

**360.** **c** The presence of tuberculate macroconidia indicates a presumptive identification of *Histoplasma capsulatum*. The identification must be confirmed using nucleic acid probes or exoantigen testing.
[QCMLS 2021: 6.5.2.1] [Mahon 2019, p592]

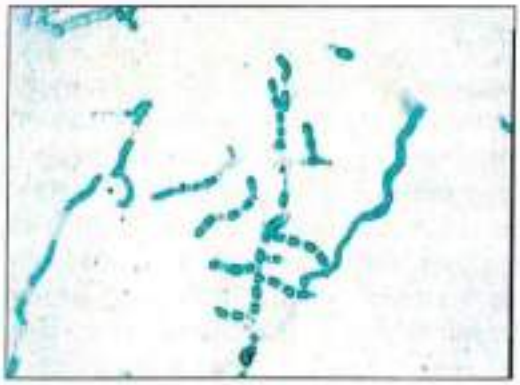


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**361. b** The characteristics listed define *Sporothrix* and differentiate it from other dimorphic fungi.
[QCMLS 2021: 6.5.2.5] [Mahon 2019, p592-597]

**362. a** Dimorphic molds include *Blastomyces, Histoplasma, Coccidioides, Paracoccidioides, Talaromyces* (formerly *Penicillium*) *marneffei, Sporothrix.*
[QCMLS 2021: 6.5.2.2] [Mahon 2019, p49, 472]

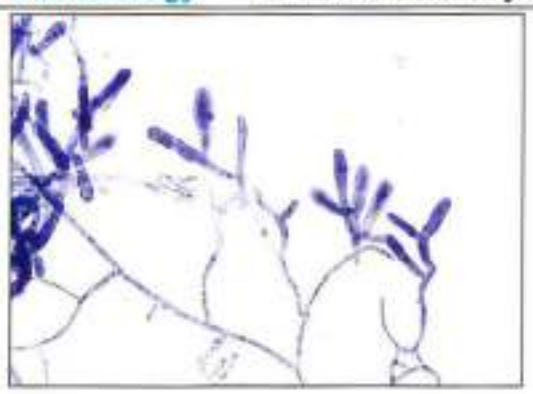


**363. a** MALDI-TOF mass spectrometry testing for identification of bacteria and fungi can be completed in minutes. This image shows morphology suspicious of *Coccidioides immitis*. MALDI-TOF is especially helpful in diagnosing slow-growing dimorphic fungi.
[QCMLS 2021: 6.5.1] [Mahon 2019, p251]



**364. c** *Talaromyces* (formerly *Penicillium*) *marneffei* is a dimorphic fungi that produces a diffusible red pigment and is endemic in Southeast Asia.
[QCMLS 2021: 6.5.2.6]

**365. d** Dermatophytes include 3 genera that are generally differentiated by their macroconidia and microconidia formation.
[QCMLS 2021: 6.5.3.3]



**366.** **a** Smooth-walled, club-shaped macroconidia are characteristic of *Epidermophyton floccosum*.
[QCMLS 2021: 6.5.3.3] [Mahon 2019, p606, 800]

**367.** **c** Hair that fluoresces yellow-green under a Woods lamp indicates the presence of a *Microsporum* species. The colony morphology and microscopic characteristics are consistent with *M. audouinii*.
[QCMLS 2021: 6.5.3.3] [Mahon 2019, p596]

**368.** **c** *Zygomycetes* grow rapidly and fill the dish with cotton candy-like growth.
[QCMLS 2021: 6.5.5] [Mahon 2019, p599-600]

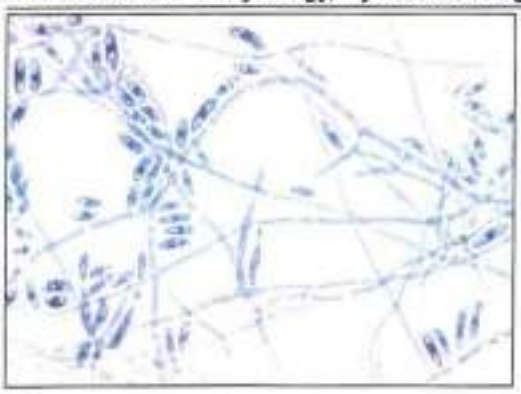
**369.** **d** Microscopic morphology is used to differentiate *Talaromyces* (formerly *Penicillium*) from *Aspergillus*. *Aspergillus* forms a swollen vesicle at the end of the conidiophore, and *Talaromyces* has no vesicle. Both molds form phialides, which produce conidia.
[QCMLS 2021: 6.5.3.1, 6.5.3.2] [Mahon 2019, p600]



**370.** **d** Elongated and tapered phialides (tenpins) are characteristic of *Paecilomyces*.
[QCMLS 2021: 6.5.3.2] [Mahon 2019, p600]



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**371.** **c** *Fusarium* frequently cause corneal infections and have sickle-shaped macroconidia.
[QCMLS 2021: 6.5.3.2] [Mahon 2019, p590-591]

**372.** **a** *Pseudallescheria boydii* is a common cause of eumycotic mycetoma. *Nocardia* can cause actinomycotic mycetomas, but *Coccidioides* and *Aspergillus* do not cause mycetomas.
[QCMLS 2021: 6.5.4] [Mahon 2019, p590]

**373.** **a** Only *Fonsecaea pedrosoi* produces cladosporium, phialophora and fonsecaea types of sporulation simultaneously.
[QCMLS 2021: 6.5.4] [Mahon 2019, p595]

**374.** **c** Valley fever, or San Joaquin fever, are all caused by *Coccidioides immitis*, which is seen in high, dusty desert, such as those found in southeast California, Arizona, and New Mexico. Sandstorms are responsible for spreading the spores.
[QCMLS 2021: 6.5.2.3] [Mahon 2019, p609, 612]

**375.** **a** Cornmeal with Tween 80 stimulates conidiation and chlamydospore production, aiding in yeast identification. Urea agar is helpful in detecting urease, which is produced by *Cryptococcus neoformans*. Brain-heart infusion is a nonselective medium tht supports the growth of saprophytic and pathogenic fungi. Potato dextrose agar is useful in demonstrating the production of pigment by some molds, such as *Trichophyton rubrum*.
[QCMLS 2021: 6.5.6.2] [Mahon 2019, p599-600, 602]



**376.** **c** *Aspergillus fumigatus*: Each of the *Aspergillus* species has a distinct appearance on media plates. *A. fumigatus* is blue-green, *A. terreus* has a cinnamon-buff colony, *A. niger* is a black colony, and *A. flavus* has a brown colony with lateral striations. *Gliocladium* is dematiaceous, and *Microsporum canis* has a brilliant yellow surface color.
[QCMLS 2021: 6.5.3.1] [Mahon 2019, p588-589]



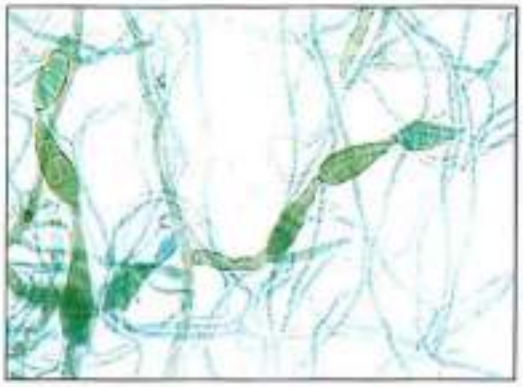
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377. **b** *Trichophyton rubrum* have microconidia and of them each has unique morphologic features. *T. rubrum* has "birds on a wire" microconidia spaced along hyphae, *T. tonsurans* has widely variable microconicia, while *T. mentagrophytes* has grapelike clusters and occasional spiral hyphae. *Microsporum canis* produces large macroconidia that are spindle shaped.
[QCMLS 2021: 6.5.3.3] [Mahon 2019, p604]

378. **a** All of the organisms mentioned, with the exception of *Malassezia*, are urease-positive, but *Cryptococcus* is the only one that is also phenol-oxidase-positive. The presence of phenol oxidase can be demonstrated on bird (Niger) seed agar, where the phenol oxidase will convert the caffeic acid in the agar into melanin pigment, yielding the characteristic brown/black color.
[QCMLS 2021: 6.5.6.1] [Mahon 2019, p598]



379. **d** *Rhizopus* is a zygomycete that produces rhizoids directly below the sporangiophores. *Mucor* does not produce rhizoids. Colonies of *Rhizopus* and *Mucor* grow rapidly and are "lid lifters".
[QCMLS 2021: 6.5.5] [Mahon 2019, p601-602]



380. **a** *Alternaria* and *Curvularia* are dematiaceous, while *Paecilomyces* and *Scopulariopsis* are brightly-colored molds. *Curvularia* macroconidia are multi-celled and crescent-shaped molds.
[QCMLS 2021: 6.5.4]



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**381. b** *Cryptococcus gattii* is similar to *C. neoformans* in appearance, but *C. neoformans* is associated with soil contaminated with pigeon droppings and is acquired by inhalation. *C. gattii* is associated with the eucalyptus tree and is found in tropical and subtropical areas. It is an emerging pathogen in the Pacific Northwest. *C. neoformans* is negative for the canavanine agar. *C. krusei* is not encapsulated and neither it nor *Rhodotorula* have association with the eucalyptus tree and are negative for the canavanine agar.
[QCMLS 2021: 6.5.6.2] [Mahon 2019, p598]

## *IV. Postanalytic Procedures*

### *A. Documentation Practices*

**382. b** Timely direct specimen results (eg, direct specimen Gram stain results), interim reports and finalized reports should be issued per laboratory policy. Reports should be issued as soon as useful information becomes available, including 'no growth' status. Most report updates are made and maintained within a laboratory information system (LIS). The LIS interfaces with the hospital information system (HIS). As results are updated or finalized within the LIS, they should be available in the HIS.
[QCMLS 2021: 6.9.2, 6.9.3, 6.9.4] [Mahon 2019, p868]

### *B. Urgent & Critical Value Reporting*

**383. a** Results that may require rapid clinical attention to avoid significant patient morbidity or mortality are classified as an urgent or critical result. The reporting of urgent/critical results includes verbally reporting results to a primary care practitioner (eg, the ordering physician) and documenting that communication. Positive blood cultures, CSF cultures, and acid-fast smears (possible *Mycobacterium tuberculosis*) are generally considered critical results. Other culture results should be reported as soon as useful information becomes available, but do not require verbal notification of a caregiver. Based on patient population and need, clinical laboratories determine their own urgent/critical reporting criteria.
[QCMLS 2021: 6.9.2] [Procop 2017, p43]

### *C. Result Review & Autoverification*

**384. b** A necessary job of the laboratory director is to provide clinicians with summaries regarding their use of the laboratory tests, positive culture rates, contamination rates in cultures that identify contaminants, and antimicrobial susceptibility results of commonly isolated bacteria in the community. This feedback is important to clinicians for reasons of quality control and patient care.
[QCMLS 2021: 6.9.4] [Procop 2017, p44]

### *D. Issuing Corrected Reports*

**385. a** A Problem Action Form (PAF) does not require employee certification number or years of experience. Also, the person filling the report should sign his/her name but it is not required that the attending physician or pathologist sign the form. Infection control is the only department that contains the recommended information for a PAF.
[QCMLS 2021: 6.9.3] [Mahon 2019, p99]

*E. Reporting to Infection Control/Prevention & Public Health*

**386. a** Working with the infection control departments within a hospital is a component of a clinical microbiologist's job. Many of the reports that are sent to the infection control department are electronic and reviewed by the epidemiologists daily. However, establishing a relationship with the epidemiologists on staff is important and can allow for a quick identification of a possible outbreak. A list of required microbiological agents that must be reported to infection control and/or a public health laboratory exist in each laboratory and can vary state by state. The reporting of these organisms is required by law.
[QCMLS 2021: 6.9.4] [Procop 2017, p44]



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Chapter 7

# Laboratory Operations

*The following items have been identified generally as appropriate for those preparing for both the MLS and MLT examinations. Items that are appropriate for the MLS examination **only** are marked with MLS ONLY.*


**527 Questions**
- 527 I. Quality Assessment/Troubleshooting
- 536 II. Safety
- 545 III. Laboratory Mathematics
- 555 IV. Manual/Automated Methodology and Instrumentation.
- 563 V. Basic Management Principles
- 570 VI. Education Principles

**572 Answers with Explanations**
- 572 I. Quality Assessment/Troubleshooting
- 578 II. Safety
- 585 III. Laboratory Mathematics
- 590 IV. Manual/Automated Methodology and Instrumentation
- 596 V. Basic Management Principles
- 600 VI. Education Principles

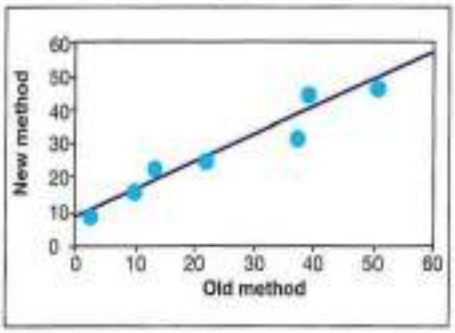

## I. Quality Assessment/Troubleshooting

### A. Preanalytical, Analytical, Postanalytical

1. Preanalytical (preexamination) variables in laboratory testing include:

   **a** result accuracy
   **b** report delivery to the ordering physician
   **c** test turnaround time
   **d** specimen acceptability

2. MLS ONLY A new methodology for amylase has been developed and compared with the existing method as illustrated in the graph. The new method can be described as:




   **a** poor correlation with constant bias
   **b** good correlation with constant bias
   **c** poor correlation with no bias
   **d** good correlation with no bias

3. Which of the following is not a potential source of postanalytical (postexamination) errors?

   **a** excessive delay in reporting or retrieving a test result
   **b** interpretation of result
   **c** verbal notification of test result
   **d** labeling the specimen at the nurses' station

4. A preanalytical error can be introduced by:

   **a** drawing a coagulation tube before an EDTA tube
   **b** mixing an EDTA tube 8-10 times
   **c** transporting the specimen in a biohazard bag
   **d** vigorously shaking of blood tube to prevent clotting

5. The most important diagnosis and therapeutic management decision tool used to interpret test results is:

   **a** statistical analysis
   **b** reference intervals
   **c** specimen acceptability
   **d** the age of a patient

6. A clean-catch urine is submitted to the laboratory in a plain sterile urine cup for routine urinalysis and culture. The routine urinalysis is performed first. Three hours later, the specimen is sent to the microbiology department for culture. The specimen should:

   **a** be centrifuged, and the supernatant cultured
   **b** be rejected due to the time delay
   **c** not be cultured if no bacteria are seen
   **d** be processed for culture only if the nitrate is positive

7. Urine samples should be examined within 1 hour of voiding because:

   **a** RBCs, leukocytes and casts agglutinate on standing for several hours at room temperature
   **b** urobilinogen increases and bilirubin decreases after prolonged exposure to light
   **c** bacterial contamination will cause alkalinization of the urine
   **d** ketones will increase due to bacterial and cellular metabolism

8. A urine specimen comes to the laboratory 7 hours after it is obtained. It is acceptable for culture only if the specimen has been stored:

   **a** at room temperature
   **b** at 4-7°C
   **c** frozen
   **d** with a preservative additive

9. Preanalytical errors are major contributors to total laboratory errors. Which of the following is not a common preanalytical error?

   **a** specimen collected in the wrong container
   **b** unlabeled specimen
   **c** QNS, hemolyzed or lipemic specimens
   **d** failure to flag a critical value on a laboratory report

10. Which of the following statements about analytical (examination) errors is true?

    **a** analytical laboratory errors are easy for ordering providers to detect
    **b** analytical errors almost never happen in a CLIA accredited laboratory
    **c** analytical errors are not obvious to providers
    **d** analytical errors can be caused by poor patient preparation

11. Laboratories should monitor and track preanalytical/preexamination indicators of quality to develop strategies to identify root causes of error and develop counter measures. Examples of preanalytical quality indicators include:

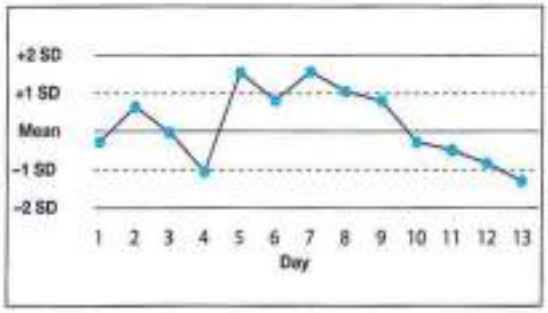
    **a** dilution errors
    **b** rate of calibration failures
    **c** compliance with PPE in the laboratory
    **d** rate of inaccurate test order entry into the laboratory information system

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12. The first procedure to be followed if the blood gas instrument is out-of-control for all parameters is:

a recalibrate, then repeat control
b repeat control on the next shift
c replace electrodes, then repeat control
d report patient results after duplicate testing

13. A Levy-Jennings quality control chart is shown, which represents control values for 13 consecutive analyses for a particular serum constituent. If the 14th value is below the –2 SD limit, which of the following should be done?




a control should be repeated to see if it will fall within the established interval
b analysis system should be checked for a deteriorating component
c analysis system should be checked for a change in reagent lot number
d no action is needed

14. MLS ONLY In quality control, ±2 standard deviations from the mean includes what percentage of the sample population?

a 50%
b 75%
c 95%
d 98%

15. Upon completion of a run of cholesterol tests, the technician recognizes that the controls are not within the 2 standard deviations confidence range. What is the appropriate course of action?

a report the results without any other action
b run a new set of controls
c run a new set of controls and repeat specimens
d recalibrate instrument and run controls

16. The following data are calculated on a series of 30 determinations of a serum uric acid control: mean = 5.8 mg/dL, 1 standard deviation = 0.15 mg/dL. If confidence limits are set at ±2 standard deviations, which of these ranges represents the allowable limits for the control?

a 5.65-5.95 mg/dL
b 5.35-6.25 mg/dL
c 5.50-6.10 mg/dL
d 5.70-5.90 mg/dL

17. What are the 3 steps of an individualized quality control plan (IQCP)?

a inventory management, instrument selection, quality control plan
b assessment of preanalytical, analytical, and postanalytical errors
c quality control plan, quality assessment, and quality improvement
d risk assessment, quality control plan, quality assessment

18. A delta check is a method that:

a determines the mean and variance of an instrument
b monitors the testing system for precision
c monitors patient samples day to day
d is determined by each laboratory facility

19. On Monday a patient's hemoglobin determination is 11.3 g/dL (113 g/L), and on Tuesday it measures 11.8 g/dL (118 g/L). The standard deviation of the method used is ±0.2 g/dL (2 g/L). Which of these statements can be concluded about the hemoglobin values given?

a one value probably resulted from laboratory error
b there is poor precision; daily quality control charts should be checked
c the second value is out of range and should be repeated
d there is no significant change in the patient's hemoglobin concentration

20. Instrument calibration verification is the process of:

a assaying quality control materials in the same manner as patient samples to confirm that the calibration of the instrument is stable throughout the reportable range
b determining the closeness of agreement between independent test results obtained under stipulated conditions
c assaying calibration materials in the same manner as patient samples to confirm that the calibration of the instrument is stable throughout the reportable range
d a one-time process completed to confirm instrument performance characteristics before the interment is used for patient testing

21. Specimen stability refers to insignificant increase or decrease in the concentration of an analyte in a stored specimen. Conditions to be considered when conducting a stability experiment would not include:

a time between collection and centrifugation
b effect of temperature and humidity on specimens
c specimen transportation conditions (time, temperature, tube orientation, agitation, light exposure)
d maximum storage time is defined by the date of the earliest-expiring reagent

22. A laboratory procedure manual should be available to personnel at the workbench and must include the following elements:

a clinical significance, frequency of test performance, quality control processes, calculations, and test interpretation
b step-by-step performance instructions, reference intervals, reagent storage location, specimen collection requirements
c limitation of test performance (interfering substances), corrective action if QC fails, critical values, and reagent preparation
d test principle, criteria for specimen rejection, list of individuals authorized to perform, and calibration procedures

23. The main goal of a quality control (QC) strategy is to reduce the risk of harm to a patient if an erroneous result is used for clinical decision making. A QC strategy should be customized to the test and testing situation. The core focus of a QC strategy centers around:

a accuracy, precision, and reproducibility of test systems
b a measurement procedure's performance capability relative to the quality required, and validation data
c interlaboratory quality control data
d critical control point quality control

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24. What is the first step that should be taken in response to an out-of-control condition?

**a** repeat testing of patient specimens since the previous in control situation
**b** periodically review mean, SD and CV to ensure an appropriate target value and SD are used
**c** review performance of proficiency testing
**d** discontinue patient testing/reporting

*C. Point-of-care Testing (POCT)*

25. Which of these attributes is the advantage for adding point-of-care testing?

MLS ONLY
**a** higher test accuracy
**b** lower costs
**c** faster TAT
**d** more skilled test personnel

26. A benefit of microassays, such as point-of-care methods, include:

**a** increased analytical reliability
**b** reduced sample volume
**c** increased diagnostic specificity
**d** reduced numbers of repeated tests

27. The primary advantage of point-of-care testing (POCT), also known as bedside testing or near patient testing is:

**a** POCT can be performed by nonlaboratorians at a lower cost
**b** the turn-around time for test results is faster
**c** POCT bypasses costly regulatory requirements that apply to testing in a central lab
**d** testing is easier to perform, which reduces the cost of training

*D. Compliance*

28. An ICD-10 code is related to:

MLS ONLY
**a** patient charges
**b** diagnosis
**c** lab accreditation
**d** test methodology

29. The use of security systems such as firewalls and data encryption for electronic transmission of patient data from a laboratory information system to a remote location are required for:

**a** LOINC
**b** HIPAA
**c** ICD-9
**d** CLIA

30. Medicare and Medicaid Services (CMS) was established un the Social Security Act of 1965. Which of these statements related to CMS payment for laboratory services is true?

**a** Medicaid is administered by the Federal government
**b** Medicare provides medical coverage to individuals 65 and older at no cost to the individual
**c** laboratories must have a current CLIA certificate to submit claims to Medicare or Medicaid
**d** Medicare part A (inpatient coverage) and part B (outpatient coverage) includes prescription drug coverage

31. Medical billing codes are used to classify a patient's treatment, diagnosis, and associated medical supplies. Which of the following is not a primary medical billing code?

a HCPCS
b HIPAA
c CPT
d ICD

*E. Regulation*

32. Which action by the phlebotomist will comply with the College of American Pathologists (CAP) Patient Safety Goal "to improve patient and sample identification at the time of specimen collection" and The Joint Commission Patient Safety Goal to "improve the accuracy of patient identification"?

a match the name and room number on the patient's ID bracelet to the name and room number on the preprinted collection label
b match the name and medical record number on the patient's ID bracelet to the name and medical record number on the preprinted collection label
c verify patient information by stating the patient's name when approaching the patient
d label the collection tubes prior to the blood draw at the patient's bedside

33. Employees are guaranteed the right to engage in self-organization and collective bargaining through representatives of their choice, or to refrain from these activities by which of the following?

a Civil Rights Act
b Freedom of Information Act
c Clinical Laboratory Improvements Act (CLIA)
d National Labor Relations Act

34. Which of the following organizations was formed to encourage the voluntary attainment of uniformly high standards in institutional medical care?

a Centers for Disease Control (CDC)
b Health Care Finance Administration (HCFA)
c The Joint Commission (TJC), formerly known as JCAHO
d Federal Drug Administration (FDA)

35. The process by which an agency or organization uses predetermined standards to evaluate and recognize a program of study in an institution is called:

a regulation
b licensure
c accreditation
d credentialing

36. Clinical Laboratory Improvement Amendments of 1988 (CLIA) was established to provide oversight to:

a research labs performing testing on human specimens
b waived point-of-care testing by nonlaboratory personnel
c CAP-accredited labs
d all US facilities or sites performing patient testing to diagnose, prevent, or treat disease

37. According to CLIA, who is responsible for classifying lab test complexity?

a the medical director whose name is on the CLIA certificate
b FDA (Food and Drug Administration)
c CMS (Centers for Medicare and Medicaid Services)
d AMA (American Medical Association)

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38. Which statement about proficiency testing (PT) is true?

MLS ONLY

**a** results can be compared to another hospital prior to submission if that hospital is in your system
**b** results between 2 technologists can be averaged
**c** CAP requires duplicate testing to ensure good instrument performance
**d** it is necessary to assess results even if a PT challenge is ungraded

39. Which of the following is part of The Joint Commission's National Patient Safety Goals?

MLS ONLY

**a** communication of critical results
**b** documentation of lab QC
**c** trending of instrument problems
**d** reconciliation of lab orders and results in the medical record

40. If your lab performs a test for which there is no commercially available control of proficiency test material, which of the following is acceptable as an alternative method for determining analytic testing reliability?

MLS ONLY

**a** perform the test in duplicate
**b** you do not have to do anything if there is nothing available
**c** make an internal lab control from a previous negative and positive
**d** have 2 technologists perform the test independently

41. CAP requires refrigerator temperatures to be recorded:

**a** daily
**b** weekly
**c** monthly
**d** periodically

42. A paper or electronic report of lab results must include:

**a** the name of the person who collected the specimen
**b** the test price
**c** a pathologist's signature
**d** the name and address of the testing laboratory

43. CAP requires that glassware cleaning practices include periodic testing for:

MLS ONLY

**a** chemical residues
**b** silicates
**c** detergents
**d** heavy metals

44. HIPAA is a federal law that requires:

**a** confidentiality of patients' health care information between 2 organizations
**b** reporting of errors in laboratory results
**c** access to patient records when there is a lawsuit
**d** unannounced inspections by accreditation agencies

45. Your friend calls and asks you to access his test results. Your doing so would violate which of the following?

**a** CAP
**b** The Joint Commission (TJC)
**c** HIPAA
**d** CLIA

46. An individualized quality control plan (IQCP) does not require:

**a** risk assessment
**b** revalidation of test performance
**c** quality control plan
**d** quality assessment

47. A quality management system (QMS) is a system for designing, implementing, maintaining, and managing quality in a laboratory. A fundamental element of a QMS is training and competence assessment of testing personnel. Training is not required:

**a** for newly hired, transferred or promoted personnel
**b** when new testing platforms are implemented
**c** when there is a change in the medical director of the lab
**d** when an employee demonstrates repeated performance issues

48. Which of the following require laboratories to establish a chemical hygiene plan?

**a** CAP (College of American Pathologists)
**b** ACS (American Chemical Society)
**c** OSHA (Occupational Safety and Health Act)
**d** CLIA (Clinical Laboratory Improvements Act)

49. Package inserts may be used:

**a** instead of a typed procedure
**b** as a reference in a procedure
**c** at the bench but not in the procedure manual
**d** if initialed and dated by the laboratory director

50. Auto-verification of test results requires all of the following to be established by the laboratory except:

**a** patient results entered into the LIS via an instrument interface
**b** patient results evaluated based on validated rules defined in the LIS
**c** successful quality control testing obtained prior to releasing patient results
**d** review of results by a qualified technologist or technician

51. Validation of calculated test results performed by a laboratory information system must be performed:

**a** every 6 months
**b** annually
**c** biennially
**d** only upon initial LIS installation

52. Proficiency testing (PT) and alternate assessment procedures (AAP) are valuable tools in the quality improvement process. PT/AAP processes are a sequence of activities that enables the laboratory to monitor and assess PT performance to meet mandatory regulatory and accreditation requirements for external assessment of examination methods. Which PT/AAP activity is allowable?

**a** sharing PT results within a multi-laboratory system before result submission
**b** interlaboratory communication after PT testing is complete
**c** interlaboratory communication after PT results are submitted
**d** sending PT samples to another laboratory for reflexing testing using the same process as used for patient testing

53. When PT materials are not available, the laboratory needs to develop alternate assessment procedure (AAP) that includes all except which element?

**a** determine the alternative assessment method
**b** determine the frequency for performing the AAP
**c** grade the target value and the maximum tolerable deviation from that target value
**d** define the criteria for acceptable performance

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54. The FDA is responsible for regulating companies who manufacture, repackage, relabel, or import medical devices sold in the United States. Which of the following statements related medical devices and biologics used in a clinical laboratory is true?

a laboratories must report instrument malfunctions directly to the manufacturer, not to the federal government
b laboratory-developed assays are exempt from FDA medical device requirements
c laboratory tests performed with analyte-specific reagents require a specific statement to be included in the patient report with the test result
d Medicare does not reimburse testing performed with analyte-specific reagents

55. The False Claims Act prohibits the billing of fraudulent claims for medical services including laboratory testing. Which of the following statements is true?

a laboratories are allowed for to bill for testing if ordered by an authorized individual
b unintentional billing errors does not put a healthcare institution at risk of losing funding
c unintentional billing errors caused by a mistake is considered fraud
d billing for a service not performed is considered billing waste

56. Which of the following agencies uniformly regulates the packaging and shipping of specimens and dangerous goods?

a Department of Transportation (DOT)
b International Air Transport Association (IATA)
c United States Postal Service (USPS)
d DOT, IATA and USPS have different rules regarding shipping of biological specimens and infectious materials

57. Title VII of the Civil Rights Act of 1964 made it illegal to discriminate against someone on the bases of race, color, religion, national origin, and sex. The Civil Rights Act:

a ends racism
b eliminates health disparities
c eliminates racial disparities
d accomodates religious practices

58. Laboratories are required to perform and document analytical validation of each test method or instrumental system before use in patient testing. Which of the following are required elements of a test validation experiment?

a units of measure, reportable range and linearity
b specimen collection container(s), precision, and frequency that the test will be performed
c accuracy, reference intervals and analytic specificity including interfering substances
d analytic sensitivity, linearity and instrument selection

59. Test precision is verified by:

a comparing results from analyzing the same specimens on multiple analyzers over time
b repeat measurements of samples of multiple concentrations within run and between runs performed over a period of time
c comparing results to a definitive or reference method for the same analyte
d comparing results of multiple specimens analyzed by multiple individuals and multiple instruments

60. A reference interval is the interval between and including the lower and upper reference limits. Determining the reference intervals that will be reported with a test result is a required element of test method validation. Which of the following is not true when establishing reference values for a new analyte is being considered?

a list of analytical interferences and sources of biological variability from literature
b establishment of selection or exclusion criteria (eg, age, sex, health)
c determine the appropriate number of reference individuals in consideration of desired confidence limits
d lower and upper limits must be established for each analyte

## II. Safety

### A. Safety Programs & Practices

61. A technician is asked to clean out the chemical reagent storeroom and discard any reagents not used in the past 5 years. How should the technician proceed?

**a** discard chemicals into biohazard containers where they will later be autoclaved
**b** pour reagents down the drain, followed by flushing of water
**c** consult SDS sheets for proper disposal
**d** pack all chemicals for incineration

62. Using a common labeling system for hazardous material identification such as HMIS® or NFPA 704, the top red quadrant represents which hazard?

**a** reactivity
**b** special reactivity
**c** health
**d** flammability

63. If the HMIS® or NFPA 704 hazardous material identification system has a number 4 in the left blue quadrant, it represents a:

**a** high health hazard
**b** low health hazard
**c** high reactivity hazard
**d** low reactivity hazard

64. Which chemical is a potential carcinogen?

**a** potassium chloride
**b** formaldehyde
**c** mercury
**d** picric acid

65. Compressed gas cylinders should:

**a** be stored with flammable materials
**b** be transported by rolling or dragging
**c** have safety covers removed when pressure regulators are unattached
**d** be secured upright to the wall or other stable source

66. MLS ONLY A chemical that is extremely volatile, flammable, and capable of forming explosive peroxides upon long-term contact with atmospheric oxygen, is:

**a** ethyl alcohol
**b** ethyl acetate
**c** diethyl ether
**d** xylene

67. The HMIS® or NFPA 704 hazardous material identification system rating for a slightly toxic chemical would be:

**a** 1 in the yellow quadrant
**b** 4 in the blue quadrant
**c** 1 in the blue quadrant
**d** 4 in the yellow quadrant

68. A chemical that causes immediate visible destruction or irreversible alterations of human tissue at the contact site is best classified as:

**a** carcinogenic
**b** toxic
**c** ignitable
**d** corrosive

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69. MLS ONLY Labels on shipped chemicals from manufacturers, importers or distributors are required to include information on:

**a** physical properties of the chemical
**b** accident instructions
**c** appropriate hazard warnings
**d** exposure limits

70. When hazardous chemicals are transferred from the original appropriately labeled container(s) to a secondary container for immediate use by the person performing the transfer, it:

**a** must be labeled with an emergency response phone number(s)
**b** must be labeled with the identity or contents of the hazardous chemical(s)
**c** must be labeled with hazard warnings related to the effect on involved target organs
**d** does not require labeling

71. MLS ONLY Which hazardous chemical combinations are incompatible and should not be stored together?

**a** acetone and xylene
**b** chlorine and ammonia
**c** ethanol and acetone
**d** sodium and potassium

72. MLS ONLY A gallon of xylene waste should be:

**a** flushed down the sink
**b** allowed to evaporate in an open room
**c** disposed of with nonincinerated regulated medical waste
**d** disposed of as an EPA hazardous waste through a licensed waste hauler

73. MLS ONLY A technologist, who has been routinely working with hazardous chemicals, begins to notice symptoms of persistent headaches after exposure to these chemicals. What is the first action the technologist should take?

**a** seek independent medical consultation and evaluation
**b** continue to perform work assignment to see if symptoms persist
**c** acquire involved SDS to investigate signs and symptoms
**d** report situation to supervisor

74. MLS ONLY When an employee reports signs and symptoms of a chemical exposure, the employer should suggest a medical consultation and evaluation, which is paid by the:

**a** employee using the employee's personal benefit time
**b** employer using the employee's personal benefit time
**c** employer on work time without loss of pay
**d** employee on work time without loss of pay

75. MLS ONLY When initial or baseline chemical exposure monitoring required by OSHA for substances like formaldehyde or xylene is performed and the results are within permissible exposure limits, repeat monitoring should be performed:

**a** when procedures or equipment surrounding use of the specific chemical change
**b** annually
**c** twice a year
**d** every 2 years

76. MLS ONLY An example of personal protective equipment (PPE) for handling hazardous chemicals is:

**a** eyewash or safety shower
**b** fume hood
**c** latex or vinyl gloves
**d** neoprene or nitrile gloves

77. One of the elements of a written laboratory chemical hygiene plan is to:

a require employees who handle chemicals to have annual medical evaluations
b prohibit use of carcinogens
c designate a laboratory chemical hygiene officer
d perform chemical monitoring every 6 months for OSHA regulated substances

78. The purpose of the OSHA Hazard Communication, is to require employers to establish a program ensuring personnel are provided with information regarding the workplace dangers of:

a bloodborne pathogens
b environmental hazards
c general safety hazards
d hazardous chemicals

79. This symbol indicates which of the following hazards?



a flammable
b electrical
c radiation
d biohazard

80. When working with sharp equipment and objects, use a:

a double-glove technique with specimen handling gloves
b mechanical device
c paper towel or gauze as a barrier
d 2-handed technique

81. For safe use and handling of liquid nitrogen:

MLS ONLY

a use chemically resistant gloves
b shield all skin and use a face shield
c store cylinders away from ventilation
d store cylinders in a horizontal position in a cool dry place

82. Incident reports for occupational injury or illness should:

a include information on the employee's past medical history
b be filed only for incidents involving serious injury or illness
c be filed for all incidents including near miss incidents
d not be retained after review by a safety committee or officer

83. All laboratory instruments should:

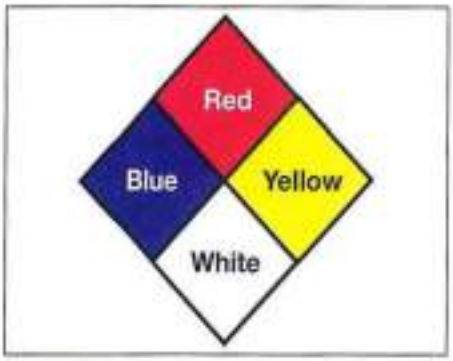
a have repairs conducted while connected to facility wiring
b be grounded or double insulated
c have safety checks performed initially and then every 6 months
d be connected to multiple outlet adapters

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84. If areas of the laboratory are designated as "clean" or "contaminated," it is appropriate for a technologist to:

**a** clean technical area bench tops after spills and on a weekly basis
**b** wear a lab coat in the break or lunch room
**c** apply lip balm in a contaminated area
**d** touch a "contaminated" area phone with ungloved hands if hands are washed afterward

85. What type of identification system does this symbol represent?




**a** transmission-based precautions
**b** physical environmental hazards
**c** chemical hazardous materials
**d** radiation hazards

86. Flammable and combustible liquids in containers ≥5 gallons should be stored in:

**a** a flammable safety cabinet vented to room air
**b** a nonexplosion proof refrigerator
**c** a fume hood
**d** an approved safety can

87. After receiving appropriate training, the first step in using a fire extinguisher is to:

**a** sweep the flow of the hose from side to side
**b** pull the pin
**c** squeeze the top handle or lever
**d** aim the hose at the base of the fire

88. To help prevent electrical fires in healthcare facilities:

**a** use multiple outlet or gang plug adapters
**b** change circuit breakers annually
**c** tape over worn wiring with certified electrical tape
**d** use only Underwriters Laboratories (UL) or other safety-agency-rated electrical equipment

89. In addition to keeping the load close to your body and tightening your abdominal muscles when lifting heavy boxes of supplies, it is important to bend at the:

**a** waist; lift with legs and buttocks
**b** knees and hips; lift with legs and buttocks
**c** knees and hips; lift with arms and back
**d** waist; twist your body when lifting

90. The best way to prevent or relieve symptoms of carpal tunnel syndrome is to:

IBLS ONLY

**a** raise arms and bend wrists downward
**b** redesign facilities
**c** bend back and neck slightly forward
**d** maintain wrists in a neutral position

91. Class C fires involve:

a grease and oil
b xylene and alcohol
c paper, wood and plastics
d electrical equipment

92. MLS ONLY A laboratory employee identifies arm and neck pain after performing repetitive movements during his/her work assignment. What is the best first action to be taken?

a report to and discuss issue with supervisor
b continue to perform work assignment and see if it improves
c make an appointment with his/her personal physician
d change or adjust his/her workstation

93. An example of personal protective equipment (PPE) is:

a a biological safety cabinet
b an emergency safety shower
c an eyewash station
d a lab coat

94. Gloves worn in the laboratory for specimen processing must be removed and hands washed when:

a answering the telephone in the technical work area
b carrying a specimen outside the technical work area through "clean" areas
c answering the telephone in a designated "clean" area
d after handling specimens from known isolation precaution patients

95. Safety glasses, face shields or other eye and face protectors must be worn when:

a working with caustic or toxic materials
b present in technical work area
c viewing microbiology culture plates
d processing specimens using a splash barrier

96. To prevent injury, a safe lab work practice is to:

a secure long hair and jewelry
b store well-wrapped food in the supply refrigerator
c wear contact lenses for eye protection
d wear comfortable, rubber-bottomed, open-weaved shoes

97. Safe handling and disposal of laboratory generated infectious waste require:

a disinfection of all waste
b thorough mixing of infectious and noninfectious waste
c separation of infectious and noninfectious waste
d incineration of all waste

98. Which of the following is the best choice for decontaminating bench tops contaminated by the AIDS virus?

a sodium hypochlorite bleach
b formalin
c a quaternary ammonium compound
d 100% alcohol

99. The safest method of disposing of hypodermic needles is:

a recap the needle with its protective sheath prior to discarding
b cut the needle with a special device before disposal
c discard the needle in an impermeable container without other handling immediately after use
d drop the needle in the waste basket immediately after use


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100. Precautions for health care workers dealing with patients or patient specimens include:

a mouth pipetting when specimens lack a "Precaution" label
b reinserting needles into their original sheaths after drawing blood from a patient
c wearing a mask and disposable gown to draw blood
d prompt cleaning of blood spills with a disinfectant solution such as sodium hypochlorite

101. Infection rate is highest for laboratory professionals exposed to blood and body fluids containing:

a hepatitis A
b hepatitis B
c CMV
d HIV

102. Which of the following forms of exposure places a technologist at the highest risk for infection with human immunodeficiency virus (HIV)?

a aerosol inhalation (eg, AIDS patient's sneeze)
b ingestion (eg, mouth pipetting of positive serum)
c needlestick (eg, from AIDS contaminated needle)
d splash (eg, infected serum spill onto intact skin)

103. Which disinfectant inactivates HIV and HBV?

a alcohol
b iodine
c phenol
d sodium hypochlorite

104. Filters generally used in biological safety cabinets to protect the laboratory worker from particulates and aerosols generated by microbiology manipulations are:

a fiberglass
b HEPA
c APTA
d charcoal

105. What is the single most effective method to prevent nosocomial spread of infection?

a wear mask, gown and gloves
b require infectious patients to mask
c wear an N95 respirator mask
d perform frequent and appropriate hand hygiene

106. Contaminated needles and syringes without safety self-sheathing devices should be:

a sheared by a needle cutter or bent
b re-capped using a 2-handed technique
c discarded directly into an appropriate sharps container
d removed from the syringe/needle holder

107. Use of "standard" (universal) precautions minimizes exposure to:

a bloodborne pathogens
b chemical hazards
c radiation hazards
d environmental hazards

108. After an accidental needle stick with a contaminated needle, the first action should be to:

a apply antiseptic ointment to the wound
b seek immediate medical assistance
c bandage the wound
d thoroughly wash the wound with soap and water

109. What is the most likely mode of transmission for bloodborne pathogens in laboratory acquired infections?

a parenteral inoculation of blood
b contact with intact skin
c airborne transmission
d fecal-oral transmission

110. Which infectious agent is considered to be the primary occupational health hazard regarding transmission of bloodborne pathogens?

a human immunodeficiency virus
b hepatitis B
c tuberculosis
d methicillin-resistant *Staphylococcus aureus*

111. When processing specimens for mycobacterial testing, what specific engineering control must be used?

a horizontal laminar flow hood
b barrier protection only
c biological safety cabinet
d fume hood

112. Hepatitis B vaccine is:

a administered as a single 1-time injection
b required for all healthcare employees
c must be provided by the employer free of charge
d recommended only when an exposure incident occurs

113. When cleaning up a small (5 mL) blood spill on the countertop, the first step after donning appropriate personal protective equipment is to:

a flood the area with an appropriate intermediate to high-level disinfectant
b absorb the spill with disposable absorbent material
c evacuate the area for 30 minutes
d clean the area with an aqueous detergent solution

114. The most effective disinfectant recommended for bloodborne pathogens is:

a sodium hypochlorite
b isopropyl alcohol
c chlorhexidine gluconate
d povidone-iodine

115. MLS ONLY Which of the following microbial agents do not respond to the general rules regarding microbial inactivation and decontamination?

a *Mycobacterium tuberculosis*
b transmissible spongiform encephalopathy agents (prions)
c agents of bioterrorism (smallpox, *Bacillus anthracis*)
d *Coccidioides immitis*

116. When processing patient blood specimens and handling other potentially infectious material, the best choice of gloves is:

a reusable utility gloves
b latex gloves only
c single-use and disposable gloves
d cut-resistant gloves


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117. While processing patient specimens, a technologist splashes a few small drops of a bronchial wash specimen on his/her gloves. The first action should be to:

**a** wash the gloves with antiseptic/soap and water
**b** continue to wear the gloves until grossly contaminated or leaving the area
**c** wash the gloves with an appropriate disinfectant
**d** change gloves and wash hands with antiseptic/soap and water

118. This symbol represents:



**a** a biohazard
**b** a radiation hazard
**c** a chemical hazard
**d** an environmental hazard

119. Regulated medical waste refers to:

IBLS ONLY

**a** chemical waste
**b** infectious waste
**c** radioactive waste
**d** all waste from healthcare facilities

120. A laboratory safety program includes engineering controls such as:

**a** biosafety hoods, chemical fume hoods and approved safety policies
**b** appropriate gloves and gowns
**c** sound dampening materials and radiation shielding
**d** good hand hygiene practices

121. The first step to be taken when attempting to repair a piece of electronic equipment is:

**a** check all the electronic connections
**b** reset all the printed circuit boards
**c** turn the instrument off
**d** replace all the fuses

122. Which of the following should be considered when packaging and transporting biological specimens and microorganisms?

**a** specimen and/or organism type and lability and hazardous materials
**b** containers, labels and packaging materials
**c** laws and regulations depending on mode of transport and whether the shipment is local, interstate, or international
**d** current OSHA certification for shipping biological specimens

123. Needles, scalpels, broken glass, and other sharp objects should be handled with extreme caution. Safe sharps practice does not include the following element:

**a** placing used microtome blade into a labeled, puncture-resistant container
**b** disposing of glass bottles in a closed cardboard box to capture shards if broken
**c** use needles with a safety mechanism built into the needle
**d** autoclaving discarded needles to destroy potentially infective agents

124. Laboratories are required to ensure compliance with applicable national, federal, state, and local laws and regulations. One of these requirements relates to public health surveillance notifiable conditions. The requirements vary depending on location. Examples of notifiable conditions include:

**a** infectious diseases, including sexually transmitted disease
**b** environmental exposure, eg, blood lead levels at defined levels
**c** mouth pipetting by laboratory personnel
**d** arthropod vector and animal-transmitted diseases

*B. Emergency Procedures*

125. A technologist spilled 10 gallons of formaldehyde on the floor. After determining the chemical poses a significant health hazard, the first action step would be to:

**a** notify emergency assistance
**b** control the spill with appropriate absorbent material
**c** evacuate the area
**d** don appropriate personal protective equipment

126. According to OSHA, what type of warning sign should be posted in an area where an immediate hazard exists and where special precautions are necessary?

**a** red, black and white "Danger" sign
**b** yellow and black "Caution" sign
**c** green and white "Safety Instruction" sign
**d** orange and black "Biohazard" sign

127. For fire safety and prevention:

**a** fire drills should be announced and practiced in advance
**b** hallways and corridors should be clear and free of obstruction at all times
**c** only one exit is necessary in laboratories that contain an explosion hazard
**d** hazard evaluations only need to be done prior to initiation of clinical operations

128. A fire occurs in the laboratory. The first course of action is to:

**a** evacuate the entire area
**b** pull the fire alarm box
**c** remove persons from immediate danger
**d** contain the fire by closing doors

129. An electrical equipment fire breaks out in the laboratory. Personnel have been removed from immediate danger, the alarm has been activated. What is the next action to be taken?

**a** evacuate the facility
**b** contain the fire by closing doors
**c** extinguish fire with type A extinguisher
**d** lock all windows and doors in the immediate area

130. A technologist splashed a corrosive chemical in his/her eyes. To prevent permanent injury, the first action should be to:

**a** bandage the eyes and seek immediate emergency medical assistance
**b** flush eyes with a chemical of opposite pH to neutralize the injury
**c** use the eyewash station to flush eyes with water for 15 minutes
**d** seek immediate emergency medical assistance

131. A technologist spilled concentrated hydrochloric acid on his/her clothing and skin, affecting a large portion of the body. After removing involved clothing, the next first aid treatment step would be to:

**a** seek immediate emergency medical assistance
**b** use emergency safety shower and flush body with water
**c** apply burn ointment to affected skin
**d** pour baking soda on the skin and bandage

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## III. Laboratory Mathematics

### A. Concentration, Volume, and Dilutions

132. An automated CK assay gives a reading that is above the limits of linearity. A dilution of the serum sample is made by adding 1 mL of serum to 9 mL of water. The instrument now reads 350 U/L. The correct report on the undiluted serum should be:

**a** 2850 U/L
**b** 3150 U/L
**c** 3500 U/L
**d** 3850 U/L

133. A standard solution is a chemical solution that has a precisely known concentration. Standard solution concentration is commonly expressed with which of the following units of measure:

**a** g/L
**b** ml/mL
**c** mg/%
**d** mol/L

134. A glucose determination was read on a spectrophotometer. The absorbance reading of the standard was 0.30. The absorbance reading of the unknown was 0.20. The value of the unknown is:

**a** 2/3 of the standard
**b** 3/5 of the standard
**c** the same as the standard
**d** 1.5x the standard

135. A technician is asked by the supervisor to prepare a standard solution from the stock standard. What is the glassware of choice for this solution?

**a** graduated cylinder
**b** volumetric flask
**c** acid-washed beaker
**d** graduated flask

136. How many mL of red blood cells are to be used to make 25 mL of a 4% red cell suspension?

**a** 0.25 mL
**b** 0.5 mL
**c** 1 mL
**d** 2 mL

137. The volume of 25% stock sulfosalicylic acid needed to prepare 100 mL of 5% working solution is:

**a** 1.25 mL
**b** 5 mL
**c** 20 mL
**d** 50 mL

138. To prepare 25 mL of 3% acetic acid, how much glacial acetic acid is needed?

MLS ONLY

**a** 0.75 mL
**b** 1.5 mL
**c** 3.0 mL
**d** 7.5 mL

**139.** How many grams of sodium chloride are needed to prepare 1 L of 0.9% normal saline?

**a** 0.9
**b** 1.8
**c** 9.0
**d** 18.0

**140.** MLS ONLY To prepare 40 mL of a 3% working solution, a technician would use what volume of stock solution?

**a** 0.9 mL
**b** 1.2 mL
**c** 1.5 mL
**d** 3.0 mL

**141.** MLS ONLY A technician is preparing a 75% solution. What volume of stock solution should be used to prepare 8 mL?

**a** 4.5 mL
**b** 6.0 mL
**c** 7.5 mL
**d** 9.4 mL

**142.** MLS ONLY A new method is being evaluated. A recovery experiment is performed with the results shown in this table:

| Sample mixture | Observed concentration |
|---|---|
| 0.9 mL serum sample + 0.1 mL $H_2O$ | 89 mEq/L |
| 0.9 mL serum sample + 0.1 mL analyte standard at 800 mEq/L | 161 mEq/L |

The percent recovery of the added analyte standard is:

**a** 55%
**b** 81%
**c** 90%
**d** 180%

**143.** When 0.25 mL is diluted to 20 mL, the resulting dilution is:

**a** 1:20
**b** 1:40
**c** 1:60
**d** 1:80

**144.** A serum glucose sample is too high to read, so a 1:5 dilution using saline (dilution A) is made. Dilution A was tested and is again too high to read. A further 1:2 dilution is made using saline (dilution B). To calculate the result, the dilution B value must be multiplied by:

**a** 5
**b** 8
**c** 10
**d** 20

**145.** In performing a spinal fluid protein determination, the specimen is diluted 1 part spinal fluid to 3 parts saline to obtain a result low enough to measure. To calculate the protein concentration, the result must be:

**a** multiplied by 3
**b** multiplied by 4
**c** divided by 3
**d** divided by 4

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146. How many mL of anti-D reagent are needed to prepare 5 mL of a 1:25 dilution?

MLS ONLY

a 0.1
b 0.2
c 0.25
d 0.5

147. If 0.5 mL of a 1:300 dilution contains 1 antigenic unit, 2 antigenic units would be contained in 0.5 mL of a dilution of:

a 1:150
b 1:450
c 1:500
d 1:600

148. A 2% saline erythrocyte suspension contains how many mL of packed erythrocytes per 5 mL of isotonic saline solution?

MLS ONLY

a 0.1
b 0.2
c 0.5
d 1.0

149. A 600 mg/dL glucose solution is diluted 1:30. The concentration of the final solution in mg/dL is:

MLS ONLY

a 2
b 20
c 180
d 1800

150. How many mL of 30% bovine albumin are needed to make 6 mL of a 10% albumin solution?

MLS ONLY

a 1
b 2
c 3
d 4

151. Which of the following is the formula for calculating the dilution of a solution? (V = volume, C=concentration)

a $V1 + C1 = V2 + C2$
b $V1 + C2 = V2 + C1$
c $V1 \times C1 = V2 \times C2$
d $V1 \times V2 = V1 \times C2$

152. A colorimetric method calls for the use of 0.1 mL of serum, 5 mL of reagent and 4.9 mL of water. What is the dilution of the serum in the final solution?

a 1:5
b 1:10
c 1:50
d 1:100

153. Dilution is the process by which the concentration or activity of a given solution is decreased by the addition of a solvent. Which of the following dilution factors would represent 1 mL of serum added to 4 mL of water?

a 1:3
b 1:4
c 1:5
d 1:6

154. Which of the following is the formula for calculating a percent (w/v) solution?

a grams of solute/volume of solvent × 100
b grams of solute × volume of solvent × 100
c volume of solvent/grams of solute × 100
d (grams of solute × volume of solvent)/100

155. A solution contains 20 g of solute dissolved in 0.5 L of water. What is the percentage of this solution?

a 2%
b 4%
c 6%
d 8%

156. How many grams of sulfosalicylic acid (MW = 254) are required to prepare 1 L of a 3% (w/v) solution?

a 3
b 30
c 254
d 300

157. How many mL of a 3% solution can be made if 6 g of solute are available?

a 100 mL
b 200 mL
c 400 mL
d 600 mL

158. The nanometer is a measurement of:

a wavelength of radiant energy
b specific gravity
c density
d intensity of light

### *B. Molarity, Normality*

159. To make 1 L of 1.0 N NaOH from a 1.025 N NaOH solution, how many mL of the NaOH should be used? MLS ONLY

a 950.0
b 975.6
c 997.5
d 1025.0

160. The sodium content (in grams) in 100 g of NaCl (atomic weights: Na = 23.0, Cl = 35.5) is approximately:

a 10
b 20
c 40
d 60

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**161.** Given the results shown in this table, calculate the molar absorptivity:

MLS ONLY

| test | result |
|---|---|
| absorbance | 0.500 |
| light path | 1.0 cm |
| concentration | 0.2 mol/L |

**a** 0.4
**b** 0.7
**c** 1.6
**d** 2.5

**162.** Which of the following is the formula for calculating the gram equivalent weight of a chemical?

**a** MW × oxidation number
**b** MW/oxidation number
**c** MW + oxidation number
**d** MW – oxidation number

**163.** 80 g NaOH (MW = 40) are how many moles?

**a** 1
**b** 2
**c** 3
**d** 4

**164.** A serum potassium (MW = 39) is 19.5 mg/100 mL. This value is equal to how many mEq/L?

**a** 3.9
**b** 4.2
**c** 5.0
**d** 8.9

**165.** Which of the following is the formula for calculating the number of moles of a chemical?

**a** g/GMW
**b** g × GMW
**c** GMW/g
**d** (g × 100)/GMW

**166.** A 1 molal solution is equivalent to:

**a** a solution containing 1 mole of solute per kg of solvent
**b** 1000 mL of solution containing 1 mole of solute
**c** a solution containing 1 GEW of solute in 1 L of solution
**d** a 1 L solution containing 2 moles of solute

**167.** Which of the following is the formula for calculating the molarity of a solution?

**a** number of moles of solute/L of solution
**b** number of moles of solute × 100
**c** 1 GEW of solute × 10
**d** 1 GEW of solute/L of solution

**168.** What is the molarity of a solution that contains 18.7 g of KCl (MW = 74.5) in 500 mL of water?

MLS ONLY

**a** 0.1
**b** 0.5
**c** 1.0
**d** 5.0

169. 25 g NaOH (MW = 40) are added to 0.5 L of water. What is the molarity of this solution if an additional 0.25 L of water are added?

MLS ONLY

a 0.25 M
b 0.50 M
c 0.75 M
d 0.83 M

170. What is the normality of a solution that contains 280 g NaOH (MW = 40) in 2000 mL of solution?

MLS ONLY

a 3.5 N
b 5.5 N
c 7.0 N
d 8.0 N

171. How many g of $H_2SO_4$ (MW = 98) are in 750 mL of 3N $H_2SO_4$?

MLS ONLY

a 36 g
b 72 g
c 110 g
d 146 g

172. How many mL of 0.25 N NaOH are needed to make 100 mL of a 0.05 N solution of NaOH?

a 5 mL
b 10 mL
c 15 mL
d 20 mL

173. A pH of 7.0 represents a $H^+$ concentration of:

MLS ONLY

a 70 mEq/L
b 10 µmol/L
c 7 nmol/L
d 100 nmol/L

### *C. Standard Curves*

174. When the exact concentration of the solute of a solution is known and is used to evaluate the concentration of an unknown solution, the known solution is:

a standard
b normal
c control
d baseline

### *D. Mean, Median, Mode, and Confidence Intervals*

175. The mean value of a series of hemoglobin controls was found to be 15.2 g/dL, and the standard deviation was calculated at 0.20. Acceptable control range for the laboratory is ±2 standard deviations. Which of the following represents the allowable limits for the control?

a 14.5-15.5 g/dL
b 15.0-15.4 g/dL
c 15.2-15.6 g/dL
d 14.8-15.6 g/dL


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176. An index of precision is statistically known as the:

a median
b mean
c standard deviation
d coefficient of variation

177. The term used to describe reproducibility is:

a sensitivity
b specificity
c accuracy
d precision

178. The ability of a diagnostic test to accurately measure only the analyte of interest in the presence of other substances is the:

a specificity
b sensitivity
c precision
d reproducibility

179. If the correlation coefficient ($r$) of 2 variables is 0:

MLS ONLY

a there is complete correlation between the variables
b there is an absence of correlation
c as one variable increases, the other increases
d as one variable decreases, the other increases

180. Which of the following is the formula for standard deviation?

a square root of the mean
b square root of (sum of the squared differences from the mean)/(N–1)
c square root of the variance
d square root of (mean)/(sum of squared differences)

181. The acceptable limit of error in the chemistry laboratory is 2 standard deviations. If you run the normal control 100 times, how many of the values would be out of the control range due to random error?

a 1
b 5
c 10
d 20

182. A mean value of 100 and a standard deviation of 1.8 mg/dL are obtained from a set of glucose measurements on a control solution. The 95% confidence interval in mg/dL would be:

a 94.6-105.4
b 96.4-103.6
c 97.3-102.7
d 98.2-101.8

183. The results shown in this table are obtained from a set of automated white blood cell counts performed on 40 samples: MLS ONLY

| statistic | value |
|---|---|
| standard deviation | 153.2/µL |
| mean value | 12,450/µL |

The coefficient of variation is:

**a** 0.01%
**b** 1.2%
**c** 2.5%
**d** 8.1%

184. The 5 sodium control values shown in this table in unit (mEq/L) are obtained:

140, 135, 138, 140, 142

Calculate the coefficient of variation.

**a** 1.9%
**b** 2.7%
**c** 5.6%
**d** 6.1%

185. The statistical term for the average value is the:

**a** mode
**b** median
**c** mean
**d** coefficient of variation

186. The most frequent value in a collection of data is statistically known as:

**a** mode
**b** median
**c** mean
**d** standard deviation

187. The middle value of a data set is statistically known as the:

**a** mean
**b** median
**c** mode
**d** standard deviation

188. Which of these formulas is used to calculate the arithmetic mean?

**a** square root of the sum of values
**b** sum of values × number of values
**c** number of values/sum of values
**d** sum of values/number of values

189. Given this set of values:

100, 120, 150, 140, 130

What is the mean?

**a** 100
**b** 128
**c** 130
**d** 640


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190. Which of these formulas is used to calculate the coefficient of variation?

**a** (standard deviation × 100)/standard error
**b** (mean × 100)/standard deviation
**c** (standard deviation/mean) × 100
**d** (variance × 100)/mean

191. A cholesterol QC chart has the data shown in the table for the normal control:

| statistic | value |
|---|---|
| X (mean of data) | 137 mg/dL |
| #x | 1918 mg/dL |
| 2 SD | 6 mg/dL |
| N | 14 |

The coefficient of variation for this control is:

**a** 1.14%
**b** 2.19%
**c** 4.38%
**d** 9.49%

*E. Sensitivity, Specificity, and Predictive Value*

192. The precision of an instrument is validated by:

MLS ONLY

**a** running the same sample multiple times
**b** performing serial dilutions
**c** processing unknown specimens
**d** monitoring normal and abnormal controls

193. The extent to which measurements agree with the true value of the quantity being measured is known as:

**a** reliability
**b** accuracy
**c** reproducibility
**d** precision

194. Diagnostic specificity is defined as the percentage of individuals:

**a** with a given disease who have a positive result by a given test
**b** without a given disease who have a negative result by a given test
**c** with a given disease who have a negative result by a given test
**d** without a given disease who have a positive result by a given test

195. Prior to implementing a new lab test, the analytical measurement range (AMR) must be verified. This is to verify a value that can be:

MLS ONLY

**a** directly measured on a specimen without any dilution or concentration
**b** reported after specimen pretreatment
**c** reported up to a 1:100 dilution
**d** sent out to a reference lab for verification

196. The predictive value of a positive test is defined as:

MLS ONLY

**a** (true-positives + true-negatives)/true-positives × 100
**b** true-positives/(true-positives + false-positives) × 100
**c** (true-positives + true-negatives)/true-negatives × 100
**d** true-negatives/(true-negatives + false-positives) × 100

197. The reliability of a test to be positive in the presence of the disease it was designed to detect is known as:

**a** accuracy
**b** sensitivity
**c** precision
**d** specificity

198. Which of the following parameters of a diagnostic test will vary with the prevalence of a given disease in a population?

**a** precision
**b** sensitivity
**c** accuracy
**d** specificity

199. The target in the figure illustrates a set of results that show a high degree of:

MLS ONLY



**a** accuracy
**b** precision
**c** sensitivity
**d** specificity

200. Limit of detection (LoD) is one of several guidelines for the evaluation and verification of detection capability claims of clinical laboratory procedures. LoD is best described as:

**a** a consensus or certified value based on experimental work under the auspices of a scientific group
**b** a situation in which measurement results are simply reported as less than an imposed threshold
**c** the measurand concentration at which precision of a measurement procedure, under stated experimental conditions, meets a stated performance requirement
**d** measured quantity value for which the probability of falsely claiming the absence of an analyte in a material, given a probability of falsely claiming its presence or minimum detectable value of the net state variable

201. Limit of quantitation (LoQ) is best described as:

**a** lowest quantity intended to be measured (concentration) in a material that can be quantitatively determined with stated accuracy under stated conditions
**b** lowest quantity intended to be measured (concentration) at which the measurement procedure displays a linear relationship with actual analyte content
**c** closeness of agreement between measured lower quantity values obtained by replicate measurement
**d** lowest measurement result that is likely to be observed with a stated probability for a blank sample

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## IV. Manual/Automated Methodology and Instrumentation

### A. Basic Laboratory Equipment

**202.** Automated analyzers generally incorporate mechanical versions of manual techniques. Modern instruments are packed in a wide array of configurations. The most common configuration is:

- **a** continuous flow analyzers
- **b** multiple specimen and multiple channel analyzers
- **c** random access analyzers
- **d** total automation analyzers

**203.** Unequivocal positive identification of each specimen can be achieved with analyzers that utilize barcode readers. One advantage of using barcode labels is:

- **a** elimination of manual specimen labeling
- **b** reduction of the number of lost specimens
- **c** reduction in specimen identification errors
- **d** reduction of specimen aliquoting errors

**204.** The analytical measurement range (AMR):

- **a** confirms that the calibrations settings are valid
- **b** establishes the range of values that can be measured with dilution
- **c** can be measured using linearity materials of any matrix
- **d** establishes the range of values that can be measured without dilution

**205.** In most compound light microscopes, the ocular lens has a magnification of:

- **a** 10x
- **b** 40x
- **c** 50x
- **d** 100x

**206.** The best way to lower the light intensity of the microscope is to:

- **a** lower the condenser
- **b** adjust the aperture diaphragm
- **c** lower the rheostat
- **d** raise the condenser

**207.** The advantage to using phase microscopy in urinalysis is to:

- **a** provide higher magnification
- **b** enhance constituents with a low refractive index
- **c** allow constituents to stain more clearly
- **d** provide a larger field of view

**208.** Köhler illumination is:

- **a** a method to ensure optimal contrast and resolution
- **b** a method to magnify objects
- **c** an indication of the type of light source in use
- **d** one method of phase contrast magnification

**209.** Removing the immersion oil from a microscope lens is:

- **a** best accomplished with alcohol pads
- **b** never done
- **c** only required when you are dismantling the objective
- **d** important to prevent oil seeping behind the lens over time

210. What does the black ring on the lower end of a microscope objective lens indicate?

**a** the kind of microscope to use
**b** the magnification of the objective
**c** nothing; it is simply decorative
**d** the objective is used with immersion oil

211. For safe operation of a centrifuge:

**a** clean with soap/detergent when maintenance is performed or spills occur
**b** open the centrifuge cover when it is in the process of slowing down
**c** leave liquid specimen tubes uncovered during centrifugation
**d** ensure proper balance is maintained

212. A centrifuge head has a diameter of 60 cm and spins at 3000 RPM. What is the maximum achievable G force?

($G = 0.0001 \times \text{radius in cm} \times (\text{RPM})^2$)

**a** 1.8 G
**b** 2,700 G
**c** 27,000 G
**d** 90,000 G

213. Which of the following is the best guide to consistent centrifugation?

**a** potentiometer setting
**b** armature settings
**c** tachometer readings
**d** rheostat readings

214. In the proper use of cobalt-treated anhydrous $CaCl_2$, the desiccant should be:

**a** changed when it turns pink
**b** changed when it turns blue
**c** kept in the dark
**d** kept in the cold

215. Which electrochemistry measuring device is used in instruments for blood gas and pH analysis?

**a** potentiometry and amperometry
**b** amperometry and bichromatic spectrophotometry
**c** conductometry and electrophoretic electrode
**d** steric fluorometry and potentiometry

### *B. Spectrophotometry & Photometry*

216. MLS ONLY Absorbance (A) of a solution may be converted to percent transmittance (%T) using the formula:

**a** 1 + log %T
**b** 2 + log %T
**c** 1 – log %T
**d** 2 – log %T

217. MLS ONLY Which of the following is the formula for calculating the unknown concentration based on Beer's law? (A = absorbance, C = concentration)

**a** (A unknown/A standard) × C standard
**b** C standard × A unknown
**c** A standard × A unknown
**d** (C standard)/(A standard) × 100

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218. The methodology based on the amount of energy absorbed by a substance as a function of its concentration and using a specific source of the same material as the substance analyzed is:

**a** flame emission photometry
**b** atomic absorption spectrophotometry
**c** emission spectrography
**d** x-ray fluorescence spectrometry

219. Which of the following wavelengths is within the ultraviolet range?

**a** 340 nm
**b** 450 nm
**c** 540 nm
**d** 690 nm

220. One means of checking a spectrophotometer wavelength calibration in the visible range is by using a:

**a** quartz filter
**b** diffraction grating
**c** quartz prism
**d** didymium filter

221. In spectrophotometry, the device that allows for a narrow band of wavelengths is the:

**a** hollow cathode lamp
**b** monochromator
**c** refractometer
**d** photodetector

222. What is the first step in preparing a spectrophotometer for an assay?

MLS ONLY

**a** adjust wavelength selector
**b** zero with deionized water
**c** read standard absorbance
**d** place a cuvette in the well

223. In a double-beam photometer, the additional beam is used to:

**a** compensate for variation in wavelength
**b** correct for variations in light source intensity
**c** correct for changes in light path
**d** compensate for variation in slit-widths

224. The source of radiant energy in atomic absorption spectrophotometry is:

MLS ONLY

**a** hollow anode lamp
**b** hollow cathode lamp
**c** halogen vapor lamp
**d** deuterium lamp

225. A spectrophotometer is being considered for purchase by a small laboratory. Which of the following specifications reflects the spectral purity of the instrument?

MLS ONLY

**a** photomultiplier tube
**b** dark current
**c** band width
**d** galvanometer

226. A chemistry assay utilizes a bichromatic analysis. This means that absorbance readings are taken at:

MLS ONLY

**a** 2 wavelengths so that 2 compounds can be measured at the same time
**b** 2 wavelengths to correct for spectral interference from another compound
**c** the beginning and end of a time interval to measure the absorbance change
**d** 2 times and then are averaged to obtain a more accurate result

227. A technologist is asked to write a procedure to measure the Evan blue concentration on a spectrophotometer. The technologist is given 4 standard solutions of Evan blue:

Std A = 0.8 mg/dL
Std B = 1.6 mg/dL
Std C = 2.4 mg/dL
Std D = 4.0 mg/dL

The first step is to:

**a** calculate the slope of the calibration curve
**b** determine the absorbance of the 4 standards
**c** find the wavelength of the greatest % transmittance for Evan blue
**d** find the wavelength of the greatest absorbance for Evan blue

228. Which of the following is used to verify wavelength settings for narrow bandwidth spectrophotometers?

**a** didymium filter
**b** prisms
**c** holmium oxide glass
**d** diffraction gratings

229. In a spectrophotometer, light of a specific wavelength is isolated from the light source by the:

**a** double beam
**b** monochromator
**c** aperture
**d** slit

### *C. Mass Spectrometry*

230. A mass spectrometer detects which property of ionized molecules?

**a** column retention time
**b** charge to mass ratio
**c** mass to charge ratio
**d** fluorescence

### *D. Osmometry*

231. Osmometry is a technique for measuring the concentration of solute particles that contribute to the osmotic pressure of a solution. The most common method used in a clinical laboratory to measure osmolality is:

**a** dew point pressure
**b** boiling point
**c** freezing point depression
**d** osmotic pressure

### *E. Electrophoresis*

232. In electrophoretic analysis, buffers:

**a** stabilize electrolytes
**b** maintain basic pH
**c** act as a carrier for ions
**d** produce an effect on protein configuration

233. On electrophoresis, distorted zones of protein separation are usually due to:

MLS ONLY
**a** presence of therapeutic drugs in serum sample
**b** dirty applicators
**c** overloading of serum sample
**d** prestaining with tracer dye

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*F. Chromatography*

**234.** Chromatography is based on the principle of:

**a** differential solubility
**b** gravity
**c** vapor pressure
**d** temperature

**235.** Gel filtration chromatography is used to separate:

MLS ONLY

**a** polar and nonpolar compounds
**b** compounds on the basis of molecular weight and size
**c** isomers of the same compound
**d** compounds on the basis of different functional groups

**236.** An $R_f$ value of 0.5 in thin-layer chromatography means:

MLS ONLY

**a** solute moves twice as far as solvent front
**b** solute moves half the distance of solvent front
**c** solute moves with solvent front
**d** solvent moves half the distance of solute

**237.** An HPLC operator notes that the column pressure is too high, is rising too rapidly and the recorder output is not producing normal peaks. The most probable cause of the problem is:

MLS ONLY

**a** not enough sample injected
**b** bad sample detector
**c** effluent line obstructed
**d** strip chart motor hanging up

**238.** To be analyzed by gas liquid chromatography a compound must:

**a** be volatile or made volatile
**b** not be volatile
**c** be water-soluble
**d** contain a nitrogen atom

**239.** A true statement about column chromatography methods, including high-performance liquid and gas chromatography, is that it:

**a** all utilizes a flame ionization detector
**b** requires derivation of nonvolatile compounds
**c** can be used to separate gases, liquids or soluble solids
**d** can be used for adsorption, partition, ion-exchange and steric-exclusion chromatography

**240.** In thin-layer chromatography, the $R_f$ value for a compound is the:

MLS ONLY

**a** ratio of distance moved by compound to distance moved by solvent front
**b** rate of movement of compound through the adsorbent
**c** difference in distance between the compound spot and solvent front
**d** distance moved by compound from the origin

**241.** The selectivity of an ion-selective electrode is determined by the:

**a** properties of the membrane used
**b** solution used to fill the electrode
**c** magnitude of the potential across the membrane
**d** internal reference electrode

### G. Electrochemistry

**242.** An ion-selective electrode (ISE) measures the:

**a** activity of one ion only
**b** concentration of one ion only
**c** activity of one ion much more than other ions present
**d** activity of only $H^+$ ions

### H. Fluorometry

**243.** Which of the following statements about fluorometry is true?

**a** a compound fluoresces when it absorbs light at one wavelength and emits light at a second wavelength
**b** the detector in a fluorometer is positioned at 180° from the excitation source
**c** fluorometry is less sensitive than spectrophotometry
**d** an incandescent lamp is commonly used in a fluorometer

### I. Nephelometry

**244.** Nephelometers measure light:

**a** scattered at a right angle to the light path
**b** absorbed by suspended particles
**c** transmitted by now-particulate mixtures
**d** reflected back to the source from opaque suspensions

**245.** The measurement of light scattered by particles in the sample is the principle of:

**a** spectrophotometry
**b** fluorometry
**c** nephelometry
**d** atomic absorption

### J. Flow Cytometry

**246.** What is the immunologic method utilized in the flow cytometer?

**a** latex agglutination
**b** enzyme linked immunoassay
**c** immunofluorescence
**d** radioimmunoassay

**247.** In flow cytometry, labeled cells:

**a** scatter the light and absorb fluorescence
**b** absorb fluorescence and emit electronic impulses
**c** scatter the light and emit fluorescence
**d** absorb both fluorescence and light

### K. Molecular Methods

**248.** When quantifying the amount of genomic DNA in a sample by spectrophotometry, an OD 260 of 1.0 corresponds to what concentration of DNA?

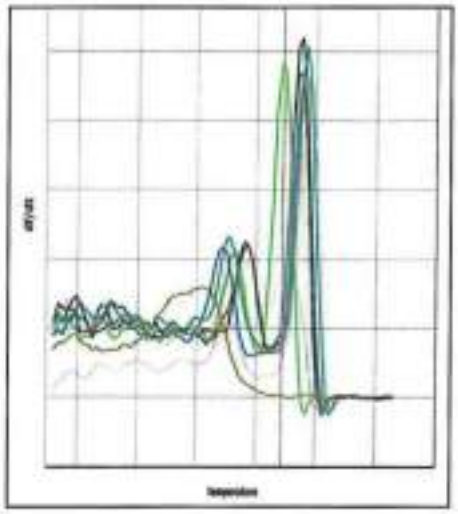
**a** 10 µg/mL
**b** 20 µg/mL
**c** 50 µg/mL
**d** 100 µg/mL

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**249.** MLS ONLY An RNA sample is isolated from peripheral blood cells of a patient. When performing spectrophotometric analysis to determine the yield of RNA in the sample, the 1:40 dilution of the 0.5 mL sample gives an OD 260 reading of 0.03125 and an OD 280 reading of 0.01760. What is the total amount of RNA contained in the 0.5 mL sample?

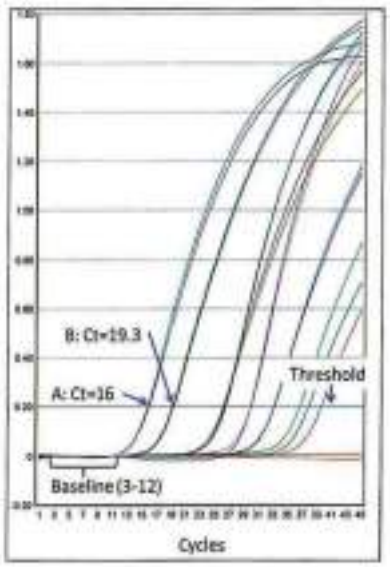
**a** 50 µg
**b** 25 µg
**c** 12.5 µg
**d** 5 µg

**250.** MLS ONLY What process does the fluorography represent?



**a** amplification plot
**b** fragment analysis
**c** melt curve analysis
**d** pyrosequencing

**251.** MLS ONLY The real-time PCR curves represent dilutions of a single sample. Assuming PCR efficiency of 100%, what dilution is represented between aliquot A and aliquot B?




**a** 2-fold
**b** 5-fold
**c** 10-fold
**d** 50-fold

252. A common use for pulsed-field gel electrophoresis is:

MLS ONLY

a DNA fingerprinting
b mitochondrial DNA typing
c epidemiological typing of bacterial strains
d tumor cell phenotyping

253. Which method for quantifying DNA would be most appropriate when testing small FFPE tissue sections by next generation sequencing?

MLS ONLY

a agarose gel
b intercalating dyes
c real-time PCR
d UV adsorption

254. Enzyme-multiplied immunoassay techniques (EMIT) differ from all other types of enzyme immunoassays in that:

a lysozyme is the only enzyme used to label the hapten molecule
b no separation of bound and free antigen is required
c inhibition of the enzyme label is accomplished with polyethylene glycol
d antibody absorption to polystyrene tubes precludes competition to labeled and unlabeled antigen

255. Which of the following statements about immunoassays using enzyme-labeled antibodies or antigens is correct?

a inactivation of the enzyme is required
b the enzyme label is less stable than an isotopic label
c quantitation of the label can be carried out with a spectrophotometer
d the enzyme label is not an enzyme found naturally in serum

256. Which of the following immunoassay labels offer the greatest detection limit?

a fluorescence
b electrochemiluminescence
c radioactivity
d chemiluminescence

### *L. Automated Microbiology Processors*

257. Which of the following is true regarding laboratory identification of mycobacteria?

a rapid testing for tuberculosis is not available because the organism is slow-growing
b automated systems can continuously monitor culture tubes for 6 weeks and report positive results for culture
c specimens must be kept at a low pH because the organisms are acid-fast
d postive automated tests for tuberculosis no longer need to be reported because of low incidence in the United States

258. Which of the following methods is not utilized by automated systems to screen urine for bacteriuria?

a chromatography
b flow cytometry
c particle filtration
d photometry

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259. A patient presents with fever and chills after a dental procedure, and bacteremia is suspected. Which of the following is true of automated systems that can detect bacteria in the patient's blood?

a results are generally available in 2 to 3 days
b specimens must be kept at 4°C from the time of collection until being run on the instrument
c antibiotics taken by the patient before their dental procedure may be removed by the device
d flow cytometry is used to determine the bacterial count

### *M. Hematology Instrumentation*

260. Which of the following values is calculated from the red blood cell indices in an automated hematology analzyer?

a red blood cell count
b hematocrit
c mean corpuscular volume
d red cell distribution width

261. A hematology analyzer counts red blood cells by which method?

a impedance
b chromogenic
c photometric
d turbidimetric

262. A specimen analyzed on an automated hematology instrument has a platelet count of 19 × $10^3$/µL (19 × $10^9$/L). The first procedure to follow is:

a report the count after the batch run is completed
b request a new specimen
c review the stained blood smear
d notify the laboratory manager

263. The electrical resistance method of cell counting requires:

a equal-sized particles
b a conductive liquid
c 2 internal electrodes for current
d 3 apertures for counting

264. On initial start-up of the automated hematology analyzer, one of the controls is slightly below the range for the mean cell volume (MCV). Which of the following is indicated?

a call for service
b adjust the MCV up slightly
c shut down the instrument
d repeat the control

### *V. Basic Management Principles*

265. MLS ONLY The laboratory manager receives a complaint from the ICU about turnaround times for coagulation tests. The first step in problem solving should be:

a gather data on current times by shift
b talk to staff about various solutions
c perform root cause analysis
d draw a process map to send to the ICU explaining why it takes so long

**266.** A technologist repeatedly misses tubercle bacilli when examining stained smears for acid-fast bacilli. What plan of action should the supervisor first take to correct this problem? MLS ONLY

- **a** issue a written warning
- **b** send the employee to a workshop to improve his/her knowledge
- **c** review the diagnostic criteria with the employee and monitor progress
- **d** reassign employee to another part of the laboratory

**267.** Which of the following is considered to be a variable cost in a clinical laboratory? MLS ONLY

- **a** overtime pay
- **b** health insurance premiums
- **c** FICA
- **d** pension contributions

**268.** Direct, indirect and overhead costs incurred during the production of tests per unit time are classified as: MLS ONLY

- **a** total costs
- **b** actual costs
- **c** standard costs
- **d** controllable costs

**269.** An advantage of reagent lease/rental agreements is: MLS ONLY

- **a** less time spent by a laboratory manager justifying new instrumentation
- **b** increased flexibility to change to a new platform as technology changes in workload
- **c** flexibility in reagent usage from 1 manufacturer to another
- **d** less expenditures over life expectancy of instrument

**270.** The number of hours used to calculate the annual salary of a full-time employee is: MLS ONLY

- **a** 1920
- **b** 1950
- **c** 2080
- **d** 2800

**271.** The overtime budget for the laboratory is $38,773, but $50,419 has already been spent. What percent over budget does this represent? MLS ONLY

- **a** 30%
- **b** 70%
- **c** 77%
- **d** 100%

**272.** Matching the content and requirements of the task with the skills, abilities and needs of the worker is a function of: MLS ONLY

- **a** leadership
- **b** job design
- **c** recruitment
- **d** reward systems

**273.** The most important part of any effective behavior modification system is: MLS ONLY

- **a** direct verbal feedback to employees
- **b** increase salary structure
- **c** job enrichment
- **d** tactful discipline


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**274.** MLS ONLY Disciplinary policy is generally developed as a series of steps with each step more strict than the previous. Normally, the first step in the process is to:

- **a** send the employee a warning letter
- **b** send the employee a counseling memo
- **c** counsel the employee verbally
- **d** dismiss less serious infractions

**275.** MLS ONLY A supervisor notices that a technologist mouth pipets liquids when making reagents. The supervisor's best course of action is to:

- **a** allow the technologist to continue this practice as long as it is not done when dealing with specimens
- **b** discuss this problem with the employee immediately
- **c** order a mechanical device (bulb pipet) for employee to use
- **d** compliment the employee on his rapid pipetting technique

**276.** MLS ONLY On repeated occasions, the day shift supervisor has observed a technologist on the night shift sleeping. Which of the following is the most appropriate initial course of action for the day supervisor?

- **a** ignore the repeated incidents
- **b** discuss the incidents with the technologist's immediate supervisor
- **c** notify the personnel department
- **d** advise the laboratory director

**277.** MLS ONLY A workload reporting system is an important part of laboratory management because it:

- **a** tells exactly how much should be charged per test
- **b** keeps personnel busy in their free time
- **c** counts only tests done and specimens received in the laboratory without inflating these figures by adding in quality control and standardization efforts
- **d** helps in planning, developing, and maintaining efficient laboratory services with administrative and budget management

**278.** Which one of the following questions can be legally asked on an employment application?

- **a** Are you eligible to work in the US?
- **b** What is your date of birth?
- **c** Is your wife/husband employed full-time?
- **d** Do you have any dependents?

**279.** MLS ONLY Which of the following topic areas can be discussed with a prospective employee during a job interview?

- **a** have you ever been arrested
- **b** number of dependents
- **c** previous employment that the applicant disliked
- **d** if they are a US citizen

**280.** MLS ONLY In general, the largest category of the operating expenses of laboratories are:

- **a** labor or labor related
- **b** reagents and supplies
- **c** equipment replacement and maintenance
- **d** safety supplies and disposables

**281.** MLS ONLY Legal pre-employment questions on an application are:

- **a** medical history of an employee
- **b** place of birth
- **c** felonies unrelated to job requirements
- **d** name and address of person to notify in case of emergency

**282.** A requirement for a job is to be able to stand for long periods of time. If you have a current employee who comes to work with a doctor's letter stating that they can no longer stand to do their job due to a disability, what is the manager's responsibility?

MLS ONLY

- **a** find the employee a desk job
- **b** make a reasonable accommodation for them to alternate sitting and standing if possible
- **c** you are not legally required to do anything
- **d** let the employee go

**283.** Which of the following is a tool that can be used to follow the progression of a project?

MLS ONLY

- **a** Pareto analysis
- **b** fishbone diagrams
- **c** Gantt charts
- **d** FIFO (first in, first out)

**284.** Which of the following is an indirect cost?

MLS ONLY

- **a** maintenance agreements
- **b** office supplies
- **c** PT test material
- **d** IT support

**285.** Which of the following activities is not under the direction or control of the lab manager?

MLS ONLY

- **a** number of employees
- **b** direct test costs
- **c** skill mix
- **d** military leave

**286.** An advanced beneficiary notice (ABN) is required when:

MLS ONLY

- **a** a test may not be covered by insurance
- **b** when there is no CPT code associated with a test
- **c** when an HMO submits any lab test
- **d** when there is automatic reflex testing for a screen

**287.** A new clinic in the community is sending a very large number of additional chemistry tests to the laboratory. The existing chemistry instrument is only 2 years old and works well; however, there is a need to acquire a high-throughput instrument to manage new testing. Which one of the following is the appropriate "justification category"?

MLS ONLY

- **a** replacement
- **b** volume increase
- **c** reduction of FTEs
- **d** new service

**288.** A general term for the formal recognition of professional or technical competence is:

MLS ONLY

- **a** regulation
- **b** licensure
- **c** accreditation
- **d** credentialing

**289.** Providing the performance feedback of employees should be done:

MLS ONLY

- **a** annually
- **b** semiannually
- **c** as needed in the judgment of management
- **d** in the form of immediate feedback and at regular intervals


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**290.** MLS ONLY Which of the following method is not an acceptable means to assess the competence of testing personnel?

- **a** a quiz to test the knowledge of background information of a given procedure
- **b** direct observation of patient testing and/or instrument maintenance
- **c** acknowledgement that an individual's training is complete
- **d** successful completion of a problem-solving exercise

**291.** MLS ONLY To be effective, constructive feedback should be:

- **a** specific to the behavior
- **b** related to general laboratory performance
- **c** focused on the person, not the behavior
- **d** repeatedly discussed for reinforcement

**292.** MLS ONLY Higher levels of employee motivation occur when the supervisor:

- **a** collaborates to set goals to be accomplished
- **b** provides all the details of the task
- **c** constantly monitors progress
- **d** immediately corrects every error

**293.** MLS ONLY Several complaints have been received from parents of children in the pediatric wing about the anxiety that venipuncture causes their children. An informal staff meeting with the phlebotomists reveals that they feel both parents and pediatric nurses are less than supportive and frequently make the task of venipuncture in children worse with their own anxiety. The best course of action would be to:

- **a** have pediatric nurses do venipuncture on children as they are more familiar with the children
- **b** limit physicians to only one draw per day on children
- **c** prepare written pamphlets for parents and in-service education for nursing personnel
- **d** take no action as parents will always overreact where their children are concerned

**294.** MLS ONLY A major laboratory policy change that will affect a significant portion of the laboratory personnel is going to take place. In order to minimize the staff's resistance to this change, the supervisor should:

- **a** announce the policy change the day before it will become effective
- **b** discuss the policy change in detail with all personnel concerned, well in advance of implementation
- **c** announce only the positive aspects of the policy change in advance
- **d** discuss only the positive aspects of the policy with those concerned

**295.** MLS ONLY When employees are going to be responsible for implementing a change in procedure or policy, the manager should:

- **a** make the decision and direct the employees to implement it
- **b** solicit the employee input but do what he/she thinks should be done
- **c** involve the employees in the decision-making process from the very beginning
- **d** involve only those employees in the decision-making process who would benefit from the change

**296.** MLS ONLY The best way to motivate an ineffective employee would be to:

- **a** confirm low performance with subjective data
- **b** set short-term goals for the employee
- **c** transfer the employee to another department
- **d** ignore failure to meet goals

**297.** A technologist has an idea that would possibly decrease the laboratory turnaround time for reporting results. In order to begin implementation of this idea, he/she should:

MLS ONLY

**a** encourage the staff to utilize the idea
**b** discuss it with his/her immediate supervisor
**c** try out the idea on himself/herself on an experimental basis
**d** present the idea to the laboratory director

**298.** The ability to make good decisions often depends on the use of a logical sequence of steps that include:

**a** defining problem, considering options, implementing decisions
**b** obtaining facts, considering alternatives, reviewing results
**c** defining problem, obtaining facts, considering options
**d** obtaining facts, defining problem, implementing decision

**299.** Delegation is a process in which:

MLS ONLY

**a** interpersonal influence is redefined
**b** authority of manager is surrendered
**c** power is given to others
**d** responsibility for specific tasks is given to others

**300.** What action should be taken when dealing with a long-term problem?

MLS ONLY

**a** ignore the problem
**b** seek more information
**c** base decision on available information
**d** refer the problem to another level of management

**301.** Which of the following actions will facilitate group interactions at staff meetings?

MLS ONLY

**a** adhering strictly to an agenda
**b** treating every problem consistently
**c** encouraging input from all staff
**d** announcing the assignments for upcoming projects

**302.** As information is reported upward through an organization, the amount of detail communicated will generally:

MLS ONLY

**a** decrease to facilitate the flow of information
**b** increase to allow consideration of all options
**c** remain the same to ensure consistency in reporting
**d** remain the same to ensure goal accomplishment

**303.** To sustain the highly motivated employee the evaluation should include:

MLS ONLY

**a** performance feedback
**b** retraining opportunities
**c** quality of performance discussions
**d** competency-based tasks

**304.** Communication is enhanced by:

MLS ONLY

**a** a planned strategy that includes listening skills and ensuring an understanding with questions
**b** relying on e-mail, memos and voice mail to communicate new information
**c** formal, hierarchical patterns instead of informal networking patterns
**d** assumptions if there are questions about the intent of the message


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**305.** Identify the first step a laboratory manager must take in the selection of a laboratory information system. MLS ONLY

- **a** write a request for proposal (RFP)
- **b** select a computer vendor
- **c** select an LIS team
- **d** decide on services needed

**306.** Laboratory results can be sent to a hospital's electronic medical record by: MLS ONLY

- **a** autofax
- **b** HL-7 interface
- **c** internet routing
- **d** backup server

**307.** CODE 128, ISBT 128, CODE 39 and interleaved 2 of 5 symbologies are used by laboratory information systems to create which of the following? MLS ONLY

- **a** barcode labels
- **b** worklists
- **c** instrument download files
- **d** patient reports

**308.** A standard electronic file format recommended for transmitting data from the laboratory information system to an electronic medical record is: MLS ONLY

- **a** Health Level 7
- **b** ISBT 128
- **c** FTP
- **d** SNOMED

**309.** The Hematology laboratory is evaluating new instruments for purchase. The supervisor wants to ensure that the instrument they select has bidirectional interface capabilities. The instrument specification necessary to meet this requirement is: MLS ONLY

- **a** 9600 baud rate
- **b** on-board test selection menu
- **c** HL-7 file format
- **d** host query mode

**310.** The Chemistry department requests that a new test be defined in the LIS to run on the existing analyzer. The new test set up is completed by the LIS coordinator. A few days later, the accessioning department receives a request for the new test but an error is displayed when they try to place the order. All other tests can be successfully ordered. The most likely cause of the error is the: MLS ONLY

- **a** instrument interface for the chemistry analyzer is down
- **b** test is not defined on the chemistry worklist
- **c** database is not properly updated with the new test information
- **d** ADT interface with the hospital system is down

**311.** The process of testing and documenting changes made to a laboratory information system is known as: MLS ONLY

- **a** validation
- **b** quality engineering
- **c** customization
- **d** hazard analysis

312. MLS ONLY Performance of laboratory information system back-up procedures does not include which of the following?

**a** creating an exact copy of LIS data
**b** off-site storage of the data media
**c** shutting down the LIS and bringing it back up
**d** completion at regularly defined intervals

313. MLS ONLY A large hospital has implemented an outreach program, which will involve processing samples received with a variety of barcode labels that are not compatible with the hospital's LIS. The laboratory does not have the staff to re-label every specimen received from their outreach clients. Which of the following solutions would be the most appropriate for the LIS coordinator to consider?

**a** mandate that the outreach clients modify their LIS to print compatible barcode labels
**b** purchase a middleware product to manage samples containing different barcode formats
**c** install a new LIS in the hospital that is compatible with the outreach clients' systems
**d** manually process all samples received from the outreach clients

314. MLS ONLY Research has shown that one of the most vital ways to promote a culture of safety in healthcare is to:

**a** hire an outside consulting firm to examine the current state and make recommendations
**b** encourage front line staff to engage leadership in the process
**c** encourage open reporting and transparent communication of mistakes
**d** observe and track compliance of hand hygiene policies

315. MLS ONLY Equity, diversity and inclusion strategies will help an employer hire:

**a** the most qualified candidates regardless of an individual's background
**b** staff into an environment where a variety of views, beliefs and values are integrated
**c** employees who value equality
**d** a variety of staff to meet quotas

## *VI. Education Principles*

316. MLS ONLY An effective program of continuing education for medical laboratory personnel should first:

**a** find a good speaker
**b** motivate employees to attend
**c** determine an adequate budget
**d** identify the needs

317. MLS ONLY The objective, "The student will be able to perform daily maintenance on the Hematology analyzer" is an example of which behavioral domain?

**a** psychomotor
**b** affective
**c** intellectual
**d** cognitive

318. MLS ONLY The first step in the development of long-term objectives for a laboratory continuing education program must include:

**a** total cost of the program
**b** total number of hours in the program
**c** a list of topics to be covered
**d** a statement of competencies to be achieved


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**319.** Given the following objective:

MLS ONLY "After listening to the audioconference, the student will be able to describe the interaction between T and B lymphocytes in the immune system, to the satisfaction of the instructor."

Which of the following test questions reflects the intent of this objective?

**a** how are T and B lymphocytes separated *in vitro*?
**b** how many T lymphocytes does a normal person have in peripheral blood?
**c** what are the morphological characteristics of B lymphocytes?
**d** how are antibodies produced after a viral infection?

**320.** MLS ONLY A course of instruction is being planned to teach laboratory employees to recognize, troubleshoot and correct simple malfunctions in selected laboratory instruments. In writing the objectives for this course, which one of the following would be most appropriate?

**a** learn how to repair 9 of 10 simple instrument malfunctions
**b** correctly answer 9 of 10 test questions dealing with simple instrument malfunctions
**c** recognize, detect and correct 9 of 10 simple instrument malfunctions
**d** document corrective action procedure for 9 of 10 simple instrument malfunctions

**321.** MLS ONLY In planning an instructional unit, the term "goal" has been defined as a:

**a** plan for reaching certain objectives
**b** set of specific tasks
**c** set of short- and long-term plans
**d** major purpose or final desired result

**322.** MLS ONLY The goal of training is to ensure that an employee gains the skills to apply knowledge to assigned work. Which of the following should not be considered when selecting the appropriate time to train?

**a** new employees for job processes and procedures
**b** all employees when a new instrument is put into testing production
**c** when laboratory workload is sufficiently low that turnaround time is not affected
**d** when training needs are identified

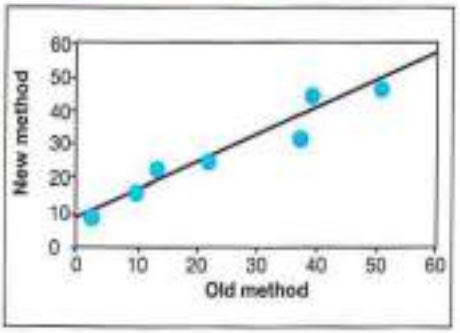
**323.** MLS ONLY Staff competency needs to be assessed at which of the following times?

**a** when there is a temporary decrease in test volume that increases capacity to perform competency assessment
**b** before training a new employee
**c** when an employee's demonstrated competence does not meet criteria
**d** periodically based on staffing adequacy

## I. Quality Assessment/Troubleshooting

### A. Preanalytical, Analytical, Postanalytical

1. **d** Preanalytical (ie, preexamination) variables include all steps in the process prior to the analytical phase of testing, starting with the physician order. Examples include accuracy of transmission of physician orders, specimen transport and preparation, requisition accuracy, quality of phlebotomy services, specimen acceptability rates, etc. The variables chosen should be appropriate to the laboratory's scope of care.
[QCMLS 2021: 7.6.1.2]




2. **b** The y-intercept, which should be close to 0 if the methods are comparable, provides an assessment of constant bias.
[QCMLS 2021: 7.4.2.3]

3. **d** A specimen labeling error is classified as preanalytical (preexamination) error.
[QCMLS 2021: 7.6.2]

4. **d** Vigorous mixing can cause hemolysis.
[QCMLS 2021: 7.6.1.2] [CLSI GP41 Collection of Diagnostic Venous Blood Specimens, p37; 4.4 Hemolysis; Clinical and Laboratory Standards Institute document H3-A6 Procedures for the Collection of Diagnostic Blood Specimens by Venipuncture 6e, p16]

5. **b** Reference intervals represent an important diagnosis and therapeutic management decision tool used to interpret test results.
[QCMLS 2021: 7.6.1.8] [CLSI C28-A3c Defining, Establishing, and Verifying Reference Intervals in the Clinical Laboratory, vii]

6. **b** It is common practice to share samples between the microbiology department and urinalysis. Ideally, the culture is set up first to prevent contamination. If that is not feasible, the sample should be aliquotted using aseptic technique, and refrigerated until it can be cultured.
[QCMLS 2021: 7.6.1]

7. **c** Evaluate each statement. **a** is incorrect because these cells don't agglutinate. **b** is partially correct, but urobilinogen decreases in light. **c** is true, bacterial overgrowth does lead to an alkaline urine. **d** is false, ketones are produced by fat metabolism in the patient.
[Strasinger 2014, p31]

8. **b** Storage must inhibit bacterial growth but not kill the bacteria. Freezing and additives are not acceptable. The most commonly used method of preservation is refrigeration.
[Strasinger 2014, p31]

9. **d** All are examples of preanalytical (preexamination) error except for **d**, which is a postanalytical (postexamination) error.
[Plaut 2009]

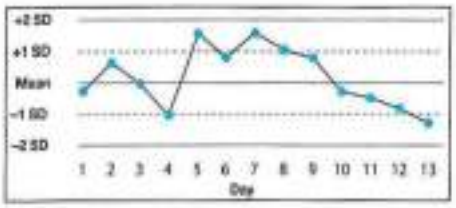
10. **c** Providers are completely dependent on the laboratory for the detection and correction of analytical errors.
[QCMLS 2021: 7.6.1]

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11. **d** Dilution errors, calibration failures, and PPE compliance occur in the analytical (examination) phase of testing. Only order entry occurs in the preanalytical (preexamination) phase.
[CAP 2009, GEN 20316]

## *B. Quality Control*

12. **a** If multiple controls are out of range and the instrument and reagents are verified, recalibration or calibration verification is required before subsequent control analysis.
[QCMLS 2021: 7.5.2.1]




13. **b** Repeating a QC measurement on a new sample of QC material may establish that the alert was caused by a deteriorated QC material rather than a method problem.
[QCMLS 2021: 7.5.2.1]

14. **c** The probability of an observation having a value within ±2 standard deviations of the mean in a normal distribution is 95.5%.
[QCMLS 2021: 7.3.1.1]

15. **c** Repeating a QC measurement on a new sample of QC material may establish that the alert was caused by a deteriorated QC material rather than a method problem.
[QCMLS 2021: 7.5.1.1]

16. **c** Standard deviation is a measure of the dispersion of data around the mean.
[QCMLS 2021: 7.3.1.1]

17. **d** Risk assessment, quality control plan, and quality assessment comprise the 3 steps of an individualized quality control plan.
[QCMLS 2021: 7.5.2.2] [Developing an IQCP, A Step-by-Step Guide. https://www.cms.gov/Regulations-and-Guidance/Legislation/CLIA/Individualized_Quality_Control_Plan_IQCP.html, p201]

18. **c** The delta check method compares current results from automated analyzers with the result from the most recent, previous values for the same patient.
[QCMLS 2021: 7.6.1.1]

19. **d** Standard deviation is a reflection of variance around the mean. In this case, ± 2 SD is 0.4 g/dL below the mean to 0.4 g/dL above the mean or an acceptable range of 0.8 g/dL. The difference in the patient's hemoglobin determinations on Monday (11.3) and Tuesday (11.8) is 0.5 g/dL, which is within the acceptable ± 2 SD range of 0.8 g/dL.
[QCMLS 2021: 7.3.1.1]

20. **c** Calibration verification involves the use of several specimens with known concentrations (patient samples, commercial calibrators, proficiency testing material, controls) to ensure the validity of the calibrator over a wider range of results. Calibration is analogous to "zeroing" a scale, and calibration verification is analogous to then testing the scale using a series of known weights. Calibration verification must be carried out at least once every 6 months and more frequently under some circumstances, eg, after major preventive maintenance.
[QCMLS 2021: 7.4.2.1]

21. **d** Specimen stability refers to the length of time that a stored specimen will continue to produce reliable results, and may be assessed for specimens stored at room temperature, at refrigerator temperature, and/or for frozen specimens. To determine, patient samples are tested immediately and at intervals under the defined storage conditions. Maximum storage time is defined as the last time before significant, predefined, variation is noted. For example, one may choose to use ±2 SD or ±3 SD as defining significant variation.
[QCMLS 2021: 7.4.2.7]

22. **b** The elements of a written laboratory procedure are test principle; patient preparation; specimen collection, labeling, handling & transport; specimen storage, preservation & stability; criteria for specimen acceptability; referral instructions; reagents; procedures for microscopic examinations; step-by-step test procedure; reportable range; test calibration; quality control (QC) procedures; steps to be taken when calibration or QC fails; limitations of procedure; reference range; critical values; results reporting & calculations (if any); and references. Frequency of test performance, reagent preparation, and list of individuals authorized to perform are not specified in the laboratory procedure manual.
[QCMLS 2021: 17.3]

23. **a** A measurement procedure's performance capability relative to the quality required and validation data, interlaboratory quality control data, and critical control point quality control are all particular methods used in QC, but measuring accuracy, precision and reproducibility of tests/test systems are the core foci of QC in the laboratory.
[QCMLS 2021: 7.5.1.5]

24. **d** The results of "out-of-control" runs should not generally be reported.
[QCMLS 2021: 7.5.2]

### C. Point-of-care Testing (POCT)

25. **c** The key advantage to POCT is faster turnaround time.
[QCMLS 2021: 7.7.12]

26. **b** Point-of-care testing devices use a small volume of sample and are therefore particularly useful in neonates, small babies, and those with increased risk from phlebotomy.
[QCMLS 2021: 7.7.12]

27. **b** Laboratory testing performed by nonlaboratory personnel outside of the clinical laboratory, close to or near the patient is defined at point-of-care testing (POCT). Performing laboratory testing at the point of care eliminates the time required to transport specimens to the laboratory, and thus reduces the total turnaround time for testing. POCT is typically performed by nonlaboratorians and nearly always subject to regulatory and accreditation standards that vary, depending on the location and test complexity.
[QCMLS 2021: 7.7.12]

### D. Compliance

28. **b** The official CMS system for assignment of codes for clinical conditions and diseases associated with consequent medical procedures, including laboratory testing.
[QCMLS 2021: 7.1.4] [www.cms.gov/Medicare/Coding/ICD10/index.html]

29. **b** HIPAA patient confidentiality requirements state that patient data is secure from unauthorized access at all levels of communication of that data.
[QCMLS 2021: 7.1.8]

30. **c** A laboratory must have a current CLIA certificate to submit claims to Medicare or Medicaid. Medicare part A helps cover inpatient medical care. Medicare part B helps cover outpatient services & inpatient physician services according to a fee schedule (fee for service). Prescription drug coverage is paid under part D. Medicaid is a federal program that is administered by states.
[QCMLS 2021: 7.1.3]

31. **b** The Health Insurance Portability and Accountability Act (HIPAA) was developed in 1996 and became part of the Social Security Act and does not involve medical billing; The primary purpose of the HIPAA rules is to protect health care coverage for workers who lose or change jobs. It protects records that can be retrieved by personal identifiers and prohibits disclosures without written consent unless a specified exception applies. The International Coding or Classification of Disease (ICD) is a coding system specific to diagnosis or a medical problem used to establish medical necessity. Current procedural terminology (CPT) is the coding system that describes procedure(s) performed. The Health Care Procedural Coding System (HCPCS) describes blood products, medical goods, or services.
[QCMLS 2021: 7.1.4, 7.1.8]



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*E. Regulation*

32. **b** The Joint Commission requires 2 patient identifiers when providing care, treatment, or service.
[QCMLS 2021: 7.6.1.1]

33. **d** National Labor Relations Act (NLRA) includes the right to form or join unions, freedom to bargain collectively with the employer and the right to engage in group activity.
[QCMLS 2021: 7.1.10]

34. **c** The CDC, HCFA and FDA are organizations created by the US federal government. The American College of Surgeons (ACS) adopted a one page document titled "The Minimum Standard for Hospitals" in 1919. In 1950 multiple medical professional organizations joined ACS as corporate members to create the Joint Commission on Accreditation of Hospitals to provide voluntary accreditation. Today "The Joint Commission" sets standards by which healthcare is measured and accredits institutions and laboratories worldwide.
[QCMLS 2021: 7.8.2] [TJC history 2021]

35. **c** Accreditation is the approval of an institution or program based on a review, by one or more independent examiners, that finds that specific requirements or predetermined standards are met.
[QCMLS 2021: 7.1.1]

36. **d** CLIA regulations include federal standards applicable to all U.S. facilities or sites that test human specimens for health assessment or to diagnose, prevent, or treat disease. Exceptions to the CLIA regulations exist for certain testing, including employment-related drug testing by SAMSHA certified laboratories, testing performed for forensic purposes (criminal investigations), and research or surveillance testing performed on human specimens in which patient-specific results are not reported (if the results are not used for diagnosis or treatment decisions).
[QCMLS 2021: 7.1.1] [https://www.cdc.gov/clia/about.html]

37. **b** The FDA categorizes diagnostic tests by their complexity—from the least to the most complex: waived tests, moderate complexity tests, and high complexity tests. Diagnostic tests are categorized as waived based on the premise that they are simple to use, and there is little chance the test will provide wrong information or cause harm if it is done incorrectly.
[QCMLS 2021: 7.1.1.1] [42 CFR 493.17]

38. **d** CAP All Common Checklist: "The laboratory assesses its performance on PT challenges that were ungraded."
[QCMLS 2021: 7.5.3.1] [CAP 2020, COM.01100]

39. **a** National Patient Safety Goal (NPSG) 02.03.01: The [organization] measures, assesses, and, if needed, takes action to improve the timeliness of reporting and the timeliness of receipt of critical tests and critical results and values by the responsible licensed caregiver.
[QCMLS 2021: 7.6.2.3] [NPSG 2009, 02.03.01]

40. **d** For tests for which CAP does not require enrollment in PT, the laboratory must at least semi-annually 1) participate in external PT, or 2) exercise an alternative performance assessment system for determining the reliability of analytical testing. Appropriate alternative performance assessment procedures include participation in an external PT program; participation in an ungraded/educational PT program; split sample analysis with another laboratory, split sample analysis with an established in-house method, use of assayed materials, clinical validation by chart review, or other suitable and documented means as determined by the medical laboratory director.
[QCMLS 2021: 7.5.3.2] [CAP 2020, COM.10500]

41. **a** CAP General Checklist: "Refrigerator/freezer temperatures checked and recorded daily using a calibrated thermometer."
[QCMLS 2021: 7.5.1.4] [CAP 2020, GEN.41042]

42. **d** CAP General Checklist: The paper or electronic report includes the following elements:
- Name and address of testing laboratory (see note below)
- Patient name and identification number, or unique patient identifier
- Name of physician of record, or legally authorized person ordering test, as appropriate
- Date and time of specimen collection, when appropriate
- Date of release of report (if not on the report, this information should be readily accessible)
- Time of release of report, if applicable (if not on the report, this information should be readily accessible)
- Specimen source, when applicable
- Test result(s) (and units of measurement, when applicable)
- Reference intervals, as applicable (see note below)
- Conditions of specimen that may limit adequacy of testing.

[QCMLS 2021: 7.6.2.2] [CAP 2020, GEN.41096]

43. **c** CAP General Checklist: There are written procedures for handling and cleaning glassware, including methods for testing for detergent removal and actions to be taken if detergent residue is detected.
[QCMLS 2021: 7.5.1.4] [CAP 2020, GEN.41770]

44. **a** HIPAA protects health insurance coverage for workers and their families when they change or lose their jobs, and also addresses the security and privacy of health data.
[QCMLS 2021: 7.1.8]

45. **c** HIPAA protects health insurance coverage for workers and their families when they change or lose their jobs, and also addresses the security and privacy of health data.
[QCMLS 2021: 7.1.8]

46. **b** IQCP provides a framework for customizing a QC program for a laboratory's test systems and unique environment. An IQCP requires: Risk Assessment (RA), Quality Control Plan (QCP), and Quality Assessment (QA).
[QCMLS 2021: 7.5.2.2]

47. **c** Newly hired, transferred or promoted personnel, and employees who demonstrate repeated performance issues require training under QMS, as well as when new testing platforms are implemented. A change in the medical director of the laboratory would not necessitate retraining of existing employees.
[QCMLS 2021: 7.5.1.2, 7.4.2.10] [CLIA QMSO3 Training and Competence Assessment 4e, p22]

48. **c** OSHAs Occupational Exposure to Hazardous Chemicals in Laboratories standard (29 CFR 1910.1450), specifies the mandatory requirements of a chemical hygiene plan to protect laboratory workers from harm due to hazardous chemicals. The plan includes the policies, procedures and responsibilities that protect workers from the health hazards due to the hazardous chemicals used in a particular work environment.
[QCMLS 2021: 7.8.1.1] [29 CFR 1910.1450]

49. **b** CAP All Common Checklist: "The use of manufacturer inserts is not acceptable in place of a procedure manual."
[QCMLS 2021: 7.4.2.9] [CAP 2020, COM.10000]

50. **d** Patient results can be automatically released to the patient record as long as they meet selected criteria defined and validated by the users. Auto-verification does not apply to manually entered results and results cannot be released unless QC has been performed and was within limits prior to releasing.
[QCMLS 2021: 7.4.2.8] [CLSI 2006, AUTO 10-A]

51. **c** The CAP requires documentation that calculated values that generate a patient report are reviewed every 2 years, or when a system change is made that may affect the calculations. This requirement applies to values calculated by the LIS or middleware.
[QCMLS 2021: 7.4.2.8] [CAP 2020, GEN.43450]

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52. **c** The sample must not be sent out of the laboratory, even if under normal circumstances, confirmatory testing might be sought outside the laboratory. There must be no communication with other laboratories that do not share the same CLIA certificate prior to submitting PT results. After the results have been submitted, however, even successful PT performance can be a source of useful information. If bias, drift, or shift implied by the reported SDs, Cembrowski & colleagues recommend a Westgard-like method of evaluating the data.
[QCMLS 2021: 7.5.3.1]

53. **c** Proficiency testing providers have a large data set of enrollee's results to develop the target value and the maximum tolerable deviation from that target value. Laboratories using AAP procedures do not have that luxury, and must determine their own unique criteria for acceptable performance.
[QCMLS 2021: 7.5.3.2]

54. **c** The FDA considers analyte-specific reagents (ASRs) to be medical devices and therefore subject to its regulation; under this authority, it published "the ASR rule" in 1997. Reports of test results must include the statement, "This test was developed, and its performance characteristics determined by (laboratory name). It has not been cleared or approved by the US Food and Drug Administration."
[QCMLS 2021: 7.1.2.1]

55. **a** The False Claims Act prohibits the submission of false or fraudulent claims for medical services to a federal government health care program. Many billing errors are simply unintentional mistakes. Ignoring mistakes, choosing not to fix them, and failing to prevent them from recurring could be considered a false claim by government payers such as Medicare and Medicaid, and the health care organization risks the loss of funding from these programs and significant financial penalties. Waste refers to practices that result in unnecessarily increased costs because of mismanagement or overuse of medical services.
[QCMLS 2021: 7.1.5]

56. **d** The Department of Transportation (DOT), United States Post Office (USPS), and International Air Transport Association (IATA) all have different requirements for packaging and shipping of biological specimens, infectious materials, and hazardous materials.
[QCMLS 2021: 7.1.6]

57. **d** Title VII of the Civil Rights Act of 1964 (Title VII) states that it is illegal to discriminate against someone on the basis of race, color, religion, national origin, sex, sexual orientation, or gender identity. The law requires that employers reasonably accommodate applicants' and employees' sincerely held religious practices.
[QCMLS 2021: 7.1.11.2.1]

58. **c** Method validation consists of experiments to prove both the clinical value of a test and performance characteristics. Verification involves the following elements: precision, accuracy, reportable range, reference intervals, analytical sensitivity, and analytical specificity. Validation has all the components of verification and also has greater statistical power (usually involves larger sample numbers), and assessment of additional parameters (clinical sensitivity, clinical specificity, and positive predictive value).
[QCMLS 2021: 7.4.1, t7.2]

59. **a** Precision refers to the reproducibility of a result. Precision is verified by replicating measurements on the same specimens, both simultaneously (within run precision) and consecutively (between run precision).
[QCMLS 2021: 7.4.2.2]

60. **d** Reference ranges or reference intervals are established based on where 95% of healthy individuals would fall. Reference intervals must be stratified based on population-dependent variables such as age, sex, and in some cases, ethnicity. A reference population is a group of healthy individuals demographically comparable to the patient population. CLSI recommends testing at least 120 such individuals, applying exclusion criteria as appropriate, and controlling preanalytical variables. The reference interval may be taken as the central 95% of values or as all values falling within +2 SD and –2 SD. Some analytes (eg, serum prostate specific antigen) have no relevant lower limit.
[QCMLS 2021: 7.3.1.8, 7.3.1.9]

## *II. Safety*

### *A. Safety Programs & Practices*

61. **c** SDS is an OSHA required document that provides information such as physical data (eg, melting point, boiling point, flash point), toxicity, health effects, first aid, reactivity, storage, disposal, protective equipment, and spill/leak procedures.
[QCMLS 2021: 7.8.1.1, 7.8.10]

62. **d** Flammability appears in the top red quadrant. Reactivity is the right yellow quadrant, special reactivity is the lower white quadrant. and health is the left blue quadrant.
[QCMLS 2021: 7.8.11]

63. **a** The correct answer is high health hazard. The ratings range for both systems is 0-4 with 0 being no hazard and 4 being the most severe hazard. The other choices are incorrect due to the hazard type or number rating.
[QCMLS 2021: 7.8.11]

64. **b** Formaldehyde is a carcinogen. The other choices are hazardous chemicals, but are not considered to be potential or actual carcinogens.
[QCMLS 2021: 7.8.11]

65. **d** Gas cylinders must be secured upright to the wall or other stable source. The other distractors are incorrect practices. Compressed gas tanks should be stored away from flammable materials, have safety covers on when pressure regulators are unattached and transported chained to a hand cart or dolly.
[QCMLS 2021: 7.8.4.5]

66. **c** Diethyl ether forms an explosive peroxide upon long-term contact with atmospheric oxygen; while the other choices do not.
[QCMLS 2021: 7.8.4.5]

67. **c** The ratings range for both systems is 0-4, with 0 being no hazard and 4 being the most severe hazard. The left blue quadrant represents a health hazard and a toxic chemical causes a health hazard. The rating of 1 would be a low or slight health hazard. The other choices are incorrect due to the hazard type or rating.
[QCMLS 2021: 7.8.1.1]

68. **d** Corrosive materials cause immediate damage to human tissue, such as a burn. Carcinogens and toxic substances are health hazards, but generally do not cause immediate tissue damage. Ignitable chemicals are both flammable and combustible and only will cause tissue damage if accidental flame or explosion occurs.
[QCMLS 2021: 7.8.1.1]

69. **c** Hazard warnings related to effect on involved target organs must be included on the label. The other distractors are not required to be listed on the label of hazardous chemicals.
[QCMLS 2021: 7.8.1.1]



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70. **d** Substances transferred to a secondary container for immediate use do not require labeling. The other choices are incorrect and are only required on the original containers or secondary containers that are not used immediately or by a different person.
[QCMLS 2021: 7.8.1.1]

71. **b** Chlorine and ammonia combine to form an extremely toxic hydrochloric gas. The other choices represent compatible storage combinations.
[QCMLS 2021: 7.8.6]

72. **d** Xylene must be disposed of as an EPA hazardous waste through a licensed waste hauler. Flushing down the sink and deliberate evaporation are prohibited as unsafe disposal and work practices. Xylene is not a regulated medical waste. However, if regulated medical waste and chemicals are mixed together in processing, the resulting waste should be disposed of as incinerated regulated medical waste.
[QCMLS 2021: 7.8.1.2]

73. **d** The first course of action should be to report the situation to a supervisor, which always is the first step in any potential workplace exposure of any kind. It is the employer's responsibility to provide a medical consultation and evaluation free of charge to employees, who are involved in potential workplace exposures. Early recognition, evaluation and treatment is key to preventing further exposure and exacerbation of symptoms. Acquiring the SDS would be part of the supervisor-directed investigation and secondary to reporting the situation to supervisor.
[QCMLS 2021: 7.8.3] [OSHA 2004]

74. **c** The medical consultation and evaluation of a chemical exposure should be paid by the employer on work time without loss of pay.
[QCMLS 2021: 7.8.3]

75. **a** If results of initial or baseline chemical exposure monitoring required by OSHA are within permissible exposure limits, repeat monitoring should be performed when procedures or equipment surrounding use of the specific chemical change. If the baseline monitoring was not within permissible exposure or action limits initially, periodic subsequent monitoring would be required. However, since the permissible limits in this case were not exceeded, subsequent monitoring only needs to be performed again if changes are made. Therefore, the other distractors are incorrect.
[QCMLS 2021: 7.8.3]

76. **d** Neoprene or nitrile gloves, which are chemically resistant, are an example of personal protective equipment. Latex or vinyl gloves are not chemically resistant. Eyewash stations, safety showers, and fume hoods are engineering controls and not personal protective equipment.
[Gile 2004, p54]

77. **c** A written laboratory plan must designate a laboratory chemical hygiene officer. It should not usually require annual medical evaluations, prohibit carcinogen use, or schedule biannual chemical monitoring.
[QCMLS 2021: 7.1.9.1]

78. **d** The OSHA Hazard Communication requires that personnel are provided with information regarding workplace dangers of hazardous chemicals.
[QCMLS 2021: 7.8.1.1]



79. **a** Healthcare workers must be knowledgeable about chemical safety signage as they may be using chemicals as preservatives in specimens for transport. The flammable sign is from the Department of Transportation Hazardous Materials Warning Signs.
[QCMLS 2021: 7.8.1.1]

80. **b** It is recommended to use a mechanical device when working with sharp equipment and objects. Double-glove technique and use of gauze or paper towel as a barrier provide inadequate protection against needlestick injuries. The two-handed technique leads to more recapping injuries than does using a one-handed technique.
[QCMLS 2021: 7.8.4.2]

81. **b** When working with liquid nitrogen, one should shield all skin and use a face shield. Cryogenic gloves, such as Zetex gloves, should be used, because chemically-resistant gloves are not protective against liquid nitrogen. Cylinders should be stored near good ventilation in an upright position, not horizontally.
[QCMLS 2021: 7.8.4.5]

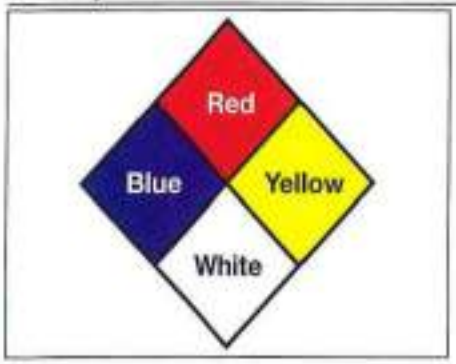
82. **c** Incident reports should be filed for all incidents, including near miss incidents. All incidents require an incident report be filed, not only those resulting in serious injury. The employee's medical history is confidential, and should not be required. Incident reports need to be maintained for a defined period of time even after they have been reviewed.
[QCMLS 2021: 7.1.9]

83. **b** All laboratory instruments should be grounded or double insulated. Repairs should not be conducted while connected to facility wiring. Safety checks should be performed initially, annually and whenever repairs are made. Multiple outlet adapters are unsafe and should not be used in the laboratory.
[QCMLS 2021: 7.8.7]

84. **d** A technologist may touch a "contaminated" area phone with ungloved hands provided hands are washed afterward. Bench tops are to be cleaned at the end of each shift as well as after spills; weekly cleaning is inadequate. Personal protective equipment, such as a lab coat, should be removed prior to leaving a "contaminated" area. Cosmetics and lip balm must not be applied in a "contaminated" area.
[QCMLS 2021: 7.8.11.1, 7.8.9.1]




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**85.** **c** The symbol in the figure represents a common labeling system for chemical hazardous material identification, such as HMIS® or NFPA 704. All other choices are incorrect. Physical environmental hazards and transmission-based precautions do not use the NFPA hazard label at all, and there is more appropriate signage for radiation. If a chemical is radioactive, there would be a radioactive symbol in the bottom white area.
[QCMLS 2021: 7.8.11]

**86.** **d** Containers of 5 gallons or more or flammable or combustible liquids should be stored in an approved safety can. Flammable safety cabinets should not be vented to room air. Flammable and combustible liquids that require refrigeration are only to be stored in an explosion-proof refrigerator. Hazardous substances, particularly of that volume, should not be stored in a fume hood.
[QCMLS 2021: 7.8.4.5]

**87.** **b** The first step is to pull the pin. The mnemonic "PASS" indicates the correct sequence of actions: 1) **p**ull the pin, 2) **a**im the hose, 3) **s**queeze the lever, and 4) **s**weep the flow.
[QCMLS 2021: 7.8.5]

**88.** **d** Only UL or other safety agency-rated electrical equipment should be used in healthcare facilities. Multiple outlet or gang plug adapters are not to be used. No requirement exists to change circuit breakers annually. Worn wiring must be replaced rather than repaired with tape.
[QCMLS 2021: 7.8.7]

**89.** **b** When lifting heavy objects, one should bend at the knees and hips, lift with legs and buttocks. One should not bend at the waist, lift with the back, or twist because these motions are more likely to result in injury.
[QCMLS 2021: 7.8.14]

**90.** **d** Maintain wrists in a neutral position to prevent or relieve carpal tunnel syndrome symptoms. Ergonomically incorrect positions include arms raised and wrists bent downward, and back bent and neck forward. The first course of action would be to evaluate posture, body mechanics and adjustment of chair and keyboard prior to facility redesign.
[QCMLS 2021: 7.8.14]

**91.** **d** Class C fires involve electrical equipment. Fires involving paper, wood, and plastics are class A, while fires involving flammable liquids (including grease, oil, xylene, and alcohol) would be class B fires.
[QCMLS 2021: 7.8.5]

**92.** **a** Arm and neck pain after repetitive motion in the workplace should be reported to and discussed with the supervisor. Pain should never be ignored, and early intervention with ergonomic injuries is important. The scenario appears to be a work-related issue, and the employee is entitled to an employer-provided medical evaluation, if indicated. Changing or adjusting the workstation is a secondary action, which should be done in consultation with the supervisor and a trained ergonomics specialist.
[QCMLS 2021: 7.8.14] [CLSI 2007]

93. **d** A lab coat is an example of personal protective equipment (PPE). Safety cabinets, eyewash stations, and safety showers are examples of engineering controls, not PPE.
[QCMLS 2021: 7.8.4.1]

94. **c** Gloves must be removed before answering the telephone in a designated "clean" area, and hands washed afterward. Gloves do not need to be removed in the technical work area. It is essential to keep gloves on during specimen transportation, even through "clean" areas, such as the hallways. Care should be taken not to touch anything in the clean areas. Gloves do not need to be changed after handling isolation precaution patient specimens unless known contamination has occurred, which is the same for any patient specimen under "standard precautions."
[QCMLS 2021: 7.8.9.1]

95. **a** Safety glasses or face shields must be worn when working with caustic or toxic materials. It is not required practice to use eye and face protection when viewing culture plates, processing specimens using a splash barrier, or being present in technical work area unless working with caustic or toxic materials.
[QCMLS 2021: 7.8.11]

96. **a** Long hair and jewelry should be secured to avoid contamination with biohazards or physical injury. The other choices are not safe or acceptable work practices: food should not be stored in supply refrigerators at all, contact lenses should not be worn in laboratories, and open weaved shoes are a spill hazard.
[QCMLS 2021: 7.8.11.1]

97. **c** Knowledgeable personnel separate waste into designated categories (eg, chemical, routine, infectious) at the point of generation to reduce disposal costs and minimize employee exposure to hazardous materials.
[QCMLS 2021: 7.8.10]

98. **a** A 10% solution of bleach is an effective and economical disinfectant, which inactivates HBV in 10 minutes and HIV in 2 minutes.
[QCMLS 2021: 7.8.9] [Henry 2017, p9-10]

99. **c** The simplest method of disposing of needles is to dispose of the entire collection device into a container reserved for sharps.
[QCMLS 2021: 7.8.9.2]

100. **d** Blood spills must be cleaned up and decontaminated by personnel using the proper PPE.
[QCMLS 2021: 7.8.9.1]

101. **b** Hepatitis B infection is a global public health problem and is one of the most common infectious diseases in the world.
[QCMLS 2021: 7.8.9.1]

102. **c** The 3 modes of HIV transmission are through intimate sexual contact, contact with blood, and perinatal. Needlestick injury falls under the second mode, and poses, on average, a 0.3% risk of transmission.
[QCMLS 2021: 7.8.9.2]

103. **d** A 10% solution of bleach is an effective and economical disinfectant, which inactivates HBV in 10 minutes and HIV in 2 minutes.
[QCMLS 2021: 7.8.9.1] [Henry 2017, p9-10]

104. **b** Microbiological hazards are contained using a biological safety cabinet with the air exhausting through a HEPA filter.
[QCMLS 2021: 7.8.9.3] [Henry 2017, p8-10]

105. **d** Performing frequent and appropriate hand hygiene is the most effective method to prevent nosocomial infection. The other choices are methods for preventing spread of infection, but are not the *most* effective method.
[QCMLS 2021: 7.8.11.1]

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106. **c** Needles without self-sheathing devices discard directly into an appropriate sharps container, to avoid undo contact with the sharp. Cutting or bending needles are a prohibited practices. Needle recapping or removal is not preferred, but if required, it must be done using mechanical devices or for recapping only using a one-handed technique.
[QCMLS 2021: 7.8.9.2]

107. **a** "Standard" precautions are designed to minimize exposure to bloodborne pathogens.
[QCMLS 2021: 7.8.9.3]

108. **d** The primary action after an accidental needlestick is to thoroughly wash the wound with soap and water. The other choices are secondary, not primary actions.
[QCMLS 2021: 7.8.9.2]

109. **a** Parenteral inoculation of blood is the most likely mode of bloodborne pathogen transmission. The other choice are unlikely transmission modes for bloodborne pathogens.
[QCMLS 2021: 7.8.9.2]

110. **b** Hepatitis B is the bloodborne pathogen most likely to be transmitted in an occupational setting.
[QCMLS 2021: 7.8.9.3]

111. **c** Mycobacteria must be processed using a biological safety cabinet. Horizontal laminar flow hoods are used for pharmacy and other "clean room" functions. Barrier protection does not provide adequate respiratory protection, and fume hoods are used for chemical, not biohazard, control.
[QCMLS 2021: 7.8.9.3]

112. **c** Hepatitis B vaccine must be provided by the employer free of charge, but it is a series of 3, not a single injection. The hepatitis vaccine is recommended for high-risk employees upon initial employment, not just when an exposure occurs, yet it is not a mandatory vaccination.
[QCMLS 2021: 7.8.9.2]

113. **b** The first step in cleaning a small spill is to absorb the spill with disposable absorbent material. Immediately flooding the area with disinfectant should be avoided, as it may create aerosolization. It is unnecessary to evacuate the area for 30 minutes due to the size and nature of the spill. The area should be first cleaned with a disinfectant, not a detergent.
[QCMLS 2021: 7.8.9.1]

114. **a** Sodium hypochlorite is a disinfectant, but alcohol, chorhexidine and povidone-iodine are antiseptics, not disinfectants.
[QCMLS 2021: 7.8.9.1]

115. **b** Transmissible spongiform encephalopathy agents (prions), such as Creutzfeldt-Jakob disease, do not respond to the general rules of inactivation and decontamination. These agents are not disinfected or inactivated through conventional means for other microbial agents, as do *M. tuberculosis*, smallpox, *Bacillus anthracis*, or *Coccidiodes immitis*.
[QCMLS 2021: 7.8.9.1]

116. **c** Single-use disposable gloves are the best choice for processing blood specimens or other potentially infectious material. Reusable utility gloves are not ideal or the best choice, but could be used if properly decontaminated after use and inspected for punctures or tears prior to reuse. Latex gloves should be avoided because of potential sensitization to the latex and the development of a latex allergy.
[QCMLS 2021: 7.8.11.1]

117. **d** When gloves become contaminated by even a small amount of potentially infective material, one should change gloves and wash hands with antiseptic soap and water. Continuing to wear contaminated gloves, or washing the gloves, are inappropriate practices.
[QCMLS 2021: 7.8.11.1]



118. **a** This symbol indicates a biohazard.
[QCMLS 2021: 7.8.9.1]

119. **b** Infectious waste is referred to as "regulated medical waste."
[QCMLS 2021: 7.8.9.1]

120. **c** PPE (gloves and gowns), hand hygiene and safety policies are examples of administrative controls that protect the user but not the environment.
[QCMLS 2021: 7.8.4.2] [CLSI 2004, GP26:A3]

121. **c** Safety guidelines indicate that before attempting to solve any trouble shooting steps for any equipment, it is necessary to turn the power off for the instrument. This step is necessary to prevent any possible electric shock.
[QCMLS 2021: 7.8.7]

122. **d** The US Department of Transportation (DOT), United States Post Office (USPS) and the International Air Transport Association have various requirements for packaging and shipping of biological specimens, infectious agents, and hazardous materials. OSHA regulates workplace exposures, but does not certify shippers of hazardous materials.
[QCMLS 2021: 7.1.6] [Tietz 2015, p82, 86]

123. **d** Broken, cracked, chipped, and/or flawed glassware should be removed from service to prevent injury and disposed of in a puncture-proof container. Sharps pose both physical and biological hazards. Safety-engineered mechanical devices and safe work practices should be used to prevent accidental injury and/or exposure. Reusable instruments should be decontaminated by using a combination of chemical and autoclaving methods before cleaning in a washer and routine sterilization. Disposable needles and blades should be discarded in designated sharps containers, and are not autoclaved.
[QCMLS 2021: 7.8.8, 7.8.9.1]

124. **c** Public health surveillance programs require reporting of certain conditions to identify and control sources of infection or disease, prevent disease, and describe health trends. Examples include various infectious diseases and lead contamination. Reporting requirements vary by state. Notifications must be documented and disclosures of personal health information (PHI) must be tracked. Mouth pipetting is a prohibited laboratory work practice but not a notifiable condition.
[QCMLS 2021: 7.6.2.4]

### *B. Emergency Procedures*

125. **c** The primary response should be to evacuate the area. The other choices are secondary responses after evacuating the area.
[QCMLS 2021: 7.8.12]

126. **a** A "Danger" sign is required where immediate hazards are present. The other choices are incorrect and used for the other purposes as listed in the choices. The word *immediate* is the key to choosing the correct answer.
[QCMLS 2021: 7.8.1.1]



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127. **b** Hallways and corridors must be clear and free of obstruction at all times for fire safety and prevention. Fire drills should not be announced or practiced in advanced. Laboratories larger than 1000 square feet and/or contain explosion hazards require 2 exits.
[QCMLS 2021: 7.8.5]

128. **c** Persons must be removed from immediate danger as the first course of action; the other choices are secondary actions. The sequence of actions after removing persons from immediate danger is to pull the fire alarm, contain the fire, and evacuate the area, if required.
[QCMLS 2021: 7.8.5]

129. **b** The next correct step is to contain the fire by closing the doors. Evacuation of the facility is the last action to be taken and only if indicated. A type A extinguisher is for paper, wood and plastics fires, not electrical fires, and fire extinguishers only should be used by those who have received appropriate training. It is unnecessary and unacceptable to lock windows and especially doors, which would block access by firefighters.
[QCMLS 2021: 7.8.5]

130. **c** The first action that should be taken is to use the eyewash station to flush eyes with water for 15 minutes. Seeking medical assistance and bandaging the eyes are secondary, not primary actions. Flushing eyes with any chemical is incorrect and completely unsafe.
[QCMLS 2021: 7.8.11.1]

131. **b** After removing clothing, use the emergency safety shower and flush body with water to stop the effects of acid. Seeking medical assistance is a secondary, not a primary first aid action step. Applying baking soda or burn ointment are incorrect and not appropriate first aid actions.
[QCMLS 2021: 7.8.11.1]

## *III. Laboratory Mathematics*

### *A. Concentration, Volume, and Dilutions*

132. **c** To correct for having used a dilution, multiply the answer obtained times the reciprocal of the dilution made.
[Campbell 1997, p114]

133. **d** To prepare a standard solution, a known mass of solute is dissolved, and the solution is diluted to a precise volume.
[Tietz, 2018 p199]

134. **a** The value of the unknown is a ratio of the absorbance reading of the unknown to the absorbance reading of the standard.
[Campbell 1997, p126]

135. **b** Volumetric flasks are used to measure exact volumes and are primarily used in preparing solutions of known concentrations.
[Tietz 2015, p116-117]

136. **c** The most commonly used equation for preparing suspension solutions is V1 × C1 = V2 × C2. A 4% red cell suspension contains 4 mL of red cells per 100 mL (1 dL) of solution. Therefore, (25) × (4) = (100) × (*x*). Solve for *x*.
[Kaplan 2003, p35]

137. **c** The most commonly used equation for preparing suspension solutions is V1 × C1 = V2 × C2. In this case, (100) × (5) = (*x*) × (25). Solve for *x*.
[Kaplan 2003, p35]

138. **a** V1 × C1 = V2 × C2. A 3% acetic acid solution contains 3 mL of acetic acid per 100 mL (1 dL) of solution. Therefore, (25) × (3) = (100)(*x*). Solve for *x*.
[Kaplan 2003, p35]

**139.** **c** This is a ratio calculation. 0.9% normal saline contains 0.9 g NaCl in 100 mL solution, ie, 0.9/100 = *x*/1000. Solve for *x* to determine how much NaCl is needed to prepare 1 L (1000 mL).
[Kaplan 2003, p35]

**140.** **b** V1 × C1 = V2 × C2. A 3% solution contains 3 mL solution A per 100 mL (1 dL) of solution B. Therefore, (40) × (3) = (100)(*x*). Solve for *x*.
[Kaplan 2003, p35]

**141.** **b** This is a ratio calculation. 75/100 = *x*/8. Solve for *x*.
[Kaplan 2003, p34]

**142.** **c** Percentage recovery = Standard recovered/standard added x 100. The standard recovered = (measured in spiked serum – measured in diluted serum) or 161–89=72. In this example the concentration of standard added is 80 (1:10 dilution of 800 mEq/mL). The amount of standard recovered = (161 – 89 =72) / 80 (standard added) * 100 = 90%.
[QCMLS 2021: 7.4.2.4] [Tietz 2018, Chapters 12 & 37]

**143.** **d** Simple dilutions are ratios of 2 volumes, which involve a single substance diluted with one other substance. In this case, 0.25 mL solution A is added to 19.75 mL solution B (ratio 0.25/19.75), for a total volume of 20 mL. This represents a dilution of 0.25/20. To convert a 0.25/20 dilution to a 1-in-something dilution, set up a ratio-proportion calculation: 0.25 is to 20 as 1 is to *x*, and solve for *x*.
[Campbell 1997, p94]

**144.** **c** To correct for having used a dilution, multiply the answer obtained times the reciprocal of the dilution made.
[Campbell 1997, p114]

**145.** **b** To correct for having used a dilution, multiply the answer obtained times the reciprocal of the dilution made.
[Campbell 1997, p114]

**146.** **b** 1/25 = *x*/5. Solve for *x*.
[Kaplan 2003, p35]

**147.** **a** This is a ratio calculation. 1:300 dilution equals 1 antigenic unit in 0.5 mL. Therefore, 2 antigenic units in 0.5 mL equals a 1:150 dilution.
[Kaplan 2003, p34]

**148.** **a** This is a ratio calculation. A 2% saline erythrocyte suspension contains 2 mL of an erythrocyte suspension per 100 mL total solution. Therefore 2/100 = *x*/5. Solve for *x*.
[Kaplan 2003, p34]

**149.** **b** The original concentration of a solution × dilution made = concentration of resulting solution.
[Campbell 1997, p96]

**150.** **b** C1 × V1 = C2 × V2. Therefore (30) × (*x*) = (10)×(6). Solve for *x*.
[Kaplan 2003, p34]

**151.** **c** The most common equation for preparing dilutions is V1 × C1 = V2 × C2, where V1 is the volume, C1 is the concentration of solution 1, and V2 and C2 are the volume and concentration of the diluted solution.
[Kaplan 2003, p34]

**152.** **d** Simple dilutions are ratios of 2 volumes, which involve a single substance diluted with one other substance. In this case, 0.1 mL solution A is added to 9.9 mL solution B (ratio 0.1/9.9), for a total volume of 10 mL. This represents a dilution of 0.1/10.
To convert a 0.1/10 dilution to a 1-in-something dilution, set up a ratio-proportion calculation: 0.1 is to 10 as 1 is to *x*, ie, 0.1/10=1/*x*, and solve for *x*.
[Campbell 1997, p93]



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153. **c** A given dilution is expressed as the amount of a solute (analyte) in a specified volume. For example, a dilution factor of 5 means a 1:5 (one to five) volume-to-volume dilution containing one volume in a total of five volumes (one volume plus four volumes). When a dilution is performed, the following equation is used to determine the volume ($V_2$) necessary to dilute a given volume ($V_1$) of solution of a known concentration ($C_1$) to the desired lesser concentration ($C_2$): C1 x V1 = C2 x V2.
[Tietz 2018, Chapter 12]

154. **a** The most frequently used expression, concentrations of m/v are reported as grams percent (g%) or g/dL, as well as mg/dL and µg/dL. When percent concentration is expressed without a specified form, it is assumed to be weight per unit volume.
[Kaplan 2003, p35]

155. **b** The concentration of a weight/unit volume solution is expressed as grams/100 mL. Therefore, 20 g in 0.5 L = 4 g in 100 mL = 4% solution.
[Campbell 1997, p135]

156. **b** The concentration of a weight/unit volume solution is expressed as g/100 mL. A 3% (w/v) solution contains 3 g in 100 mL; therefore, 1 L contains 30 g.
[Campbell 1997, p136]

157. **b** This is a ratio calculation. 3/100 = 6/*x*. Solve for *x*.
[Campbell 1997, p136]

158. **a** Nanometer is a unit for wavelength.
[Kaplan 2003, p85]

## *B. Molarity, Normality*

159. **b** C1 × V1 = C2 × V2. Therefore 1 × 1 = 1.025 × *x*. Solve for *x*.
[Kaplan 2003, p34]

160. **c** This is a ratio calculation. The atomic weight of NaCl = 23.0+35.5 = 58.5. The percent of sodium content in NaCl is 23/58.5 = 0.3932%. Multiply the percent of Na content or 0.3932 × 100 grams = 39.32, or approximately 40 grams.
[Kaplan 2003, p35]

161. **d** Absorbance = (molar absorptivity coefficient) × (light path) × (concentration). Therefore, (molar absorptivity) = (absorbance)/(light path) × (concentration).
[Kaplan 2003, p38]

162. **b** By definition, a gram equivalent weight of an element or compound is the mass that will combine with or replace 1 mole of hydrogen.
[Campbell 1997, p143]

163. **b** Moles = grams/molecular weight.
[Campbell 1997, p138]

164. **c** Electrolyte equivalents can be calculated from the equation: mg/dL × 10 = 10 mg/L. Because mg/mEq weight is the millimolar weight in mg/valence, mg/L /mg/mEq = mEq/L.
[Kaplan 2003, p36]

165. **a** Moles = grams/molecular weight.
[Campbell 1997, p138]

166. **a** Molality is the number of moles of solute per 1 kg of solvent.
[Campbell 1997, p142]

167. **a** Molarity is a number that expresses the number of moles of substance in 1 L of solution.
[Campbell 1997, p138]

168. **b** Molarity (M) equals (grams/GMW)/L.
[Campbell 1997, p139]

**169.** **d** Molarity (M) equals (grams/GMW)/L.
[Campbell 1997, p139]

**170.** **a** A 1 Normal solution contains 1 gram equivalent of solute in 1000 mL of solution.
[Campbell 1997, p144]

**171.** **c** A 1 Normal solution contains 1 gram equivalent of solute in 1000 mL of solution.
[Campbell 1997, p144]

**172.** **d** C1 × V2 = C2 × V2. Therefore (0.25) × (*x*) = (100) × (0.05). Solve for *x*.
[Campbell 1997, p144]

**173.** **d** pH = log(1/[$H^+$]) = b – log(a). The relationship between a and b is such that b may be any value and the [$H^+$] will always be the same. To work out these problems, assign b the value that is the smallest whole number that is equal to or greater than the pH. In this problem, b = 7. Therefore, 7 = 7 – log(a); log(a) = 7 – 7; log(a) = 0; a = 1; [H+] = (a) × (10 – b). In this problem, [$H^+$] = 1 × 10–7, or 100 nmol/L.
[Campbell 1997, p196]

## *C. Standard Curves*

**174.** **a** Definition of a standard.
[Tietz 2015, p38]

## *D. Mean, Median, Mode, and Confidence Intervals*

**175.** **d** Standard deviation is a measure of the dispersion of data around the mean.
[QCMLS 2021: 7.3.1.1]

**176.** **d** Precision is the closeness of agreement among replicate measurements, or reproducibility. The coefficient of variation, a more useful measure of reproducibility, is the measure of relative random error expressed as a percentage.
[QCMLS 2021: 7.3.1.2]

**177.** **d** Precision is the reproducibility of analytical results, or the degree to which results of multiple analyses of the same specimen agree.
[QCMLS 2021: 7.3.1.2]

**178.** **a** Analytic specificity refers to the ability of the assay to accurately measure the analyte in the presence of other substances. Analytic specificity refers to the ability to detect small quantities of an analyte.
[QCMLS 2021: 7.3.1.3]

**179.** **b** A correlation coefficient, or *r*-value, of 0 indicates that there is no correlation between the methods, while an *r*-value of +1 and –1 indicate a perfect positive and negative correlation between methods, respectively.
[QCMLS 2021: 7.4.2.3]

**180.** **b** Standard deviation is a reflection of variation calculated based on the average distance of an individual point from the mean. 1) Determine the mean of a set of data points. 2) Calculate the distance of each point from the mean and square each distance measurement. 3) Sum the squared distances from the mean and divide the sum by the number of data points minus 1 (N – 1). 4) Take the square root.
[QCMLS 2021: 7.3.1.1]

**181.** **b** The probability of an observation having a value within ±2 standard deviations of the mean in a normal distribution is 95.5%. Therefore, 5 control values out of 100 would be out of control due to random error.
[QCMLS 2021: 7.3.1.1]

**182.** **b** The probability of an observation having a value within ±2 standard deviations of the mean in a normal distribution is 95.5%. Therefore, 100 ± 2 × 1.8 = 95% confidence interval.
[QCMLS 2021: 7.3.1.1]

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183. **b** The coefficient of variation % = (standard deviation/mean) × 100.
[QCMLS 2021: 7.3.1.2]

184. **a** The mean = sum of values/number of values; standard deviation = square root of the sum of (observed values – mean) squared / number of samples; coefficient of variation % = (standard deviation/mean) × 100. Therefore, (2.64/139) × 100 = 1.9%.
[QCMLS 2021: 7.3.1.2]

185. **c** The *mean* is the most widely recognized descriptive statistic. The mean of a set of data can be calculated in several ways, but in each case the result is, generally, an indication of the central point of the data.
[QCMLS 2021: 7.3.1.1]

186. **a** The *mode* is the most frequently occurring value in a set of data.
[QCMLS 2021: 7.3.1.1]

187. **b** The *median* is the value in the middle of a data set in which all the values are ranked from lowest to highest. This descriptive statistic is useful in sets of data that are heavily skewed or unevenly distributed, where the mean can be a misleading statistic.
[QCMLS 2021: 7.3.1.1]

188. **d** The *arithmetic mean* is the quantity that is most familiar and is ordinarily meant when we refer to the mean or average of a set of data. It is calculated as the sum of values divided by the number of values.
[QCMLS 2021: 7.3.1.1]

189. **b** The mean = sum of values/number of values.
[QCMLS 2021: 7.3.1.1]

190. **c** Imprecision is commonly expressed as the standard deviation. Coefficient of variation (CV) is preferred when the SD increases in proportion to concentration. CV, a measure of relative random error usually expressed as a percentage: CV (%) = (standard deviation/mean) × 100.
[QCMLS 2021: 7.3.1.1]

191. **b** The coefficient of variation % = (standard deviation/mean) × 100. In this example: CV=(3/137) x100 = 2.19%.
[QCMLS 2021: 7.3.1.1]

## *E. Sensitivity, Specificity, and Predictive Value*

192. **a** Validation is the confirmation by objective evidence that the requirement for the specified use of the instrument/procedure is consistently fulfilled.
[QCMLS 2021: 7.4.2.2]

193. **b** Accuracy or trueness is the closeness of agreement with the true value.
[[QCMLS 2021: 7.4.2.3]

194. **b** Specificity (true negative rate) is defined as negativity in the absence of disease.
[QCMLS 2021: 7.3.1.3]

195. **a** GEN 42085: The analytical measurement range (AMR) is the range of analyte values that a method can directly measure on the specimen without any dilution, concentration, or other pretreatment not part of the usual assay process.
[QCMLS 2021: 7.4.2.6.1]

196. **b** The predictive value of a positive test indicates the probability that a laboratory result outside the reference interval reflects the true presence of disease.
[QCMLS 2021: 7.3.1.4]

197. **b** A method used for screening must have a high degree of sensitivity to detect everyone with the disease.
[QCMLS 2021: 7.3.1.3]

198. **c** Accuracy is the comparison of a result with the true value.
[QCMLS 2021: 7.3.1.2]



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199. **b** Precision describes the reproducibility of a method. The narrower the distribution of results, the smaller the standard deviation.
[QCMLS 2021: 7.3.1.2]

200. **c** Analytical sensitivity refers to the lowest analyte concentration that is reliably detectable. It includes limit of detection (LoD), which refers to the lowest concentration that can be distinguished from background.
[QCMLS 2021: 7.4.2.5]

201. **a** Functional sensitivity, also called limit of quantitation (LoQ), is the lowest analyte concentration reliably quantified with an acceptable coefficient of variation.
[QCMLS 2021: 7.4.2.5]

## *IV. Manual/Automated Methodology and Instrumentation*

### *A. Basic Laboratory Equipment*

202. **c** Random access analyzers.
[Tietz 2015, p255]

203. **c** Reduction in identification errors.
[QCMLS 2021: 7.4.2.8] [Tietz 2015, p255 256]

204. **d** The analytical measurement range (AMR) is the range of analyte values that a method can directly measure on the specimen without any dilution, concentration, or other pretreatment not part of the usual assay process.
[QCMLS 2021: 7.4.2.6.1] [CAP 2017, p7-9]

205. **a** The ocular lens has a magnification of 10x. This multiplied by the magnification of the objectives 10x (low power), 40x (high power), etc, equals the total magnification.
[QCMLS 2021: 7.7.3]

206. **c** Adjusting the condenser of the diaphragm of the microscope also affects image resolution. Adjusting the main light source only changes the light intensity.
[Strasinger 2018, p107]

207. **b** The diffracted light in phase microscopy enhances slight variations in the refractive indices of constituents with low refractive indices. Staining is not required to enhance low refractive index constituents when using phase microscopy.
[Strasinger 2014, p107]

208. **a** The first step in achieving correct illumination, optimal contrast, alignment, and resolution of microscopic images is to ensure the microscope has been adjusted by using the technique known as Köhler Illumination.
[Nikon 2018]



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209. **d** Proper cleaning of microscope lenses is an important part of overall microscope maintenance. Objectives should be cleaned using lens paper, whenever they are not in use, to prevent the immersion oil from seeping inside the lens.
[Nikon 2018]

210. **d** There are various colored rings on microscope objectives. Color coded bands indicate several features of the objective. The black ring closest to the specimen end of the oil immersion lens is a feature which helps with easy identification of the objective. This helps prevent the user from using oil with the incorrect lens and provides the guidance at a glance.
[Nikon 2018]

211. **d** Ensure proper balance is established and maintained during centrifuge operation. A disinfectant should be used, not soap and water, for cleaning. The cover should not be opened until the centrifuge comes to a complete stop. All specimen tubes need to be covered to prevent aerosol formation.
[QCMLS 2021: 7.7.2]

212. **b** To determine the G-factor, use 0.0001 × 30 cm × $(3000)^2$. Remember that the radius is half the diameter.
[QCMLS 2021: 7.7.2]

213. **c** The devices listed are electronic, but the only device that measures rpm is a tachometer. RPM must be consistent to produce adequate G-force.
[QCMLS 2021: 7.7.2]

214. **a** Dry cobalt-treated $CaCl_2$ is blue. When it turns pink, it should be changed.
[QCMLS 2021: 7.8.6]

215. **a** Blood gas analyzers measure pH using a potentiometric method with ion-selective electrode technology, partial pressure of oxygen ($PO_2$) using amperometric methods based on the principles of the Clark electrode, and partial pressure of carbon dioxide ($PCO_2$) using modified potentiometric methods bases on the principles of the Severinghaus electrode.
[QCMLS 2021: 7.7.10, t7.8]

## *B. Spectrophotometry & Photometry*

216. **d** Absorbance, A, is a measure of the amount of light stopped, or absorbed, by a solution, and the absorbance of light is a logarithmic function. Hence, the A scale is a logarithmic scale. On the other hand, transmittance, T, is a measure of the amount of light allowed to pass through a solution. Because the following relationship is true, A = light stopped and T = light passed through, A and T are inversely related. They are also logarithmically related, because the absorption of light is a logarithmic function.
[Campbell 1997, p212]

217. **a** This is a ratio calculation: (absorbance of the unknown)/(concentration of the unknown) = (absorbance of the standard)/(concentration of the standard).
[QCMLS 2021: 7.7.4]

218. **b** Definition of atomic absorption.
[QCMLS 2021: 7.7.5] [Tietz 2015, p138-139]

219. **a** Visual vs ultraviolet wavelength.
[QCMLS 2021: 7.7.4] [Tietz 2015, p134]

220. **d** Visual wavelength calibration.
[QCMLS 2021: 7.7.4] [Tietz 2015, p137-138]

221. **b** Spectrophotometer wavelength.
[QCMLS 2021: 7.7.4] [Tietz 2015, p137-138]

222. **a** For maximal light absorption of the measured chromogen.
[QCMLS 2021: 7.7.4]

223. **b** The additional beam corrects for variation in light intensity. The other choices do not describe double beam-in-space.
[QCMLS 2021: 7.7.4] [Tietz 2015, p133-134]

224. **b** Hollow cathode lamp emitting the line spectrum of the pure metal of the measured element.
[QCMLS 2021: 7.7.5] [Tietz 2015, p133-136]

225. **c** Bandwidth refers to the range of wavelengths that pass through the exit slit.
[QCMLS 2021: 7.7.4]

226. **b** Definition of bichromatic analysis.
[QCMLS 2021: 7.7.4]

227. **d** The first step is to find the wavelength for the maximum absorbance of the measured chromogen.
[QCMLS 2021: 7.7.4]

228. **c** Wavelength accuracy is an essential performance parameter to be tested for the spectrophotometer. For the narrow-spectral bandwidth instruments, holmium oxide glass may be scanned over the range of 280-650 nm. The material shows very sharp absorbance peaks at the well defined wavelengths. Prism and diffraction gratings are monochromatic devices. The didymium filter is used with broader bandwidth instruments.
[QCMLS 2021: 7.7.4] [Tietz 2015, p137-138]

229. **b** Light from the lamp or light source of a photometer is reduced to a specific wavelength by the monochromator. In spectrophotometers, a diffraction-grate is used as a monochromator.
[QCMLS 2021: 7.7.4]

## *C. Mass Spectrometry*

230. **c** Mass spectrometers detect mass-to-charge ratios of ionized molecules.
[QCMLS 2021: 7.7.9]

## *D. Osmometry*

231. **c** Osmolality is best measured using freezing point depression. Osmolality can be measured by dew point, boiling point, osmotic pressure and freezing point however, freezing point is the most common method used in clinical laboratories.
[QCMLS 2021: 7.7.6] [Tietz 2015, p421]

## *E. Electrophoresis*

232. **c** Electrophoresis involves the migration of a charged molecule or particle in a liquid medium under the influence of an electric field. The 3 roles of the buffer are 1) to carry the applied current; 2) to establish the pH for the electrophoresis procedure and 3) to determine the electrical charge on the solute.
[QCMLS 2021: 7.7.8] [Tietz 2015, p172-173]

233. **c** A commonly encountered problem in electrophoresis is perceived holes in staining patterns. This is due to the analyte being present in too high a concentration. The corrective action is to apply a less concentrated sample.
[QCMLS 2021: 7.7.8]

## *F. Chromatography*

234. **a** Chromatography functions based on differential solubility of various molecules.
[QCMLS 2021: 7.7.9] [Tietz 2015, p184]

235. **b** Gel filtration chromatography separates molecules by size exclusion.
[QCMLS 2021: 7.7.9] [Tietz 2015, p188]



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**236.** **b** The retention (retardation), factor ($R_f$) in thin-layer chromatography is the ratio traveled by a compound as compared to the solvent front; therefore, if $R_f = 0.5$, the solute moves half the distance as the solvent front.
[QCMLS 2021: 7.7.9] [Tietz 2015, p189-190]

**237.** **c** Backflow causes increased pressure and bad chromatography.
[QCMLS 2021: 7.7.9] [Tietz 2015, p193-194]

**238.** **a** Samples to be analyzed by gas-liquid chromatography must be volatile or become volatile upon heating. The vaporized sample then flows, with the inert carrier gas, through the column.
[QCMLS 2021: 7.7.9] [Bishop 2005, p111]

**239.** **d** Steric-exclusion chromatography is based upon smaller molecules being trapped by the porous column material, but larger molecules are carried along by the mobile phase.
[QCMLS 2021: 7.7.9] [Bishop 2005, p108-109]

**240.** **a** $R_f = D_s / D_t$. In thin-layer chromatography, the retention (retardation) factor ($R_f$) is the distance the leading edge of a sample component moves ($D_s$) divided by total distance the solvent moves ($D_t$); eg, sample moves 5 cm and solvent moves 10 cm, $R_f = 0.5$.
[QCMLS 2021: 7.7.9] [Tietz 2017, p293]

**241.** **a** Ion-selective electrodes portend their selectivity properties through the use of particular membranes, as the selected ions interact with the membranes used in ion-selective electrodes.
[QCMLS 2021: 7.7.8] [Tietz 2015, p197-198]

## *G. Electrochemistry*

**242.** **c** Ion-selective electrodes (ISE) are a potentiometric method of analysis. ISEs respond to individual ions in a sample.
[QCMLS 2021: 7.7.8]

## *H. Fluorometry*

**243.** **a** Fluorescence occurs when a molecule absorbs light at one wavelength and emits light of a longer wavelength. The detector in a fluorometer is usually at right angles to the incident light source. A xenon or mercury vapor lamp emits enough UV light to be useful as a light source in a fluorometer.
[QCMLS 2021: 7.7.4]

## *I. Nephelometry*

**244.** **a** A nephelometer measures the concentration of suspended particles by employing a light beam and a detector set at 90 degrees from the source, measuring the intensity of scattered light.
[QCMLS 2021: 7.7.7]

**245.** **c** Nephelometric methods are based upon light scatter being proportional to the number of particles in suspension, such as antigen-antibody complexes, which are physically larger than uncomplexed molecules.
[QCMLS 2021: 7.7.4]

## *J. Flow Cytometry*

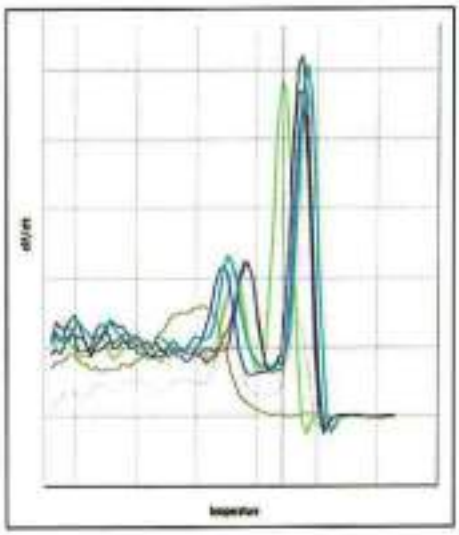
**246.** **c** Flow cytometry uses fluorescent-labeled monoclonal antibodies to identify cells of interest by binding to specific components within or on the surface of the cells.
[QCMLS 2021: 8.5.15]

**247.** **c** In flow cytometry, a laser beam hits cells as they pass through the instrument in single file. The amount of light scatter is measured from each cell at 2 different angles, and is used to identify the cell type on the basis of size and granularity. Cells are also identified on their ability to emit fluorescence after they have been incubated with fluorescent-labeled monoclonal antibodies that bind to specific surface markers.
[QCMLS 2021: 8.5.15] [Stevens 2017, p194]

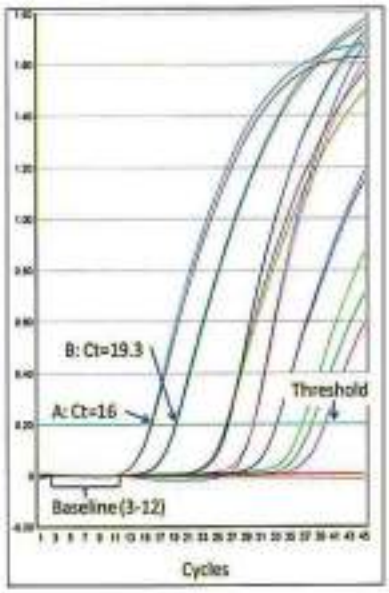
## K. Molecular Methods

**248. c** The absorptivity constant for DNA is 50 µg/mL. To determine concentration, multiply the spectrophotometer reading in absorbance units by the constant for DNA.
[Buckingham 2012, p81]

**249. b** The absorptivity constant for RNA is 40 µg/mL (OD260 × RNA constant × dilution factor × volume of sample = (0.03125 × 40 × 40) × 0.5 =50 × 0.5 = 25 µg.
[Buckingham 2012, p81]



**250. c** This is a plot of the derivative of the fluorescence data vs temperature of melt curve of PCR products generated from different serotypes of enterovirus.
[Buckingham 2012, p198]




**251. c** Serial 2-fold dilutions have a Ct difference of 1, while 10-fold dilutions are separated by 3.3 Ct.
[Patrinos 2005, p116]

**252. c** PFGE after digestion of bacterial chromosomes with restriction enzymes is used to determine similarity between bacterial isolates.
[Buckingham 2012, p325]

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253. **b** Agarose gels and standard real-time PCR require too much of the small amount of DNA contained in small tissue sections. Fluorescent intercalating dyes are more sensitive than absorbance for low-concentration samples and allow for more specific measurement of double-stranded DNA.
[Buckingham 2012, p84]

254. **b** Enzyme-multiplied immunoassay is a homogeneous enzyme immunoassay used to quantitate drugs, hormone, etc. Unlike the heterogeneous assays, the free-labeled reactant does not have to be separated from the bound-labeled reactant.
[QCMLS 2021: 7.7.11]

255. **c** Enzymes are one of the several labels used in labeled immunoassays. The catalytic property of the enzyme is used to detect and quantitate the immunological reaction. Alkaline phosphatase and horseradish peroxidase are examples of enzymes used. The product of the enzymatic activity is monitored spectrophotometrically.
[Bishop 2005, p152]

256. **b** Chemiluminescent labels are based on the emission of light produced during a chemical reaction. These labels are very useful because they provide very low levels of detection ($2 \times 10^{-20}$ mol/L) with little or no background interference. Electrochemiluminescence differs from chemiluminescence in that the reactive species that produce the chemiluminescent reaction are electrochemically generated from stable precursors at the surface of an electrode. With its use, detection limits of 200 fmol/L and a dynamic range extending over 6 orders of magnitude can be obtained.
[Clarke 2006, p122]

## *L. Automated Microbiology Processors*

257. **b** Direct nucleic acid amplification assays can detect *Mycobacterium tuberculosis* directly from a patient specimen, with results available in less than 8 hours. Automated, continuously monitoring culture systems for mycobacteria are available. Tubes are incubated for 6 weeks, and positive tubes stained for acid-fast bacilli.
[QCMLS 2021: 6.2.1.3]

258. **a** The laboratory approach to bacteriuria may include such automated screening methods as photometry, flow cytometry, and particle filtration.
[QCMLS 2021: 6.7.1.5]

259. **c** Automated systems exist to identify bacterial diseases of the blood. Many allow detection of positive blood cultures in 6-24 hours; some incorporate antimicrobial removal device that absorbs any antibiotic present in patient's blood.
[QCMLS 2021: 6.7.1.5]

## *M. Hematology Instrumentation*

260. **b** The hematocrit is calculated from the mean cell volume (MCV) and red blood cell count (RBC). Automated instruments determine the hematocrit using the directly measured MCV and RBC count with the formula: MCV (fL) × RBC ($\times 10^{12}$/L)/1000.
[QCMLS 2021: 4.10.7.2]

261. **a** Most hematology analyzers use one of two basic principles of operation: electronic impedance and optical scatter.
[QCMLS 2021: 4.10.1.1]

262. **c** Low platelet counts should be verified with a review and estimate from a stained peripheral blood smear.
[McKenzie 2020, p1070]

263. **b** With the Coulter principle of particle counting, cells suspended in a conductive diluent traverse a small aperture, and cause an increase in electrical resistance to the flow of current between 2 electrodes. These voltage changes (pulses) can be detected and measured. The number of pulses detected is proportional to the number of cells passing through the aperture.
[QCMLS 2021: 4.10.1.2]

264. **d** Repeat of one out-of-range control is the first appropriate course of action.
[McKenzie 2015, p912]

## *V. Basic Management Principles*

265. **a** Evaluation of turn-around times is an important part of the laboratory's quality assurance program and is a good way to assess the laboratory's performance in the overall testing process. The laboratory manager is responsible for determining the overall testing process and schedules that will be the most cost effective and provide reliable test results within a clinically appropriate time frame.
[Garcia 2004, p382]

266. **c** Guidelines when appraising a poor performer include reviewing the performance standard for the task and then list specific changes and time frame in which the behavior (improvement) must be corrected. An improvement plan may be added to the appraisal form to assist the employee improve their performance. The improvement plan can include attending workshops or continuing education programs or working with an experienced technologist.
[QCMLS 2021: 7.4.2.10]

267. **a** Variable costs are indirect or direct costs that vary in direct proportion to test volume.
[QCMLS 2021: 7.2.1]

268. **b** A cost is an expense that includes the money spent on supply, overhead and labor to produce a product or service.
[QCMLS 2021: 7.2.1]

269. **b** The decision to lease or buy equipment is based on the least costly alternative given the expense to provide this service and the anticipated revenue for offering the service.
[QCMLS 2021: 7.2.2]

270. **c** FTE (full-time equivalent) is 2080 hours per year (8 hours/day × 5 days/week × 52 weeks/year).
[QCMLS 2021: 7.2.2]

271. **a** Converting a fraction into a percentage: 50,419/38,773 × 100 = 130%; 130% – 100% = 30% over budget.
[QCMLS 2021: 7.3.1.5]

272. **b** A job description is a written delineation of the title, duties, responsibilities and reporting relationships of a position and the requisite qualifications needed.
[QCMLS 2021: 7.5.1.2]

273. **a** Communication between employers and employees is key in the retention of employees as well as in giving a successful performance appraisal. Feedback lets employees know what is expected of them and what is expected from the employer. They will be more satisfied with their job and likely be more productive as well.
[QCMLS 2021: 7.4.11.3]

274. **c** The first step in progressive counseling is verbal counseling.
[QCMLS 2021: 7.4.11.3]

275. **b** The first step in progressive counseling is verbal counseling.
[QCMLS 2021: 7.4.11.3]



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276. **b** The first step in progressive counseling is verbal counseling.
[QCMLS 2021: 7.4.11.3]

277. **d** A workload recording system is important in understanding the resource requirements for each patient "sector" and is used to determine staffing needs.
[QCMLS 2021: 7.2.2]

278. **a** Questions that could be discriminatory in nature (eg, nationality, marital status, dependents, religion, affiliations, sexual orientation, or physical or mental disabilities) should be avoided on the employment application or during the interview process, unless the question is relevant or pertinent to the particular job; otherwise, it can be considered illegal.
[QCMLS 2021: 7.4.11.1]

279. **c** Questions that could be discriminatory in nature (eg, nationality, marital status, dependents, religion, affiliations, sexual orientation or physical or mental disabilities) should be avoided on the employment application or during the interview process, unless the question is relevant or pertinent to the particular job; otherwise, it can be considered illegal.
[QCMLS 2021: 7.4.11.1]

280. **a** Labor comprises approximately 70% of a healthcare organization's budget.
[QCMLS 2021: 7.2.2]

281. **d** Questions that could be discriminatory in nature (eg, nationality, marital status, dependents, religion, affiliations, sexual orientation, or physical or mental disabilities) should be avoided on the employment application or during the interview process, unless the question is relevant or pertinent to the particular job; otherwise, it can be considered illegal.
[QCMLS 2021: 7.4.11.1]

282. **b** An employer is required to make reasonable accommodations unless it creates an undue hardship (financially or logistically) to the business.
[QCMLS 2021: 7.4.11.1]

283. **c** A Gantt chart, commonly used in project management, shows activities (tasks or events) displayed against time. A list of the activities appears at the left of the chart and at the top is a suitable time scale. Each activity is represented by a bar; the position and length of the bar reflects the start, duration and end of the activity.
[Lock 2007, p187; www.gantt.com]

284. **d** Indirect costs are all costs that are not directly related to the test, but are a part of total laboratory expenses.
[QCMLS 2021: 7.2.1]

285. **d** Manageable and unmanageable cost components for a laboratory manager. While it is under a manager's control to schedule and train employees, "acts of God" (eg, military leave, pregnancy, injury) are not under their control and affect their ability to schedule employees.
[QCMLS 2021: 7.4.11.1]

286. **a** ABN (advanced beneficiary notice) is a notification given to Medicare beneficiaries prior to receiving a service that may not be covered by insurance and that the beneficiary may be required to assume financial responsibility.
[QCMLS 2021: 7.1.3]

287. **b** The decision of whether or not to acquire new equipment requires analysis and supporting documentation to include: purpose, importance, projected client demand and utilization, estimated useful life, estimated total acquisition cost, estimated associated yearly cast out-flows, and estimated yearly cash inflows or savings.
[QCMLS 2021: 7.2.2]

288. **d** Credentialing or certification is demonstration of qualification or competence in a particular area of practice. It recognizes an individual as meeting predetermined criteria set by the administering organization, usually a professional association.
[QCMLS 2021: 7.5.1.2]

289. **d** Performance appraisals are done to provide ongoing performance feedback.
[QCMLS 2021: 7.4.2.11.3]

290. **c** Personnel competence can be assessed by a quiz, direct observation of tasks, or a problem solving exercise. Acknowledgement that an individual's training is complete is not sufficient.
[QCMLS 2021: 7.4.10] [CLSI 2016, p39-43]

291. **a** It is important to reward job excellence and identify unacceptable behavior in an employee evaluation.
[QCMLS 2021: 7.4.2.11.3]

292. **a** All employees need feedback to sustain motivation.
[QCMLS 2021: 7.4.2.11.3]

293. **c** Patient education can alleviate anxiety.
[Hudson 2004, p100]

294. **b** Informative communication shares knowledge about laboratory processes and policies.
[QCMLS 2021: 7.4.2.11.3]

295. **c** The employee moved into a participating situation.
[Hudson 2004, p79]

296. **b** Short-term goals should be achievable, which enhances employee motivation.
[Hudson 2004, p73-77]

297. **b** Scalar principle (the chain of command) is the accepted, demonstrably effective way to assess and implement change in a large organization.
[QCMLS 2021: 7.5.1.2]

298. **c** Problem solving includes 3 steps: define problem, write down the facts, examine solution (consider options).
[Hudson 2004, p184-186]

299. **d** Delegation should be used as an opportunity for subordinate learning. When a task is delegated, the authority transfers to the one to whom the task is delegated.
[Hudson 2004, p80]

300. **b** A collaborative approach can be used for a long-term problem. To work together to solve an issue, both views of the issue are important.
[Hudson 2004, p92]

301. **c** Team development constant communication between team members should be fostered by leaders.
[QCMLS 2021: 7.1.11.3]

302. **a** Downward communication is used primarily to state objectives, disseminate policies or changes, provide directives and convey general information to subordinates. Upward communication flows up through the ranks to top management and is used to report and convey information.
[Turnbull 2005]

303. **a** All employees need feedback to sustain motivation.
[QCMLS 2021: 7.4.2.11.3]

304. **a** Personal communication should be planned: P = plan your communication, L = listen to others, A = avoid assumptions/ensure communication is understood, N = network.
[Lab Medicine 2008b, p261-264]

305. **c** The first step in choosing a laboratory information system is the selection of the members of an LIS team.
[Hudson 2004, p212-214]

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306. **b** HL-7 interface is the standard for healthcare information management.
[QCMLS 2021: 7.4.2.8]

307. **a** LIS systems use barcode labels for positive sample identification for many of the processes in the lab. Each LIS vendor will specify which of several barcode label formats are compatible with their system.
[Kasoff 2009]

308. **a** Health Level 7 (HL-7) is the organization that specifies requirements for electronic data transmission formatting. It is considered to be the industry standard.
[QCMLS 2021: 7.4.2.8]

309. **d** The Host Query mode of an instrument is a 2-way communication process whereby the instrument reads the specimen barcode and queries the LIS for the orders. The tests to be run are sent to the instrument (ie, bidirectional communication). Once results are obtained, they are sent back to the LIS.
[Kasoff 2009]

310. **c** Test definitions, as well as all types of stored master files, must be contained in the database.
[Cowan 2005, p1-20]

311. **a** Validation of the LIS is required by various accrediting agencies in order to prove that the system is performing as expected. This must be done at implementation and over time as changes are made to the LIS.
[QCMLS 2021: 7.4.2.8]

312. **c** System back-up procedures are mandated by accrediting agencies and include creating an exact copy of sensitive LIS data at regularly defined intervals and storage of the backed-up data at a separate location.
[Cowan 2005, p59-86]

313. **b** Middleware applications have become readily available, are capable of supporting changes in laboratory workflow, and provide enhanced data management tools without having to upgrade or change the LIS.
[Vail 2008, p26-36]

314. **c** In a broad context, a culture of safety in healthcare pertains to core values and behaviors aligned with elevating safety over other goals, while having vital commitment to the culture of safety by the entire team (leadership and frontline team members). Transparency, accountability, honest and frequent communication, while following improvement models and continuously learning are keys to effective leadership at all levels to ensure a patient-centered culture of safety. Research has shown that one of the most vital ways to promote a patient-focused culture of safety is to encourage open reporting and transparent communication of any mistakes, potential problems, and/or error laden workflows that are noticed. Resilient learning organizations utilize each mistake to improve patient care.
[QCMLS 2021: 7.8.13.1]

315. **b** Laboratory leaders should develop engagement strategies incorporating effective, inclusive communications that articulate and promote the value of each team member to the success of laboratory outcomes and in benefit to patients. Inclusive community-based engagement is beneficial to a diverse laboratory team and elevates the laboratory's visibility and value.
[QCMLS 2021: 7.1.11.3]

## VI. Education Principles

316. **d** Continuing education for medical laboratory personnel is undertaken for many reasons (ie, improving job performance; mandates by regulatory agencies; satisfaction of intellectual curiosity or enhancement of professional skills).
[QCMLS 2021: 7.4.2.10]

317. **a** Psychomotor objective: The learner must perform the task.
[Hudson 2004, p248-249]

318. **d** The employer may identify an educational need during the annual competency evaluations.
[Hudson 2004, p241-243]

319. **d** The intent of objective: What does the instructor want the student to do at the conclusion of the unit.
[Hudson 2004, p248]

320. **c** An objective is a specific statement of what is expected of the learner after a period of instruction. There are 3 questions that educational objectives must address. What does the instructor want the student to do? Under what conditions should the learner accomplish the objective? What is the criterion that signifies achievement?
[Hudson 2004, p248]

321. **d** A goal targets the purpose of an educational unit. The objective specifies what a learner is expected to know or do.
[Hudson 2004, p247-248]

322. **c** Training involves an organized documented process to provide instructions to help a new employees attain or regain a required level of knowledge and skill required to perform a task or procedure. When a new method or new instrument is introduced to the laboratory, a process is undertaken that includes assessment of performance, writing a procedure, integration into the information system, training, and education.
[QCMLS 2021: 7.4.1, 7.4.2.10]

323. **b** Competency assessment confirms a person's ability to use his or her skill and knowledge to perform assigned duties correctly. It must be performed and documented for testing personnel for each test or test system that an individual is authorized to perform before reporting test results. Then, competency assessment should be performed semiannually during the first year of testing, then annually; when methodology or instruments are changed; when new test platforms are introduced; with newly hired, transferred, or promoted personnel; and when an employee demonstrates repeated performance issues. A single mistake by an employee would not trigger a need for staff competency assessment.
[QCMLS 2021: 7.4.2.10]

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Chapter 8

# Immunology

*The following items have been identified generally as appropriate for those preparing for both the MLS and MLT examinations. Items that are appropriate for the MLS examination **only** are marked with MLS ONLY.*


**601 Questions**
- 601 I. Principles of Immunology
- 616 II. Diseases of the Immune System
- 621 III. Transplantation
- 622 IV. Infectious Disease Serology
- 626 V. Serologic Procedures
- 635 VI. Test Results

**640 Answers with Explanations & Citations**
- 640 I. Principles of Immunology
- 647 II. Diseases of the Immune System
- 650 III. Transplantation
- 651 IV. Infectious Disease Serology
- 654 V. Serologic Procedures
- 660 VI. Test Results


## I. Principles of Immunology

### A. Immune System Physiology

1. A DPT vaccination is an example of:

   **a** active humoral-mediated immunity
   **b** passive humoral-mediated immunity
   **c** cell-mediated immunity
   **d** immediate hypersensitivity

2. Cells known to be actively phagocytic include:

   **a** neutrophils, monocytes, basophils
   **b** neutrophils, eosinophils, monocytes
   **c** monocytes, lymphocytes, neutrophils
   **d** lymphocytes, eosinophils, monocytes

3. Normal serum constituents that can rapidly increase during infection, injury or trauma are referred to as:

   **a** haptens
   **b** acute phase reactants
   **c** opsonins
   **d** chemotaxins

4. The acute phase reactant that has the fastest response time and can rise 100× is:

   **a** alpha-1 antitrypsin
   **b** haptoglobin
   **c** c-reactive protein
   **d** ceruloplasmin

5. Acquired immunity can be characterized as having what qualities?

   **a** nonadaptive and nonspecific
   **b** internal and external mechanisms
   **c** sensitivity and short acting
   **d** specificity and memory

6. Skin, lactic acid in sweat, pH balance, mucous, and the motion of cilia represent which type of immunity?

   **a** natural, innate
   **b** acquired
   **c** adaptive
   **d** auto

7. Substances that are antigenic only when coupled to a protein carrier are:

   **a** opsonins
   **b** haptens
   **c** adjuvants
   **d** allergens

8. A haptenic determinant will react with:

   **a** both T cells and antibody
   **b** T cells but not antibody
   **c** neither T cells nor antibody
   **d** antibody but not T cells

9. Antibodies composed of IgG immunoglobulin:

   **a** occur during the primary response to antigen
   **b** are larger molecules than IgM antibodies
   **c** can cross the placenta from mother to fetus
   **d** can be detected in saline crossmatches

10. Cells that are precursors of plasma cells and also produce immunoglobulins are:

    **a** macrophages
    **b** B lymphocytes
    **c** T lymphocytes
    **d** monocytes

11. Antibodies are produced by:

    **a** killer cells, T cells
    **b** marrow stem cells
    **c** mast cells
    **d** B cells

12. Polyclonal B-cell activation:

    **a** inhibits antibody production
    **b** requires the participation of T-helper cells
    **c** results from the activation of suppressor T cells
    **d** can induce autoantibody production

13. Macrophages are characterized by:

    **a** surface receptors for C3b complement
    **b** surface CD3 expression
    **c** *in vitro* synthesis of immunoglobulin
    **d** large amounts of rough endoplasmic reticulum

14. Macrophage phagocytosis of bacteria is enhanced by which of the following:

    **a** opsonin
    **b** antigen
    **c** hapten
    **d** secretory piece

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15. Humoral antibodies are produced by which cells?

a macrophages
b T lymphocytes
c B lymphocytes
d neutrophils

16. In hybridoma technology, the desirable fused cell is the:

a myeloma-myeloma hybrid
b myeloma-lymphocyte hybrid
c lymphocyte-lymphocyte hybrid
d lymphocyte-granulocyte hybrid

17. Which test is used to evaluate the cellular immune system in a patient?

a skin test for commonly encountered antigens
b determination of isohemagglutinin titer
c immunoelectrophoresis of serum
d lymphocyte proliferation to mitogen/antigen

18. T cells are incapable of:

a collaborating with B cells in antibody responses
b secretion of immunoglobulins
c secretion of cytokines
d producing positive skin tests

19. T lymphocytes are incapable of functioning as:

a cytotoxic cells
b helper cells
c phagocytic cells
d regulatory cells

20. Nonspecific killing of tumor cells is carried out by:

a cytotoxic T cells
b helper T cells
c natural killer cells
d antibody and complement

21. When a natural killer cell comes in contact with a cell expressing MHC class I proteins what action will result?

a perforins released
b granzymes released
c inhibition of killing
d cytotoxic mechanisms

22. Some MHC class III genes code for?

a antigens
b antibodies
c lymphocytes
d complement

23. Which of the following is an important cellular mediator of immune complex tissue injury?

a mast cell
b neutrophil
c basophil
d eosinophil

24. Which of the following mediators is released during T-cell activation?

a immunoglobulins
b complement C3
c histamine
d cytokines

25. MHC Class I includes which molecules?

a complement
b HLA A, B, C
c cytokines
d HLA DP, DQ, DR

26. C3b and Fc receptors are present on:

a B lymphocytes
b monocytes
c B lymphocytes and monocytes
d neither B lymphocytes nor monocytes

27. T lymphocytes that possess the CD8 surface marker mediate which of the following T-cell functions?

a delayed-type hypersensitivity
b regulatory
c cytotoxic
d helper

28. A normal control for a B lymphocyte enumeration assay should have a value of what percentage of total lymphocytes counted?

a 21%
b 48%
c 76%
d 89%

29. An immunofluorescence test using reagent antibody directed against the CD3 surface marker would identify which of the following cell types in a sample of human peripheral blood?

a all mature T lymphocytes
b T helper lymphocytes only
c cytotoxic T lymphocytes only
d T regulatory cells only

30. Which of the following cytokines is produced by an activated CD8 cytotoxic T cell?

a interleukin 13 (IL-13)
b interleukin 5 (IL-5)
c stromal cell-derived factor 1 (SDF-1)
d interferon gamma (IFNγ)

31. What is the location of B-cell maturation?

a thymus
b spleen
c bone marrow
d liver

32. What is the location of T-cell maturation?

a bone marrow
b gut-associated lymphoid tissue (GALT)
c thymus
d kidneys


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33. A living microbe with reduced virulence that is used for vaccination is considered:

**a** dormant
**b** virulent
**c** attenuated
**d** denatured

*B. Immunoglobulins*

34. Measurement of serum levels of which of the following immunoglobulins could serve as a screening test for multiple allergies?

**a** IgA
**b** IgE
**c** IgG
**d** IgM

35. IgM antibodies are frequently hemolytic because of:

**a** their dimeric structure
**b** the molecule's 5 antigen-binding sites
**c** their sedimentation coefficient of 7-15 S
**d** their efficient ability to fix complement

36. To which of the following classes do the antibodies that cause hemolytic disease of the newborn belong?

**a** IgA
**b** IgE
**c** IgG
**d** IgD

37. The first isotype of immunoglobulin made by the fetus that may be elevated in cases of in-utero infection is:

**a** IgA
**b** IgG
**c** IgM
**d** IgD

38. The immunoglobulin classes most commonly found on the surface of circulating B lymphocytes in the peripheral blood of normal persons are:

**a** IgM, IgA
**b** IgM, IgG
**c** IgM, IgD
**d** IgM, IgE

39. Antibody class/isotype and antibody subclass are determined by major physiochemical differences and antigenic variation found primarily in the:

**a** constant region of heavy chain
**b** constant region of light chain
**c** variable regions of heavy and light chains
**d** constant regions of heavy and light chains

40. The ratio of kappa to lambda light chain-producing cells in normal individuals is:

**a** 1:1
**b** 2:1
**c** 3:1
**d** 4:1

41. Which of the following immunoglobulin isotypes is associated with a secretory component (transport piece)?

**a** IgA
**b** IgD
**c** IgE
**d** IgG

42. The immunoglobulin class typically found to be present in saliva, tears and other secretions is:

**a** IgG
**b** IgA
**c** IgM
**d** IgD

43. Treatment of IgG with papain results in how many fragments from each immunoglobulin molecule?

**a** 2
**b** 3
**c** 4
**d** 5

44. The immunoglobulin isotype associated with immediate hypersensitivity or atopic reactions is:

**a** IgA
**b** IgM
**c** IgD
**d** IgE

45. Which of the following immunoglobulins is the most efficient at agglutination?

**a** IgG
**b** IgA
**c** IgM
**d** IgE

46. Antibodies to which of the following immunoglobulins is known to have produced anaphylactic reactions following blood transfusion?

**a** IgA
**b** IgD
**c** IgE
**d** IgG

47. The key structural difference that distinguishes immunoglobulin subclasses is the:

**a** number of domains
**b** stereometry of the hypervariable region
**c** the sequence of the constant regions
**d** covalent linkage of the light chains

48. Immunoglobulin idiotypic diversity is best explained by the theory of:

**a** somatic mutation
**b** germ line recombination
**c** antigen induction
**d** clonal selection


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49. Which class of immunoglobulin is thought to function as an antigenic receptor site on the surface of immature B lymphocytes?

a IgD
b IgE
c IgA
d IgG

50. The IgM molecule is a:

a dimer
b trimer
c tetramer
d pentamer

51. Which of the following immunoglobulins is present in the highest concentration in normal human serum?

a IgM
b IgG
c IgA
d IgE

52. Which of the following statements about immunoglobulins is true?

a immunoglobulins are produced by T lymphocytes
b IgA class is determined by the gamma heavy chain
c IgA class exists as serum and secretory molecules
d there are only 2 subclasses of IgG

53. Membrane-bound immunoglobulin molecules:

a have an additional amino-terminal sequence of about 40 residues
b are not anchored in a transmembrane configuration
c are anchored by a hydrophobic sequence of about 26 residues
d are anchored by a hydrophilic region

54. The area of the immunoglobulin molecule referred to as the hinge region is located between which domains?

a $V_H$ and $V_L$
b $C_{H1}$ and $C_{H2}$
c $C_{H2}$ and $C_{H3}$
d $C_{H3}$ and $V_L$

55. Antibody idiotype is dictated by the:

a constant region of heavy chain
b constant region of light chain
c variable regions of heavy and light chains
d constant regions of heavy and light chains

56. Antibody allotype is determined by the:

a constant region of heavy chain
b constant region of light chain
c variable regions of heavy and light chains
d constant regions of heavy and light chains

57. Which IgG subclass is most efficient at crossing the placenta?

a IgG1
b IgG2
c IgG3
d IgG4

58. The J-chain is associated with which of the following immunoglobulins?

a IgA
b IgG
c IgE
d IgD

59. Bence Jones proteins are:

a immunoglobulin catabolic fragments in the urine
b monoclonal light chains
c whole immunoglobulins in the urine
d Fab fragments of a monoclonal protein

60. Which 2 immunoglobulin classes have a J-chain associated with the structure?

a IgG, IgA
b IgA, IgE
c IgG, IgM
d IgM, IgA

*C. Antigen-Antibody Interactions*

61. Avidity may be defined as the:

a degree of hemolysis
b titer of an antigen
c dilution of an antibody
d strength of a reacting antibody

62. The strength of a visible reaction is known as:

a prozone reaction
b absorption
c avidity
d elution

63. Which of the following describes an antigen-antibody precipitation reaction of non-identity?

a precipitin lines cross, forming double spurs
b precipitin lines fuse, forming a single spur
c no precipitin lines are formed
d precipitin lines fuse, forming a single arc

64. Which test has the greatest sensitivity for antigen detection?

a precipitin
b agglutination
c ELISA
d complement fixation

65. Excess antigen in precipitation gel reactions will:

a have no effect on the precipitate reaction
b not dissolve precipitate after formation
c enhance the precipitate reaction
d dissolve the precipitate after formation

66. Soluble immune complexes are formed under the condition of:

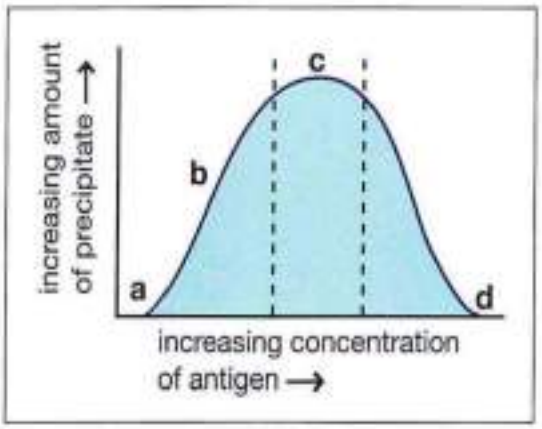
a antigen deficiency
b antigen excess
c antibody excess
d complement

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67. The visible serological reaction between soluble antigen and its specific antibody is:

**a** sensitization
**b** precipitation
**c** agglutination
**d** opsonization

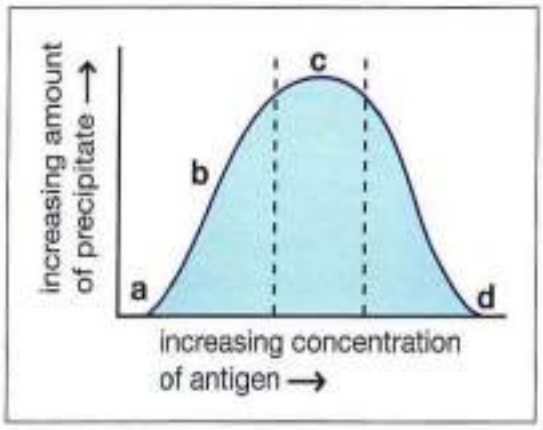
68. The curve is obtained by adding increasing amounts of a soluble antigen to fixed volumes of monospecific antiserum:




The area on the curve for equivalence precipitate is:

**a** a
**b** b
**c** c
**d** d

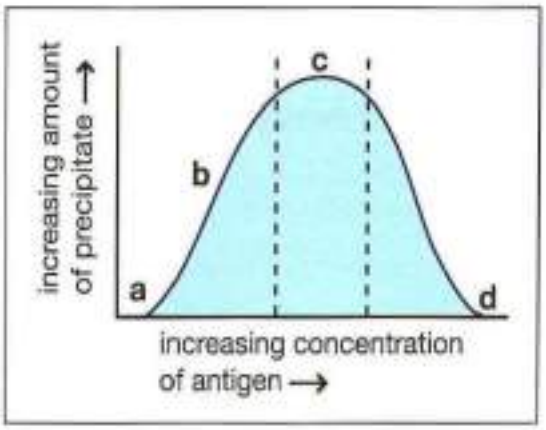
69. The curve is obtained by adding increasing amounts of a soluble antigen to fixed volumes of monospecific antiserum.




The area on the curve where no precipitate formed due to antigen excess is:

**a** a
**b** b
**c** c
**d** d

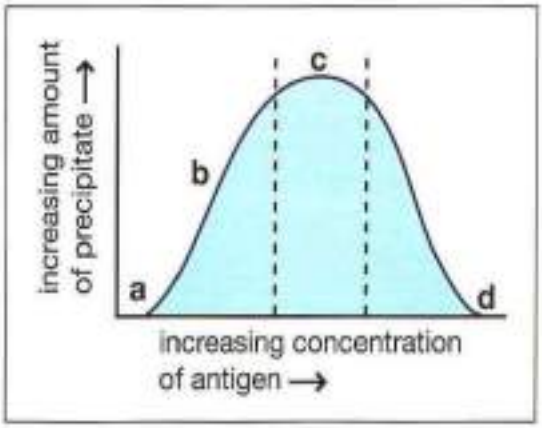
70. The curve is obtained by adding increasing amounts of a soluble antigen to fixed volumes of monospecific antiserum.




The area on the curve for prozone is:

**a** a
**b** b
**c** c
**d** d

71. The curve is obtained by adding increasing amounts of a soluble antigen to fixed volumes of monospecific antiserum.

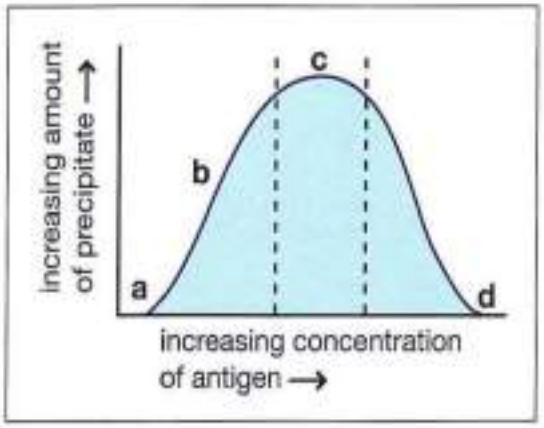



The area on the curve where soluble antigen-antibody complexes have begun to form is:

**a** a
**b** b
**c** c
**d** d

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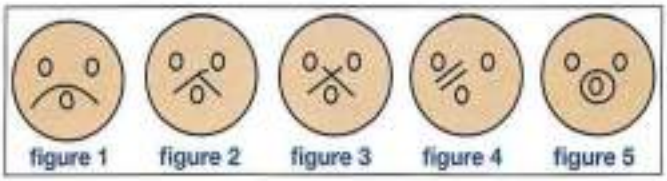
72. The curve is obtained by adding increasing amounts of a soluble antigen to fixed volumes of monospecific antiserum.




The area in which the addition of more antigen would result in the formation of additional precipitate is.

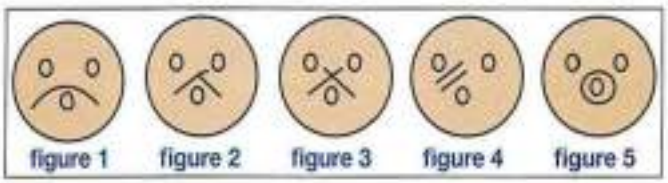
**a** a
**b** b
**c** c
**d** d

73. Which of the figures demonstrates a reaction pattern of identity?



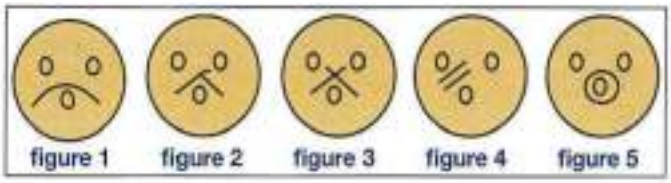

**a** figure 1
**b** figure 2
**c** figure 3
**d** figure 4

74. Which of the figures demonstrates a reaction pattern of partial dentity?



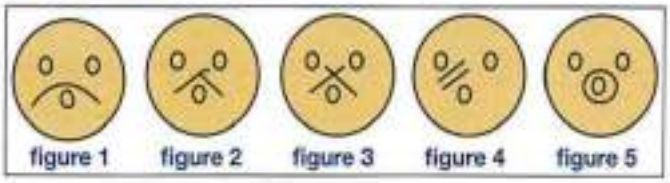

**a** figure 1
**b** figure 2
**c** figure 3
**d** figure 4

75. Which of the figures demonstrates a reaction pattern of nonidentity?



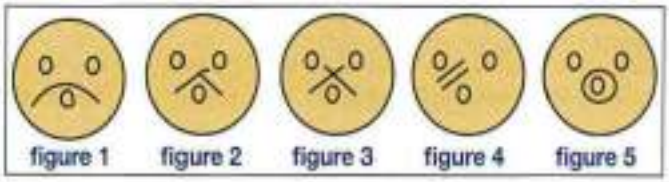

**a** figure 1
**b** figure 2
**c** figure 3
**d** figure 5

76. Which of the figures demonstrates a reaction pattern showing 2 different antigenic molecular species?




**a** figure 1
**b** figure 2
**c** figure 3
**d** figure 4

77. A nonspecific precipitin reaction is demonstrated in which of the figures?




**a** figure 1
**b** figure 3
**c** figure 4
**d** figure 5

78. A series of 8 tubes are set-up with 0.79 mL of diluent in each. A serial dilution is performed by adding 10 µL of serum to the first tube and then transferring 10 µL through each remaining tube. What is the serum dilution of tube 7?

**a** $1{:}2.431 \times 10^{11}$
**b** $1{:}2.621 \times 10^{11}$
**c** $1{:}1.920 \times 10^{13}$
**d** $1{:}2.097 \times 10^{13}$

79. MLS ONLY The enzyme control tube in an ASO hemolytic assay exhibits no cell lysis. What is the most likely explanation for this?

**a** incorrect pH of buffer
**b** low ionic strength buffer
**c** oxidation of the enzyme
**d** reduction of the enzyme


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80. This pattern of agglutination is observed in an antibody titration.

| tube # | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1+ | 2+ | 4+ | 4+ | 3+ | 3+ | 2+ | 1+ | 1+ | 0 | 0 |

This set of reactions most likely resulted from:

**a** faulty pipetting technique
**b** postzoning
**c** prozoning
**d** the presence of a high-titer, low-avidity antibody

81. Which of the these substances will not stimulate an immune response unless it is bound to a larger molecule?

**a** antigen
**b** adjuvant
**c** hapten
**d** antibody

82. Blood is drawn from a patient for serological tests for a viral disease shortly after the time of onset and again 4 weeks later. The results of the tests are considered diagnostic if the:

**a** first antibody titer is 2x the second
**b** first and second antibody titers are equal
**c** first antibody titer is 4x the second
**d** second antibody titer is at least 4x the first

83. The best method to detect infections due to rubella, Epstein-Barr and human immunodeficiency viruses is:

**a** antigen detection by EIA
**b** cell culture
**c** antigen detection by Western blot
**d** antibody detection by EIA

84. Immunoassays are based on the principle of:

**a** separation of bound and free analyte
**b** antibody recognition of homologous antigen
**c** protein binding to isotopes
**d** production of antibodies against drugs

85. What kind of antigen-antibody reaction would be expected if soluble antigen is added to homologous antibody?

**a** precipitation
**b** agglutination
**c** complement fixation
**d** hemagglutination

**86.** The best interpretation of the cold agglutinin titer results show in this table is:

| tube # | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| dilution | 1:1 | 1:2 | 1:4 | 1:8 | 1:16 | 1:32 | 1:64 | 1:128 | 1:256 | 1:512 |
| 4°C | + | + | + | + | + | + | + | + | 0 | 0 |
| 37°C | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

**a** positive, 1:128
**b** negative
**c** invalid because 37°C reading is negative
**d** repeat the 4°C readings

**87.** In an indirect ELISA method designed to detect antibody to the rubella virus in patient serum, the conjugate used should be:

**a** anti-human IgG conjugated to an enzyme
**b** anti-rubella antibody conjugated to an enzyme
**c** rubella antigen conjugated to an enzyme
**d** anti-rubella antibody conjugated to a substrate

**88.** In immunofixation electrophoresis:

**a** the antibody reacts with the antigen and then the complex is electrophoresed
**b** the antigen is electrophoresed into an antibody containing gel
**c** the antigen is electrophoresed and then monospecific antisera is reacted with it
**d** the antigen is electrophoresed, transferred to nitrocellulose and then antibody reacts with it and an EIA is performed

**89.** A single test that could reliably detect neonatal infection in the absence of clinical signs is:

**a** serum immunoelectrophoresis
**b** differential leukocyte count
**c** CD4 cell counts
**d** quantitative serum IgM determination

**90.** The serological test that can be modified to selectively detect only specific IgM antibody in untreated serum is:

**a** Ouchterlony
**b** enzyme immunoassay
**c** hemagglutination inhibition
**d** passive hemagglutination

**91.** What is the immunologic method utilized in the flow cytometer?

**a** latex agglutination
**b** enzyme linked immunoassay
**c** immunofluorescence
**d** radioimmunoassay

### *D. Complement*

**92.** The assembly of the complement "membrane attack unit" is initiated with the binding of:

**a** C1
**b** C3
**c** C4
**d** C5


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93. Which of the following is most likely to activate the alternative pathway of complement activation?

a lipopolysaccharides
b glycoproteins
c haptens
d IgG complexed with antigen

94. Which of the following is the larger residual split portion of C3?

a C3a
b C3b
c C4
d Clq

95. Which of the following activities is associated with C3b?

a opsonization
b anaphylaxis
c vasoconstriction
d chemotaxis

96. Which of the following is the "recognition unit" in the classical complement pathway?

a C1q
b C3a
c C4
d C5

97. Which of the following is the "membrane attack complex" of complement activation?

a C1
b C3
c C4, C2, C3
d C5b, C6, C7, C8, C9

98. Which of the following releases histamine and other mediators from basophils?

a C3a
b properdin factor B
c Clq
d C4

99. Which of the following is not a mechanism involved in the complement cascade?

a apoptosis
b opsonization
c inflammation
d cytolysis

100. The activation mechanism of the alternative complement pathway is initiated differently from that of the classical pathway because, in the alternative pathway:

a antigen-antibody complexes containing IgM or IgG are required
b endotoxin alone cannot initiate activation
c C1 component of complement is involved
d antigen-antibody complexes containing IgA or IgE may initiate activation

101. Which of the following is cleaved as a result of activation of the classical complement pathway?

a properdin factor B
b Clq
c C4
d C3b

102. The component associated only with the alternative pathway of complement activation is:

a C4
b Clq
c properdin factor B
d C3a

103. Which of the following complement components is a strong chemotactic factor as well as a strong anaphylatoxin?

a C3a
b C3b
c C5a
d C4a

104. The C3b component of complement:

a is undetectable in pathological sera
b is a component of the C3 cleaving enzyme of the classical pathway
c is cleaved by C3 inactivator into C3c and C3d
d is not part of the alternative pathway

105. Components of the complement system most likely to coat a cell are:

a C1 and C2
b C3 and C4
c C6 and C7
d C8 and C9

106. Potent chemotactic activity is associated with which of the following components of the complement system:

a C1q
b C5a
c C3b
d IgG

## *II. Diseases of the Immune System*

### *A. Autoimmunity*

107. Hereditary angioedema is characterized by:

a decreased activity of C3
b decreased activity of C1 esterase inhibitor
c increased activity of C1 esterase inhibitor
d increased activity of C2

108. Systemic lupus erythematosus (SLE) patients often have which of the following test results?

a high titers of ANA
b decreased serum immunoglobulin levels
c high titers of anti-smooth muscle antibodies
d high titers of anti-mitochondrial antibody

109. Systemic lupus erythematosus patients (SLE) with active disease often have which of the following test results?

a high titers of anti-microsomal antibodies
b high titers of anti-smooth muscle antibodies
c marked decrease in serum $CH_{50}$
d decreased serum immunoglobulin levels

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110. Which of the following is decreased in serum during the active stages of systemic lupus erythematosus (SLE)?

a anti-nuclear antibody
b immune complexes
c complement (C3)
d anti-DNA

111. The ability of the immune system to recognize self-antigens versus nonself antigens is an example of:

a specific immunity
b tolerance
c cell-mediated immunity
d humoral immunity

112. Tissue injury in systemic rheumatic disorders such as systemic lupus erythematosus is thought to be caused by:

a cytotoxic T cells
b IgE activity
c deposition of immune complexes
d cytolytic antibodies

113. High titers of anti-thyroid peroxidase antibodies are most often found in:

a Graves disease
b systemic lupus erythematosus
c chronic hepatitis
d thyroid disease

114. Which of the following is an organ-specific autoimmune disease?

a myasthenia gravis
b rheumatoid arthritis
c Addison disease
d progressive systemic sclerosis

115. In chronic active hepatitis, high titers of which of the following antibodies are seen?

a anti-mitochondrial
b anti-smooth muscle
c anti-DNA
d anti-parietal cell

116. In primary biliary cirrhosis, which of the following antibodies is seen in high titers?

a anti-mitochondrial
b anti-smooth muscle
c anti-DNA
d anti-parietal cell

117. Anti-glomerular basement membrane antibody is most often associated with this condition:

a systemic lupus erythematosus
b celiac disease
c chronic active hepatitis
d Goodpasture syndrome

118. In pernicious anemia, which of the following antibodies is characteristically detected?

a anti-mitochondrial
b anti-smooth muscle
c anti-DNA
d anti-parietal cell

119. An example of an organ specific disease with autoimmune antibodies is:

a Wegener granulomatosus
b rheumatoid arthritis
c Hashimoto thyroiditis
d systemic lupus erythematosus

120. Which is a recognized theory of the origin of autoimmunity?

a enhanced regulatory T-cell function
b diminished helper T-cell activity
c production of antibodies that cross-react with tissue components
d deficient B-cell activation

*B. Hypersensitivity*

121. After a penicillin injection, a patient rapidly develops respiratory distress, vomiting and hives. This reaction is primarily mediated by:

a IgG
b IgA
c IgM
d IgE

122. In skin tests, a wheal and flare development is indicative of:

a immediate hypersensitivity
b delayed hypersensitivity
c anergy
d Arthus reaction

123. Which immunologic mechanism is usually involved in bronchial asthma?

a immediate hypersensitivity
b antibody mediated cytotoxicity
c immune complex
d delayed hypersensitivity

124. Antihistamines like Benadryl®:

a depress IgE production
b block antigen binding to surface IgE
c bind histamine
d block $H_1$ histamine receptors

125. Delayed hypersensitivity may be induced by:

a contact sensitivity to inorganic chemicals
b transfusion reaction
c anaphylactic reaction
d bacterial septicemia

126. The most rapid immediate hypersensitivity reaction is associated with:

a transfusion
b anaphylaxis
c contact dermatitis
d serum sickness

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127. A 2-year-old patient is suspected to have a peanut allergy because of breathing difficulties (anaphylaxis), which progressed to wheezing within a few minutes upon consumption of peanut butter. What *in vitro* diagnostic test (eg, solid-phase enzyme immunoassay [EIA] or indirect ELISA) would confirm this type of allergy?

**a** total serum IgG and allergen specific IgG
**b** total serum IgA and allergen specific IgA
**c** total serum IgE and allergen specific IgE
**d** total serum IgM and allergen specific IgM

*C. Immunoproliferative Diseases*

128. A monoclonal spike of IgG, Bence Jones proteinuria, and bone pain are usually associated with:

**a** Burkitt lymphoma
**b** Bruton disease
**c** severe combined immunodeficiency disease
**d** multiple myeloma

129. The hyperviscosity syndrome is most likely to be seen in monoclonal disease of which of the following immunoglobulin classes?

**a** IgA
**b** IgM
**c** IgG
**d** IgD

130. Patients suffering from Waldenström macroglobulinemia demonstrate excessively increased concentrations of which of the following?

**a** IgG
**b** IgA
**c** IgM
**d** IgD

131. Cells from a patient with hairy cell leukemia have immunologic features of:

**a** mast cells and B lymphocytes
**b** B lymphocytes and T lymphocytes
**c** granulocytes and monocytes
**d** B lymphocytes and monocytes

132. Which T-cell malignancy may retain "helper" activity with regard to immunoglobulin synthesis by B cells?

**a** Hodgkin lymphoma
**b** acute lymphocytic leukemia (ALL)
**c** Sézary syndrome
**d** chronic lymphocytic leukemia (CLL)

133. Which of the following is an important marker for the presence of immature B cells in patients with acute lymphocytic leukemia (ALL)?

**a** terminal deoxynucleotidyl transferase (TdT)
**b** adenosine deaminase
**c** G6PD
**d** purine nucleoside phosphorylase

*D. Immunodeficiency*

134. Which of the following are true statements about selective IgA deficiency?

a associated with a decreased incidence of allergic manifestations
b high concentration of secretory component in the saliva
c associated with an increased incidence of autoimmune diseases
d found in approximately 1 out of every 50 persons

135. Which of the following is the most common humoral immune deficiency disease?

a Bruton agammaglobulinemia
b IgG deficiency
c selective IgA deficiency
d Wiskott-Aldrich syndrome

136. Which of the following is a true statement about Bruton agammaglobulinemia?

a it is found only in females
b there are normal numbers of circulating B cells
c there are decreased to absent concentrations of immunoglobulins
d the disease presents with pyogenic infections 1 week after birth

137. Immunodeficiency with thrombocytopenia and eczema is often referred to as:

a DiGeorge syndrome
b Bruton agammaglobulinemia
c ataxia telangiectasia
d Wiskott-Aldrich syndrome

138. Which of the following has been associated with patients who have homozygous C3 deficiency?

a undetectable hemolytic complement activity in the serum
b systemic lupus erythematosus
c no detectable disease
d a lifelong history of life-threatening infections

139. Hereditary deficiency of early complement components (C1, C4 and C2) is associated with:

a pneumococcal septicemia
b small bowel obstruction
c lupus erythematosus like syndrome
d gonococcemia

140. Hereditary deficiency of late complement components (C5, C6, C7 or C8) can be associated with which of the following conditions?

a pneumococcal septicemia
b small bowel obstruction
c systemic lupus erythematosus
d a systemic *Neisseria* infection if exposed

141. Combined immunodeficiency disease with loss of muscle coordination is referred to as:

a DiGeorge syndrome
b Bruton agammaglobulinemia
c ataxia telangiectasia
d Wiskott-Aldrich syndrome

142. A patient with a B-cell deficiency will most likely exhibit:

a decreased phagocytosis
b increased bacterial infections
c decreased complement levels
d increased complement levels


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143. A marked decrease in the CD4 lymphocytes and decrease in the CD4/CD8 ratio:

**a** is diagnostic for bacterial septicemia
**b** may be seen in most hereditary immunodeficiency disorders
**c** is associated with a viral induced immunodeficiency
**d** is only seen in patients with advanced disseminated cancer

## *III. Transplantation*

### *A. Graft-versus-host Disease*

144. A patient underwent renal transplant, receiving a kidney from an unrelated donor. This type of transplant is termed:

**a** allograft
**b** syngraft
**c** autograft
**d** xenograft

145. Which of the following is a hyperacute reaction that is not part of chronic cell-mediated transplant rejection?

**a** narrowing and occlusion of graft blood vessels
**b** reaction of T and B cells to graft antigen
**c** antibodies to MHC antigens on white cells
**d** arteriosclerosis of the graft arterial wall

146. Graft-versus-host disease is:

**a** initiated by the recipient
**b** a minor concern in bone marrow transplant
**c** asymptomatic in most cases
**d** initiated by the donor

### *B. HLA Typing*

147. Incompatibility by which of the following procedures is an absolute contraindication to allotransplantation?

**a** MLC (mixed lymphocyte culture)
**b** HLA typing
**c** Rh typing
**d** ABO grouping

148. Bone marrow transplant donors and their recipients should be preferentially matched for which antigen system(s)?

**a** ABO-Rh
**b** HLA
**c** CD4/CD8
**d** $P1^{A1}$

149. A 28-year-old man is seen by a physician because of several months of intermittent low back pain. The patient's symptoms are suggestive of ankylosing spondylitis. Which of the following laboratory studies would support this diagnosis?

**a** a decreased synovial fluid $CH_{50}$ level
**b** low serum $CH_{50}$ level
**c** positive HLA-B27 antigen test
**d** rheumatoid factor in the synovial fluid

150. HLA typing of a family yields the results in this table:

| source | locus A | locus B |
|---|---|---|
| father | (8,12) | (17,22) |
| mother | (7,12) | (13,27) |

On the basis of these genotypes, predict the possibility of ankylosing spondylitis in this percentage of their children.

**a** 25% of their children
**b** 50% of their children
**c** 75% of their children
**d** 100% of their children

151. HLA-B8 antigen has been associated with an increased incidence of which of the following pairs of diseases?

**a** ankylosing spondylitis and myasthenia gravis
**b** celiac disease and ankylosing spondylitis
**c** myasthenia gravis and celiac disease
**d** Reiter disease and multiple sclerosis

### *C. Tumor Immunology*

152. Alpha-fetoprotein, an oncofetal antigen, is least likely to be found in:

**a** pregnancy
**b** hepatocellular carcinoma
**c** cirrhosis
**d** breast carcinoma

153. Tumor-associated transplantation antigens (TAAs) are not found in:

**a** fetal cells
**b** tumor cells
**c** bacterial cells
**d** viral cells

154. Immunoediting describes the theory that our immune system prevents many cancers. Which of the following is not one of its 3 phases (called the 3 'e's)?

**a** efficiency
**b** equilibrium
**c** escape
**d** elimination

## *IV. Infectious Disease Serology*

### *A. Clinical Significance & Epidemiology of Viral Pathogens*

155. The presence of HBsAg, anti-HBc and often HBeAg is characteristic of:

**a** early acute phase HBV hepatitis
**b** early convalescent phase HBV hepatitis
**c** recovery phase of acute HBV hepatitis
**d** past HBV infection

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156. From the test results in the table, it can be concluded that patient #3 has:

| patient | HBsAg | anti-HBc IgM | anti-HAV IgM |
|---|---|---|---|
| 1 | – | – | + |
| 2 | + | + | – |
| 3 | – | + | – |

**a** recent acute hepatitis A
**b** acute hepatitis B
**c** acute hepatitis C (non-A/non-B hepatitis)
**d** chronic hepatitis B

157. The disappearance of HBsAg and HBeAg, the persistence of anti-HBc, the appearance of anti-HBs, and often of anti-HBe indicate:

**a** early acute HBV hepatitis
**b** early convalescent phase HBV hepatitis
**c** recovery phase of acute HBV hepatitis
**d** carrier state of acute HBV hepatitis

158. An example of a live attenuated vaccine used for human immunization is:

**a** rabies
**b** tetanus
**c** hepatitis B
**d** measles

159. What assay would confirm the immune status to hepatitis B virus?

**a** HBsAg
**b** anti-HBs
**c** IgM anti-HBcAg
**d** hepatitis C Ag

160. The following procedure has been routinely used for detection of hepatitis B surface antigen (HBsAg) because of its high level of sensitivity:

**a** hemagglutination
**b** counterimmunoelectrophoresis
**c** radial immunodiffusion
**d** ELISA

161. Which of the following is the best indicator of an acute infection with the hepatitis A virus?

**a** the presence of IgG antibodies to hepatitis A virus
**b** the presence of IgM antibodies to hepatitis A virus
**c** a sharp decline in the level of IgG antibodies to hepatitis A virus
**d** a rise in both IgM and IgG levels of antibody to hepatitis A virus

162. Which serological marker of HBV (hepatitis B virus) infection indicates recovery and immunity?

**a** viral DNA polymerase
**b** HBe antigen
**c** anti-HBs
**d** HBsAg

**163.** The profile that matches the typical test profile for chronic active hepatitis due to hepatitis B virus is:

| profile | HBsAg | IgM anti-HBc | anti-HBc | anti-HBs |
|---|---|---|---|---|
| a | + | – | + | – |
| b | + | + | – | – |
| c | – | – | + | – |
| d | – | – | – | + |

**a** profile a
**b** profile b
**c** profile c
**d** profile d

**164.** A 26-year-old nurse developed fatigue, a low-grade fever, polyarthritis and urticaria. She had cared for a patient with hepatitis 2 months earlier. Which of the following findings are likely to be observed in this nurse?

**a** a negative hepatitis B surface antigen test
**b** elevated AST and ALT levels
**c** a positive rheumatoid factor
**d** a positive Monospot™ test

**165.** The classic antibody response pattern following infection with hepatitis A is:

**a** increase in IgM antibody→decrease in IgM antibody→increase in IgG antibody
**b** detectable presence of IgG antibody only
**c** detectable presence of IgM antibody only
**d** decrease in IgM antibody→increase in IgG antibody of the IgG3 subtype

**166.** The 20 nm spheres and filamentous structures of HBV are:

**a** infectious
**b** circulating aggregates of HBcAg
**c** circulating aggregates of HBsAg
**d** highly infectious when present in great abundance

**167.** The enzyme-linked immunosorbent assay (ELISA) technique for the detection of HBsAg:

**a** requires radiolabeled Clq
**b** is quantitated by degree of fluorescence
**c** uses anti-HBs linked to horseradish peroxidase
**d** uses beads coated with HBsAg

**168.** The antigen marker most closely associated with transmissibility of HBV infection is:

**a** HBsAg
**b** HBeAg
**c** HBcAg
**d** HBV

**169.** Chronic carriers of HBV:

**a** have chronic symptoms of hepatitis
**b** continue to carry HBV
**c** do not transmit infection
**d** carry HBV but are not infectious


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170. Hepatitis C differs from hepatitis A because it:

**a** has a highly stable incubation period
**b** is associated with a high incidence of icteric hepatitis
**c** is associated with a high incidence of the chronic carrier state
**d** is seldom implicated in cases of posttransfusion hepatitis

171. The initial immune response following fetal infection with rubella is the production of which class(es) of antibodies?

**a** IgG
**b** IgA
**c** IgM
**d** both IgG and IgA

*B. Stages of Infection of Treponema pallidum & Borrelia burgdorferi*

172. MLS ONLY Which laboratory technique is most frequently used to diagnose and follow the course of therapy of a patient with secondary syphilis?

**a** flocculation
**b** precipitation
**c** complement fixation
**d** indirect immunofluorescence

173. Which of the following is true of the first stage of infection with *Borrelia burgdorferi*?

MLS ONLY
**a** a generalized rash develops within 4-6 hours of a bite by a deer tick
**b** the patient may be asymptomatic except for the rash
**c** once developed, the rash persists for 7-10 days
**d** serologic testing is often positive within 1 week after the tick bite

174. MLS ONLY Which of the following is characteristic of the second stage of infection with *Borrelia burgdorferi*?

**a** spread to brain and spinal cord via cerebrospinal fluid
**b** spread to multiple organ systems via the bloodstream
**c** 3-4 week latency after tick bite
**d** involvement of the liver, gallbladder, and pancreas

175. Which of the following is characteristic of the late stage of infection with *Borrelia burgdorferi*?

MLS ONLY

**a** spread to brain and spinal cord via cerebrospinal fluid
**b** development in patients refractory to antibiotic therapy
**c** arthritis, peripheral neuropathy, or encephalomyelitis
**d** resistance to antibiotic therapy

176. MLS ONLY Which of the following is consistent with CDC recommendations for confirmation of infection with *Borrelia burgdorferi*?

**a** perform screening EIA with comercially-prepared antibody coated slides
**b** if asymptomatic but screening positive, perform both IgM and IgG western blot
**c** confirm diagnosis if 1 of 3 critical IgM bands reactive on nitrocellulose strip
**d** confirm diagnosis if half of the 10 critical IgG bands reactive on nitrocellulose strip

177. MLS ONLY Which of the following statements is most accurate regarding polymerase chain reaction (PCR) confirmation of infection with *Borrelia burgdorferi*?

**a** PCR detects specific *Borrelia* antibodies
**b** no cross-reactivity issues exist with PCR because of its specificity
**c** PCR is recommended to routine positive screening tests
**d** PCR can utilize either a fluorescent or enzyme marker

### C. Tuberculosis Infection

**178.** Purified protein derivative is used to assess the presence of infection with *Mycobacterium tuberculosis* in the:
MLS ONLY

**a** Mantoux test
**b** RAST test
**c** serum sickness test
**d** Laurell test

**179.** The Mantoux test for *Mycobacterium tuberculosis* is based on a:
MLS ONLY

**a** Type III hypersensitivity reaction
**b** Type II hypersensitivity reaction
**c** Type I hypersensitivity reaction
**d** Type IV hypersensitivity reaction

**180.** The Mantoux skin test to identify infection with *Mycobacterium tuberculosis* is:
MLS ONLY

**a** a cell-mediated response
**b** an IgG response
**c** an IgE response
**d** an IgM response

**181.** The Quanti-FERON-TB Gold In-Tube (QFT-GIT) test to support diagnosis of latent tuberculosis measures release of interferon gamma (IFNγ) by:
MLS ONLY

**a** T cells
**b** macrophages
**c** neutrophils
**d** B cells

**182.** A cytokine that is classically associated with Th1 cells is:
MLS ONLY

**a** interleukin-4
**b** interferon gamma
**c** interleukin-5
**d** interferon alpha

## V. Serologic Procedures

### A. ANA

**183.** Anti-nuclear antibody tests are performed to help diagnose:

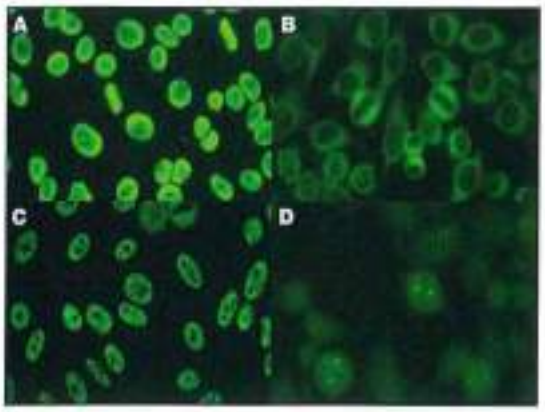
**a** acute leukemia
**b** lupus erythematosus
**c** hemolytic anemia
**d** Crohn disease

**184.** In the anti-double-stranded DNA procedure, the substrate most commonly utilized is:

**a** rat stomach tissue
**b** mouse kidney tissue
**c** *Crithidia luciliae*
**d** *Toxoplasma gondii*


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185. Which of the ANA patterns shown in this image would be associated with high titers of antibodies to the Sm antigen?




a image A
b image B
c image C
d image D

186. Sera to be tested for IFA-ANA 6 days after drawing is best stored at:

a room temperature
b 5°C ± 2°C
c –70°C in a constant temperature freezer
d –20°C in a frost-free self-defrosting freezer

187. Antibodies directed at native DNA are most frequently associated with which pattern of fluorescence in the IFA-ANA test?

a rim
b diffuse
c speckled
d centromere

188. The technologist observes apparent homogenous staining of the nucleus of interphase cells while performing an IFA-ANA, as well as staining of the chromosomes in mitotic cells. This result is:

a indicative of 2 antibodies, which should be separately reported after titration
b expected for anti-DNA antibodies
c inconsistent; the test should be reported with new reagent
d expected for anti-centromere antibodies

189. The result of an anti-nuclear antibody test is a titer of 1:320 with a peripheral pattern. Which set of data best correlate with these results?

a anti-dsDNA titer 1:80, and a high titer of antibodies to Sm
b anti-mitochondrial antibody titer 1:160, and antibodies to RNP
c anti-Scl-70, and antibodies to single-stranded DNA
d high titers of anti-SS-A and anti-SS-B

190. A positive ANA with the pattern of anti-centromere antibodies is most frequently seen in patients with:

a rheumatoid arthritis
b systemic lupus erythematosus
c CREST syndrome
d Sjögren syndrome

191. In the indirect fluorescent anti-nuclear antibody test, a homogenous pattern indicates the presence of antibody to:

a RNP
b Sm
c RNA
d DNA

192. In the indirect fluorescent anti-nuclear antibody test, a speckled pattern may indicate the presence of antibody to:

a histone
b Sm
c RNA
d DNA

193. Anti-RNA antibodies are often present in individuals having an anti-nuclear antibody immunofluorescent pattern that is:

a speckled
b rim
c diffuse
d nucleolar

194. Anti-extractable nuclear antigens are most likely associated with which of the following anti-nuclear antibody immunofluorescent patterns?

a speckled
b rim
c diffuse
d nucleolar

195. In an anti-nuclear antibody indirect immunofluorescence test, a sample of patient serum shows a positive, speckled pattern. Which would be the most appropriate additional test to perform?

a anti-mitochondrial antibody
b immunoglobulin quantitation
c screen for Sm and RNP antibodies
d anti-DNA antibody using *Crithidia luciliae*

196. In the anti-nuclear antibody test, what are the fixed HEp-2 cells?

a unlabeled antigen
b labeled antigen
c labeled antiglobulin
d unlabeled antiglobulin

### *B. Thyroid Antibodies*

197. Which of the following is true of Hashimoto thyroiditis?

a it is more common in male patients
b it is also known as chronic lymphocytic thyroiditis
c anti-thyroid-stimulating hormone receptor (TRAb) is usually present
d the goiter is firm rather than rubbery

198. Which of the following is true of Graves disease?

a it is more common in male patients
b it is also known as chronic lymphocytic thyroiditis
c the presence of anti-thyroid-stimulating hormone receptor (TRAb) is diagnostic
d the goiter is firm rather than rubbery

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199. Which of the following thyroid antibodies causes hyperthyroidism by binding to the TSH receptor?

a anti-thyroglobulin (Tg)
b anti-thyroperoxidase (TPO)
c anti-thyroid-stimulating hormone receptor (TRAb)
d anti-thyroid-stimulating hormone (TSH)

*C. Rheumatoid Factor*

200. Rheumatoid factors are immunoglobulins with specificity for allotypic determinants located on the:

a Fc fragment of IgG
b Fab fragment of IgG
c J chain of IgM
d secretory of component of IgA

201. Rheumatoid factor in a patient's serum may cause a false:

a positive test for the detection of IgM class antibodies
b negative test for the detection of IgM class antibodies
c positive test for the detection of IgG class antibodies
d negative test for the detection of IgG class antibodies

202. Rheumatoid factors are defined as:

a antigens found in the sera of patients with rheumatoid arthritis
b identical to the rheumatoid arthritis precipitin
c autoantibodies with specificity for the Fc portion of the immunoglobulin (IgG) molecule
d capable of forming circulating immune complexes only when IgM-type autoantibody is present

203. False-positive rheumatoid factor in agglutination and nephelometric methods can be due to elevated levels of:

a cryoglobulin
b histidine-rich-glycoprotein
c aspartame
d C1q

204. Rheumatoid factor is most often of which of the following classes:

a IgE
b IgA
c IgM
d IgG

205. Rheumatoid factor consists of antibodies that bind the:

a Fc portion of the IgG molecule
b Fab portion of the IgG molecule
c Fc portion of the IgM molecule
d Fab portion of the IgM molecule

206. Which of the following is not commonly associated with the effects of rheumatoid factor?

a joint inflammation
b immune complex deposition
c capillary endothelial space widening
d complement inhibition

*D. Direct Detection Methods for Pathogens*

**207.** In the direct immunofluorescence assay for *Legionella pneumophila*, patient sample affixed to the slide may be detected with:

**a** primary antigen with fluorescent conjugate
**b** primary antibody with fluorescent conjugate
**c** secondary antigen with fluorescent conjugate
**d** secondary antibody with fluorescent conjugate

**208.** An agglutination test wherein the antigen is a natural component of the infectious entity is a(n):

**a** indirect agglutination procedure
**b** direct agglutination procedure
**c** passive agglutination procedure
**d** reverse passive agglutination procedure

**209.** A suspected anthrax lesion is submitted to the lab for preparation and testing. The test of choice to perform on this sample is:

**a** capture immunoassay
**b** indirect immunoassay
**c** direct immunoassay
**d** sandwich immunoassay

*E. Labeled Immunoassays*

**210.** The specificity of an immunoassay is determined by the:

**a** label used on the antigen
**b** method used to separate the bound from free antigen
**c** antibody used in the assay
**d** concentration of unlabeled antigen

**211.** In the indirect immunofluorescence method of antibody detection in patient serum, the labeled antibody is:

**a** human anti-goat immunoglobulin
**b** rheumatoid factor
**c** goat anti-human immunoglobulin
**d** complement

**212.** A substrate is first exposed to a patient's serum, then after washing, anti-human immunoglobulin labeled with a fluorochrome is added. The procedure described is:

**a** fluorescent quenching
**b** direct fluorescence
**c** indirect fluorescence
**d** fluorescence inhibition

**213.** The primary advantage of labeled immunoassays compared to unlabeled immunoassays is:

**a** rapidity
**b** quality
**c** sensitivity
**d** cost

**214.** Which of the following pairs of considerations do not pertain to ELISA testing?

**a** competitive vs noncompetitive
**b** heterogeneous vs homogeneous
**c** luminescent vs fluorescent
**d** radioactive vs colorimetric


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215. A classic ELISA test is performed on a sample from a 67-year-old male patient who is suspected of having ingested large amounts of drug X. The ELISA procedure is performed as follows:

1. incubation of patient sample in antibody-coated microliter plate wells
2. plate washing
3. addition of peroxidase-linked drug X conjugate
4. incubation followed by plate washing
5. addition of hydrogen peroxide
6. evaluation of color change

Which of the following is true regarding this procedure?

**a** hydrogen peroxide is the substrate
**b** the conjugate contains the antibody
**c** the test is homogeneous
**d** the test is noncompetitive

216. A physician suspects the presence of anti-IFNγ autoantibodies in a patient. A serum sample is sent to your laboratory to measure the presence or absence of anti-IFNγ autoantibodies by an enzyme-immunoassay. Which of the following would be used as the detecting reagent in the kit?

**a** horseradish peroxidase-conjugated recombinant human IFNγ
**b** horseradish peroxidase-conjugated mouse Ab to goat Ab
**c** horseradish peroxidase-conjugated goat Ab to human IFNγ
**d** horseradish peroxidase-conjugated mouse Ab to human Ab

### *F. Nontreponemal Syphilis Testing*

217. Cholesterol is added to the antigen used in flocculation tests for syphilis to:

**a** destroy tissue impurities present in the alcoholic beef heart extract
**b** sensitize the sheep RBCs
**c** decrease specificity of the antigen
**d** increase sensitivity of the antigen

218. Flocculation tests for syphilis detect the presence of:

**a** reagin antibody
**b** antigen
**c** hemolysin
**d** Forssman antigen

219. Flocculation tests for syphilis use antigen composed of:

**a** *Treponema pallidum*
**b** reagin
**c** cardiolipin and lecithin
**d** charcoal

220. A serological test for syphilis that depends upon the detection of cardiolipin-lecithin-cholesterol antigen is:

**a** FTA-ABS
**b** RPR
**c** MHA-TP
**d** TPI

221. The serological test for syphilis recommended for detecting antibody in cerebrospinal fluid is:

**a** non-treponemal antibody
**b** CSF-VDRL
**c** FTA-ABS
**d** MHA-TP

222. Biological false-positive VDRL reactions are frequently encountered in patients with:

**a** lupus erythematosus
**b** acquired immune deficiency syndrome (AIDS)
**c** gonorrhea
**d** tertiary syphilis

### *G. Treponemal Syphilis Testing*

223. In the direct fluorescent antibody test for primary syphilis, spirochetes are detected by addition of labeled antibody to:

**a** *Treponema pallidum*
**b** cardiolipin
**c** human immunoglobulin
**d** nonpathogenic treponemes

224. In the FTA-ABS test, the presence of a beaded pattern of fluorescence along the treponeme indicates:

**a** positive identification of *Treponema pallidum*
**b** presumptive diagnosis of active syphilis
**c** presence of non-treponemal antibody (NTA)
**d** false-positive reaction

225. For diagnosis of late latent or tertiary syphilis, the most appropriate assay is:

**a** RPR
**b** VDRL
**c** FTA-ABS
**d** FTA-ABS IgM

226. The air temperature throughout the serology laboratory is 20°C. How will this affect VDRL and RPR test results?

**a** no effect—the acceptable test range is 20-24°C
**b** weaken reactions so that false-negatives occur
**c** strengthen reactions so that positive titers appear elevated
**d** increase the number of false-positives from spontaneous clumping

### *H. Cytokine Testing*

227. IL-4 can stimulate B cells to produce IgE, a major contributor to:

MLS ONLY

**a** Type I hypersensitivity
**b** Type II hypersensitivity
**c** Type III hypersensitivity
**d** Type IV hypersensitivity

228. Cytokines involved in innate immune regulation include all but:

MLS ONLY

**a** IL-6
**b** IL-3
**c** IL-1
**d** TNF-alpha

229. Multiplex bead assays for the detection of cytokines are efficient quantifiers of numerous cytokines at one time. One complication of such testing is:

MLS ONLY

**a** the cross-reactivity of fluorescent probes
**b** the short half-life of certain cytokines
**c** the lack of test sensitivity
**d** inadequate cytokine-specific antibodies

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## I. Target Amplification

**230.** MLS ONLY The procedure for compatibility testing in organ transplant medicine is very similar for the donor and the recipient. One additional procedure that the recipient must undergo that is not relevant to donor testing is:

- **a** ABO typing
- **b** HLA typing
- **c** CMV testing
- **d** anti-HLA antibody testing

**231.** MLS ONLY PCR is extremely sensitive, but testing may be complicated by:

- **a** an inability to detect nucleic acids from nonviable organisms
- **b** poor quality amplification
- **c** contamination of samples yielding false-positive results
- **d** lack of specificity

**232.** MLS ONLY Methods of target amplification in molecular testing include all but:

- **a** reverse transcriptase-based amplification
- **b** strand displacement amplification
- **c** signal amplification
- **d** primer amplification

## J. Nucleic Acid Sequencing

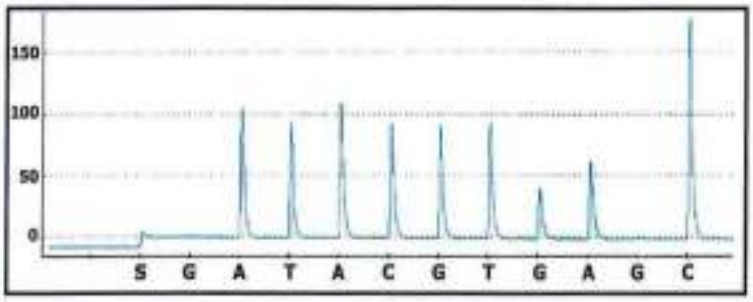
**233.** MLS ONLY What type of short read files with quality scores are used for storing next generation sequencing read data?

- **a** FASTA
- **b** FASTQ
- **c** RRBS
- **d** WGBS

**234.** MLS ONLY In dideoxy chain termination sequencing (Sanger method), what does a heterozygous nucleotide position look like on an electropherogram?

- **a** 1 peak twice the height of those around it
- **b** 2 peaks in the same position, one twice the height of the other
- **c** 2 peaks of equal height at the same position
- **d** 3 peaks of equal height at the same position

**235.** MLS ONLY What is the sequence of the DNA shown on this pyrogram?




- **a** allele 1: ATACGTGCC allele 2: ATACGTGCC
- **b** allele 1: ATACGTGCC allele 2: ATACGTACC
- **c** allele 1: ATACGTGAGC allele 2: ATACGTGAGC
- **d** allele 1: ATACGTGACC allele 2: ATACGTGACC

*K. Hybridization Techniques*

**236.** Which condition has the highest stringency for DNA probe hybridization?

**a** low temperature, low salt concentration
**b** high temperature, low salt concentration
**c** high temperature, high salt concentration
**d** low temperature, high salt concentration

*L. Other*

**237.** Anti-phospholipid antibodies associated with autoimmune disorders tend to have immunoglobulin (IgG) that belongs to which of the following subclasses?

**a** IgG1 and IgG3
**b** IgG2 and IgG4
**c** IgG1 and IgG4
**d** IgG2 and IgG3

**238.** The IF staining pattern on ethanol-fixed leukocytes slides shows a perinuclear or nuclear staining pattern. This pattern is typically due to:

**a** C-ANCA
**b** LKM
**c** P-ANCA
**d** GBM

**239.** The most commonly used serological indicator of recent streptococcal infection is the antibody to:

**a** streptolysin O
**b** hyaluronidase
**c** NADase
**d** DNA

**240.** A heterophile antigen is best described as?

**a** an auto-antigen
**b** existing in an unrelated animal
**c** resulting from an amnestic response
**d** an adjuvant to increase immune response

**241.** In laser flow cytometry, applying a voltage potential to sample droplets as they stream past the light beam and using charged deflector plates results in:

**a** an emission of red fluorescence from cells labeled with fluorescein isothiocyanate
**b** an emission of green fluorescence from cells labeled with rhodamine
**c** a 90° light scatter related to cell size
**d** the separation of cells into subpopulations based on their charge

**242.** What should be adjusted to make sure emission of fluorescence from a single fluorophore does not bleed into multiple channels?

**a** compensation
**b** threshold
**c** voltage
**d** fluid speed

**243.** What does light emitted as FSC measure?

**a** cell granularity/complexity
**b** cell size
**c** cell surface marker fluorescence
**d** nucleic acid marker fluorescence


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244. What type of biological sample is best measured by flow cytometry?

a single cell suspension
b a piece of tissue
c intact BM core biopsy
d plasma

245. The process of centering the sample core within the sheath fluid is known as:

a acoustic focusing
b hydrodynamic focusing
c liquid focusing
d isoelectric focusing

246. To which filters do these definitions correspond: 1) transmit light in the specified wavelength, 2) transmit light equal to or longer than specified wavelength, 3) transmit light equal to or shorter than specified wavelength

a longpass, shortpass, bandpass
b shortpass, bandpass, longpass
c bandpass, longpass, shortpass
d bandpass, shortpass, longpass

247. The electronic signal produced by the detectors is proportional to:

a the amount of light striking them
b the speed of fluid flow
c the presence of clumps in the sample
d the viscosity of the fluid

## *VI. Test Results*

### *A. Interpretation*

248. A 25-year-old woman is seen by a physician because of Raynaud phenomenon, myalgias, arthralgias and difficulty in swallowing. There is no evidence of renal disease. An ANA titer is 1:5120 with a speckled pattern with mitotic cells. Which of the following are also likely to be found in this patient?

a high-level nDNA antibody and a low $CH_{50}$ level
b high-level Sm antibody
c high-titer rheumatoid factor
d high-level ribonucleoprotein (RNP) antibody

249. In assessing the usefulness of a new laboratory test, sensitivity is defined as the percentage of:

a positive specimens correctly identified
b false-positive specimens
c negative specimens correctly identified
d false-negative specimens

250. Which of the following is most useful in establishing a diagnosis in the convalescence phase of a viral infection?

a slide culture
b serological techniques
c shell vial
d culture on McCoy media

251. A 16-year-old boy with infectious mononucleosis has a cold agglutinin titer of 1:2000. An important consideration of this antibody's clinical relevance is the:

**a** thermal range
**b** titer at 4°C
**c** specificity
**d** light chain type

252. The serum hemolytic complement level ($CH_{50}$):

**a** is a measure of total complement activity
**b** provides the same information as a serum factor B level
**c** is detectable when any component of the classical system is congenitally absent
**d** can be calculated from the serum concentrations of the individual components

253. A patient's serum is being analyzed in a sandwich assay. This patient has received mouse monoclonal antibody therapy, and shows a false-positive reaction in the sandwich assay. This is due to:

**a** the mouse antibody in the patient's serum reacting to the antigen
**b** the presence of human anti-mouse antibody activity
**c** antibody to a mouse virus
**d** production of a monoclonal gammopathy of unknown significance after the antibody treatment

254. Calculate the absolute count for B lymphocytes from the flow cytometric data in this table:

| test | result |
|---|---|
| absolute WBC | 8930 |
| total lymphocytes | 30% |
| B lymphocytes | 40% |
| T lymphocytes | 58% |

**a** 1072
**b** 2679
**c** 3572
**d** 6251

255. In flow cytometry, labeled cells:

**a** scatter the light and absorb fluorescence
**b** absorb fluorescence and emit electronic impulses
**c** scatter the light and emit fluorescence
**d** absorb both fluorescence and light


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256. The total number of T cells expected from this data from a peripheral blood sample is:

| test | result |
|---|---|
| WBC | $10.0 \times 10^3/\mu L$ ($10.0 \times 10^9/L$) |
| neutrophils | 68% |
| lymphocytes | 25% (40% T cells) |
| monocytes | 4% |
| eosinophils | 2% |
| basophils | 1% |

**a** 200
**b** 1000
**c** 2000
**d** 2500

257. A peripheral blood total leukocyte count is $10.0 \times 10^3/\mu L$ ($10.0 \times 10^9/L$). The differential reveals 55% neutrophils, 2% eosinophils, 40% lymphocytes and 3% monocytes. Assuming a lymphocyte recovery of 85-95%, what is the expected number of T cells in a normal individual?

**a** 750/µL
**b** 2500/µL
**c** 4000/µL
**d** 8000/µL

258. Given this hematologic data, calculate the absolute CD4:

| test | result |
|---|---|
| WBC | $5.0 \times 10^3/\mu L$ ($5.0 \times 10^9/L$) |
| lymphocytes | 15% |
| CD4 | 8% |

**a** 40
**b** 60
**c** 400
**d** 750

259. Which is the correct interpretation of this hematologic data?

| test | result |
|---|---|
| WBC | $5.0 \times 10^3/\mu L$ ($5.0 \times 10^9/L$) |
| lymphocytes | 15% |
| CD4 | 8% |

**a** CD4% and absolute CD4 count are normal
**b** consistent with an intact immune system
**c** consistent with a viral infection such as HIV
**d** technical error

260. A patient's abnormal lymphocytes are positive for CD2 antigen, lack C3 receptors, and are negative for surface immunoglobulin. This can be classified as a disorder of:

**a** T cells
**b** B cells
**c** monocytes
**d** natural killer cells

*B. Confirmatory Testing*

261. A patient returns from a camping trip, where he vacationed for several days in a tick-endemic environment. One month after his return home, he visited his healthcare provider with complaints of fatigue, numbness and history of rash surrounding an insect bite that occurred immediately on his return home from vacation. Suspecting Lyme disease, his clinician ordered an ELISA to test for Lyme antibody, results of which were equivocal on two different tests. His health care provider then ordered follow-up testing with:

**a** western blot
**b** Southern blot
**c** northern blot
**d** eastern blot

262. On a follow-up visit to her healthcare provider, a young woman is given the results of her FTA-ABS test. She has been symptomatic for a couple of months before this visit, with fever, swollen cervical lymph nodes, fatigue, and rash affecting the palms of her hands and the soles of her feet, but she is now feeling better. Earlier test strategies that one would expect the health care provider to have requested should not include which of the following?

**a** RPR
**b** TPHA
**c** VDRL
**d** RAST

263. Hepatitis B, a disease arising from a DNA virus, is diagnosed in its acute phase via serology for HBsAg, HBcIgM and total alpha-HBc. Once a patient reaches chronic HBV infection, effectiveness of therapy may be evaluated using:

**a** HBV DNA testing with realtime PCR
**b** alpha-HBe testing
**c** alpha-HBs testing
**d** IgM alpha-HBc

264. Mumps is diagnosed by swollen salivary glands, and it spreads from the infected person to others through contact with saliva or respiratory secretions. Enzyme immunoassays (EIA) is used to detect mump-specific antibodies to confirm current infection. Which of the following results confirm mumps infection?

**a** detection of IgM antibodies within 3-4 days of onset of symptoms; detection of IgG within 7-10 days and 4-fold increase in IgG between specimens collected in acute & convalescence phases
**b** detection of IgM antibodies within 5 days and rash; the detection of IgG within 10-14 days apart
**c** detection of IgM antibodies with acute infection; detection of 4-fold increase in IgG between acute & convalescent
**d** detection of IgG at >6 months of age demonstrating post-birth infection

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*C. Disease State Correlation*

**265.** A 37-year-old male presents to his health care provider with recent onset of fatigue, fever, myalgias, nausea, vomiting and diarrhea, complaining that he feels like he is suffering from unremitting flu. Visible on his right forearm is a tattoo that he states he had since age 17. In the last 2 days, he had noted onset of right upper quadrant abdominal pain and dark-colored urine. His diagnosis is most likely:

**a** hepatitis B
**b** hepatitis D
**c** hepatitis C
**d** hepatitis A

**266.** A 17-year-old female presents to her health care provider with symptoms of sore throat, cervical lymphadenopathy, fever, fatigue, and myalgias. Which of the following should be excluded from her differential diagnosis?

**a** group A strep pharyngitis
**b** infectious mononucleosis
**c** human immunodeficiency virus
**d** rotavirus

**267.** Which of the following best correlates with a diagnosis of systemic lupus erythematosus (SLE)?

**a** HEp-2 substrate IFA homogeneous pattern and antibodies to dsDNA (correct answer)
**b** HEp-2 substrate IFA centromere pattern and antibodies to centromere B
**c** HEp-2 substrate IFA speckled pattern and antibodies to Scl-70
**d** HEp-2 substrate IFA speckled pattern and antibodies to SSB

**268.** An HLA allele that is classically associated with ankylosing spondylitis is:

**a** HLA-DQ2
**b** HLA-B27
**c** HLA-DQ8
**d** HLA-DR2

**269.** A 54-year-old female previous smoker presents with a recent history of fever, swollen joints, and morning stiffness. Laboratory results indicate elevated C-reactive protein, positive cyclic citrullinated peptide of 205 units (cut-off: 20 units), speckled ANA pattern (titer 1:320), and negative rheumatoid factor. What is the most likely diagnosis?

**a** reactive arthritis
**b** rheumatoid arthritis
**c** systemic sclerosis
**d** Sjögren syndrome

## I. Principles of Immunology

### A. Immune System Physiology

1. **a** Vaccines stimulate the host to produce antibodies against a specific antigen to prevent disease.
[QCMLS 2021: 8.1.5]

2. **b** Lymphocytes and basophils are not phagocytic.
[QCMLS 2021: 8.1.1.1]

3. **b** This is the definition of acute phase reactants.
[QCMLS 2021: 8.1.1.1] [Stevens 2017, p34-37]

4. **c** CRP response time is typically 4-6 hours with levels a 100× to a 1000× higher and peaking around 48 hours.
[QCMLS 2021: 8.1.1.1] [Stevens 2017, p35-36]

5. **d** Specificity and memory are hallmarks of acquired immunity.
[QCMLS 2021: 8.1.1.2] [Stevens 2017, p4]

6. **a** The list of components in the stem are all naturally occurring and do not require the activation of the immune system to be present or functional.
[QCMLS 2021: 8.1.1.1] [Stevens 2017, p4]

7. **b** This is the definition of a hapten.
[QCMLS 2021: 8.1.3.2] [Stevens 2017, p19-20]

8. **d** In the immune response to a hapten carrier complex, the hapten portion of the molecule binds to B cells, while the carrier portion binds to T cells, and the B cell is stimulated to produce hapten-specific antibody.
[Abbus 2007, p225-226]

9. **c** Biological functions of immunoglobulins; IgG crosses the placenta.
[QCMLS 2021: 8.1.2.4] [Stevens 2017, p65]

10. **b** Biological functions of lymphocytes; B cells make antibody, and become plasma cells.
[QCMLS 2021: 8.1.1.2] [Stevens 2017, p10]

11. **d** The function of B cells is to produce antibodies.
[QCMLS 2021: 8.1.1.2] [Stevens 2017, p7]

12. **d** Autoimmunity, theories of how it develops.
[QCMLS 2021: 8.1.2.5] [Stevens 2017, p234-237]

13. **a** Macrophages have surface receptors for C3b.
[QCMLS 2021: 8.1.1.1] [Stevens 2017, p6]

14. **a** Macrophage phagocytosis is enhanced by opsonins.
[Stevens 2017, p37-39]

15. **c** B cells make humoral antibodies.
[QCMLS 2021: 8.1.1.2] [Stevens 2017, p7]

16. **b** Monoclonal antibodies are produced by cells that are a fused hybrid of a mouse spleen cell with a mouse myeloma cell. The spleen cell confers the antibody specificity, and the tumor cells give it the ability to keep reproducing.
[Stevens 2017, p72-73]

17. **a** Skin tests are used to determine whether the delayed-type hypersensitivity response mediated by T cells is functioning properly. All other tests listed evaluate humoral antibody responses.
[QCMLS 2021: 8.5.1] [Stevens 2017, p227-228]

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18. **b** While T cells help B cells in the process of antibody production, they are not capable of secreting immunoglobulins themselves. Immunoglobulins are produced only by B cells and plasma cells.
[QCMLS 2021: 8.5.1]

19. **c** Phagocytosis is mediated by macrophages and neutrophils, not by lymphocytes.
[QCMLS 2021: 8.5.1]

20. **c** Natural killer cells do not specifically bind to tumor antigens, as do T lymphocytes and antibodies, and can kill tumor cells without having had prior exposure to them.
[QCMLS 2021: 8.1.1.2] [Stevens 2017, p39-41]

21. **c** NK cells distinguish healthy cells from infected or cancerous cells by recognizing the major histocompatibility complex (MHC) Class I, which is expressed on all healthy cells.
[QCMLS 2021: 8.1.1.2] [Stevens 2017, p39-41]

22. **d** MHC Class I codes for HLA-A, B, and C. MHC Class II codes for HLA-DR, DQ, and DP. MHC Class III codes for complement and cytokines.
[QCMLS 2021: 8.1.1.3] [Stevens 2017, p264-265]

23. **b** The tissue damage resulting from type III hypersensitivity is caused by the deposition of immune complexes, which recruit neutrophils to the tissues. The neutrophils release their lysosomal enzymes, resulting in inflammation and damage to the surrounding tissues.
[QCMLS 2021: 8.2.2.4]

24. **d** While B cells are involved in humoral immunity through the production of antibodies, T cells mediate their responses through the release of soluble proteins called cytokines.
[QCMLS 2021: 8.2.2.4]

25. **b** MHC Class I codes for HLA-A, B, C. MHC Class II codes for HLA-DR, DQ, DP. MHC Class III codes for complement and cytokines.
[QCMLS 2021: 8.1.1.3] [Stevens 2017, p264-265]

26. **c** Receptors for the C3b component of complement and for the Fc portion of immunoglobulin are found both on B cells and on monocytes, and are thought to play a role in the clearance of immune complexes.
[QCMLS 2021: 8.1.1.1]

27. **c** Cytotoxic T cells, which are capable of destroying targets such as tumor cells and virus-infected cells, bear the CD8 surface marker, while the other cell types listed are positive for the CD4 surface marker.
[QCMLS 2021: 8.1.1.7] [Stevens 2017, p48-49]

28. **a** Normal peripheral blood should contain approximately 80% T lymphocytes and 20% B lymphocytes.
[QCMLS 2021: 8.1.1.2]

29. **a** CD3 is a marker used to identify T lymphocytes. It is present on the surface of all mature T cells, regardless of T-cell subset.
[QCMLS 2021: 8.1.1.7]

30. **d** IL-13 and IL-5 are examples of T-helper type 2 (Th2) cytokines, which means they are produced by Th2 CD4 T cells. SDF-1 is a chemokine (not a cytokine) produced by stromal cells. IFNγ is a potent effector cytokine that is produced by activated CD8 cytotoxic T cells. Activated T-helper type 1 (Th1) CD4 T cells can also produce IFNγ.
[QCMLS 2021: 8.1.5.3]

31. **c** B cells develop from common lymphoid progenitors in the bone marrow until the mature B-cell stage after which they leave the marrow for secondary lymphoid organs. B cells do not develop in the thymus or other organs, such as liver and kidneys.
[QCMLS 2021: 8.1.1.2]

32. **c** T cells leave the bone marrow early in their development, and then move to the thymus for further development. T cells develop primarily in the thymus. B cells do not develop in the thymus or other organs, such as liver and kidneys.
[QCMLS 2021: 8.1.1.2]

33. **c** Attenuated vaccines consist of live, whole bacterial cells or viral particles that are treated in such a way that they have reduced virulence within the host but retain the ability to induce an immune response. Denatured vaccines are not live, and toxoids are chemically modified toxins from a pathogen. Dormant and virulent microbes are not used for vaccination.
[https://en.wikipedia.org/wiki/Attenuated_vaccine]

*B. Immunoglobulins*

34. **b** Biological functions of immunoglobulins: IgE is the antibody involved in multiple allergies.
[QCMLS 2021: 8.1.2.4] [Stevens 2017, p69]

35. **d** Biological functions of immunoglobulins: IgM binds complement well and is hemolytic.
[QCMLS 2021: 8.1.2.4] [Stevens 2017, p66]

36. **c** Biological functions of immunoglobulins: IgG antibody crosses the placenta and is involved in hemolytic disease of the newborn.
[QCMLS 2021: 8.1.2.4] [Stevens 2017, p222-223]

37. **c** Neonate IgG comes from the mother, but if there is elevated IgM (the first immunoglobulin made by the fetus), an *in utero* or neonatal infection is indicated.
[QCMLS 2021: 8.1.2.4] [Stevens 2017, p67]

38. **c** IgM and IgD are the isotypes of immunoglobulin that are found on most circulating B cells. They are in effect the B-cell receptor.
[QCMLS 2021: 8.1.2.5, 18.5] [Stevens 2017, p62]

39. **a** The constant region of the heavy chain determines the biological function of the immunoglobulin and defines the immunoglobulin into 1 of 5 subclasses.
[QCMLS 2021: 8.1.2.3]

40. **b** The immunoglobulin molecule is made up of 1 or more units (# of units depending on heavy chain type) composed of 2 heavy chains and 2 light chains. The light chains can be either the kappa or lambda type. About 65% of the human immunoglobulin molecules have kappa chains and 35% have lambda chains.
[QCMLS 2021: 8.1.2.2]

41. **a** A secretory IgA molecule is composed of 2 units of 2 heavy chains and 2 light chains. These chains are joined by the J-chain, and are protected from the harsher environment where they are secreted by an additional chain called the transport piece.
[QCMLS 2021: 8.1.2.4] [Stevens 2017, p67-68]

42. **b** Immunoglobulin A (IgA) is the most abundant immunoglobulin in saliva, tears, and other mucosal secretions and plays an important role in mucosal immunity.
[QCMLS 2021: 8.1.2.4] [Stevens 2017, p67-68]

43. **b** Papain cleaves IgG into 2 Fab fragments and 1 Fc fragment.
[QCMLS 2021: 8.1.2.4] [https://vlab.amrita.edu?sub=3&brch=70&sim=1349&cnt=1]

44. **d** Immunoglobulin function: IgE is the immunoglobulin involved with allergy.
[QCMLS 2021: 8.2.2.1] [Stevens 2017, p69]

45. **c** Immunoglobulin function: IgM is best at agglutination.
[QCMLS 2021: 8.1.3.7] [Stevens 2017, p66-67]

46. **a** Immunoglobulin deficiency; patients with IgA deficiency can have an anaphylactic reaction during transfusions.
[QCMLS 2021: 8.2.4.3] [Stevens 2017, p330]



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47. **c** Immunoglobulin subclasses differ from each other in their Fc regions; this is the reason that the different classes have different biological function. The Fc region is the region that is crystallizable after papain cleavage. It varies in sequence in the different classes of immunoglobulin.
[QCMLS 2021: 8.1.2.2] [Stevens 2017, p63]

48. **b** The diversity sequence of the variable region, which is expressed as many idiotypic differences, is what allows so many different antigens to be bound by antibody. This huge diversity in the variable region develops due to VDJ recombinant events for the heavy chain and VJ recombinant events for the light chain.
[QCMLS 2021: 8.1.2.4] [Stevens 2017, p70-71]

49. **a** The main function of IgD is in B-cell development; it is the isotype of surface immunoglobulin on immature B cells.
[QCMLS 2021: 8.1.2.4] [Stevens 2017, p68-69]

50. **d** Immunoglobulin structure: IgM is a pentamer.
[QCMLS 2021: 8.1.2.4] [Stevens 2017, p67]

51. **b** IgG is highest in concentration in normal sera.
[QCMLS 2021: 8.1.2.4] [Stevens 2017, p65-66]

52. **c** Immunoglobulin structure: IgA is present in serum and secretions.
[QCMLS 2021: 8.1.2.4] [Stevens 2017, p67-68]

53. **c** IgD and surface IgM are anchored in the B-cell membrane; in order to be anchored in the membrane, they must contain a hydrophobic region. The hydrophobic region is about 26 residues long.
[QCMLS 2021: 8.1.2.4] [Stevens 2017, p66-69]

54. **b** Immunoglobulin structure: hinge is between $C_{H1}$ and $C_{H2}$.
[QCMLS 2021: 8.1.2.2] [Stevens 2017, p70-71]

55. **c** Immunoglobulin structure: idiotype is in variable regions of heavy and light chains.
[QCMLS 2021: 8.1.2.4] [Stevens 2017, p70-71]

56. **d** Immunoglobulin allotype is in constant regions of heavy and light chains.
[QCMLS 2021: 8.1.2.4] [Stevens 2017, p70-71]

57. **a** Immunoglobulin IgG subclass function; IgG1 goes through placenta best.
[QCMLS 2021: 8.1.2.3] [Stevens 2017, p65-66]

58. **a** Immunoglobulin structure: J-chain associated with IgA.
[QCMLS 2021: 8.1.2.3] [Stevens 2017, p67-68]

59. **b** Definition of Bence Jones protein.
[QCMLS 2021: 8.1.2.2] [Stevens 2017, p312-314]

60. **d** IgM and IgA have a J-chain, which serves as a linkage point for disulfide bonds between 2 adjacent monomers.
[QCMLS 2021: 8.1.2.3] [Stevens 2017, p67-68]

### *C. Antigen-Antibody Interactions*

61. **d** Avidity is the sum of the interactions between all interacting antigen epitopes and Fab sites, and describes the strength of the interaction of an antibody with its antigen.
[QCMLS 2021: 8.1.3.2]

62. **c** Avidity is the sum of the interactions between all interacting antigen epitopes and Fab sites, and describes the strength of the interaction of an antibody with its antigen.
[QCMLS 2021: 8.1.3.2]

63. **a** Non-identity: each antigen forms an independent precipitin line with the corresponding antibody at an equivalence point.
[QCMLS 2021: 8.1.3.4]

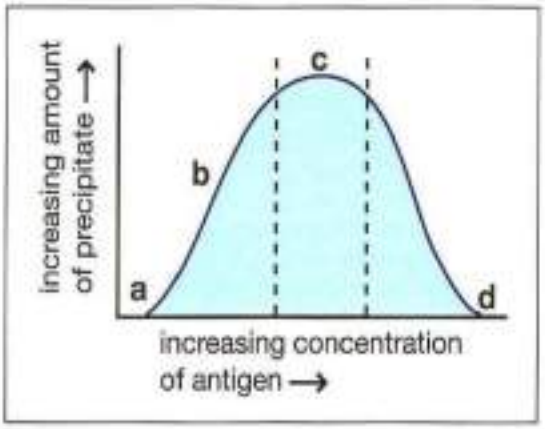
64. **c** ELISA is a solid-phase immunoassay that uses anti-immunoglobulins that are labeled with an enzyme that can be detected by the appearance of color on the addition of a substrate.
[QCMLS 2021: 8.5.13]

65. **d** Precipitation does not occur due to lack of free antibody. With excess amounts of antigen, equilibrium shifts to favor small lattice size, dissolving precipitate.
[QCMLS 2021: 8.1.3.2]

66. **b** Upon the addition of antigen to a fixed quantity of antibody, immune complexes start forming immediately. As more antigen is added, the reaction moves to antigen excess. Precipitation does not occur due to lack of free antibody.
[QCMLS 2021: 8.1.3.2]

67. **b** Precipitation takes place when antibodies and soluble antigens are mixed.
[QCMLS 2021: 8.1.3.2]

***Please consult the figure below for questions 68-72 regarding the precipitin curve.***




68. **c** The proportion of antigen to antibody in the zone of equivalence, shown as **c** in the figure, is optimal, ie, where antibody and antigen multivalent sites are equal for maximal precipitation.
[QCMLS 2021: 8.1.3.2]

69. **d** Excess of antigen results in soluble complexes.
[QCMLS 2021: 8.1.3.2]

70. **a** *Prozone*: suboptimal precipitation occurs in the region of antibody excess. *Prozone effect*: Occasionally, it is observed that when the concentration of antibody is high (ie, lower dilutions), there is no agglutination and then, as the sample is diluted, agglutination occurs. The lack of agglutination at high concentrations of antibodies is called the *prozone effect*. Lack of agglutination in the prozone is due to antibody excess resulting in very small complexes that do not clump to form visible agglutination.
[QCMLS 2021: 8.1.3.2]

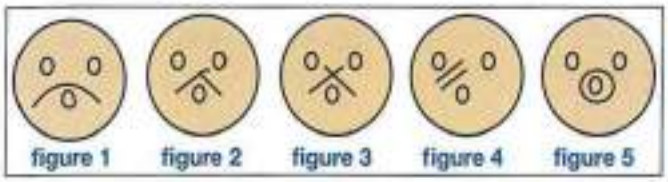
71. **d** As more antigen is added beyond the optimal concentration where antibody and antigen mulivalent sites are equal, formation of precipitate decreases, and this area of the curve is depicted on the figure as **d**. As antigen begins to exceed antibody, equilibrium begins to favor soluble complexes, and precipitation of antigen-antibody complexes is reversed.
[QCMLS 2021: 8.1.3.2]



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72. **b** As more antigen is added, formation of precipitate increases, and this area of the curve is depicted on the figure as **b**. Once optimal concentration of antigen is reached, maximal precipitation occurs in the zone of optimal concentration, or zone of equivalence (**c** in the figure) where antibody and antigen mulivalent sites are equal, and after this point, adding more antigen would not further increase the amount of precipitate.
[QCMLS 2021: 8.1.3.2]

***Please consult the figures in the illustration below for questions 73-77 regarding the precipitin reaction.***




73. **a** A continuous coalescing precipitin line forms when the 2 antigens are identical, the pattern of identity.
[QCMLS 2021: 8.1.3.4, f8.11]

74. **b** Two related antigens form independent precipitin lines that form a single spur, the pattern of partial identity.
[QCMLS 2021: 8.1.3.4, f8.11]

75. **c** Two nonrelated antigens form independent precipitin lines that cross over each other, the pattern of nonidentity.
[QCMLS 2021: 8.1.3.4, f8.11]

76. **d** When 2 parallel precipitin lines exist, it indicates that there are 2 distinct antigenic molecular species present. The identity arc indicates that the 2 antigens are identical in figure 1. Figure 2, partial identity, shares a determinant that is part of antigen 1.
[QCMLS 2021: 8.1.3.4, f8.11]

77. **d** If the antibody well is surrounded by a precipitin circle, there is a nonspecific precipitin reaction to that antibody. The identity arc indicates that the 2 antigens are identical in figure 1. Figure 2, partial identity, shares a determinant that is part of antigen 1.
[QCMLS 2021: 8.1.3.4, f8.11]

78. **d** A dilution involves the solute, the material being diluted and the diluent. 1/dilution = (amount of solute)/(total volume). In the problem, the 8th tube is diluted.
[Stevens 2017, p134-135]

79. **d** ASO is an enzyme inhibition test. In the ASO test, a serum results in inhibition of the reagent's enzymatic ability to lyse human red blood cells.
[Miller 1991, p190-191]

80. **c** *Prozone*: suboptimal precipitation occurs in the region of antibody excess. Occasionally, it is observed that when the concentration of antibody is high (ie, lower dilutions), there is no agglutination and then, as the sample is diluted, agglutination occurs. The lack of agglutination at high concentrations of antibodies is called the *prozone effect*. Lack of agglutination in the prozone is due to antibody excess resulting in very small complexes that do not clump to form visible agglutination.
[QCMLS 2021: 8.1.3.2]

81. **c** A hapten is a small molecule that can act as antigen if it binds to a larger protein molecule. Antigens, viruses and antibodies can stimulate an immune response on their own depending on the context of stimulation. Adjuvants are compounds used to stimulate the immune response to an antigen in a vaccine preparation. They do not directly serve as antigens.
[QCMLS 2021: 8.1.3.2]

82. **d** A 4-fold or greater increase in antibody titer from second of 2 serum specimens taken from a patient during the acute and convalescent phases of an infection is considered to be diagnostic.
[QCMLS 2021: 8.4.6]

83. **d** Serological tests are commonly used to detect antibodies in infections with viruses that are difficult to culture, such as rubella, HIV, and EBV. EIA is a common serological method because it is sensitive, specific, and can be automated.
[QCMLS 2021: 8.4.6] [Stevens 2017, p158-161]

84. **b** "Immunoassay" is a general term for an assay involving binding of an antibody to a specific antigen.
[QCMLS 2021: 8.5.12]

85. **a** By definition, precipitation involves combination of antigen and soluble antibody to form insoluble complexes that fall out of solution.
[QCMLS 2021: 8.1.3] [Stevens 2017, p143]

86. **a** A cold agglutinin titer is read as the last dilution showing agglutination at 4°C.
[QCMLS 2021: 8.1.3.7]

87. **a** In an indirect ELISA, patient antibody to an antigen (eg, rubella antigen) is detected by addition of an enzyme-labeled antibody to human immunoglobulin, which, in turn, binds to the patient's IgG.
[QCMLS 2021: 8.5.12.2] [Stevens 2017, p159-160]

88. **c** In immunoelectrophoresis, first the serum is separated in an agarose gel by electrophoresis; then, in a trough that is cut parallel to the plane of the electrophoresis, antiserum is placed. The antibody diffuses toward the serum proteins, and arcs of antibody-antigen precipitation occur.
[QCMLS 2021: 8.1.3.6] [Stevens 2017, p144-146]

89. **d** An elevated IgM level indicates an *in utero* or neonatal infection.
[QCMLS 2021: 8.1.2.3] [Stevens 2017, p66-67]

90. **b** Indirect labeled assays can be designed to utilize class-specific antihuman immunoglobulin, that is IgM specific.
[QCMLS 2021: 8.5.13] [Stevens 2017, p156-158]

91. **c** Flow cytometry uses fluorescent-labeled monoclonal antibodies to identify cells of interest by binding to specific components within or on the surface of the cells.
[QCMLS 2021: 8.5.15]

### *D. Complement*

92. **d** In the complement cascade, the membrane attack unit begins with C5.
[QCMLS 2021: 8.1.4] [Stevens 2017, p93, 95]

93. **a** Lipopolysaccharides initiate the alternative pathway of complement.
[QCMLS 2021: 8.1.4.2] [Stevens 2017, p19-20, 96-97]

94. **b** C3 breaks down to a small C3a, which floats away and is an anaphylatoxin. The larger C3b, which lands on the target surface and becomes part of C5 convertase, is also a powerful opsonin.
[QCMLS 2021: 8.1.4] [Stevens 2017, p94]



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95. **a** C3 breaks down to a small C3a, which floats away and is an anaphylatoxin. The larger C3b, which lands on the target surface and becomes part of C5 convertase, is also a powerful opsonin.
[QCMLS 2021: 8.1.4] [Stevens 2017, p94]

96. **a** Complement C1q is the recognition unit.
[QCMLS 2021: 8.1.4] [Stevens 2017, p94]

97. **d** Complement, C5-C9, is the membrane attack complex.
[QCMLS 2021: 8.1.4] [Stevens 2017, p95-96]

98. **a** C3a, C4a and C5a are anaphylatoxins and cause release of histamine from basophils and mast cells.
[QCMLS 2021: 8.1.4] [Stevens 2017, p102]

99. **a** Apoptosis is programmed cell death and is not involved in the complement cascade.
[QCMLS 2021: 8.1.4.2] [Stevens 2017, p279]

100. **d** IgA and IgE do not initiate the classical pathway of complement but can initiate the alternative pathway.
[QCMLS 2021: 8.1.4.2] [Stevens 2017, p92]

101. **c** Classical complement pathway: C1 is activated by binding 2 Fc fragments of immunoglobulin, then C4 is split to C4a and C4b; C4b binds, and next C2 is bound and cleaved; this forms C3 convertase, so C3 binds and is cleaved; C3b stays bound; C3a floats away; C5 next is split and C5b binds and C5a floats away. C6, C7, C8, and C9 bind sequentially and cause a hole in the membrane.
[QCMLS 2021: 8.1.4.1] [Stevens 2017, p94-95]

102. **c** Components unique to the alternative pathway include Factor B, Factor D and properdin. Properdin promotes the association of C3b with Factor B.
[QCMLS 2021: 8.1.4.1] [Stevens 2017, p92-99]

103. **c** C5a is both chemotactic and an anaphylatoxin.
[QCMLS 2021: 8.1.4.1] [Stevens 2017, p102]

104. **c** C3 is a very powerful amplifying step of the classical and alterative pathways and also forms a powerful opsinogen. Thus it must be subject to controls, and one such control is that it can be broken down to C3c and C3d.
[QCMLS 2021: 8.1.4.1] [Stevens 2017, p102-103]

105. **b** Conversion of C4 to C4a + C4b is an amplifying step with 30 molecules of C4 for every molecule of C1; C3 conversion is a large amplifying step with about 200 molecules converted for every C3 convertase. These extra molecules can bind to the cell surface.
[QCMLS 2021: 8.1.4] [Stevens 2017, p95-95]

106. **b** C5a is a chemotaxin for neutrophils, basophils, mast cells and monocytes.
[QCMLS 2021: 8.1.4.1] [Stevens 2017, p102]

107. **b** C1 esterase inhibitor is associated with hereditary angioedema.
[QCMLS 2021: 8.2.4.9] [Stevens 2017, p104]

## *II. Diseases of the Immune System*

### *A. Autoimmunity*

108. **a** Anti-nuclear antibodies are found in the sera of 95% of SLE patients.
[QCMLS 2021: 8.2.1.15]

109. **c** CH50 is a good screening test for complement deficiencies in the classical pathway. In SLE patients, the classical pathway is critical for immune complex clearance.
[QCMLS 2021: 8.2.1.15, 8.5.11.6]

110. **c** C3 becomes depleted due to the autoantibody called C3-nephritic factor (C3NeF).
[QCMLS 2021: 8.2.1.15]

111. **b** Tolerance is essential to prevent autoimmunity, ie, mounting an immune response to self-antigens, whereas all other answer choices are related to specific immune responses. There are active mechanisms to induce tolerance at different stages of development in the immune system, and particularly apply to T and B cells of the adaptive immune response, since they have antigen-specific receptors.
[QCMLS 2021: 8.2.1]

112. **c** Large soluble complexes often accumulate along the basement membrane in the kidney. There is impaired ability to process and clear immune complexes in SLE.
[QCMLS 2021: 8.2.1.15]

113. **d** TPO (thyroid peroxidase) is the microsomal antigen of the thyroid epithelial cell. TPO antibodies are positive for about 90% of patients with chronic thyroiditis.
[QCMLS 2021: 8.5.7]

114. **c** Addison disease has antibodies circulating to adrenal antigens.
[QCMLS 2021: 8.2.1.3]

115. **b** Chronic active hepatitis (CAH) has at least 2 subsets, the classic or type I which is associated with a positive ANA test and positive smooth muscle antibodies. The condition is associated with an attack on the hepatocytes.
[QCMLS 2021: 8.2.1.4]

116. **a** PBC is characterized by the presence of antimitochondrial antibodies.
[QCMLS 2021: 8.2.1.10]

117. **d** Goodpasture syndrome is an autoimmune disease mediated by circulating autoantibodies with specificity to the GBM and the alveolar basement membrane.
[QCMLS 2021: 8.2.1.6]

118. **d** Parietal cell antibodies are found in 90% of the cases with pernicious anemia. The other autoantibodies are not organ specific.
[QCMLS 2021: 8.2.1.10]

119. **c** In Hashimoto disease, the autoantibodies produced are specifically directed against the thyroid gland, whereas in the other diseases, they are not organ-specific.
[QCMLS 2021: 8.2.1.7]

120. **c** The theory of molecular mimicry states that antibodies produced against foreign antigens, such as certain microorganisms, can cross-react with self antigens to produce autoimmunity. All the other answers are incorrect because they would result in a decreased immune response.
[QCMLS 2021: 8.2.1]

*B. Hypersensitivity*

121. **d** Biological functions of immunoglobulins: IgE mediates immediate-type hypersensitivity seen in penicillin allergy.
[QCMLS 2021: 8.2.2.1] [Stevens 2017, p69]

122. **a** Immediate hypersensitivity is tested by skin tests with edema and erythema, which also can be called wheal and flare.
[QCMLS 2021: 8.2.2.1] [Stevens 2017, p219]

123. **a** Immediate hypersensitivity is involved in bronchial asthma.
[QCMLS 2021: 8.2.2.1] [Stevens 2017, p217]



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124. **d** Antihistamines block histamine binding to histamine receptors.
[QCMLS 2021: 8.2.2.1] [Stevens 2017, p215]

125. **a** Contact dermatitis is a delayed-type hypersensitivity reaction due to T-cell responses to environmental chemicals or metals. The other conditions are examples of other types of hypersensitivity.
[QCMLS 2021: 8.2.2.4]

126. **b** Anaphylactic, or type I hypersensitivity, occurs very rapidly, usually within 30 minutes after antigen exposure. While transfusion reactions and serum sickness are also examples of immediate hypersensitivity, they generally do not occur as rapidly. Contact dermatitis is a delayed hypersensitivity reaction, manifesting between 24 and 72 hours after antigen exposure.
[QCMLS 2021: 8.2.2.1]

127. **c** Type I hypersensitivity, known as immediate-type hypersensitivity, is mediated by IgE antibodies. Allergens bind IgE on the surface of mast cells, causing degranulation and release of preformed mediators, including histamine, heparin, proteases, and chemotactic factors. Common local or systemic reactions include fever, allergic asthma, hives, or anaphylaxis. Common allergic triggering substances include peanuts, cow milk, eggs, pollen, and penicillin. Type II hypersensitivity is antibody-mediated cellular cytotoxicity, mediated by IgM & IgG antibodies. Common local or systemic reactions include rheumatic fever, myasthenia gravis, Goodpasture syndrome, Graves disease, and multiple sclerosis. Common triggering factors are anti cell surface antigen reactions in the health conditions including blood transfusion reactions, erythroblastosis fetalis, various autoimmune diseases and early transplant rejection. Type III hypersensitivity is mediated IgM & IgG antibodies that form immune complexes and activate complement. Common local or systemic reactions and health conditions include systemic lupus erythematosus, arthus reaction, and serum sickness. Type IV hypersensitivity, or delayed-type hypersensitivity (DTH), is mediated by a T-cell response to antigen. Antigens are generally either contact or intracellular. Contact antigens includes inorganic chemicals, poison ivy, and nickel. Intracellular antigens can be located on bacteria, fungi, parasites, or viruses. Transplant and tumor cell rejection are also DTH. Symptoms include local swelling and inflammation after initial sensitization phase of 1-2 weeks after first contact with antigen.
[QCMLS 2021: 8.2.2]

### *C. Immunoproliferative Diseases*

128. **d** Multiple myeloma shows a monoclonal IgG spike in serum protein electrophoresis and light chains in the urine known as Bence Jones protein. In addition, the tumor cells can grow in the bone forming round lesions that are very painful.
[QCMLS 2021: 8.2.3.1, 8.1.2.2] [Stevens 2017, p315-314]

129. **b** Waldenström macroglobulinemia is a monoclonal gammopathy in which the tumor cells are making IgM. The uncontrolled secretion of such a high molecule weight compound causes a severe increase in viscosity.
[QCMLS 2021: 4.9.6.2.4, 8.2.3.4] [Stevens 2017, p314-315]

130. **c** Waldenström macroglobulinemia is a monoclonal gammopathy in which the tumor cells are making IgM. The uncontrolled secretion of such a high molecular weight compound causes a severe increase in viscosity.
[QCMLS 2021: 8.2.3.4] [Stevens 2017, p314-315]

131. **d** Hairy cell leukemia cells have surface markers such as CD19 and CD20, which are characteristic of B cells, and other markers, such as CD11c, which are found on monocytes. They also stain positive for tartrate resistant acid phosphatase, which is found in osteoclasts and macrophages.
[QCMLS 2021: 4.9.6.1.3]

132. **c** Sézary cells are cells that have the helper T-cell phenotype (CD3+, CD4+).
[QCMLS 2021: 4.9.6.2.3]

133. **a** TdT is an enzyme that adds nucleotides onto the 3' end of a DNA molecule. It is present in immature T cells and B cells, and is used to differentiate ALL from mature B-cell malignancies.
[QCMLS 2021: 4.10.12]

### *D. Immunodeficiency*

134. **c** Immunoglobulin A deficiency (IgA deficiency) is related to autoimmune diseases.
[QCMLS 2021: 8.2.4.3] [Stevens 2017, p330]

135. **c** Selective IgA deficiency is the most common immunodeficiency.
[QCMLS 2021: 8.2.4.3] [Stevens 2017, p67-68]

136. **c** Bruton agammaglobulinemia is an X-linked disease (therefore, usually found in males) in which there is a decreased immunoglobulin concentration, which becomes apparent at 2-6 months of age, *ie*, after the infant's maternally-transferred immunoglobulin has decreased due to its biological half-life. Problem is in differentiation at the pre-B-cell stage.
[QCMLS 2021: 8.2.4.2] [Stevens 2017, p330]

137. **d** Wiskott-Aldrich syndrome is an X-linked recessive defect that exhibits immunodeficiency, eczema and thrombocytopenia.
[QCMLS 2021: 8.2.4.6] [Stevens 2017, p333]

138. **d** C3 deficiency is the worst deficiency to have; it causes a lifelong history of life-threatening infections.
[QCMLS 2021: 8.2.5.2] [Stevens 2017, p103-104]

139. **c** C1, C4, C2 deficiencies are associated with a lupus erythematosus-like syndrome.
[QCMLS 2021: 8.2.5.1] [Stevens 2017, p103-104]

140. **d** C5, C6, C7, C8 deficiencies are associated with a more severe *Neisseria* infection if the patient is exposed due to less pathogen clearance.
[QCMLS 2021: 8.2.5.3] [Stevens 2017, p103-104]

141. **c** X-linked infantile agammaglobulinemia is Bruton agammaglobulinemia; DiGeorge syndrome is a T-cell deficiency; Swiss-type is a SCID; ataxia has additional muscular coordination problems.
[QCMLS 2021: 8.2.4.7] [Stevens 2017, p333]

142. **b** Antibodies are important in defense against bacterial infections. Patients with B-cell deficiencies are unable to produce adequate amounts of antibodies, and therefore exhibit increased bacterial infections. Also, B cells are not phagocytic and do not produce complement.
[QCMLS 2021: 8.2.4.1] [Stevens 2017, p54-56]

143. **c** A decrease in the number of T-helper (CD4+) cells and a consequent decrease in the ratio of CD4+:CD8+ cells is a characteristic finding of HIV infection and AIDS.
[QCMLS 2021: 8.2.8.1, 6.6.20] [Stevens 2017, p437-438]

## *III. Transplantation*

### *A. Graft-versus-host Disease*

144. **a** Allograft is transplantation between nonidentical individuals of the same species.
[QCMLS 2021: 8.3.1.3] [Mahmoudi 2016, p62]



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145. **c** White cell antigen interacts with pre-existing antibodies to repeated pregnancy or blood transfusions. It is a hyperacute reaction.
[QCMLS 2021: 8.3.5.4] [Mahmoudi 2016, p64]

146. **d** Graft-versus-host disease is, by definition, initiated by the donor, which is particularly troublesome in an immunocompromised host that has undergone bone marrow transplant.
[QCMLS 2021: 8.3.6] [Mahmoudi 2016, p65]

### *B. HLA Typing*

147. **d** ABO incompatibility results in hyperacute rejection. This reaction is mediated by anti-A or anti-B antibodies that naturally occur in individuals who lack the corresponding A or B antigen.
[QCMLS 2021: 8.3.4.1] [Stevens 2017, p266]

148. **b** Mismatches in HLA antigens between a donor and recipient of a bone marrow transplant can lead to graft-versus-host disease, in which T lymphocytes in the bone marrow graft mount an immune response against the foreign histocompatibility antigens of the immunocompromised recipient.
[QCMLS 2021: 8.3.4.2] [Stevens 2017, p266]

149. **c** More than 95% of patients with ankylosing spondylitis are positive for the HLA-B27 antigen; therefore, a positive result for this test would support the diagnosis.
[QCMLS 2021: 8.1.1.4]

150. **b** HLA-B27 is associated with ankylosing spondylitis. There is a 50% chance that the mother will transmit her B27 allele to her children.
[QCMLS 2021: 8.1.1.3]

151. **c** Individuals who are HLA-B8-positive have a 5× greater risk than HLA-B8-negative persons of developing myasthenia gravis, and a 9× greater risk of developing celiac disease over a lifetime.
[Turgeon 2009, p429]

### *C. Tumor Immunology*

152. **d** Breast carcinoma is related to the tumor marker carcinoembryonic antigen (CEA) and not alpha-fetoprotein (AFP).
[QCMLS 2021: 8.3.7.5] [Mahmoudi 2016, p71]

153. **c** Tumor associated antigens (TAA) are found on both tumor and normal human cells, and also on viral cells; but not on bacterial cells.
[QCMLS 2021: 8.3.7.5] [Rittenhouse-Olson 2018, p200-201]

154. **a** The theory of immune editing supersedes that of immune surveillance, and includes elimination (surveillance), equilibrium (tumor resistance to elimination), and escape (eventual path to malignancy).
[QCMLS 2021: 8.3.7.3] [Mahmoudi 2016, p72]

## *IV. Infectious Disease Serology*

### *A. Clinical Significance & Epidemiology of Viral Pathogens*

155. **a** The markers listed appear early during hepatitis B infection; HBsAg and HBeAg disappear prior to convalescence and recovery.
[QCMLS 2021: 8.4.9] [Stevens 2017, p412-413, 415-418]

156. **b** IgM anti-HBc may be the only marker present during the "window period" between disappearance of HBsAg and the appearance of anti-HBs in late acute hepatitis B.
[QCMLS 2021: 8.4.9] [Stevens 2017, p412-413, 415-417]

157. **c** Anti-HBs and anti-HBe are associated with recovery and development of immunity in hepatitis B, while HBsAg and HBeAg are antigens from HBV that are present during the infectious stages of disease.
[QCMLS 2021: 8.4.9] [Stevens 2017, p412-413, 415-417]

158. **d** To prevent measles, a vaccine consisting of live, weakened measles (rubeola) virus is used. The vaccine for rabies consists of killed rabies virus; the vaccine for tetanus is a toxoid; and the vaccine for hepatitis B is made up of a recombinant subunit.
[Goldsby 2003, p482]

159. **b** Antibody to the surface antigen of hepatitis B virus (anti-HBs) is the major protective antibody in hepatitis B and provides evidence of immunity against this infection.
[QCMLS 2021: 8.4.9.2]

160. **d** Of all the methods listed, ELISA is the most sensitive and the only one that is used for detection of HBsAg in the clinical laboratory.
[QCMLS 2021: 8.4.9, 8.5.13] [Stevens 2017, p415-417]

161. **b** IgM is the first antibody to be produced during an immune response, and levels decline within 6-12 months; it is therefore an indicator of a current infection.
[QCMLS 2021: 8.4.9.1] [Stevens 2017, p411-419]

162. **c** Antibody to the surface antigen of hepatitis B virus (anti-HBs) appears after the acute stage of infection during convalescence and is a marker of recovery and immunity, while the other markers listed are components of the virus itself.
[QCMLS 2021: 8.4.9.2]

163. **a** HBsAg is an indicator of active infection, either acute or chronic. While IgM anti-HBc is present in acute infection, it disappears after this stage, while IgG antibody to the core antigen persists for life. Anti-HBs is an indicator of immunity, and this has not been achieved in chronic active hepatitis.
[QCMLS 2021: 8.4.9.2] [Stevens 2017, p411419]

164. **b** The nurse's history and symptoms suggest that she has hepatitis. The liver enzymes, alanine aminotransferase (ALT) and aspartate aminotransferase (AST) are elevated in hepatitis as general indicators of liver inflammation.
[QCMLS 2021: 2.4.5] [Stevens 2017, p412-413]

165. **a** The immune response to HAV follows the classic pattern for an antibody response, with IgM appearing first, followed by a decline in IgM and appearance of IgG.
[QCMLS 2021: 8.4.9.1]

166. **c** These structures, which consist entirely of HBsAg, circulate in the serum but are not infectious since they lack the other viral components.
[Turgeon 2009, p282]

167. **c** The ELISA for HBsAg is a sandwich technique in which HBsAg in patient serum binds to anti-HBs on a solid phase; the HBsAg is then detected by the addition of an anti-HBs labeled with an enzyme.
[QCMLS 2021: 8.4.9.2, 8.5.13] [Stevens 2017, p415-418]

168. **b** HBeAg is present in patient serum during periods of active HBV replication, and is therefore a marker of high infectivity.
[QCMLS 2021: 8.4.9.2] [Stevens 2017, p415-418]

169. **b** Some patients who become infected with HBV do not develop immunity and become long-term carriers of the virus who can transmit the infection to others.
[QCMLS 2021: 8.4.9.2]



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170. **c** About 85% of persons infected with HCV will develop a chronic infection, while hepatitis A does not progress to a chronic state.
[QCMLS 2021: 8.4.9.1, 8.4.9.4] [Stevens 2017, p418-419]

171. **c** IgM is the first immunoglobulin to be produced during an immune response, and is produced by infants with congenital infections. IgG in the blood of a newborn infant is primarily of maternal origin, since it can cross the placenta, while IgA would be acquired through mother's breast milk.
[QCMLS 2021: 8.4.6.7]

### *B. Stages of Infection of Treponema pallidum & Borrelia burgdorferi*

172. **a** Nontreponemal tests are used to screen for syphilis and monitor syphilis patients during therapy. These tests, the VDRL and RPR, are based on the principle of flocculation, created by the clumping of the fine cardiolipin particles used in the tests, after binding to patient's antibody.
[QCMLS 2021: 8.4.2] [Stevens 2017, p374-375]

173. **b** *Borrelia burgdorferi* is the causitive agent of Lyme disease transmitted to humans by a deer tick bite. In the first stage of infection, most cases involve a localized rash, erythema migrans, which appears between 2 days and 2 weeks after the bite. Patients may be otherwise asymptomatic, and the rash fades within 3-4 weeks. Most first-stage patients are negative on serologic testing, because the antibody response can take up to 6 weeks.
[QCMLS 2021: 8.4.3.1]

174. **b** *Borrelia burgdorferi* may spread to multiple organ systems via the bloodstream in the second stage of infection. The first stage of infection lasts 3-6 weeks; without antibiotic treatment, neurologic or cardiac involvement may develop in the second stage.
[QCMLS 2021: 8.4.3.1]

175. **c** The late stage of *Borrelia burgdorferi* infection occurs in a subset of untreated patients months to years after infection. Symptoms are numerous and diverse, and arthritis, peripheral neuropathy or encephalomyelitis may develop. Infection normally responds well to antibiotic treatment even in the late stage.
[QCMLS 2021: 8.4.3.1]

176. **d** The Centers for Disease Control and Prevention (CDC) recommend initial EIA or IFA (using commercially prepared slides coated with antigen) followed by specific western blot testing for confirmation of disease. If screening is positive or equivocal, IgG western blot is indicated, and if signs or symptoms present for 30 days or less, IgM western blot also indicated. In the western blot, nitrocellulose strips are reacted with patient serum to assess antibody to specific *Borrelia* antigens. The CDC defines a positive reaction as at least 2 of 3 critical IgM bands reactive on the strip, and at least 5 of 10 critical IgG bands reactive on the strip.
[QCMLS 2021: 8.4.4]

177. **c** Polymerase chain reaction (PCR) for *Borrelia burgdorferi* detects genetic material from the organism, and is very specific, with no cross-reactivity issues. It is used only when the diagosis is very difficult, and not approved for routine testing.
[QCMLS 2021: 8.4.4.4]

### *C. Tuberculosis Infection*

178. **a** The Mantoux test is the name of the tuberculin skin test for *Mycobacterium tuberculosis*. It requires injection of a purified protein derivative (PPD).
[QCMLS 2021: 8.4.7.4] [Rittenhouse-Olson 2018, p150]

179. **d** People exposed to *Mycobacterium tuberculosis* can develop a T-cell immune response to that organism that can elicit a type IV delayed-type hypersensitivity reaction.
[QCMLS 2021: 8.4.7.4] [Rittenhouse-Olson 2018, p150]

180. **a** The type IV hypersensitivity reaction in the skin test for tuberculosis involves T-cell immune responses, not antibody.
[QCMLS 2021: 8.4.7.4] [Rittenhouse-Olson 2018, p150]

181. **a** The Quanti-FERON-TB Gold In-Tube (QFT-GIT) assay measures IFNγ produced by T cells following stimulation with *M. tuberculosis* antigens. IFNγ-producing T cells are critical mediators of the cellular response to *M. tuberculosis*. Macrophages do not produce IFNγ; however, they are activated by this cytokine. Neutrophils are one of the first cell types recruited to the site of inflammation or infection and are critical for control of bacterial infections. They phagocytose and destroy pathogens, generally bacteria and fungi, but do not secrete IFNγ. B cells produce antibodies and present antigens to T cells, but B cells do not produce IFNγ.
[QCMLS 2021: 8.4.7.5]

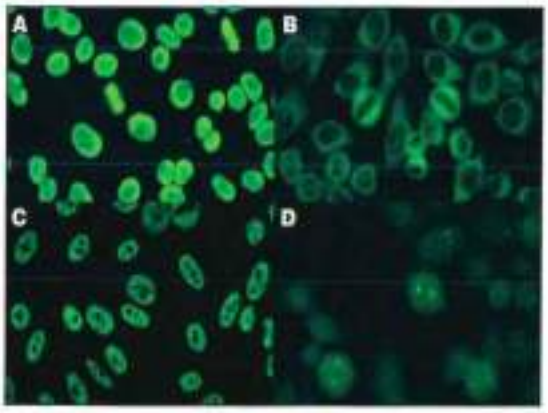
182. **b** Interferon gamma (IFNγ) is classically associated with Th1 cells. The Th1 response, which results in generation of IFNγ-producing T cells. This response is important for cell-mediated immunity, which is typically directed against intracellular pathogens such as mycobacteria. Interleukin-4 is classically associated with a Th2 response that drives antibody production. Interleukin 5 is associated with Th2 responses. Along with interleukin-4 and interleukin-13, interleukin-5 drives eosinophil and IgE production. Interferon alpha is produced by innate immune cells and is important for anti-viral defense.
[QCMLS 2021: 8.4.7.1]

## *V. Serologic Procedures*

### *A. ANA*

183. **b** ANA detects circulating antibodies to nuclear antigens in systemic rheumatic diseases, such as lupus erythematosus (SLE).
[QCMLS 2021: 8.5.6]

184. **c** The *Crithidia* substrate has giant mitochondrion containing native DNA that is free from contaminating histone antigens.
[QCMLS 2021: 8.5.6]




185. **c** Sm is characteristic of a speckled pattern.
[QCMLS 2021: 8.5.6]

186. **b** Storage at 4°C is sufficient for samples analyzed up to a week after collection. For longer periods (months or years) –20°C is preferable.
[Rose 2002, p926]



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187. **a** Antigen target in the diffuse (homogeneous) pattern is DNA.
[QCMLS 2021: 8.5.6]

188. **b** Homogeneous pattern may indicate the presence of anti-DNA antibodies for both single- or double-stranded DNA.
[QCMLS 2021: 8.5.6]

189. **a** Peripheral pattern reacts with the antigenic determinants of double-strandedness of DNA.
[QCMLS 2021: 8.5.6]

190. **c** A centromere is the specialized area of chromosome constriction during metaphase. Autoantibodies to centromere antigens are found in 22% of patients with progressive systemic sclerosis (PSS, or diffuse scleroderma) and in 90% of patients with the subset of scleroderma known as the CREST syndrome (**c**alcinosis, **R**aynaud, **e**sophageal dysfunction, **s**clerodactyly, and **t**elangiectasia).
[QCMLS 2021: 8.5.6]

191. **d** The homogeneous/rim ANA pattern can be caused by: antibodies to double and single-stranded DNA (seen in SLE in high titers and in lower titers in other rheumatic diseases).
[QCMLS 2021: 8.5.6]

192. **b** Sm antigen is a non-histone nuclear protein composed of several polypeptides of differing molecular weights. Sm causes a speckled pattern.
[QCMLS 2021: 8.5.6]

193. **d** Nucleolar pattern is characteristic of staining of the nucleolus seen as 1 or 2 large dots within each nucleus and is produced most frequently in the presence of antibody to nucleolar RNA.
[QCMLS 2021: 8.5.6]

194. **a** Rnp/Sm extractable nuclear antigens show coarse nuclear speckles.
[QCMLS 2021: 8.5.6]

195. **c** Screen for Rnp/Sm extractable nuclear antigens will show coarse nuclear speckles.
[QCMLS 2021: 8.5.6]

196. **a** In the indirect immunofluorescence test for anti-nuclear antibody, the fixed HEp-2 cells is the unlabeled antigen. The anti-globulin is the secondary antibody, and it is fluorescently labeled.
[QCMLS 2021: 8.5.6.1]

*B. Thyroid Antibodies*

197. **b** Hashimoto thyroiditis, also known as chronic lymphocytic thyroiditis, is the most common autoimmune disease, affecting 0.8% of the population. It is characterized by a rubbery goiter, TPO antibodies in 90% and Tg antibodies in 60-80%, but TRAb is not present. Hashimoto thyroiditis is 5-10 times more prevalent in females than males.
[QCMLS 2021: 8.5.7.3]

198. **d** Graves disease is common autoimmune disease characterized by a firm goiter, presence of TRAb antibodies, and female predominance. However, the presence of TRAb is not diagnosic by itself. TRAb antibody-binding assays involve competitive binding between a labeleed TRAb reagend a patient antibody for TSH receptor.
[QCMLS 2021: 8.5.7.4]

199. **c** Autoantibodies develop against proteins involved in thyroid hormone production. TRAb antibodies bind to the TSH receptor, causing hyperthyroidism from uncontrolled receptor stimulation.
[QCMLS 2021: 8.5.7]

### C. Rheumatoid Factor

**200.** **a** Antigen detected is located on the Fc portion of the IgG molecule.
[QCMLS 2021: 8.5.8.1]

**201.** **a** IgM RF is the species most commonly measured in clinical assays.
[QCMLS 2021: 8.5.8.1]

**202.** **c** Rheumatoid factor is an autoantibody to the Fc portion of the immunoglobulin molecule.
[QCMLS 2021: 8.5.8.1]

**203.** **d** C1q like RF will bind and cross link IgG; cryoglobulin: false-negative.
[QCMLS 2021: 8.5.9]

**204.** **c** Rheumatoid factor (RF) is most often of the IgM isotype.
[QCMLS 2021: 8.5.8.1] [Rittenhouse-Olson 2018, p109]

**205.** **a** Rheumatoid factor (RF) affects immunoassays because it binds to the Fc portion of any immunoglobulin, rendering a false positive result.
[QCMLS 2021: 8.5.8.1] [Rittenhouse-Olson 2018, p109]

**206.** **d** Rheumatoid factor (RF) can stimulate complement cascade; it does not inhibit it.
[QCMLS 2021: 8.5.8] [Rittenhouse-Olson 2018, p163-164]

### D. Direct Detection Methods for Pathogens

**207.** **b** In the DFA for *Legionella pneumophila*, sputum or lung washing sample is adsorbed to a plate or well. The fluorescent detection label is attached to a homologous antibody.
[QCMLS 2021: 8.5.12.1] [Stevens 2017, p158-159]

**208.** **b** A direct agglutination procedure employs and entity with a naturally occurring antigen to which a host may build antibody. An example is in ABO blood group testing, wherein a host's red cells have naturally occurring antigens.
[QCMLS 2021: 8.1.3.7] [Stevens 2017, p139]

**209.** **c** In the text for anthrax, a skin lesion suspected of containing *Bacillus anthracis* is incubated with labeled antibody, which makes the test a direct form of immunoassay.
[QCMLS 2021: 8.5.12.1] [Rittenhouse-Olson 2018, p105]

### E. Labeled Immunoassays

**210.** **c** Specificity refers to the ability of an individual antibody combining site to react with only 1 antigenic determinant or the ability of a population of antibody molecules to react with only 1 antigen.
[QCMLS 2021: 8.5.13]

**211.** **c** Fluorochrome is a labeled antihuman immunoglobulin for indirect immunofluorescent assays.
[QCMLS 2021: 8.5.12]

**212.** **c** First, react the target with an unlabeled antibody, then follow with a fluorescent dye.
[QCMLS 2021: 8.5.12]

**213.** **c** Historically, large amounts of immune complexes were required to visualize a reaction with the naked eye, making the process inherently insensitive. Addition of labels to detect smaller concentrations of immune complexes has enhanced sensitivity.
[QCMLS 2021: 8.5.13] [Rittenhouse-Olson 2018, p105]

**214.** **d** Radioactivity plays no part in enzyme immunoassays. Enzyme immunoassay can support colorimetric, chemiluminescent, and fluorescent labels.
[QCMLS 2021: 8.5.13] [Rittenhouse-Olson 2018, p107]

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**215.** **a** Peroxidase is the enzyme that acts on hydrogen peroxide, which is the substrate. The scenario describes a competitive assay because it uses patient antigen that competes with the conjugate-linked reference antigen for binding to a specific antibody, which is on the plate wells, not in the conjugate. Since plate washing occurs twice, the assay is not homogenous, because heterogeneous assays require 1 or more steps of separation, where unbound antibodies and/or unbound analyte must be washed away.
[QCMLS 2021: 8.5.13] [Rittenhouse-Olson 2018, p110]

**216.** **d** Unconjugated recombinant human IFNγ would be used as the capture reagent in the assay. If the serum sample contains anti-IFNγ autoantibodies, they will bind to the recombinant human IFNγ. To detect these bound antibodies (Ab), one needs to use a reagent that will bind to human Ab. Therefore, the correct response is horseradish peroxidase-conjugated mouse Ab to human Ab.
[QCMLS 2021: 8.5.13]

### *F. Nontreponemal Syphilis Testing*

**217.** **d** VDRL antigen contains 0.9% cholesterol.
[QCMLS 2021: 8.4.2.2]

**218.** **a** Flocculation is the aggregation of fine particles to form small clumps. This reaction occurs in nontreponemal tests for syphilis when reagin antibody reacts with the fine cardiolipin antigen particles.
[QCMLS 2021: 8.4.2.2] [Stevens 2017, p374]

**219.** **c** Flocculation tests for syphilis are nontreponemal tests that detect antibody specific for cardiolipin antigen. When this antibody, called reagin, combines with the fine cardiolipin particles, small clumps are formed in a reaction called flocculation.
[QCMLS 2021: 8.4.2.2] [Stevens 2017, p374]

**220.** **b** Nontreponemal tests for syphilis, such as the RPR, detect antibody to cardiolipin antigen complexed with lecithin and cholesterol.
[QCMLS 2021: 8.4.2.2] [Stevens 2017, p374-375]

**221.** **b** The VDRL test is the only serological test recommended for testing of spinal fluid because of the low incidence of false-positive results.
[QCMLS 2021: 8.4.2.3] [Stevens 2017, p374-375]

**222.** **a** Patients with the autoimmune disease, lupus erythematosus, frequently produce antibody against cardiolipin, the same antibody that is used to screen for syphilis by the VDRL test.
[QCMLS 2021: 8.4.2.2] [Stevens 2017, p374-375]

### *G. Treponemal Syphilis Testing*

**223.** **a** Direct detection of *Treponema pallidum* organisms in lesions from patients with primary or secondary syphilis can be performed by dark-field microscopy or fluorescent antibody testing. In direct testing, labeled antibody that is specific for *T. pallidum* binds directly to the spirochetes.
[QCMLS 2021: 8.4.2.1] [Stevens 2017, p373]

**224.** **d** Serum from patients with lupus erythematosus may produce a false-positive result in the FTA-ABS test, which appears as a beaded pattern of fluorescence.
[QCMLS 2021: 8.4.2.5]

**225.** **c** Treponemal tests, such as the FTA-ABS, remain positive throughout the course of syphilis (except for IgM, which is only positive in the early stages), while nontreponemal tests are generally nonreactive in the late stages of disease.
[QCMLS 2021: 8.4.2]

**226.** **b** It is recommended that the VDRL and RPR tests be performed at a temperature of 23-29°C; optimal agglutination does not occur at temperatures below that range, resulting in false-negative tests.
[Turgeon 2009, p219-222]

### H. Cytokine Testing

**227. a** Type I hypersensitivity reactions involve the production of IgE, stimulated by IL-4.
[QCMLS 2021: 8.2.2.1] [Rittenhouse-Olson 2018, p65]

**228. b** IL-3 is not part of innate immune regulation, but rather is a player in hematopoietic regulation. It specifically stimulates myeloid cell production.
[QCMLS 2021: 4.1.1] [Rittenhouse-Olson 2018, p65]

**229. b** Due to the variable half-life of cytokines in multiplex testing, PCR is beginning to supplant traditional testing.
[Stevens 2017, p87]

### I. Target Amplification

**230. d** As HLA, CMV, and ABO typing is required of donors and recipients, anti-HLA antibody testing is required in addition for recipients to eliminate unacceptable donors.
[QCMLS 2021: 8.3.4] [Rittenhouse-Olson 2018, p235]

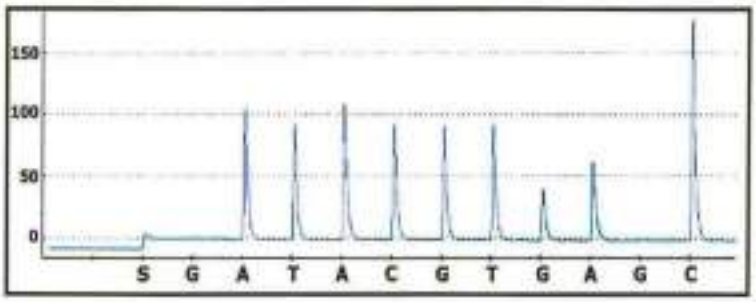
**231. c** Contamination of samples can result in amplification of contaminants during PCR. This continues to be a leading complications with this test method.
[QCMLS 2021: 8.4.8.6, 9.6] [Rittenhouse-Olson 2018, p404]

**232. d** Primer amplification is not intended as a consequence of PCR, but it may occur as a procedural confounder, ie, primer dimer.
[QCMLS 2021: 9.3.3-9.3.10] [Rittenhouse-Olson 2018, p403-404]

### J. Nucleic Acid Sequencing

**233. b** FASTA are text files containing multiple DNA sequences with some text. FASTQ files are like FASTA files, but with added information about the quality of each base in the sequence.
[Zhang 2016].

**234. c** In a clean sequencing reaction, only 1 color peak is present at each nucleotide position for homozygous positions and 2 peaks of different colors for heterozygous positions.
[QCMLS 2021: 9.3.14.1]




**235. b** The nucleotides are dispensed one at a time in the order given at the bottom of the pyrogram. Incorporation of a nucleotide onto the growing strand results in a flash of light. If there is a string of homopolymers, the flash amplitude correlates to the number of residues added.
[QCMLS 2021: 9.3.14.2]

### K. Hybridization Techniques

**236. b** High temperature and low salt concentration favors dissociation of the probe: DNA targets that are not completely base-paired along the length of the probe. This will give the lowest nonspecific (background) hybridization.
[QCMLS 2021: 9.2.2.3]

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237. **b** IgG2 and IgG4 are associated with autoimmune disorders; IgG1 and IgG3 are dominant in infections.
[Rose 2002, p974]

238. **c** C-ANCA shows a granular cytoplasmic staining, and P-ANCA shows a perinuclear or nuclear staining pattern.
[QCMLS 2021: 8.5.5]

239. **a** Patients with recent streptococcal infections produce antibodies to several enzymes produced by streptococcal bacteria. The antibodies detected most commonly in the laboratory are those directed against streptolysin O or DNase B.
[QCMLS 2021: 6.1.6.1.2] [Stevens 2017, p354]

240. **b** Heterophile antigens exist in an unrelated plant or animal but are either identical or closely related in structure so that antibody to one will cross-react with the other.
[Stevens 2017, p21]

241. **d** Laser flow cytometry is the underlying principle of cell sorting into subpopulations.
[Turgeon 2009, p174]

242. **a** The term "compensation," as it applies to flow cytometric analysis, refers to the process of correcting for fluorescence spillover, ie, removing the signal of any given fluorochrome from all detectors except the one devoted to measuring that dye. Because the fluorophores used in flow cytometry emit photons of multiple energies and wavelengths, a mathematical method called compensation was developed to address the measurement of the photons of 1 fluorophore in multiple detectors.
[QCMLS 2021: 8.5.15]

243. **b** In flow cytometry, the light scattered by cells is measured by 2 optical detectors: forward scatter (FSC) that detects scatter along the path of the laser, and side scatter (SSC) which measures scatter at a ninety-degree angle relative to the laser. FSC intensity is proportional to the diameter of the cell, and is primarily due to light diffraction around the cell. FSC signal can be used for the discrimination of cells by size. SSC, on the other hand, is from the light refracted or reflected at the interface between the laser and intracellular structures, such as granules and nucleus. SSC provides information about the internal complexity (ie, granularity) of a cell.
[QCMLS 2021: 8.5.15]

244. **a** Flow cytometry allows users to analyze single cells in a population. Single cells are passed through the path of a laser and interrogated with various visible and fluorescent light sources that allow users to assess a cell's size, granularity, and target protein composition.
[QCMLS 2021: 8.5.15.1]

245. **b** Hydrodynamic focusing is a technique that enables users of flow cytometry cells to gauge the size of particles in a flow channel, whether they are blood cells, viruses or bacteria. A typical flow channel will be something as large as 2-3 mm in diameter down to a miniscule 50 µm. As particles enter the chamber, they will typically pass through a laser beam that causes a temporary disruption in the optics. The size and nature of these disruptions are what cause an instrument to be able to measure the particle's dimensions.
[QCMLS 2021: 8.5.15]

246. **c** Bandpass filters are optical filters that allow transmission of a specific range of wavelengths, or band, while blocking other wavelengths. Many off-the-shelf bandpass filter options are available, but when an application has specific requirements for bandwidth or center wavelength not readily available, longpass and shortpass filters can be stacked creating a customized bandpass filter. Longpass filters are optical filters that reflect short wavelengths while transmitting, or passing, long wavelengths. Conversely, shortpass filters transmit short wavelengths but reflect long ones.
[QCMLS 2021: 8.5.15] [https://en.wikipedia.org/wiki/Band-pass_filter]

247. **a** A photomultiplier tube (PMT) converts a light pulse into an electrical signal of measurable magnitude. Localization of the event in the final image depends on the amount of light sensed by each PMT and thus on the pattern of PMT voltage output.
[QCMLS 2021: 8.5.15]

## *VI. Test Results*

### *A. Interpretation*

248. **d** Coarse nuclear speckles are seen in Raynaud syndrome. Autoantibodies, *eg*, anti-nRNP, are associated with certain symptoms such as Raynaud phenomenon.
[QCMLS 2021: 8.5.6]

249. **a** The ratio of the true-positives to the sum of true-positives plus the false-negatives.
[QCMLS 2021: 7.3.1.3]

250. **b** Cell culture methods are best used during the acute phase of a viral infection, when the viral titer is high. Serological techniques, which detect antibody to the virus, would be better in detecting viral infection during the convalescence phase, when the patient is recovering and the number of viral particles has decreased.
[QCMLS 2021: 8.4.6]

251. **a** It is important to determine the thermal range of reactivity, because cold antibodies are most likely to cause disease if they react with red blood cells at temperatures from 30-32°C.
[QCMLS 2021: 8.1.3.7]

252. **a** Complement, definition of the $CH_{50}$ reaction.
[QCMLS 2021: 8.5.11] [Stevens 2017, p105]

253. **b** Patients that have received mouse monoclonal antibody therapy may have a false-positive when tested with a sandwich assay due to the mouse monoclonal IgM utilized in the assay to test for a hepatitis B surface antigen.
[QCMLS 2021: 8.5.13] [Stevens 2017, p469-470]

254. **a** The absolute number of B lymphocytes would be calculated by multiplying the absolute white blood cell count by the percentage of total lymphocytes to get the total leukocyte count; then multiplying the total leukocyte count by the percentage of B lymphocytes.
$8930 \times 0.30 \times 0.40 = 1072$.
[Turgeon 2009, p65]

255. **c** In flow cytometry, a laser beam hits cells as they pass through the instrument in single file. The amount of light scatter is measured from each cell at 2 different angles, and is used to identify the cell type on the basis of size and granularity. Cells are also identified on their ability to emit fluorescence after they have been incubated with fluorescent-labeled monoclonal antibodies that bind to specific surface markers.
[QCMLS 2021: 8.5.15] [Stevens 2017, p194]



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**256.** **b** The total number of T cells is calculated by multiplying the total WBC ($10 \times 10^3$/µL) by the percent of lymphocytes (25%, or 0.25) and the percent of T cells (40%, or 0.40). The answer for this calculation is $1 \times 10^3$ or 1000.
[Turgeon 2009, p65]

**257.** **b** A normal individual would have approximately 63-84% T cells. With the values given in this example, the absolute T-cell number would range from $2.52 \times 10^3$ (2520; calculated by multiplying the total leukocyte count ($10 \times 10^3$) by the % of lymphocytes (40% or 0.40), and by the lower limit of the % of T cells (63%) to $3.36 \times 10^3$ (3360; calculated by multiplying the total leukocyte count ($10 \times 10^3$) by the % of lymphocytes (40% or 0.40), and by the upper limit of the % of T cells (84%). With a % recovery of 85-95%, values would range between 2142 (2520 × 0.85) and 3192 (3360 × 0.95).
[Turgeon 2009, p61, 65]

**258.** **b** The total number of lymphocytes can be calculated by multiplying the WBC ($5 \times 10^3$, or 5000) by the % of lymphocytes (15% or 0.15) to get 750. This number is then multiplied by the % of CD4+ cells (8% or 0.08) to get an absolute CD4 cell count of 60.
[Stevens 2017, p446-448]

**259.** **c** The patient's percent of CD4+ cells (8%) is well below the normal range of 50-60%, and the patient's absolute number of CD4+ cells (5000 × 0.15 × 0.08 = 60) is also far below the normal range of 500-1300 cells/µL peripheral blood. These findings are consistent with AIDS, which is caused by HIV.
[Stevens 2017, p446-447]

**260.** **a** CD2 and CD3 are T-cell markers, while surface immunoglobulin is a B-cell marker.
[QCMLS 2021: 4.9.6.2] [Stevens 2017, p7]

### *B. Confirmatory Testing*

**261.** **a** If a patient has clinical symptoms of Lyme disease, the first tier test is ELISA. If the results are equivocal, ELISA is repeated. If the results are still equivocal, the next tier of testing is western blot.
[QCMLS 2021: 8.4.4.3] [Rittenhouse-Olson 2018, p339-340]

**262.** **d** RAST is a method for testing for allergic reactions. The other tests are for syphilis antibody.
[QCMLS 2021: 8.4.2, 8.2.2.1] [Rittenhouse-Olson 2018, p335-336]

**263.** **a** In chronic HBV infection, the presence of anti-HBcAG or HBV DNA indicate renewed active viral replication.
[QCMLS 2021: 8.4.9] [Stevens 2017, p417]

**264.** **a** Viral culture is the gold standard for detecting mumps, although molecular testing is becoming the primary method. However, the confirmation of current infection is determined when IgM antibodies can be detected within 3-4 days of onset of symptoms (negative result does not rule out mumps infection). IgG antibodies can be detected within 7-10 days; 4-fold increase in IgG occurs between specimen collected in the acute phase, and a specimen later collected in the convalescence phase. Rubella: Antibodies to rubella are detected on hemagglutination inhibition, latex agglutination, or EIA. IgM or 4-fold rise in IgG titers in samples collected 10-14 days apart indicates acute infection. IgM antibodies appear ~5 days after onset of rash. Rubeola: EIA is used to test for measles antibodies to determine immune status. IgM antibodies present with acute infection; 4-fold increase in IgG between acute & convalescent sera. TORCH panel (**to**xoplasma, **r**ubella, **C**MV, **H**SV): Infants tested at >6 months of age for IgG antibodies demonstrate post-birth infection.
[QCMLS 2021: 8.4.6]

*C. Disease State Correlation*

**265.** **c** Hepatitis C is associated with tattoo and other needle stick history leading to bloodborne disease exposure.
[QCMLS 2021: 8.4.9.4] [Stevens 2017, p417]

**266.** **d** Rotavirus, while detected by antigen testing, is a gastrointestinal disease and is not indicated by the reported symptoms.
[QCMLS 2021: 8.6.17] [Stevens 2017, p425]

**267.** **a** The presence of anti-dsDNA antibodies in association with a homogeneous pattern using HEp-2 substrate IFA is classic for the diagnosis of SLE. HEp-2 substrate centromere (centromere B) and nucleolar (Scl-70) patterns are associated systemic sclerosis, while a speckled pattern with SS-B antibodies is frequently seen in Sjögren syndrome patients.
[QCMLS 2021: 8.2.1.15, 8.5.6]

**268.** **b** Only HLA-B27 has a strong genetic risk for ankylosing spondylitis (AS). HLA-DQ2, HLA-DQ8, and HLA-DR2 are not useful in assessing risk for AS.
[QCMLS 2021: 8.2.1.2]

**269.** **b** Of the autoimmune diseases listed, only rheumatoid arthritis (RA) has the strongest association with smoking and positivity for cyclic citrullinated protein/peptide antibodies. RA can occur with or without the presence of rheumatoid factor, and ANA can sometimes be positive. Reactive arthritis is generally due to an infection.
[QCMLS 2021: 8.2.1.13, 8.5.8]



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